

Packaging (and Wire Bonding)

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Synonyms

Electronic packaging; IC packaging; MEMS packaging

Definition

Packaging provides electrical or mechanical access to microdevices from the outside world. Packaging also provides protection to microdevices from being damaged by the elements, such as dust, moisture and chemicals from the environment.

Wire bonding is a technology that connects microdevices to the outside world electrically using small metal wires.

Overview

Historically, packaging technology was developed by the integrated circuit (IC) industry. When semiconductor microdevices such as transistors are fabricated on a semiconductor wafer, electrical signals can only be accessed through many micrometer-sized metal lines and contacts using specialized equipment such as a probe station, equipped with an optical microscope and micromanipulators with fine tungsten needles. In order to provide low cost, general access to the microdevices, semiconductor wafers are first diced into small chips (or dies) and small metal wires are used to connect the metal contacts to a *carrier case*. The carrier case has millimeter-sized pins that can be accessed using standard circuit boards.

In IC packaging, the carrier case also provides a moisture barrier. Often, failure in a microdevice is the contact between the metal wires and the metal contacts on the substrate due to moisture-induced corrosion. For

microelectromechanical system (MEMS) devices, moisture presents more of a threat than for conventional IC devices [1]. Many micromechanical MEMS components (such as a microcantilever beam) are positioned a few micrometers from the substrate. Moisture condensing on the surface of the substrate will pull down the MEMS devices due to surface tension. The micromechanical component will no longer be functional.

For microfluidic devices, packaging not only provides electrical access to the metal electrodes in the microchannels, but also provides mechanical access. Small tubing is used to connect the fluid ports on the microfluidic device to syringe pumps or other external devices.

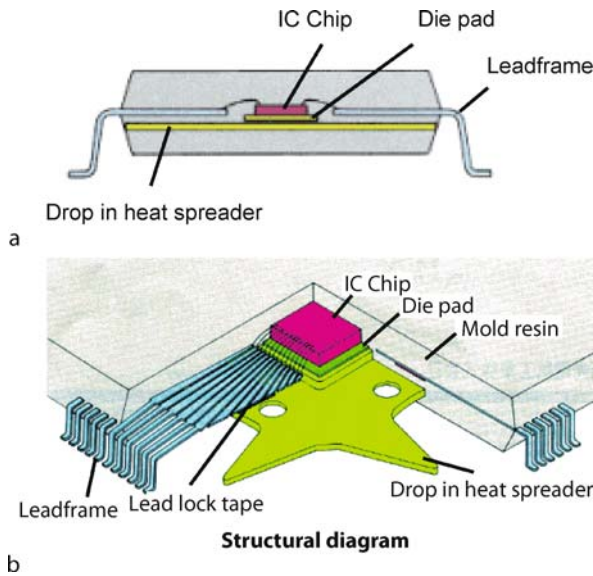
Wire bonding is a technology that uses small gold or aluminum wires with a diameter around 20 to 50 μm to build electrical connections to the MEMS/IC devices. The wires are fed through a small aperture on the tip of an ultrasound actuator (or horn). The tip presses the wire to the metal contact surface on the device substrate. Ultrasound energy is applied to vibrate the tip, making the wire rub against the metal contact surface. The vibration-induced friction can (1) create a clean interface between the wire and the metal surface and (2) provide thermal energy that bonds the wire to the metal surface.

Basic Methodology

DIP (Dual In-Line Package)

DIP technology uses moisture-resistant plastic compounds to encapsulate MEMS/IC devices [2]. As shown in Fig. 1a (Courtesy of Shinko Electric Industries Co. Ltd), the two row of pins commonly seen in IC devices that extended out from the plastic body are used for electronic connections, hence the term *dual in-line*.

After IC devices are manufactured on a substrate wafer, a dicing saw is used to break the wafers into smaller chips. The chips are then bonded to a metal frame (called a lead frame) using epoxy. The lead frame is often made of Cu – Ni – Fe alloys with a thermal expansion coefficient matched with that of the silicon substrate as well as of the plastic compound. Wire bonding is used to connect con-



Packaging (and Wire Bonding), Figure 1 (a) Cross-section of a DIP package; (b) exploded view of a QFP package

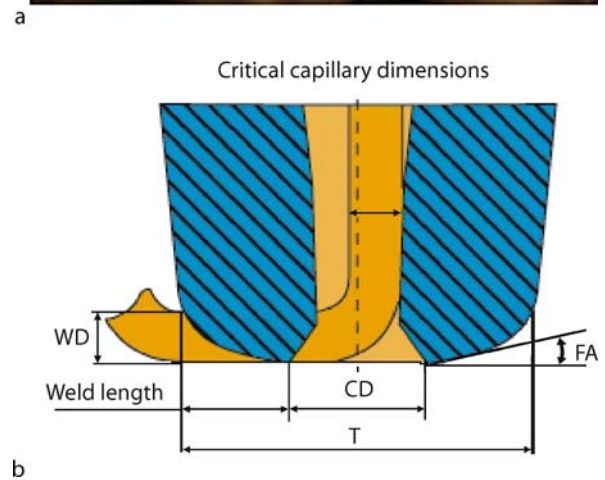
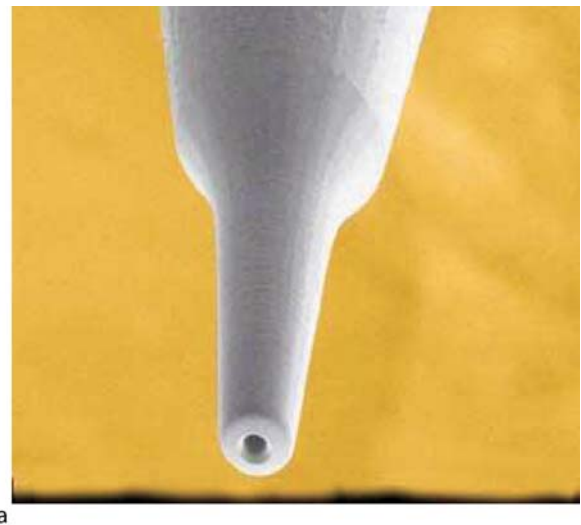
tact pads on the device to the lead frame. The device (and the lead frame) is then put into a mold. The plastic compound, with a thermal expansion coefficient matched with silicon, is molded around the device/lead frame to form a protective case.

Many other variations of DIP technology are readily available, such as SH-DIP (shrink DIP), SIP (single in-line package), SOP (small out-package) and QFP (quad flat pack, as shown in Fig. 1b (Courtesy of Shinko Electric Industries Co. Ltd)). The basic techniques are the same as DIP but the variants provide different form factors and different arrangement of the pins.

Wafer Bonding

MEMS devices need to be separated from the substrate using a wafer dicing machine before they can be packaged. However, unlike IC devices, water used in the dicing operation will destroy micromechanical components. Moreover, DIP technology cannot be used since the molding process will also destroy the micromechanical part of the MEMS devices.

The wafer bonding technique has been used to encapsulate MEMS devices before the dicing process. Wafers (silicon or glass) with recessed microcavities are used to bond to the MEMS device wafer. The microcavity provides a protection cap to the micromechanical component. The hermetic sealing between the bonded wafers will prevent water (during dicing operation) from coming into the cavity. It will also delay moisture diffusion from the envi-

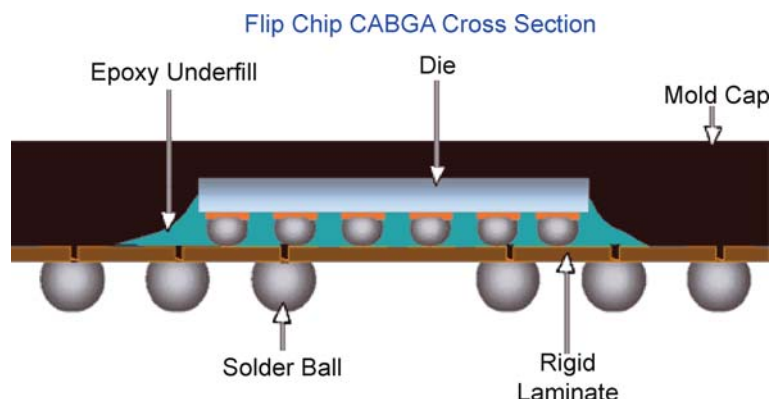


Packaging (and Wire Bonding), Figure 2 Ultrasound horn for wire bonding

ronment into the cavity. Furthermore, the strength of the sealing will need to be high enough to sustain the subsequent molding in a DIP process.

Anodic Bonding

Anodic bonding has been widely used to form a pressure reference chamber in silicon pressure sensors. Borosilicate glass wafers with a thermal expansion coefficient matched with silicon are often used to bond with silicon wafers using a strong electric field and high temperature. The glass wafer is first chemically etched or sand blasted to form small cavities. Each cavity will accommodate one or more MEMS devices. The glass wafer is made contact with a silicon surface on a hot plate at 400 °C. An electric field is applied between the glass and the silicon such that sodium ions drift away from the glass-silicon interface.



Packaging (and Wire Bonding), Figure 3 Flip-chip technology for electric interconnection

Dissolved oxygen in the glass moves towards the glass–silicon interface and forms silicon dioxide that bonds the silicon surface and glass wafer (also silicon dioxide).

Silicon–Silicon Wafer Bonding

A clean and polished silicon wafer can be bonded to another silicon wafer at room temperature [3]. Two bonding mechanisms exist: (1) hydrophilic silicon surfaces will allow hydrogen bonds to form between two wafers with thin silicon dioxide layers and (2) hydrophobic silicon surfaces will allow van der Waals interactions between two clean silicon surfaces. Thermal annealing at high temperature ($> 1000^{\circ}\text{C}$) is then required to permanently bond the silicon wafers.

Wafer Bonding with Intermediate Layers

Wafers can be bonded using intermediate layers such as metals to form hermetic seals. Polymers, in general, do not provide water-tight sealing, and are hence less important in the field of MEMS packaging. Soft metals such as indium can be bonded with silicon with a gold passivation [4]. Aluminum has been used to bond with silicon dioxide [5] and silicon nitride [6] to form hermetic seals.

Fluid Interconnects for Microfluidic Devices

In Lab-on-a-Chip applications, not only electrical, but also mechanical access (fluids) needs to be realized to the outside world. Flexible tubing such as polyimide tubes is often used to connect the small opening in microfluidic chips [7]. The sealing between the tube and the device opening has to be leak-tight. Most methods involve the use of placing a tube into the small fluid access via and sealants, such as polysiloxane [14] or polydimethylsilox-

ane [8], are then poured in to form seals. However, the sealants must not clog the tubing.

Many fluid interconnect techniques have been demonstrated to address the issues with the blocking of the tubing as a result of sealant application. These methods include the use of polymer membranes that prevent the seal from clogging [9] and the use of thermal shrink plastic tubing [10]. The published methods have been demonstrated in research and academic settings; much is still needed to be done for industry-relevant scale [15].

Electrical Interconnection

Wire bonding has been used in IC packaging for many years. Ultrasound energy is used to bond small metal wires on the metal pads on silicon wafers. As shown in Fig. 2a (Courtesy of Dr. I.W. Qin, Advanced Packaging, PennWell, Inc.), an ultrasound horn has a small opening at its tip. A small metal wire is fed through the opening, as shown in Fig. 2b (Courtesy of Dr. I.W. Qin, Advanced Packaging, PennWell, Inc.). The horn then makes contact with the surface and bonds the wire to the surface using ultrasound energy. The pitch size is often determined by the precision of the ultrasound horn placement, the size of the ultrasound horn and the size of the metal wires. As IC and MEMS I/O densities increase, it is becoming more and more challenging for wire bonding technology to meet the need of high-density I/O.

Other electric interconnection techniques such as wafer bumping and *flip-chip* bonding have emerged [11]. As shown in Fig. 3 (Courtesy of Amkor Technology, Inc.), small metal balls (Ag/Sn or Sn/Pb) are directly electroplated on the metal pads of the device chip. The chips are then “flipped over” and bonded to a printed circuit board (PCB) or a substrate with prefabricated metal connections. An epoxy underfill is then used to fill the space

between the chip and the PCB. A plastic compound is then molded over the chip for protection.

Packaging Reliability and Testing

A major characteristic of a MEMS or IC package is reliability. One often asked question is “how long will the device be operational”? From a packaging perspective, the protection and electrical/mechanical interconnection must withstand the elements. For example, for an accelerometer, the moisture from the environment must be stopped or delayed by the packaging seal so that the accelerometer sitting in the package will function properly. Furthermore, the temperature cycling often seen in automotive applications may raise concerns over the electric interconnection. The metal-to-metal contacts may form intermetallic compounds that are thermally mismatched with the surroundings and cracks will form when the package goes through temperature cycling. The metal contact resistance will then increase and eventually form an open circuit.

Many testing methods are available to evaluate the reliability of an IC/MEMS package. For example, if the dominant failure mode of a MEMS device is water condensation in the package, *autoclave* tests are available to test the reliability of the devices [1]. In an autoclave test, pressurized water is used to fill bonding imperfections and small crevasses that may exist in a package to accelerate the device failure.

Key Research Findings

In recent years, low-cost, localized “welding” has demonstrated reliability and possibilities for packaging of MEMS devices [5, 12]. One of the emphases has been on vacuum packaging for MEMS resonators and gyroscopes for high quality factor performance in a vacuum. Vacuum packaging for MEMS resonators using aluminum-to-silicon nitride bonding was shown to maintain the vacuum level in a harsh environment (130°C, 2.7 atm and 100% relative humidity) for a long period of time [6]. A specialized metallization process was also demonstrated to produce a vacuum package [13].

Future Directions for Research

Due to the diversity of MEMS devices, MEMS packaging methodology is device and application dependent. Compromise solutions between cost, reliability and process compatibility need to be considered.

Cross References

- Anodic Bonding
- Micromolding

- Techniques for Transferring Samples to Chips
- Wafer Bonding

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Parallelization

Definition

Simultaneous implementation of several chemical/biological analyses in an on-chip format generally from a single reagent source.

Cross References

- Surface-Directed Capillary Flow Systems

Parallel (Stratified) Flow

Definition

Stratified flow is that multiphase flow where the phases flow along each other in distinct strata or layers.

Cross References

► Pressure-driven Two Phase Flows

Partial Differential Equations (PDEs)

Definition

A partial differential equation (PDE) is a relation involving an unknown function of several independent variables and its partial derivatives with respect to those variables. Partial differential equations are used to formulate and solve problems that involve unknown functions of several variables, such as the propagation of sound or heat, electrostatics, electrodynamics, fluid flow, elasticity, or more generally any process that is distributed in space, or distributed in space and time. (Definition taken from wikipedia: http://en.wikipedia.org/wiki/Partial_differential_equation)

Particle Dynamics Simulation

► Molecular Dynamics Simulation Method

Particle Image Thermometry

Definition

Particle Image Thermometry (PIT) is a method to determine two dimensional temperature fields of fluids. The particles carry the temperature information and are seeded into the fluid. Thermochromic Liquid Crystals (TLCs) are often used for that.

Particle Image Velocimetry (PIV)

Definition

PIV is used to estimate two-dimensional velocity fields of fluid flows. The fluid is seeded with ► [tracer](#) particles (able to follow the flow) whose motion gives information about the motion of the fluid. Therefore two images taken at two

times are divided into ► [interrogation areas](#). The interrogation areas belonging together are correlated (image processing algorithm) which leads to a pixel displacement. With the time interval between the capture of the two pictures, the pixel displacement, and the pixel length the velocity of the considered interrogation area can be calculated.

Particle Manipulation

► Bead Based Microfluidic Platforms

Particle Manipulation Using Ultrasonic Fields

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Synonyms

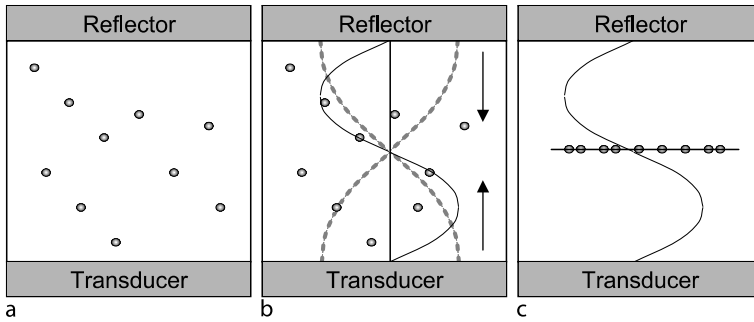
Ultrasonic separation; Ultrasonic concentration; Ultrasonic particle manipulation; Acoustic separation

Definition

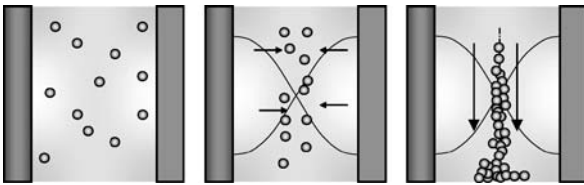
Ultrasonic fields, particularly standing waves or pseudo standing waves, can be used to manipulate particles within fluids. Typically particles in aqueous media will move to pressure nodes of the field, which can be planes, lines or points, depending on the standing wave geometry. By control of the standing wave patterns, it is possible to agglomerate particles, separate particles from the medium, wash particles from one fluid to another or fractionate populations of particles. In this article, techniques that have been used in microfluidic applications will be discussed, although many of the techniques can also be used at the macro scale.

Overview

It has been known for many years that acoustic standing waves can influence the behaviour of particles in either a liquid or gaseous medium. This effect relies on acoustic radiation forces that act on particles and other acoustic discontinuities (such as second phase fluids) within an ultrasonic field [1]. These forces are generally larger



Particle Manipulation Using Ultrasonic Fields, Figure 1 Movement of particles within an ultrasonic standing wave (USW). (a) Before a standing wave is generated, particles are evenly distributed throughout the fluid (b) a standing wave is excited by driving a transducer at a thickness resonance of the fluid layer, generating a pressure node in the centre of the fluid layer (dotted grey lines). A net force profile is established as per the solid line (in the direction of the arrows) (c) particles aggregate at the pressure node, driven by the force



Particle Manipulation Using Ultrasonic Fields, Figure 2 Enhanced sedimentation. Particles within an ultrasonic field (left) are agglomerated by the field (centre) and then tend to sediment under gravity (right)

in an ultrasonic standing wave (USW) than in a progressive wave and so it is mainly standing waves that are used for particle manipulation. The magnitude of this force is dependent upon the distance from the node of the standing wave and causes particles to move to the nodal plane. Use in microfluidic systems concentrates on active dimensions comparable with an acoustic wavelength. Acoustic frequencies of a few megahertz have wavelengths of the order of a few hundred micrometres. For example, a 3MHz acoustic wave in water has a wavelength of 500 μm . Many of the devices to be discussed here actually use sub-wavelength active dimensions and so it can be seen that ultrasonic waves of this order of frequency are compatible with the dimensional scale typically used in microfluidic systems. Consider the system represented in Fig. 1, in which particles within a fluid are bounded by an ultrasonic transducer and a reflective boundary. If the transducer is driven so as to excite a resonance frequency of the cavity (in this case, a half wavelength mode as shown by the pressure profile) a standing wave will be set up, with associated pressure maxima and minima. Particles will, in general, be transported towards pressure nodes within the field, by radiation forces. Figure 1b shows the relative magnitude and direction of the force across the depth of the chamber. The arrows show the net force direction, and the black

curve gives an indication of relative magnitude. The dotted curve shows the acoustic pressure profile (in this case a half wavelength) with a resulting node in the centre of the fluid layer.

The magnitude of the force seen by the particles is dependent on

- the radius of the particle
- the energy density in the standing wave
- the distance of the particles from the nodal plane
- the acoustic contrast factor of the particles.

This last factor is a function of the differences in mass density and compressibility of the fluid and the particle.

By considering these factors, it can be seen that particles can be either all concentrated in one place, or by making use of the different size of particle populations and a competing force such as viscous drag, fractionated by size. It is also possible, by making use of acoustic contrast factor variations to fractionate similar sized populations.

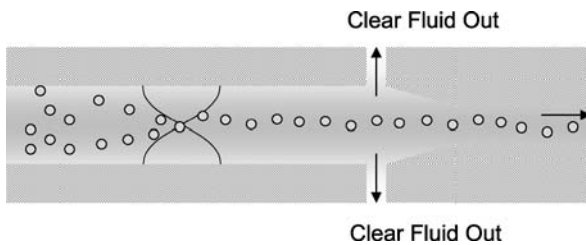
Basic Methodology

Particle Concentration or Filtration

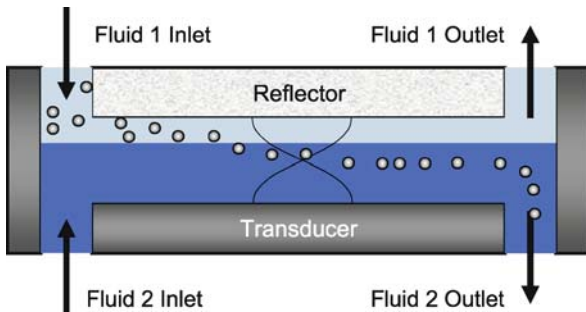
An obvious application of this technique is in a particle concentration or filtration system. There are two main approaches adopted

- Particle agglomeration
- Flow through separation.

In a particle agglomeration system (see [2]), devices are typically operated in a vertical direction (see Fig. 2). Particles are driven to the standing wave nodes by the acoustic radiation forces, and start to form agglomerates. Agglomerates form due to secondary radiation forces (see *key research findings*). As agglomerates grow, the relative importance of gravitational forces over viscous drag forces grows until the clumps of particles are able to sediment under gravity. Alternatively, once particles have been



Particle Manipulation Using Ultrasonic Fields, Figure 3 Flow through concentration based on technique described by Hawkes and Coakley [3]



Particle Manipulation Using Ultrasonic Fields, Figure 4 Particle Washing by forcing particles from one medium to another under laminar flow conditions

trapped, the field can be removed and the particle agglomerate falls.

Although as described, this technique has obvious application in batch processing, it can be modified to work as a flow-through device.

Flow-through systems are used in a continuous mode of operation in which a mix of fluid and particles is pumped through the device and the ultrasonic standing wave field concentrates the particles into a nodal plane. In this way a barrier-free filter is created.

This allows concentrated particles to be drawn off from the centre of the flow and clarified fluid to be drawn from either the upper or lower outlet, or from both as shown in Fig. 3. In this configuration, a mixture of particles and fluid are input from the left and flow through the device. In the main chamber an acoustic field concentrates the particles into a nodal plane in the centre. Clear fluid is then drawn out through the top and bottom outlets while the concentrated particles are allowed to proceed to the third exit.

This approach relies on the particles remaining at the centre of the channel due to the ordered nature of the highly laminar flow in a microchannel. The device developed by Hawkes and Coakley [3] was fabricated using traditional techniques and a microfabricated device has been developed using channels that are wet etched into silicon and Pyrex layers anodically bonded together [4].

An alternative technique that sorts particles into discrete patterns within a microchamber has been used by Neild et al. [5]. In the devices used by this group, acoustic energy is coupled into a fluid by first exciting flexural waves in a glass plate that bounds the fluid. Changing the excitation frequency, and in some cases using different transducers to excite the plate, allows particles to be aligned and manipulated within the fluid. Work described in reference [5] uses the acoustic pattern to align particles prior to further manipulation by a mechanical microgripper.

While most work has used planar resonators, it is also possible to manipulate particles in alternative geometries such as cylindrical tubes [6].

Particle Washing

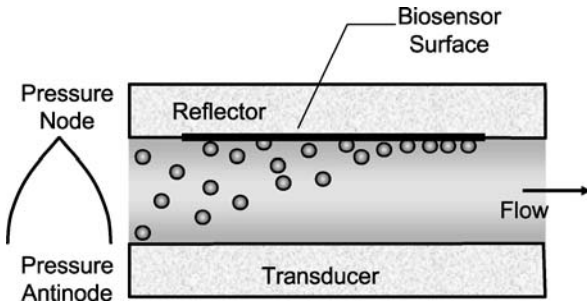
Applications have also been developed whereby particles can be driven acoustically from one fluid to another, making use of the laminar flow inherent in microfluidic systems. This precludes turbulent mixing, and so fluids can be in physical contact with minimal mixing. Figure 4 shows the principle. A fluid with particles in is fed in through fluid 1 inlet, and the acoustic force drives the particles from fluid 1 to fluid 2. Fluid 2 can then be withdrawn from outlet 2 with the particles. The ability to wash cells and particles and to move them from one medium to another is a critical element of many forms of analysis and hence an important component of a Lab-on-a-Chip toolbox.

A microfabricated device that works in a manner similar to that shown in Fig. 4 has been developed by Peterson et al. [7]. Three inlet and three outlet channels feed a single parallel sided channel anisotropically wet etched into a silicon substrate. This channel is excited at its half-wave resonance by a transducer mounted in the plane of the silicon, perpendicular to the axis of the standing wave. Similar devices have been used by the same group for a number of applications, including the separation of lipids from blood (in which the lipids move to the pressure antinode and the red cells to the pressure node) and fractionation of different size beads.

Biosensor Performance Enhancement

Ultrasonic particle manipulation can also be used to enhance biosensor performance, by using the ultrasound to drive the particles of interest to the biosensor surface. In this implementation (Fig. 5), a resonant reflector layer is used to generate a pressure node at a boundary, (effectively creating a quarter wave resonance within the fluid layer), rather than in the centre of the chamber, as previously described.

Such a device was demonstrated by Martin et al [8], who forced spores in a 135 μm deep chamber on to an antibody



Particle Manipulation Using Ultrasonic Fields, Figure 5 Enhancing biosensor performance based on technique of Martin et al [8]

coated surface and increased the capture of those spores by a factor of 70 when compared with the same system without ultrasonic actuation. This has the potential to overcome the diffusion limit on detection in microfluidic biosensor systems and significantly increase sensor sensitivity.

Fractionation

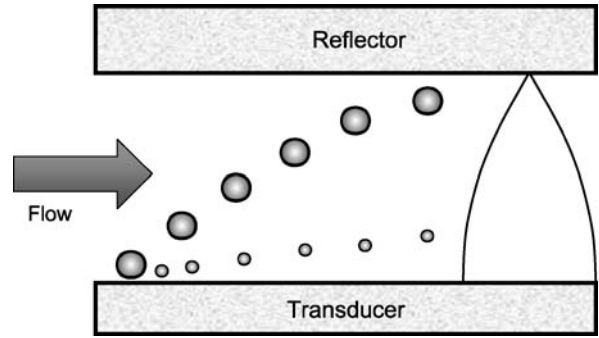
A further illustration of the use of ultrasonic fields is shown in Fig. 6. This demonstrates the principle of particle fractionation using acoustics. Different sized particles of the same material will experience different magnitudes of force within the same field. Thus larger particles, experiencing a larger force, will move more rapidly to the nodal plane. As long as the lateral flow is balanced against the vertical particle movement, the population of particles will be fractionated as shown. If the residence time in the field is too long though, all the particles will end up at the nodal plane.

Again, a quarter wave structure is used here to spread the particles across the width of the chamber, and the pressure profile is shown in the diagram. This requires the whole particle population to be input at the bottom of the channel, requiring some pre-conditioning of the particle distribution before the particles are introduced to this device, as the particles are driven from bottom to top in this case.

Key Research Findings

Calculation of Radiation Forces within the Fluid Layer

For a more detailed description of the background, including key references, see Gröschl's 1998 paper [1]. King provided the first comprehensive calculation of acoustic radiation forces on a small particle within a standing wave in 1934. This theory was developed for rigid spheres in an inviscid fluid and predicts that particles will move toward either the node or antinode of the field depending on the ratio of the fluid and particle densities. When the ratio of the incompressible particle density to the fluid density is



Particle Manipulation Using Ultrasonic Fields, Figure 6 Fractionation of particles, making use of the net larger force on larger particles

less than 0.4, the acoustic force will act towards the pressure antinode. If the density ratio is above 0.4 (which is the general case for real particles) the acoustic radiation force will act towards the pressure node of the standing wave. King's predictions have been shown to be in good agreement with experimental results for solid spheres suspended in a gas, but are subject to significant errors for spheres which are not rigid in comparison to the fluid medium, which is the case of bubbles within a resonant field. In the light of this, King's theory was extended by Yosioka and Kawasima to allow for compressible spheres in 1955. They demonstrated that the time-averaged acoustic radiation force on a spherical particle of radius a , at position x within a one dimensional standing wave of acoustic energy density ε is:

$$F(x) = 4\pi k \varepsilon a^3 \Phi(\beta\rho) \sin(2kx) \quad (1)$$

The *acoustic contrast factor*, $\Phi(\beta\rho)$ is given by:

$$\Phi(\beta\rho) = \frac{\rho_p + \frac{2}{3}(\rho_p - \rho_f)}{2\rho_p + \rho_f} - \frac{\beta_p}{3\beta_f} \quad (2)$$

where β and ρ are the compressibility and the mass density of the fluid and the particle, indicated by subscripts f and p respectively. The wave number, k is equal to $2\pi/\lambda$ where λ is the wavelength of the standing wave.

For most applications of interest, including cells in aqueous solution, the acoustic force will act towards the pressure node, however in the case of bubbles that are smaller than resonant size, and certain two-phase fluid mixtures, the bubble, or second phase fluid, will experience an acoustic force acting towards the pressure antinode.

Particle Positions Under Gravity and Other Forces

Yosioka and Kawasima also considered the direction in which an acoustic force will act on a particle in suspension. The equilibrium position of a particle within the

field will be determined by the resolution of this acoustic force and any other forces acting on the particle. Typically this would be gravity, although it applies to any force. In the case of a planar resonant field in which the positive x -axis points vertically upwards, the particle will be in equilibrium when:

$$3k\varepsilon\Phi(\beta\rho)\sin(2kx) = (\rho_p + \rho_f)g \quad (3)$$

The equilibrium position, for this case (in which the particle is small compared to the acoustic wavelength) is independent of the size of the particle, but dependent upon the acoustic contrast factor and the strength and frequency of the ultrasonic field. This offers a means of characterizing the acoustic field, the pressure amplitude of which is often difficult to determine in microsystems, by balancing acoustic and gravitational forces on an acoustically levitated particle. Particles that are not neutrally buoyant will have a minimum acoustic energy density required to hold the particle against buoyancy forces.

In many applications, including the manipulation of biological cells, particles are close to neutrally buoyant and the equilibrium position given by Eq. 3 will be very close to the pressure node. However for heavy particles, or for bubbles, this buoyancy offset may be significant.

Lateral Forces Real standing wave fields do not only exhibit pressure variations along the axis of the standing wave. They also have variations in the lateral directions leading particles to move not just to the axial pressure nodes, but also to particular points within those energy nodal planes. The practical effect of this is that instead of having a homogeneous layer of particles, the layer can organise itself into stripes, or striations, or other patterns. These inhomogeneities in the field may be caused by a number of phenomena including a decrease in acoustic energy towards the edges of axial fields, near field effects [9] due to geometric interference patterns, multi-dimensional acoustic modes within the enclosure, the coupling of structural (for example bending) modes into the fluid or inhomogeneity of the ultrasonic source.

In some cases these lateral variations provide a desirable feature of the device operation, particularly in systems designed to trap particles, while in others, the variations are not desirable and may be detrimental to the device performance, such as flow through concentrators. In real devices lateral forces may well exist due to a combination of the above causes. The magnitude and distribution of the lateral forces within resonators has been measured and shown to compare well with calculations using Gorkov's formulation for radiation forces (described by Gröschl in [1]).

Secondary Radiation Forces In addition to the axial and lateral radiation forces attributable to the primary acoustic field, secondary acoustic forces are produced between particles themselves. These particle-particle interactions, known as Bjerknes forces, aid the formation of aggregates within a standing wave, but such effects are negligible until the particles are in close proximity.

Some Practical Considerations

The Influence of Particles on the Field Resonant chambers for particle manipulation (including microfluidic chambers) are designed to provide specific acoustic characteristics. However, the chambers must work when filled with a fluid-particle mix which will have acoustic characteristics that differ from the particle-free fluid. Many applications will need to work over a range of particle concentrations that will modify the acoustic field. In cases where the particles are very different acoustically from the fluid medium, agglomerations can cause localized breakdown of the acoustic field resulting in a disruption of the ordered collection of particles at the pressure nodes.

Temperature Changes The temperature of an acoustic resonator will typically rise during use due to energy dissipation in the transducer and acoustic absorption within the layers of the resonator. The temperature rise will depend on the input power, the size of the resonator, the transducer material, and the material characteristics of the resonator layers. Hence there are significant differences in the thermal behaviour of resonators reported in the literature, with, for example, Hawkes and Coakley [3] reporting less than 1 K temperature increase in micro-scale chambers, while at the other extreme, Doblhoffdier et al. [2] required an integrated cooling circuit to avoid significant temperature rises in their rather larger chamber designed for the retention of mammalian cells.

Cell Viability Many of the uses envisaged for USW manipulation of particles are based upon manipulation of biological cells and so cell viability after treatment with ultrasound becomes an important issue. This has been reported on by several studies which have found that viability is maintained. For example Doblhoffdier et al. [2] suspended mammalian cells in a large resonator at different power levels and reported no significant loss in viability.

Hultström et al. [10] trapped cells in a 260 μm deep microfluidic chamber constructed from glass and PDMS. It was demonstrated that cell viability was not significantly impeded by 75 min exposure to a 0.85 MPa 3 MHz USW field. Indeed, tests on the cells following exposure

suggested that the exposure to ultrasound enhanced the cells' ability to proliferate. The same group has demonstrated the ability of ultrasound traps to enhance bioanalysis in a number of different microchambers and has constructed microdevices that integrate ultrasonic and dielectrophoretic manipulation.

Future Directions for Research

Radiation forces as generated by low MHz ultrasonic standing waves offer a method of manipulating particles in a fluid, with particular application at the microscale. This gives tremendous potential for trapping and manipulating larger particles and groups of particles, perhaps prior to finer manipulation from other techniques. In addition, the evidence that ultrasound can manipulate and control the position of cells without adversely affecting cell viability suggest that ultrasonic standing waves hold significant potential in the manipulation of biological cells. The field is still relatively immature and requires work on the characterisation and optimisation of chambers and defining the scale limits of manipulation in practical devices.

Cross References

- Acoustic Streaming
- Acoustics Based Biosensors
- Biosample Preparation Lab-on-a-Chip Devices
- Cavitation in Microdomains
- Cell Sorting
- Lab-on-chip Devices for Particle and Cell Separation
- Piezo/PZT in Microfluidics

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Particle Methods

► Meshless Methods

Passive Micromixers

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Synonyms

Microfluidic mixer; Micromixing chip; Micromixer; Passive mixer

Definition

A passive micromixer is one of the microfluidic devices. It utilizes no energy input except the mechanism (pressure head) used to drive the fluid flow at a constant rate. Due to the dominating laminar flow on the microscale, mixing in passive micromixers relies mainly on chaotic advection realized by manipulating the laminar flow in microchannels or molecular diffusion with increasing the contact surface and time between the different fluid flows.

Overview

Microfluidic devices have had a considerable impact in the fields of biomedical diagnostics and drug development, and are extensively applied throughout the food and chemical industries. The very small scale of the flow

channels in microfluidic systems increases the surface to volume ratio, and is therefore advantageous for many applications. However, the characteristic Reynolds number ($Re = \rho v / \eta$) of liquid flows in such microchannels is very small. For example, the Reynolds number in a typical water-based microfluidic system with a channel width of 100 μm , a liquid flow rate of 1 mm/s, a fluid density of 1 g/cm³ and a viscosity of 0.001 N s/m² is of the order of 0.1. In such low Reynolds number regimes, turbulent mixing does not take place, and hence species mixing occurs primarily as a result of diffusion, which is an inherently slow process. Consequently, the aim of microfluidic mixing schemes is to enhance the mixing efficiency such that a thorough mixing performance can be achieved within shorter mixing channels, thus reducing the characteristic size of microfluidic devices. Furthermore, developing efficient mixing schemes is essential to increase the throughput of microfluidic systems in general and to realize the concept of micro-total-analysis systems and Lab-on-a-Chip systems in particular.

In micromixers, two operating parameters are of importance: the Reynolds number Re and the Péclet number Pe . The Reynolds number, given by

$$Re = VL_h / \nu \quad (1)$$

represents the ratio of momentum to viscous friction, where V is the average flow velocity, L_h is the hydraulic diameter and ν is the kinematic viscosity. A turbulent flow can be indicated by a high Reynolds number above a critical value (around 2300 on the macroscale). In most cases of microfluidics, a laminar flow can be expected with a low Reynolds number. The Péclet number, given by

$$Pe = VL/D \quad (2)$$

represents the ratio of the mass transport due to convection and that of diffusion, where L is the mixing path and D is the diffusion coefficient. Convection is dominant at higher Péclet numbers.

The theory reported on mixing began with Einstein's investigation of Brownian movements [1]. The common mixing methods are the generation of chaotic advection and turbulence. Fluid motions vary irregularly so that quantities such as pressure and velocity show a random variation in space and time. The random movement quickly disperses the mixed components. Chaotic advection can be generated by stirring the flow, which is very effective for low Reynolds numbers. The concepts of splitting, stretching, folding and breaking up play an important role in the mixing quality.

In contrast to the passive mixing schemes presented above, in which the microchannel configuration is specifically designed to increase either the contact area or the contact time (or both) of the multiple species, active mixing schemes improve the mixing performance by applying external forces to the sample flow in order to accelerate the diffusion process. Typically, active schemes are implemented by incorporating some form of mechanical transducer within the microfluidic device using micro-fabrication techniques. It has been shown that microfluidic mixers which use embedded ultrasonic transducers to generate acoustic waves to stir the samples are capable of achieving a high mixing performance. However, the acoustic vibrations also generate considerable heat, which is undesirable since it may lead to unwanted reactions between the samples. Various microfluidic devices with embedded microelectrodes designed to mix the sample fluids dielectrophoretically have been presented. It has also been shown that the use of embedded electrode pairs to change the surface energy of the microchannel walls provides an efficient means of enhancing species mixing by inducing local instabilities in the flow stream. Compared to other active mixing schemes, the use of embedded electrode pairs has the advantage that moving components are not required, and hence the resulting microfluidic devices are not only cheaper, but also more reliable.

Besides chaotic advection, molecular diffusion is another important form of mass transfer in passive micromixers. In such scales where the flow is strictly laminar, the mixing is entirely ensured by molecular diffusion between layers of different concentrations. Consequently, when the thickness of each fluid layer is larger than the characteristic diffusion length, an effective and rapid mixing is difficult to achieve. The diffusion length is often approximated by $\sqrt{Dt}$, where t is the residence time of the species. To overcome the limitations by integrating particular designs of microchannels on the basis of different fluidic mechanisms, the most common approach is based on increasing the contact surface between the different fluids and decreasing the diffusion path between them. In general, passive micromixers can be categorized as lamination, chaotic advection and special designs, based on the arrangement of the mixed phases.

Basic Methodology

Theory of Lamination

Mixing in general follows a two-step sequence. The first is segmentation where a heterogeneous mixture of substantially homogeneous flows of the two fluids is formed by convection; the second is interdiffusion of molecules between nearby flows. Segmentation procedures, well

known from the macroscopic world, such as stirring or creation of turbulent flow, cannot be scaled down to microsystems with dimensions so small that Reynolds numbers exceeding 2000 (the limit for turbulent flow) are not feasible.

Perfect lamination can be obtained in a flow with a sufficiently low Reynolds number provided that the channel system guiding the fluid is designed in such a way that the pressure gradients in the flow channels will force the two fluids to laminate. Figure 1 shows a generic channel arrangement laid out in a free three-dimensional space. When considering symmetry it is obvious that a perfect lamination and a split into two channels is obtained. Depending on the flow direction, either horizontal (Fig. 1a) or vertical (Fig. 1b) lamination will be the result [2].

A detailed theoretical description of flow and diffusion phenomena in the laminar regime is very useful during design and testing of mixers for microsystems. The fully developed velocity profile for laminar flow in a straight channel of rectangular cross-section of dimensions $W \times H$ is easily found by numerical methods. For channels having width (W) significantly larger than height (H) the velocity profile turns out to be very close to parabolic in the direction of the short dimension and almost similar to H from the channel walls. This is also expected intuitively, as the distance to the nearest walls is the only physically important parameter for laminar flow. In Ref. [2], it was found that the peak velocity (v_{peak}) in the middle of the channel is related to the average velocity (v_{average}) (as calculated from the volume flow rate) as

$$v_{\text{peak}} = \frac{3}{2} \left(\frac{1}{1 - 0.62H/W} \right) V_{\text{average}} \quad (3)$$

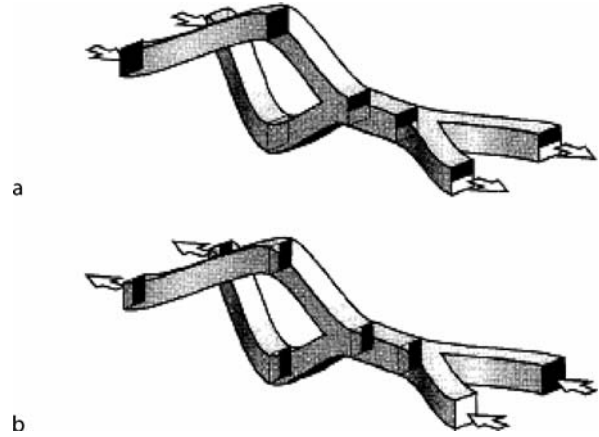
within 1.5% when $W \geq 3H$.

When solving the diffusion problem for a laminated structure, it is useful to consider Fourier expansion of the initial concentration profile, as it is well known that the time evolution of the concentration profile is given by the time evolution of each of the Fourier components. Each Fourier component κ decays exponentially during the diffusion process with a time constant τ_k given by

$$\tau_k = (2\pi)^{-2} \frac{L_k^2}{D} \quad (4)$$

where L_k is the wavelength of the Fourier component and D is the diffusivity.

This explains concisely what is also expected from quantitative arguments that a 2^n -layer structure can mix 4^{n-1} times faster than a 2-layer structure, as the wavelength



Passive Micromixers, Figure 1 A generic two-input two-output mixer including lamination and splitting shown in free three-dimensional space. (a) Horizontal lamination and (b) vertical lamination [2]

of the Fourier component of longest wavelength and thus slowest decay rate scales with the layer thickness [2].

Meanwhile, the Fourier argument makes a small but important detail of mixing by lamination very clear: The lamination has to be symmetric, i. e. the top- and bottom-most layer should be of only half thickness compared to the other layers. Otherwise, i. e. if the lamination is asymmetric, the Fourier expansion of the initial profile will contain a component with a wavelength twice the channel dimension, no matter how thin the laminated layers are. The component will sooner or later become dominant because of its slow decay, and only a minor benefit is obtained by lamination. The qualitative counterpart of this argument is that, unless the outermost layers are of half thickness, half of their content of the diffusing species has to be distributed over all of the inner layers before full mixing is achieved. When being of only half thickness, they are balanced by their neighboring layer as are all the other layers.

However, a simple cascade of two-input two-output mixers results in an asymmetric lamination. Symmetric lamination can be obtained with a modification of the first mixer into a three-input two-output mixer which builds up a $(H/4, H/2, H/4)$ type of laminate. This initial mixer can be followed by as many regular two-input two-output mixers as desired for the particular application.

Micromixers Based on Lamination

In miniature flow systems with Reynolds numbers varying from 2 to 100, flow structures can be artificially induced which assist in flow segmentation through inertia effects. Wu et al. [3] presented an analytical model of diffusive

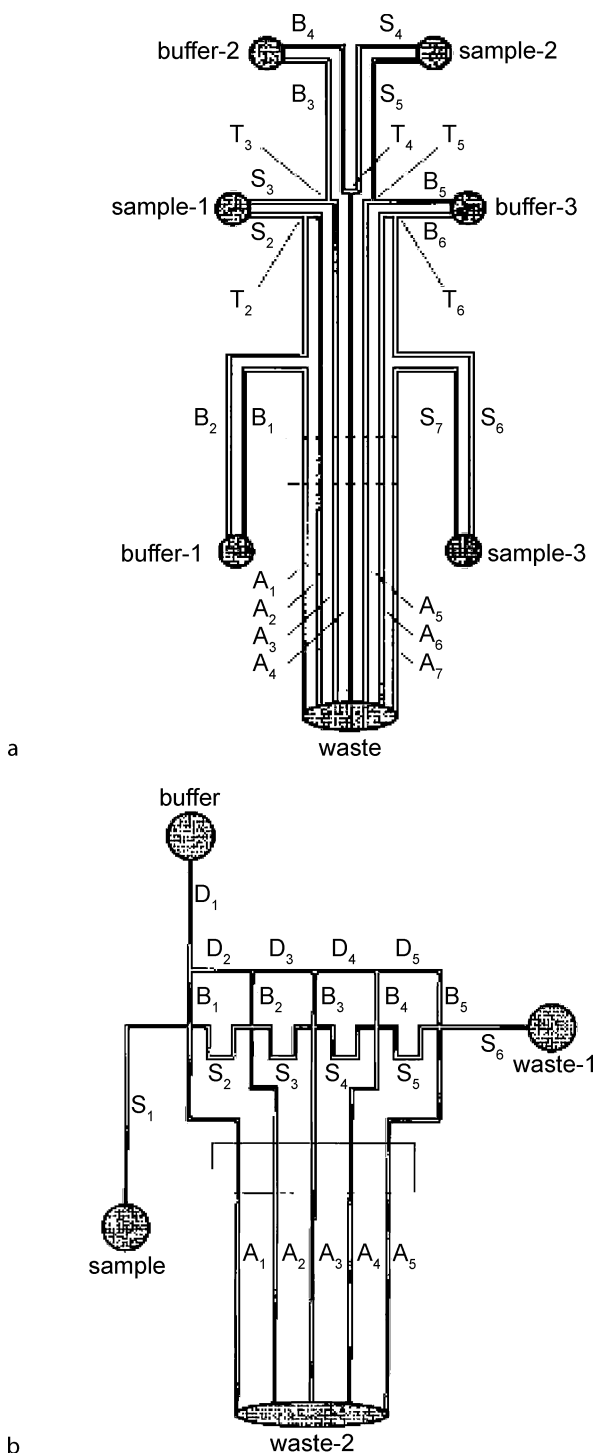
mixing in microchannels. The dimensionless analysis generalizes the solution for different channel sizes and different diffusion coefficients by varying the Peclet numbers. It could be seen that the nonlinear model described well the diffusive mixing process in the microchannels and the mixing performance was improved with lower Peclet numbers due to the stronger diffusion effect. Another lamination micromixer using inertia forces was reported in the literature, involving a coaxial mixer containing an inner core liquid stream with a low flow rate and an outer flow stream with a higher flow rate.

Jacobson et al. [4] proposed micromixers for electrokinetically driven parallel and serial mixing. To simplify the voltage control hardware, electrokinetic mixing is effected using a single voltage source with the channels dimensioned to perform the desired voltage division. In addition, the number of fluid reservoirs is reduced by terminating multiple buffer, sample or analysis channels in single reservoirs. The parallel micromixer (Fig. 2a) was designed with a series of independent T-intersections, and the serial micromixer (Fig. 2b) was based on an array of cross intersections and sample shunting. The parallel micromixer has the advantages that each set of channels is independent of the others. Consequently, a failure in one set of channels does not affect the operation of the others. The dynamic range of the serial micromixer is larger especially if the sample shunting is carried out with additional cross intersection added to the channel manifold beyond the fifth cross intersection. Wong et al. [5] presented a micro T-mixer fabricated on a silicon substrate and bonded to a Pyrex glass plate (Fig. 3). The experimental results revealed that a mixing channel with a hydraulic diameter of $67\text{ }\mu\text{m}$ and an applied pressure of 5.5 bar were sufficient to result in complete species mixing in less than 1 ms following the initial contact between the two species flows with Reynolds numbers of 400–500.

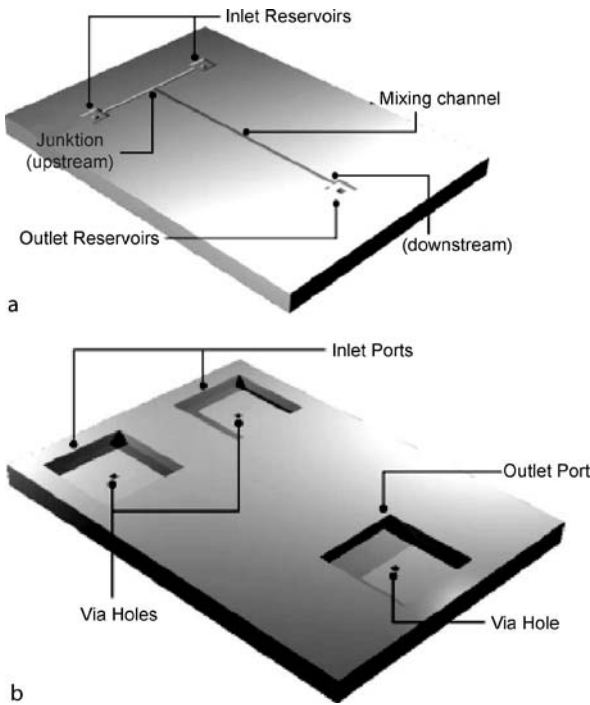
Micromixers Based on Intersecting Channels

Micromixers with intersecting channels can be used to split, rearrange and combine component streams in order to enhance their mixing. He et al. [6] proposed a picoliter-volume mixer (Fig. 4) with intersecting channels of various lengths and a bimodal width distribution. In the proposed design, all of the channels running parallel to the flow direction had a width of $5\text{ }\mu\text{m}$, while those intersecting the parallel channel network had a width of $27\text{ }\mu\text{m}$ and were aligned at an angle of 45° . The microchannels were designed such that the two species streams merged into one larger stream and then flowed together along a mixing channel of length $300\text{ }\mu\text{m}$. The results showed

that this configuration enabled a complete species mixing within a distance of $200\text{ }\mu\text{m}$, representing a considerable

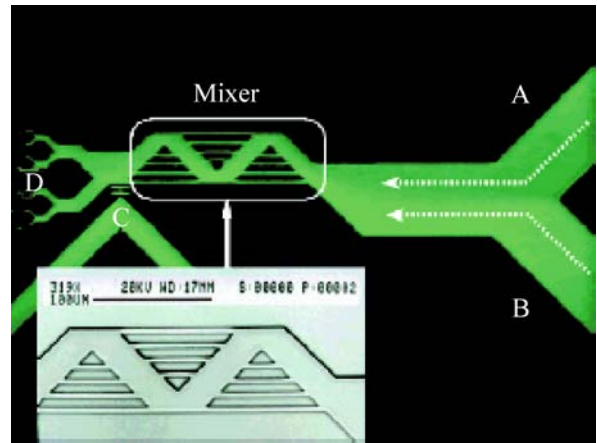


Passive Micromixers, Figure 2 (a) Schematic of the microchip for parallel electrokinetic mixing. (b) Schematic of the microchip for serial electrokinetic mixing [4]



Passive Micromixers, Figure 3 (a) The front face of the micro T-mixer test chip. (b) The back face of the micro T-mixer [5]

improvement compared to the mixing distance of $3000\ \mu\text{m}$ required to fully mix the two streams in a conventional straight channel. Melin et al. [7] presented a micromixer in which a constantly changing, time-dependent flow pattern was created within a two-sample liquid plug as the plug passed through a planar mixer chamber. It was shown that the micromixer created a larger mass transfer within the liquid plug than that achieved in many of the passive devices presented previously. The proposed mixing chamber comprised a main meandering channel with perforated walls, which formed connecting channels between parallel segments of the main meandering channel. In the mixing process, two discrete liquid samples are fed into the mixing chamber in a laminar manner. When the liquid wets one side of the first perforated wall segment, it enters the perforation, but is prevented from exiting by surface tension effects. However, as the bulk of the liquid proceeds to the next main channel segment, the other side of the perforated wall becomes wetted. As a result, the liquid/air interface at the exit of the perforation is replaced by a liquid/liquid interface, and hence the liquid flows freely through the perforation. The same phenomenon is repeated in each of the main channel segments as the liquid flows progressively through the mixing chamber. The resulting constantly changing flow lines within the liquid plug cause a self-folding effect within the

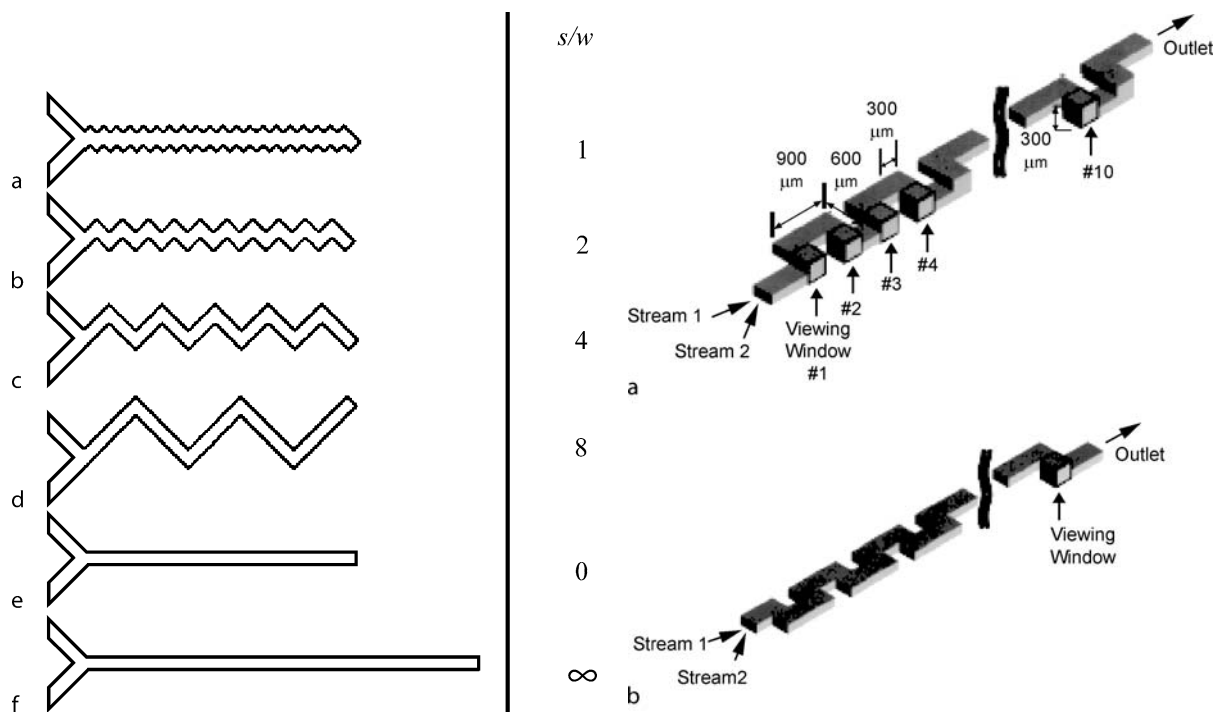


Passive Micromixers, Figure 4 Photomicrograph of a microfabricated mixer. The small image in the lower left corner is the SEM of the mixer. This mixer is about 100 by $200\ \mu\text{m}$ wide and $10\ \mu\text{m}$ in depth. Effect of mixing is evaluated by bringing in two fluids from channels A and B, and flowing to D. Channel C is not used in this work [6]

plug which yields a significant improvement in the mixing performance.

Micromixers Based on Zigzag Channels

Mengeaud et al. [8] presented a $100\ \mu\text{m}$ wide zigzag microchannel integrating a “Y” inlet junction (Fig. 5). The effect of the periodic step value (s) on the mixing efficiency was investigated in a series of experimental trials. The results indicated that for $\text{Re} = 0.26$, the mixing efficiency increased from 65% to 83.8% when the geometry ratio was increased from 1 to 8. For low values of s/w , the number of angles increases, resulting in an increase in the effective width and a reduction in the effective length. For low Reynolds number flows, it was found that the most efficient zigzag configuration, corresponding to $s = 800\ \mu\text{m}$, gave a mixing efficiency of 83.8%. For $\text{Re} = 267$, the mixing efficiency increased rapidly to 98.6% as the geometry ratio was increased to 4, but reduced slightly to 88.1% as the geometry ratio was further increased, thus indicating the existence of an optimal zigzag geometry. After the research, the zigzag passive micromixer showed a modified Tesla structure. The species streams flowed close to the angled surface as a result of the Coanda effect, and this effect was used to guide the fluid streams such that they collided with one another. Mixing cells in opposite directions were then used to repeat the transverse dispersion caused by the flow impact. In the micromixer, one of the fluids was divided into two sub-streams and one of these two sub-streams was then merged with the second fluid stream from the main channel in the micromixer. The two streams were then



Passive Micromixers, Figure 5 Different geometries of a microchannel. (a) $s/w = 1$, (b) $s/w = 2$, (c) $s/w = 4$, (d) $s/w = 8$, (e) $s/w \rightarrow 0$, and (f) $s/w \rightarrow \infty$ [8]

mixed with the second sub-stream, resulting in a strong impact around the sub-channel of the micromixer.

Micromixers Based on Three-Dimensional Serpentine Structures

The three-dimensional serpentine micromixer was designed to induce a chaotic mixing effect. It was found that the mixing efficiency of the serpentine microchannel was twice that obtained in a conventional straight microchannel. Liu et al. [9] fabricated a three-dimensional serpentine micromixer (Fig. 6) featuring a “C”-shaped repeating unit designed to induce chaotic advection. Mixing in the serpentine channel relies on the flow field being sufficiently three-dimensional, with secondary flows stretching and folding the fluid, greatly increasing the interfacial area across which diffusion occurs. The results showed that for flows with a Reynolds number of 70, the mixing efficiency in the serpentine channel was 16 times better than that in a conventional straight channel and 1.6 times better than that in a zigzag channel.

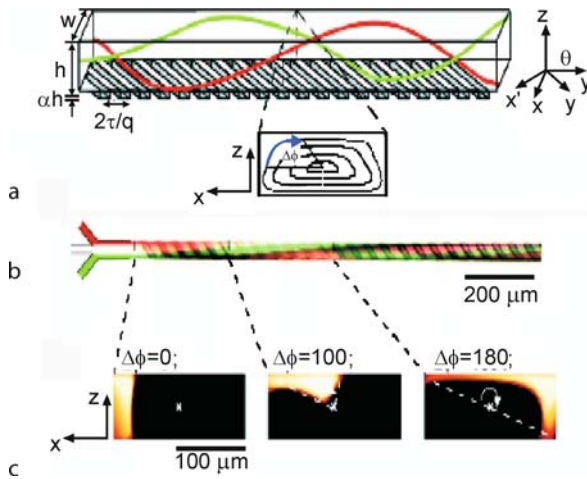
Micromixers Based on Embedded Barrier

Based on the combination of multilamination and chaotic advection effects, Kim et al. [10] developed barrier



Passive Micromixers, Figure 6 (a) Schematic of the three-dimensional serpentine channel. “Viewing windows” in the channel are labeled 1–10. (b) Schematic of zigzag channel. (c) Schematic of straight channel [9]

embedded micromixers for pressure-driven flow in which chaotic flow was induced by applying periodic perturbations to the velocity field via periodically inserted barriers along the upper surface and helical-type flow structures were induced by slanted grooves on the lower surface of the microchannel. Observations using a confocal microscope revealed cross-sectional mixing behaviors in several locations of the micromixer. The proposed design was validated experimentally at a flow rate corresponding to a Reynolds number of 2.28 (corresponding to $Pe = 1.24 \times 10^4$ with a Rhodamine diffusivity of $2.8 \times 10^{-10} \text{ m}^2 \text{ s}^{-1}$). Laser scanning around the entrance zone of the micromixer identified a bright image only in the half-zone containing Rhodamine. Bright images



Passive Micromixers, Figure 7 Three-dimensional twisting flow in a channel with obliquely oriented ridges on one wall. (a) Schematic of channel with ridges. (b) Optical micrograph showing a top view of a red stream and a green stream flowing on either side of a clear stream in the channel. (c) Fluorescent confocal micrographs of vertical cross-sections of a microchannel such as in (a) [11]

were also observed at the no-barrier zone in the first half-cycle, thus confirming the cross-sectional rotating flow effect induced by the slanted grooves. When the streams entered the barrier zone in the next half-cycle, laser scanning showed that the flow was rotated yet further. The experimental results confirmed that the barrier embedded micromixer yielded an excellent species mixing within a short channel length.

Micromixers Based on Slanted Well

Stroock et al. [11] presented chaotic mixers (Fig. 7) designed to mix streams of steady pressure-driven flows in microchannels at low Reynolds numbers. The authors considered two different microchannel designs, namely one channel with straight ridges on one wall and a second channel with a staggered herringbone pattern on one wall. The mixing performances of the two microchannels were compared with that of a smooth microchannel. In the evaluation experiments, cross-sectional images taken at the same distance (3 cm) from the junction were acquired and the mixing efficiency was assessed by computing the standard deviation σ of the fluorescence intensity in the confocal images as a function of the distance along the mixing channel length, Δy . The deviation maintained a constant value of approximately 0.5, indicating a poor mixing performance. The results revealed that the micromixer with straight ridges on one microchannel wall yielded a poor mixing performance. The standard deviation profile for a low Pe number was approximately 0, which signifies an

almost complete mixing. Others investigated the effects of the asymmetry index p and depth ratio α of the groove on the mixing performance of a staggered herringbone mixer with patterned grooves. Since the two vortices within the mixing channel are determined by the asymmetry index p , vortices with dissimilar scales were shown to provide a better mixing performance than two equally sized vortices. Furthermore, the results showed that the intensity of the vertical fluid motions at the side edges of the grooves increased with increasing groove depth α , resulting in a significant improvement in the mixing effect.

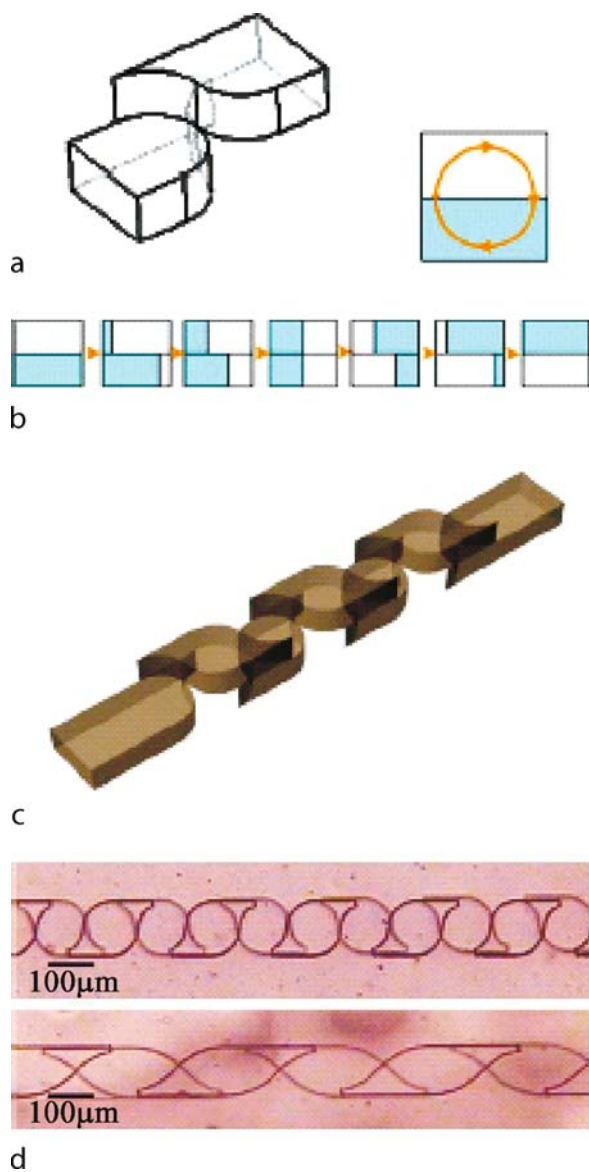
Micromixers Based on Twisted Channels

In twisted microchannel applications, the micromixer featured is a three-dimensional structure of twisted microchannels designed to induce a chaotic mixing effect within the fluid streams. The twisted microchannels were designed such that the angle of the channels' lower surfaces alternated in each longitudinal subsection of the mixing channel. The fluid flow sways from side to side as it travels along the microchannel, resulting in a chaotic advection effect. Park et al. [12] presented a breakup method based on passive rotation for enhancing the mixing efficiency of micromixers. The proposed method not only actively generated interfaces between the mixing species, but also enhanced the diffusion process at the interface. The micromixer actually incorporated two separate mixing regions, namely one region with short segments of length 200 μm and a second region with long segments of length 400 μm (Fig. 8). The results showed that this design caused a rotation of the fluids flowing through the channel. It was found that the fluids were mixed well and their interface was distorted at high Reynolds numbers. Conversely, at low Reynolds numbers, the fluids were hardly mixed and the interface was not distorted, i. e. only the respective positions of the two fluid streams were changed.

Key Research Findings

Grooved Staggered Herringbone Micromixers

Grooved staggered herringbone micromixers are based on the original design proposed by Aubin et al. [13] and have an off-centered herringbone pattern on the lower surface of the microchannel, which creates a transverse velocity component in the flow field. Such mixers have several mixing cycles, where each cycle comprises two sequential regions of grooves, i. e. two half-cycles. The direction of asymmetry of the herringbone pattern switches with respect to the centerline of the microchannel every half cycle. In order to improve the mixing efficiency, the depth of the grooves in staggered herringbone micromixers should be specified



Passive Micromixers, Figure 8 Schematic diagrams of (a) one mixer segment and (b) flows in the microchannel recirculating along the walls. (c) Schematic of mixer 1. (d) Optical micrograph of mixer 1: top, micromixer composed of short mixer segments; bottom, micromixer composed of long mixer segments [12]

as 30% of the channel height. This promotes both spatial homogenization and a reduction of the striation thickness without increasing the pressure drop across the mixer.

Surface Chemistry Technology in Microchannels

In microfluidic systems, very high pressure gradients are generally required to drive and manipulate the fluid flow. Due to the small characteristic scale of the microchan-

nels in typical microfluidic devices, surface forces dominate, and high friction effects are generated. Conventionally, microchannels are fabricated using silicon dioxide. Silicon dioxide surfaces are typically negatively charged due to their deprotonated silanol groups ($\equiv\text{Si}-\text{O}^-$). When these surfaces come in to contact with a solution containing ions, the positive ions are attracted to the surface, forming an important layer and a diffuse layer on the surface. If an electrical field is then applied, the positive ion diffuse layer moves in the direction of the applied field. The movement of the diffuse layer drags the bulk fluid into motion via momentum coupling, resulting in the so-called electro-osmotic flow (EOF) phenomenon [14].

Future Directions for Research

Mixing in electro-osmotic flow is generally diffusion-dominated. However, it has been shown that the introduction of electrically charged surface heterogeneities enhances the mixing efficiency by creating localized flow circulation regions. By selectively patterning positively charged molecules on a negatively charged channel wall, flow vortices can be induced as a result of the differences in electrostatic potential between the homogeneous and heterogeneous surfaces. It has been shown that these localized flow vortices can be exploited to yield a significant improvement in the mixing performance in a variety of microchannel configurations, including in-line, staggered, serpentine, herringbone and diagonal arrays [15]. In their numerical investigations, the authors specified the patch length and spacing parameters in such a way as to maintain a constant ratio of the heterogeneous-to-homogeneous surface areas over a channel length of 1.8 mm. The results revealed that the nonsymmetrical heterogeneous patterns, namely the staggered and diagonal patterns, generated a better mixing performance than either the symmetrical herringbone pattern or the in-line arrangement. With a theoretical mixing efficiency of 96%, the staggered configuration provided the best mixing performance, outperforming the diagonal, herringbone and serpentine configurations by 8%, 31% and 36%, respectively. Furthermore, compared to the homogeneous case, the staggered configuration provided a 61% improvement in the mixing efficiency. The use of heterogeneous surface charge patches to manipulate electrokinetic flows provides a simple, low-cost solution to mixing problems in Lab-on-a-chip systems, and is therefore likely to receive increasing attention in future studies.

Cross References

- [Microfluidic Mixing](#)
- [Fundamentals of Diffusion in Microfluidic Systems](#)
- [Passive Micromixer](#)

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Passive Mixer

► Passive Micromixers

Patch Clamp Measurements On-Chip

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Synonyms

Patch clamp technique; Voltage clamp; Patch voltage clamp technique; Cell-attached (single-channel) voltage clamp

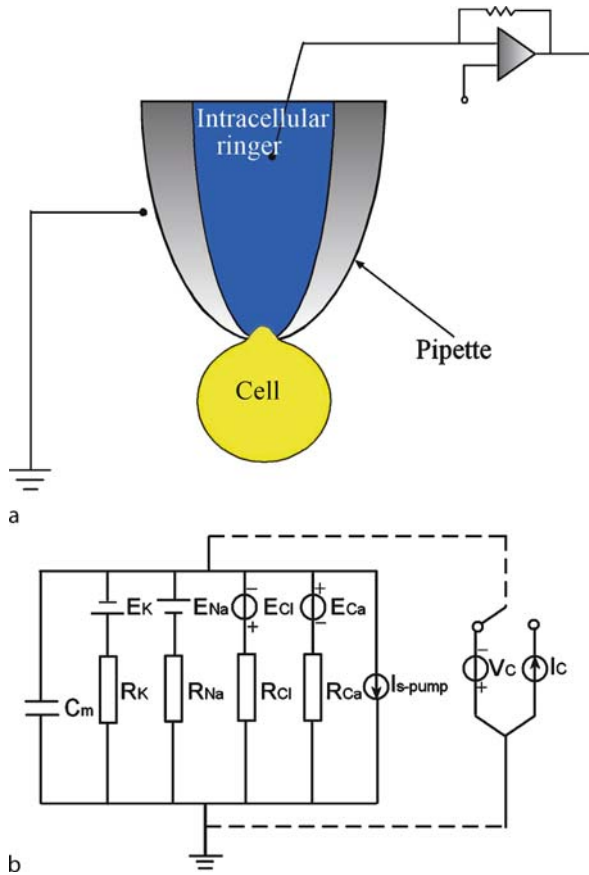
Definition

Ion channels, like molecular pores that open and close in response to stimuli, allow electrically charged atoms – such as sodium, potassium and calcium ions – to pass in and out of a cell through the cell's outer membrane by opening and closing like tiny gates. The *patch clamp technique*, a dominant technique providing valuable information for ion channels, including their electrophysiological properties, molecular structures and the mechanism of drug action in cellular electrophysiology with high resolution, involves the fabrication of a pipette with a micrometer-sized tip which is filled with a conducting solution and the tip gentle suction directed towards the cell membrane with a micromanipulator in order to form a gigaseal between the pipette tip and the cell membrane. A *gigaseal* is a tight seal between the membrane and pipette rim with a resistance in the range of gigaohms ($10^9 \Omega$).

Overview

The patch clamp technique, as a dominant technique, provides valuable information for ion channels, including their electrophysiological properties, molecular structures and the mechanism of drug action in cellular electrophysiology with high resolution. The rationale for using *patch pipettes* and the principles of patch clamp recording, including the information provided, are elaborated in detail first to understand the mechanism of the technique. The advantages and disadvantages of four configurations of this technique, i.e. on-cell patch clamp, whole cell patch clamp, inside-out patch clamp and outside-out patch clamp, are then described in the following. Two different methods for preparation of pipette on-chip and patch clamp device on-chip fabrication and electrophysiology recordings are given to provide reference for the reader. There are some simple but perspicuous figures presented to allow an easier grasp of the patch clamp technique. Finally, the key research findings about this technique are summarized and future directions for research are proposed.

Ion channels provide pores for the passive diffusion of ions across biological membranes. They are often highly selective for a particular ionic species, leading to a classification into sodium (Na^+), potassium (K^+), calcium (Ca^{2+}), chloride (Cl^-) and unspecific cation channels. The direc-



Patch Clamp Measurements On-Chip, Figure 1 Patch clamp technique and an amended equivalent circuit

tion of net ion transport, which is associated with an electric current, depends on the electrochemical gradient (established by interplay of active pumps, co-transporters and ion channels) for the relevant ionic species. Channels occur not only in the external (plasma) membrane, but also in membranes of intracellular organelles. Ion channels are essential for many of the cell's functions, such as impulse conduction in the nervous system or pace-making in the cardiac system, and their dysfunction is pivotal in many diseases. The list of human diseases known to be associated with defects in ion channels has grown considerably during the past few years. Hence, there is a great interest in ion channels not only from academia to understand the biophysics of their function but also from the biotechnology and pharmaceutical industries to develop drugs that modulate ion channels to correct any malfunction. Several methods are available today to test candidate ion channel active drugs (ICADs): electrophysiology (patch clamp), binding assays, radioactive flux assays, membrane potential-sensitive fluorescent dyes, ion-sensitive dyes and

voltage sensing based on fluorescence resonant energy transfer (FRET).

Giving an unsurpassed information content and having high sensitivity, the patch clamp technique, developed by Erwin Neher and Bert Sakmann at the Max Planck Institute, Germany (they were awarded the Nobel Prize in Medicine for this discovery; see Fig. 1), is a powerful tool for studying the electrophysiological properties of biological membranes. It is mainly used to detect currents that pass a patch of cell membrane. This *patch* can be a large one, as large as almost the whole cell membrane; or a small one, so small as to contain only several or even one ion channel. *Clamp* mainly refers to fixing the voltage across the membrane, i. e. membrane potential, at a constant level, or at different levels stepwise, under which conditions transmembrane currents are measured (i. e. voltage clamp mode). Also, the patch clamp technique can sometimes be used to observe the membrane potential when a constant current (usually zero), on which sometimes a squarewave current is superposed, is injected into the cell (i. e. current clamp mode). And furthermore, with this technique cell secretion ions can also be studied by measuring the capacitance of the cell membrane (C_m), which, used as an indicator of the area of cell membrane, may change during cell secret ion.

Basic Methodology

Mechanism

Rationale for Using Patch Pipettes

A basic limitation for any current measurement, disregarding instrumentation noise, is the Johnson or thermal noise of the signal source, which for a simple resistor is given by

$$\sigma_n = \sqrt{4kT\Delta f/R}$$

where σ_n is the root-mean-square deviation of the current, k the Boltzmann constant, T the absolute temperature, Δf the measurement bandwidth and R the resistance. From this equation, it is clear that the internal resistance of a signal source (or, more generally, the complex impedance) should be very high for low-noise current recording. Specifically, to record a current of 1 pA at a bandwidth of 1 kHz with 10% accuracy, the internal resistance of the signal source should be about 2 G Ω or higher. We now know that the input resistances of small cells can be as high as that. But in the early 1970s the conventional microelectrode techniques required large cells for reliable current measurements, and these typically had input resistances in the range 100 k Ω to 50 M Ω . Thus, it seemed impossible to reach the required resolution with

standard techniques and standard preparations. What was required was a smaller signal source.

It was clear that they should be good tools for single-channel measurements, if only the *pipette-to-membrane seal* could be made good enough. The impedance of the patch itself should be higher than required, even for a patch as large as 10 μm in diameter. An incomplete seal, however, is “seen” by the measuring amplifier in parallel to the patch, and its noise is superimposed onto the patch signal.

Principles of Patch Clamp Recording

As is known, there is a resting potential (RP) across the membrane of most cells, and based on it action potentials (AP) or local responses can occur. All these electric phenomena involve specific states or activities of different ion channels and pump proteins in the membrane. The commonly cited explanation can be found in any textbook of general physiology, so we only present an amended equivalent circuit (Fig. 1) for a membrane patch, which is useful for understanding various electric phenomena that occur in the cell membrane.

This circuit includes multiple branches but only two nodes. Channels for different ions are equivalent to voltage sources, whose electromotive forces are equal to their respective equilibrium potentials determined by the Nernst formula, and whose internal resistances depend on the permeability of the membrane for, and the diffusion coefficients as well as the concentrations of, respective ions, etc. The ion pumps can be represented by corresponding current sources, all of which can be summed up forming one current source as shown in Fig. 1. If there exist transporters in the membrane, they can be electrically modeled (omitted in Fig. 1) in the same way as pumps. The capacitor represents the effect of the lipid bilayer of the membrane together with the extracellular solution (or the bath solution under artificial conditions) and intracellular solution. All these branches are arranged in parallel.

If no external circuit acts on the membrane, and the membrane potential is constant or changes very slowly, its value can be given by the node voltage formula:

$$V_m = \frac{E_K g_K + E_{Na} g_{Na} + E_{Ca} g_{Ca} + E_{Cl} g_{Cl} - I_{s-pump}}{g_K + g_{Na} + g_{Ca} + g_{Cl}}$$

$$= \frac{\sum E_j g_j - I_{s-pump}}{\sum g_j}$$

Incidentally, if g_K is much larger than the conductance for other ions, and the contribution of the net pump current to V_m is small enough to be omitted, which is the case for most cells at rest, it can be seen that $RP = V_m \approx E_K$.

If the state of a certain kind of ion channel changes, there generally occurs a change in more than one channel current, accompanied by a change in the membrane potential, i.e. hyper- or hypo-charge of the membrane capacitance. What interests us most is what ion channels mediate a certain cellular electric activity caused by a certain factor such as alteration in V_m or a chemical (ligand) stimulus. Some ion channels are voltage dependent, that is, their permeability changes with the alteration of V_m . By holding and changing V_m artificially to measure the currents through these channels we can get the details of their voltage-dependent characteristics. On the other hand, many channels such as ligand gated ion channels and G-protein coupled ion channels are chemical dependent, and also respond to V_m alteration passively (i.e. voltage independent: the response current is linear with V_m alteration with the permeability or conductance of the ion channel remaining constant) or even actively (i.e. voltage dependent as explained above). Under normal circumstances, the activity of these channels caused by a chemical is accompanied by the change in membrane potential. To distinguish between the effect of a chemical and that of V_m alteration, it is necessary to fix V_m at a constant level to remove the influence of the V_m alteration, or at different levels stepwise to observe the amplitude and configuration of the channel current traces for analyzing its ionic mechanisms. The following is a further discussion using some concise and explicit formulae.

V_m is set at a constant level. Once V_m is set constant, the capacitive current is zero. This means that the current recorded via the recording pipette equals the transmembrane current (current carried by ions flowing through channels and ions transported by pumps); or briefly speaking, the current recorded is the transmembrane current. In this case, the current via each kind of ion channel can be expressed as

$$I_j = (V_m - E_j)g_j$$

And the active current caused by the activity of this kind of channel is

$$\Delta I_j = (V_m - E_j)\Delta g_j$$

The whole transmembrane current (including that carried by ion pump s) is

$$I = \sum (V_m - E_i)g_j + \sum I_{s-pump}$$

and the whole active transmembrane current is

$$\Delta I = \sum (V_m - E_j)\Delta g_j + \sum \Delta I_{s-pump}$$

It is clear that the term $\Sigma(V_m - E_j)\Delta g_j$ represents the current response caused by the activity of ion channels, and the term $\Sigma\Delta I_{s-pump}$, by the activity of ion pumps.

V_m is set dynamically from one definite level to another. If V_m clamped jumps from an initial level to a new one, that is, an artificially controlled step voltage $g\Delta V_m$ occurs across the cell membrane, and it causes the activity of some ion channels, the transmembrane current will change from $\Sigma(V_m - E_j)g_j$ to $\Sigma(V_m + \Delta V_m - E_j)(g_j + \Delta g_j)$ (pump currents are omitted here for simplifying the discussion), and the alteration of the transmembrane current will be

$$\Delta I = \Sigma \Delta V_m g_j + \Sigma (V_m + \Delta V_m - E_j)(\Delta g_j)$$

The term $\Sigma\Delta V_m g_j$ is the linear transmembrane current change caused by the step voltage, i.e. the passive current response. It can be removed using leak subtraction function for a better observation of the current change caused by the activity of ion channels, i.e. the active current response, which is represented by the term $\Sigma(V_m + \Delta V_m - E_j)(\Delta g_j)$.

Based on the above important concept, the reversal potential V_{rev} is easily accessible. It generally refers to the membrane potential at which the active current response to a kind of stimulus disappears, and beyond which it reappears but flows in the opposite direction. So, let $\Sigma(V_m - E_j)\Delta g_j$ be zero in case (1), or let $\Sigma(V_m + \Delta V_m - E_j)(\Delta g_j)$ be zero in case (2), and we arrive at: $V_{rev} = (\Sigma E_j \Delta g_j) / (\Sigma \Delta g_j)$ (pump currents are omitted, and note that V_{rev} refers to V_m in case (1) and to $V_m + g\Delta V_m$ in case (2)). This is a general formula for discussing reversal potential, and it is helpful in exploring the mechanism underlying a current response that has been observed.

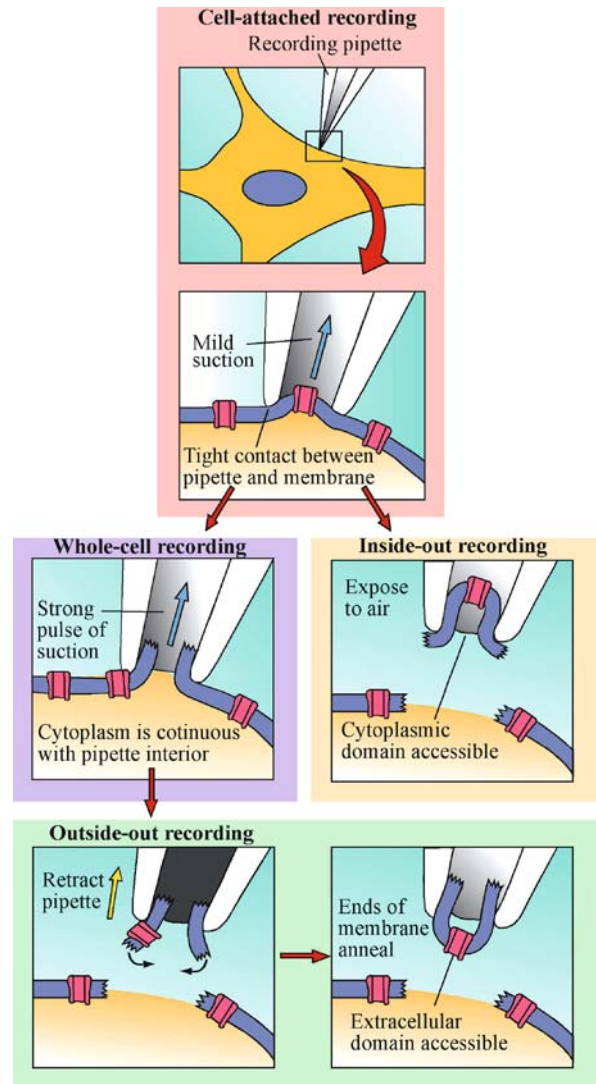
Four Configurations

Figure 2 shows the four configurations of patch clamp technique. They are discussed as following.

On-Cell Patch Clamp

A cell-attached (single-channel) voltage clamp is a form of patch clamp when the voltage of only a small part of the cell is controlled allowing current flowing through individual ion channels to be distinguished.

This technique is based on a very simple idea. A pipette with a very small opening is used to make tight contact with a tiny area, or patch, of neuronal membrane. After the application of a small amount of suction to the back of the pipette, the seal between pipette and membrane becomes so tight that no ions can flow between the pipette and the membrane. Thus, all the ions that flow when a single ion

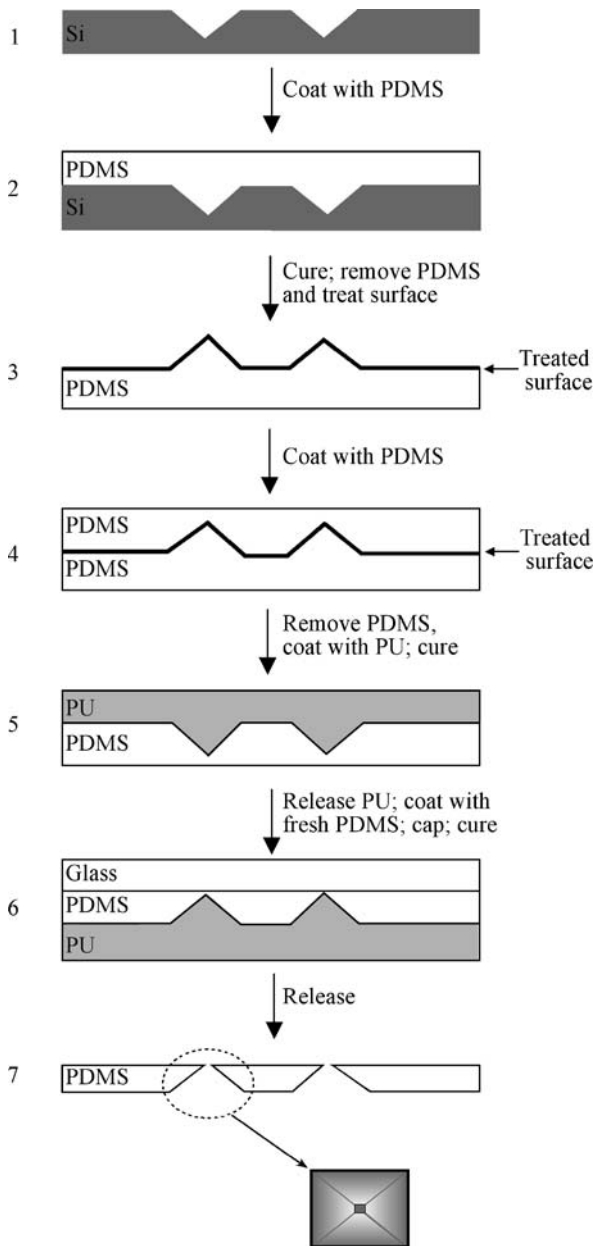


Patch Clamp Measurements On-Chip, Figure 2 Four configurations of patch clamp

channel opens must flow into the pipette. The resulting electrical current, though small, can be measured with an ultrasensitive electronic amplifier connected to the pipette. Based on the geometry involved, this arrangement usually is called the cell-attached patch clamp recording method. As with the conventional voltage clamp method, the patch clamp method allows experimental control of the membrane potential to characterize the voltage dependence of membrane currents.

Whole Cell Patch Clamp

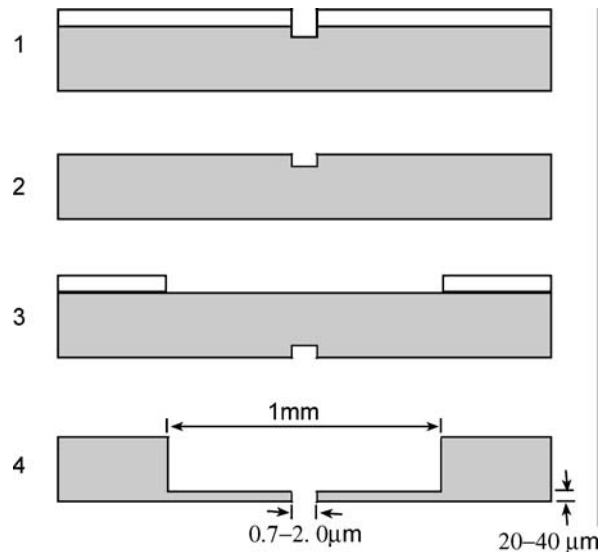
A whole cell voltage clamp is a form of patch clamp where electrical access is gained to the whole of the cell (usually



Patch Clamp Measurements On-Chip, Figure 3 Multi-pipette tip on-chip fabrication

by disrupting the membrane underneath the pipette by suction), allowing the electrical activity of the whole cell to be measured.

If the membrane patch within the pipette is disrupted by briefly applying strong suction, the interior of the pipette becomes continuous with the cytoplasm of the cell. This arrangement allows measurements of electrical potentials and currents from the entire cell and is therefore called the whole cell recording method. The whole cell configura-



Patch Clamp Measurements On-Chip, Figure 4 The steps in fabricating a silicon substrate for use as a planar patch clamp. Circles ($2\text{ }\mu\text{m}$ diameter) were defined in a photoresist (white layer) using optical lithography and then etched to a depth of $20\text{--}40\text{ }\mu\text{m}$ using deep reactive ion etching. The SU-8 resist was then stripped off with an acid bath and the wafer was inverted for back-side processing. A 1 mm diameter well, centered on the $2\text{ }\mu\text{m}$ pore, was defined using optical lithography. Deep reactive ion etching was used to etch the 1 mm well to a depth such that a small pore was formed connecting the base of the well to the other side of the wafer. A $0.5\text{--}1\text{ }\mu\text{m}$ thick SiO_2 film was then deposited over the surface of the entire wafer (using PECVD). This film electrically isolates the front side of the wafer from the back and reduced the pore diameter to $1.5\text{--}1.8\text{ }\mu\text{m}$. The deposition conditions could be varied to deposit thicker oxides and thereby reduce pore diameters as small as $0.7\text{ }\mu\text{m}$

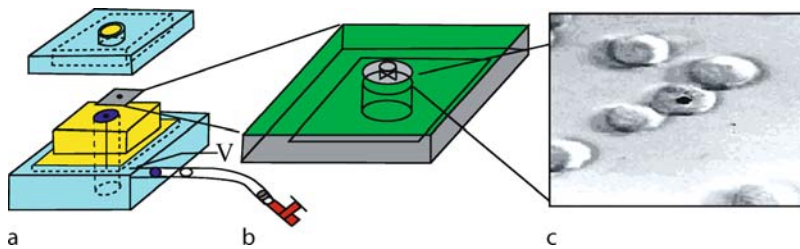
tion also allows diffusional exchange between the pipette and the cytoplasm, producing a convenient way to inject substances into the interior of a *patched* cell.

This is a powerful technique for studying ionic currents at the level of the whole cell membrane.

Inside-Out Patch Clamp

Once a tight seal has formed between the membrane and the pipette, small pieces of membrane can be pulled away from the cell without disrupting the seal; this yields a preparation that is free of the complications imposed by the rest of the cell. Simply retracting a pipette that is in the cell-attached configuration causes a small vesicle of membrane to remain attached to the pipette. By exposing the tip of the pipette to air, the vesicle opens to yield a small patch of membrane with its (former) intracellular surface exposed. This arrangement is called the inside-out patch recording configuration.

This is a true single-channel patch clamp mode allowing the measurement of single-channel currents with the



Patch Clamp Measurements On-Chip, Figure 5 Assembly of the patch clamp device. (a) Aligning and sealing the processed silicon chip (grey wafer) between an upper and lower chamber (yellow and blue) electrically isolates one side of the chip from the other (except through the pore in the silicon), and enables independent experimental manipulation of each side. The top and bottom chambers are composed of cured PDMS (yellow) coupled to machined polycarbonate (light blue) supports. The bottom polycarbonate support interfaces the lower PDMS vertical channel to a syringe which is used to apply negative pressure. (b) Except for the area immediately surrounding the pore on the topside and the 1 mm well on the underside of the chip, the entire top and bottom surfaces of the silicon wafer were stamped with SU-8 photoresist (green) to enable sealing to the PDMS layers and reduce the chip capacitance to as low as 15 pF. (c) A single RIN m5F cell is drawn onto the pore (black spot) following suction. A few seconds after landing on the pore, an individual cell tends to center itself over the pore. This image is of a chip that was recycled after being previously used to patch clamp a cell

added benefit of making it possible to change the medium to which the intracellular surface of the membrane is exposed. Thus, the inside-out configuration is particularly valuable when studying the influence of intracellular molecules on ion channel function, like calcium or ATP.

Outside-Out Patch Clamp

If the pipette is retracted while it is in the whole cell configuration, a membrane patch is produced that has its extracellular surface exposed. This arrangement is called the outside-out recording configuration.

This is a true single-channel recording mode. The patch can be superfused on the extracellular side with solutions containing materials such as neurotransmitters, agonists like GABA or glutamate to activate receptor channels in the patch. This range of possible configurations makes the patch clamp method an unusually versatile technique for studies of ion channel function.

Examples of Patch Clamp On-Chip Design and Fabrication

Multi-Pipette Tip On-Chip Fabrication Based on Polydimethylsiloxane (PDMS)

To scale up fabrication of the planar PDMS, as a typical example, a 4×4 on-chip array described in literature is given here.

The 4×4 array of $400 \mu\text{m}$ deep, self-terminated pits spaced 1 mm apart was anisotropically etched in a silicon wafer. The strategy was to use the precision of silicon micromachining to create a master structure with good ordering that would define a shape similar to the tip of a glass micropipette in a thin sheet of molded PDMS. That shape was then transferred through two PDMS molding steps into an array of pyramids in polyurethane (see Fig. 3). An

important step in forming the PDMS in step 4 of the figure was the treatment of the first PDMS mold so that the two PDMS layers could be separated. The final PDMS electrode partition was formed by pouring freshly prepared PDMS onto the polyurethane mold and then capping the polyurethane pyramids with a glass slide (step 6). The contact between the tip of the polyurethane pyramid and the glass slide excluded the PDMS to form small apertures. The size and shape of the aperture depended on the amount of contact with the glass slide. The authors chose polyurethane because it deforms sufficiently when contacting the glass slide to make holes $4 - 20 \mu\text{m}$ in size. For patch clamp measurements the PDMS planar electrode array was plasma-oxidized then mounted in the recording system shown in Fig. 6. To electrically isolate each electrode, the partition was sealed onto the chamber bottom over an array of openings that each contained an Ag–AgCl wire and suction tubing. The Ag–AgCl wire was connected to the multiplexer chip mounted on the circuit board under the chamber. The multiplexer then connected to the input of the amplifier. The tubing connected to a suction manifold that permitted individual or global suction.

The electrode resistance values were measured to confirm that the electrodes were electrically isolated and that the apertures were open. Fig. 6 shows a *macro-patch* recording obtained when a devitellinized oocyte, expressing Shaker potassium channels, was dropped onto an aperture. The ionic currents carried by a population of ~ 80 Shaker channels were measured in response to positive step changes in membrane potential. The overall success rate for forming a gigaseal on a xenopus oocyte using a planar PDMS patch electrode was 13% (10 out of 75). This is lower than the $\sim 40\%$ success rate for the conventional patch electrodes with or without PDMS coating and surface treatment.

In general, a planar electrode has many advantages over the conventional patch pipette electrode. First, the planar electrode can be easily scaled to an array of electrodes to permit multi-cell or multi-site measurements. Second, the planar electrode has the potential for integrating microfluidic lines that would permit exchange of the electrode solution, something rarely achieved with a micropipette. The amplifier electronics can also be integrated into the substrate of the electrode for improved noise performance. Third, a planar electrode can be mounted over a fluorescence microscope to simultaneously record electrical and optical signals.

Also, micromolding a planar electrode from PDMS has advantages over the other microfabricated planar electrodes under development. First, the fabrication is simple and economical so that the electrodes are disposable, therefore eliminating the difficult step of cleaning the device. Second, microfluidic lines can be micromolded in the same structure or added by bonding PDMS layers. Third, PDMS is optically transparent so that alignment to underlying components can be done visually. The transparency of the material may also be advantageous when integrating an optical recording system. Fourth, the PDMS surface can be chemically modified to facilitate seal formation.

However, the probe nature of the conventional patch pipette still has its advantages. Most importantly, it can penetrate the top layers of tissue to permit patch clamp measurement on a cell within a tissue. Second, the electrode can be pulled away from a cell to make an *excised patch* recording, useful for applying modifying agents to the patch of membrane. Third, the pipette can be moved quickly between lines of solution flow for quick changes in modifying agents. Integrated microfluidics may negate the need for these last two points but more complicated micromachining may be required to fabricate planar patch electrodes that can penetrate tissue.

The need to plasma-treat the PDMS surface just before use may be a disadvantage of the planar electrode. This would add a plasma system to the required equipment for patch clamp experiments. However, the planar electrodes do not require use of the high-quality optical microscope and micromanipulators currently required for patch clamp recording. In addition, conventional patch pipettes commonly require several processing steps before use including heat polishing and coating.

Patch Clamp Device On-Chip Fabrication

As another typical example, a method for silicon chip-based patch clamp electrodes integrated with PDMS microfluidics is given in the following.

Silicon Chip Microfabrication. Up to 80 silicon chips to be used for patch clamping experiments were parallel fabricated in a single sequence of steps (Fig. 4) using stand-alone processing tools. Patterns for pores were defined using a photoresist in concert with a custom-designed mask and a mask aligner (operating with 365 and 405 nm radiation lines and in constant power mode). The patterns were developed to etch out the pores using a plasmatherm deep reactive ion-etching system. In the region surrounding the pore, the silicon wafer was thinned to about 20–40 μm for the micropores (1.5–2 μm diameter) and 10–20 μm for the nanopores (0.7–1 μm diameter). Each wafer was fabricated with a small array of usable pores. In the following, only one device was used for any given experiment, and so the chips were diced into 8×8 mm sections, each containing a single pore. The chips were acid cleaned, rinsed using copious amounts of 18 M Ω H₂O and dried under flowing nitrogen. The chips were then coated with a 0.5–1.0 μm thick insulating film of SiO₂ using plasma-enhanced chemical vapor deposition (PECVD). The top side of the chip was exposed for 20–40 min and the bottom side was exposed for 10–20 min. This procedure shrunk the pore diameters by about 0.4 μm . For the nanopores, the oxide was deposited at lower pressure. Deposition at lower pressure was necessary in order to increase the oxide density, deposit a thicker layer of SiO₂, and to conserve the circular symmetry of the nanopores. At this point, the tip fabrication process was essentially complete.

PDMS Microfluidic Fabrication. Fluidic platforms for both the front and back sides of the wafer were prepared from micromolded PDMS. For most measurements, the chambers simply allowed for separate chemical access to the front and back sides of the micropore. The minimum microfluidic channel widths were 100 μm and the typical channel heights were 100 μm . A mold master was photolithographically patterned into a spin-coated film of photoresist on silicon as described in the literature. The microfluidic channels that were used for cell focusing consisted of two layers of micromolded PDMS that formed one final microchannel network component. The bottom layer of the PDMS microchannel network defined the focusing channels which interfaced the silicon substrate, and the top layer interfaced the focusing channel to larger channels that had inlet and outlet ports. Modified syringe needles were pierced through the top layer to connect the input and output ports with Teflon tubing for the introduction and extraction of cells and solutions. The focusing channel was designed to allow for optical access to the micropore region when the syringes were in place.

Device Assembly. The molded PDMS fluidic chambers were pressure bonded to the silicon chips (Figs. 5a,b). This chemically isolated one side of the pore from the other,

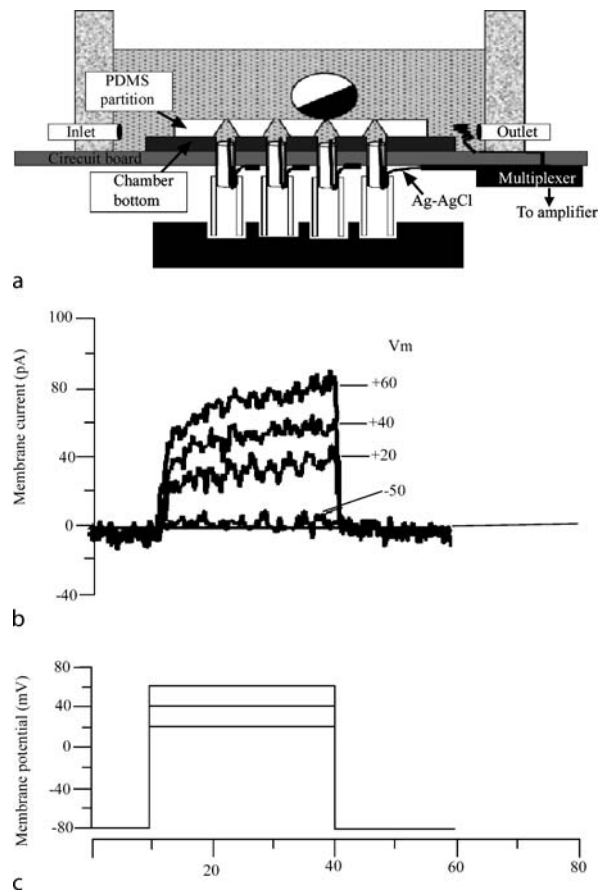
thereby creating a device that enabled independent experimental control of each side of the pore (or membrane sealed over the pore). To prevent electrical leak around the silicon chip, SU-8-5 photoresist was applied around the edges of the surface of the chip with a cotton swab (Fig. 5b) or by transferring the SU-8-5 with a PDMS stamp, and then cured. This coating not only permitted a quick, greaseless method for sealing the chip surface to the PDMS surfaces of the experimental cell-chip chamber (Fig. 5a), but it also reduced the chip capacitance. Chip capacitance values as small as 15 pF were achieved when the stamping technique was utilized. However, for the devices that were integrated with PDMS microchannels, little or no SU-8 resist was applied, since the PDMS channels themselves prevented solution leak around the silicon chip and minimized stray capacitance. In order to recycle the same batch of chips several times, the SU-8 resist and oxide layers were sequentially stripped away using standard oxidizing solutions. First, an oxidizing clean (2:1 concentrated $\text{H}_2\text{SO}_4\text{:H}_2\text{O}_2$ (30 %)) was used to remove the SU-8-5 resist. Next, the oxide was stripped using a buffered oxide etching (BOE) solution. The oxide layer was then re-deposited using PECVD.

Electrophysiology recordings. To record currents from sealed cells, the silicon chip assembly was used as a chamber for a voltage clamp setup (Fig. 6). Each half of the chamber (top and bottom) was connected to the electrodes of a headstage. The top chamber was connected to the input of the patch clamp amplifier and the bottom chamber was connected either to either the ground or the command voltage depending on the amplifier used.

The silicon chip device, coupled to a voltage clamp setup was prepared for cells by adding external and internal solution to the top and bottom chamber, respectively. An amount of 50 μl of cell solution (about 5×10^4 cells) was introduced into the top chamber with a syringe that had a 2 cm piece of Teflon tubing on its tip. The cells were tracked by viewing the top surface of the chip with a microscope and by monitoring the electrical resistance across the pore. Once the cells covered 20–30 % of the chip surface, gentle suction was applied from the bottom chamber side. The progress and quality of the seal of a cell over the pore was monitored visually through the microscope and by measuring the capacitance and resistance across the pore. Once the seal reached maximum resistance, the suction was released and electrophysiological recordings from the cell were made.

Key Research Findings

The inherent complexity of the technique limits its uses in high-throughput screening for drug discovery and drug



Patch Clamp Measurements On-Chip, Figure 6 Planar PDMS patch electrode array recording of Shaker potassium channel currents. (a) Schematic of planar patch electrode array recording system. The 400 μm thick, oxidized PDMS partition is sealed onto the chamber bottom with vacuum grease. The chamber bottom contains an array of openings, each containing tubing and an Ag–AgCl wire. The tubing connects to a suction manifold and the Ag–AgCl wire connects to a multiplexer chip on the circuit board under the chamber bottom. The multiplexer connects each contact to the amplifier electronics. Another Ag–AgCl wire in the bath solution connects to the ground of the amplifier. Bath solution is exchanged through solution lines into and out of the bath chamber. A devitalized oocyte is dropped onto the aperture to make a patch clamp recording. (b) Current traces recorded from a xenopus oocyte expressing inactivation-removed Shaker potassium channels. Each trace represents a step change in membrane potential from rest, 80 mV. Voltage steps are to 50, 20, 40 and 60 mV, as noted, before returning to 80 mV. Leakage currents were subtracted from the raw data by a P/4 protocol (Penner, 1995).

safety. To meet high-throughput screening requirements, many efforts have been made to improve the patch clamp system. Planar patch clamp electrodes, scalable and easy to use, have been fabricated using silicon oxide-coated nitride membranes, deep reactive ion etched silicon holes coated with PECVD oxide, polyimide films, track-etched quartz, silicon oxide nozzles, glass substrates and oxygen

plasma-treated PDMS. Microfluidics have also been integrated with patch clamp systems using PDMS or glass. Those devices have made possible fast fluid exchange for single electrodes.

Some patch clamp setups with more advantages have been developed to be used more easily. For example, a lateral-based patch clamp setup is compatible with high-resolution microscopy (with a working distance in the 100–300 μm range). The idea is to trap a cell onto a lateral microchannel in a manner which makes it easier in terms of fabrication and fluidic integration. Niels Fertig has made use of state-of-the-art microstructuring techniques to define three-dimensional micro-openings in planar substrates, with the geometry of an inverted patch clamp pipette tip.

Future Directions for Research

The on-chip-based patch clamp technique broadens the possibilities of ion channel analysis by transferring the classic patch clamp principle onto a microstructured chip format. The planar patch clamp technique enables the automation and parallelization of ion channel measurements. This is an example of the beneficial integration of microtechnological processes in the field of cellular electrophysiology. The chip-based patch clamp technique not only improves ease of use as well as throughput capability compared to conventional patch clamping, but also opens new experimental possibilities for the biophysical characterization of ion channels such as intracellular perfusion or optical accessibility of the chips for combined optical and electrical recordings from ion channels. The automated and parallel patch clamp methods should be further developed in the near future for academia and industry.

Cross References

► Ion Channel

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Patch Clamp Technique

► Patch Clamp Measurements On-Chip

Patch Voltage Clamp Technique

► Patch Clamp Measurements On-Chip

PCR Lab-on-Chip Devices

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Synonyms

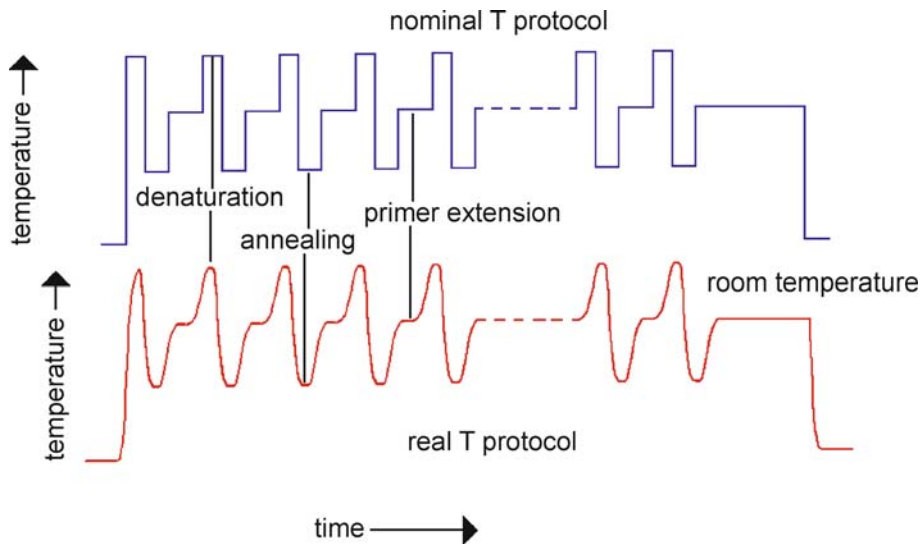
Microthermocyclers; Chip thermocyclers

Definition

PCR Lab-on-Chip devices are microfluidic instruments for the biochemical amplification of DNA by thermocycling using the polymerase chain reaction at the microscale.

Overview

PCR Lab-on-Chip devices are miniaturized fluidic systems for fast DNA amplification. The polymerase chain reaction (PCR) is a molecular biological technique for addressing and exponentially amplifying a DNA fragment. The PCR process is based on temperature cycles with an enzymatic amplification step. It works without the use of a living cell. For PCR, a liquid mixture of an analyte



PCR Lab-on-Chip Devices, Figure 1 Temperature protocol for PCR with repeated thermal addressing of process steps; nominal and real temperature tracks (double-strand separation, primer annealing, primer extension)

containing DNA with an amplification buffer solution containing nucleoside triphosphates (building units for DNA synthesis), primers (starter molecules), catalyst (the enzyme polymerase) and some additives is applied. The exponential increase of concentration of the sequence of interest is due to the doubling during each temperature cycle. The synthesis of a daughter strand of DNA on a mother single strand (template) occurs by template-directed condensation of nucleotides. The primers are applied for addressing a selected sequence part of DNA template molecules from the analyte sample carrying the genetic information of interest. Very small amounts of interesting DNA are required. Under optimized conditions, a single template DNA molecule is sufficient for starting the process. DNA fragments up to about 10,000 base pairs can be amplified by PCR.

The addressed sequence is identically replicated to a large extent during the thermocycling process. The miniaturization by transferring the technology from classic block thermocyclers into Lab-on-Chip devices is associated with a miniaturization of sample and with a large reduction of process time. In addition, Lab-on-Chip PCR allows the integration of sample preparation and post-amplification treatments including labelling and/or separation in order to realize a completely automated process chain in a miniaturized system. DNA microanalysis systems can be built, in which the micro PCR is the key operation for getting the molecular information carrier in a concentration suited for convenient signal readout.

Amplification of DNA by Lab-on-Chip PCR is mainly used for genotyping. In connection with a reverse tran-

scription process using RNA-dependent DNA polymerase, this technique can also be applied for the characterization of RNA, in particular for the identification and measurement of messenger RNA. As a result, the Lab-on-Chip PCR devices can be also applied for gene expression studies.

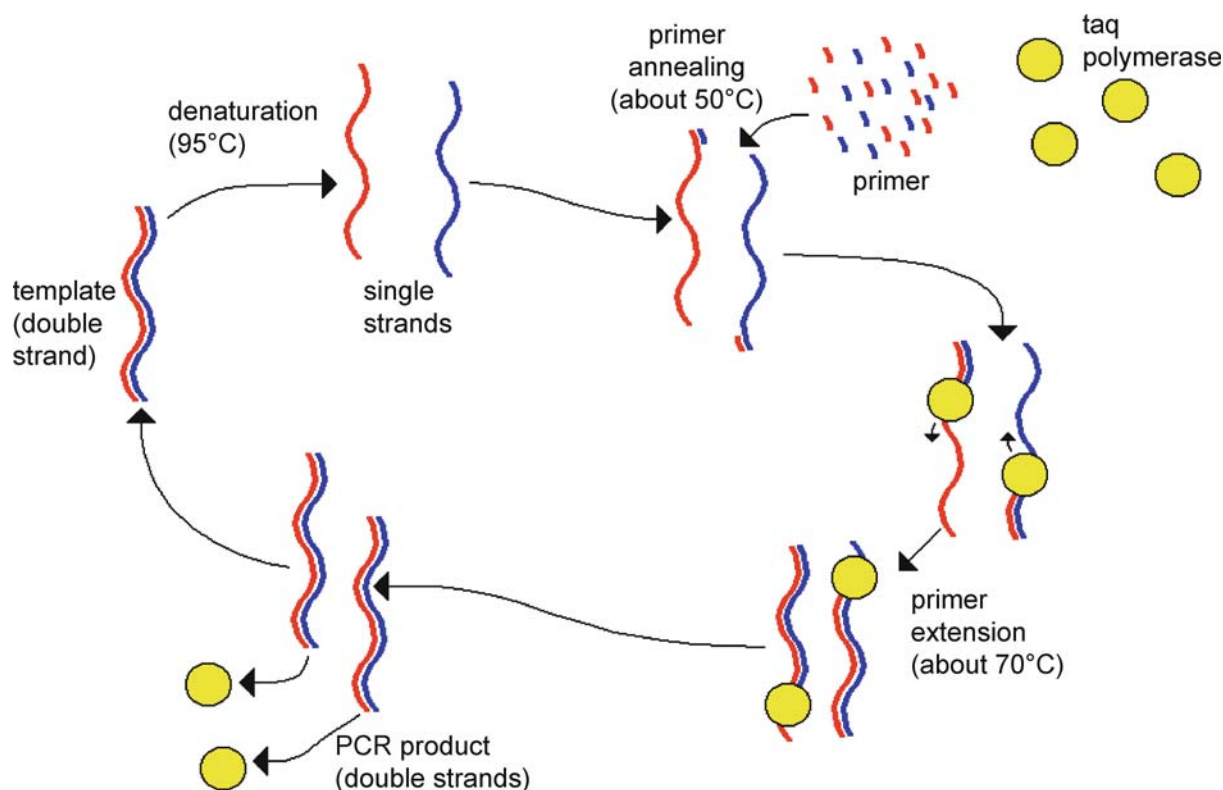
The application of PCR chip devices is connected with several important advantages with respect to conventional systems:

- low sample volumes
- very fast process steps
- low power consumption, suited for point-of-care diagnostics
- well reproducible temperature protocols
- processing of larger sample series under the same conditions
- low space requirements of devices
- integration of functions like reverse transcription process (for gene expression studies), on-line measurement of double-strand DNA concentration or real-time measurement by multi-sensor arrangements.

Hence, miniaturized PCR devices are very interesting for a broader application of PCR in laboratories as well as for point-of-care diagnostics, screenings or investigations for food safety and environmental protection.

Basic Methodology

The basic technique is the PCR invented by K. Mullis in 1983. This technique allows one to replicate DNA identically without any living cell. The PCR has developed



PCR Lab-on-Chip Devices, Figure 2 Principal molecular processes of PCR during thermocycling

into a common technique used in medical and biological research and diagnostics as well as in forensic, archaeological and other investigations. It is the key method for sequencing of genes, for the diagnosis of hereditary diseases, for paternity testing and for molecular identification of infectious microorganisms.

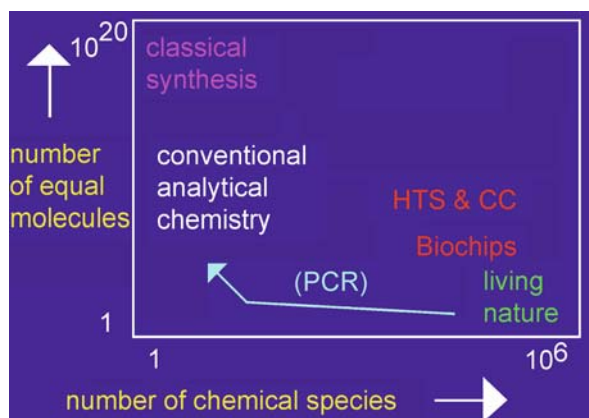
The PCR technique succeeds by the use of the DNA-dependent DNA polymerase from bacteria living in hot sources (taq-polymerase). This enzyme is able to elongate primers (oligonucleotide starter molecules) in a complementary pattern to a template DNA single strand at a temperature of about 70 °C. This high temperature is necessary in order to create a temperature-controlled three-step process. The whole process includes the following steps:

1. the initialization step for denaturation of DNA double strand into two single strands at about 95 °C by thermal disruption of the hydrogen bonds between the complementary bases in the double strand;
2. the addressing of a short well-known sequence pattern by base-pairing of primers (*primer annealing*) at suitable temperatures in the range between about 45 and 65 °C depending on primer length, AT/CG ratio and desired selectivity; and

3. the elongation of the primers with the complementary order of nucleotides by the synthesis of the new DNA strand by the enzymatic activity of the taq-polymerase (*primer extension*).

Two suitable primers are applied in order to generate both new complementary single strands of a selected DNA sequence in the replication during one temperature cycle. The repeated application of temperature (Fig. 1) cycles leads to a fast increase of the concentration of the selected DNA sequence by doubling the DNA in each cycle (Fig. 2). As a result, the selected sequence can be amplified by a factor of about 1000 in 10 temperature cycles, by a factor of about 1 million in about 20 temperature cycles and about 1 billion in 30 temperature cycles. This amplification is high enough for all characterization techniques like fluorescence detection, electrophoretic characterization, sequencing or chip hybridization. PCR bridges the gap between the molecular diversity in living nature using small numbers of key molecules and the analytical tools of chemistry and sensorics (Fig. 3).

Classic PCR is performed in Eppendorf tubes, microtiter-plates or similar repositories by conventional thermocyclers with thermostatted blocks. The typical time required



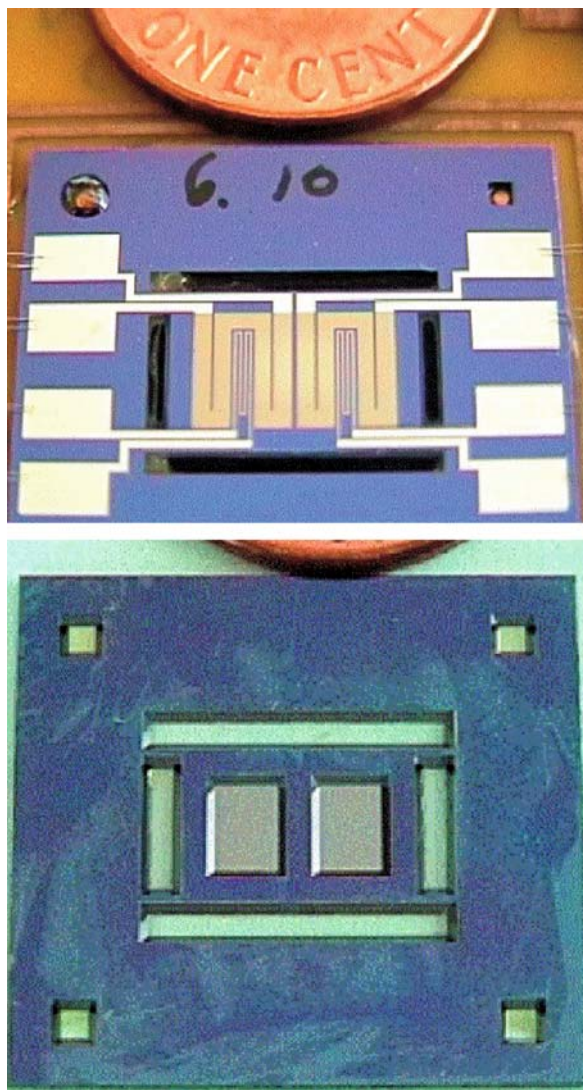
PCR Lab-on-Chip Devices, Figure 3 Schematic of PCR as gap bridging technology between the biomolecular and chemical worlds: number of molecular species (types) and number of molecules of the same type in chemical and biological systems

is in the range of about 90 min up to about 3 h. Faster working thermocyclers need powerful heaters as well as coolers in order to realize high heating and cooling rates.

The temperature control can be realized by alternating mechanical contact to heating/cooling plates, by fluidic switching between heating and cooling liquids or by Joule heating using thick-film or thin-film microheaters. The latter method is frequently combined with passive cooling by heat conduction or convective cooling. In several chip devices, microheaters and thermal microsensors are prepared in close proximity in order to realize efficient feedback loops in electrical heating.

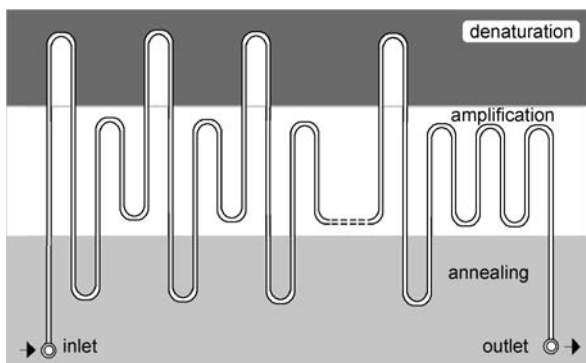
Despite the high amplification factor, the volumes can be small that have to be operated in thermocycling for PCR. So, DNA amplification can be done with volumes in the lower μl or in the nl range. The reduction of sample volume gives the possibility to apply chip devices or other miniaturized systems for the PCR. The reduction of size allows a crucial enhancement of heating and cooling rate and at the same time a reduction of power consumption. Power reduction and process rate enhancement become possible due to the lowering of parasitic heat capacitances and by efficient transfer of heat inside the system. So, miniaturized PCR devices can work at high process speed with low power requirement. Therefore, they are particularly applicable for hand-held devices, mobile analytical systems and microautomated systems.

Microtechnology offers two different strategies for Lab-on-a-Chip PCR. The first is the stationary concept. This concept corresponds to the classic thermocycler. The core element is a miniaturized chamber for PCR liquid equipped with heaters and temperature sensors (Fig. 4). Fast PCR is realized by fast heating and cooling the whole



PCR Lab-on-Chip Devices, Figure 4 Example of a silicon chip thermocycler (IPHT Jena, Germany): microlithographically patterned thin-film transducers at the back side (above), liquid chambers and air chambers for heat transfer control at the front side (below)

system. The maximum process rate is mainly determined by the ratio of heating and cooling power to the heat capacity of chamber and sample. The second strategy is based on a microflow-through concept. The device contains a long microchannel, which is conducted in meanders through three temperature zones (Fig. 5). Thus, the three regions of the device remain at constant temperature, and the temperature cycling is realized by changing contact of the liquid with the three different temperature zones. The maximum process rate is mainly determined by the heat capacity of the streaming liquid itself and by the heat conduc-



PCR Lab-on-Chip Devices, Figure 5 Principle of construction of a flow-through chip thermocycler

tivity between sample and walls. In principle, the time for single temperature cycles can be reduced down to a few seconds. The whole PCR process with 20–40 cycles can be done in time scales between a few minutes and about half an hour.

The acceleration of the PCR process is a very important advantage of PCR Lab-on-a-Chip systems. A second advantage is in the possibility to integrate further functions into the microdevice. Thus, steps of sample preparation as well as separation and detection supporting process steps can be integrated. In this way, miniaturized DNA analysis systems are constructed, in which the Lab-on-Chip amplification by PCR is the central process.

Key Research Findings

The motivation for miniaturization of PCR was driven by several factors. First, PCR should be made faster in order to get results more quickly. Second, larger sets of samples should be processed. Third, DNA-based diagnostics should be evolved from a laboratory technique requiring experienced specialists to an automated on-site diagnostic method operated after brief instructions by anybody. Finally, the amplification by PCR should be integrated in a diagnostic chain (μ -TAS concept) and implemented in a miniaturized transportable system. The power consumption of the thermocycling process should be as low as possible.

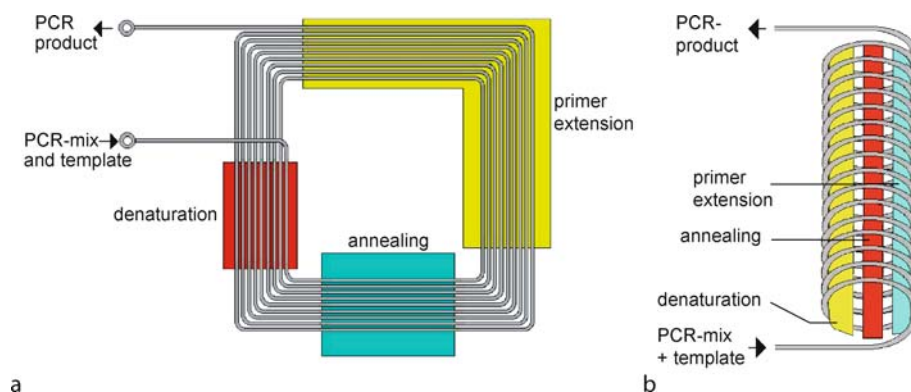
The simplest way for meeting the requirements mentioned above and realizing micro-PCR on-a-Chip consists in a transfer of the conventional PCR technique into a microchamber chip. Therefore, microchambers for liquid volumes between a few μ l and several hundred nl were etched anisotropically in silicon wafers. A lithographically micropatterned metallic thin-film system at the backside of the wafer was applied in order to control temperature

by Joules heating and thermoresistive temperature measurement. Metal film thicknesses of about 1 μ m or several hundred nm were sufficient for achieving all temperatures necessary for PCR. The bottom membrane of the chamber carrying the metal transducers was built by the residual silicon thickness after time-controlled chamber etching. The high heat conductivity of the material guaranteed a fast heat transfer between the thin-film transducers and the liquid.

The cooling steps of PCR could be realized by direct contact of the bottom membrane with a cooling element (e.g. a Peltier cooler) or by indirect cooling using the passive heat transfer from the silicon chamber to ambient. For the second method, the fine tuning between temperature homogeneity in the core region of the PCR chamber and temperature gradients between core region (with PCR chamber) and the outer frame region of the microsystems were of particular importance. The problem was solved by the definition of spatially restricted areas (*heat conduction beams*) for passive heat transfer between core and frame by etching holes into the silicon. Optimized chips supplied high temperature homogeneity in the core region of PCR chips and heating and cooling rates of 40–80 K/s. Cycle times could be downscaled to a few seconds. The time required for the whole PCR process could be reduced down to 30–10 min.

An important advantage of microchamber chips, beside the good dynamics, is the flexibility. Temperature/time protocols can easily be adapted and varied. Thus, this technique is useful for the development and optimization of protocols.

Microtiterplates (MTPs) are the most commonly used chamber arrays for PCR. The open chambers have to be covered by oil or sealed by membranes in order to suppress evaporation of water during the process. A fast working PCR in arrays can be realized, if small chamber volumes are applied and the wall thickness of the plate material is thin enough for fast heat transfer. Such chamber arrays can be prepared by in situ forming of array plates by means of a thermostatted block suitable as metal master. A further miniaturization of chamber volumes is possible by using nanotiterplates (NTPs). Microlithography allows the preparation of chambers in the lower μ l range as well as over the whole nl and pl range. There are technologies for preparation in silicon as well as in glass. Polymer carriers are made in many cases by replication from microstructured silicon or glass master chips. A wide spectrum of materials is available for NTPs. The thermostating can be realized by the conventional incorporation of the sealed NTPs in metal block devices or by integration of thermal transducers in the carrier systems. The application of micropatterned thin-film heaters



PCR Lab-on-Chip Devices, Figure 6 Micro-flow-through PCR in tube arrangements: (a) spiral-like arrangement (two-dimensional), (b) helical arrangement (three-dimensional)

and thin-film sensors for direct thermostating of NTPs is compatible with well-established microsystem technologies using glass and silicon substrates. It is more difficult in the case of polymer materials. The high conductivity of silicon is responsible for the attractiveness of using these materials for all processes connected with fast temperature changes. Silicon carriers are of particular interest for NTPs used in PCR. The comparatively high affinity of silicon against proteins is a disadvantage and demands special surface covering or chemical surface modifications (see below).

The power consumption for heating and cooling of samples can be reduced if the temperatures in the device are kept constant in three different regions and the sample is moved between these three regions. This movement can be operated at the surface of a thermostating chip, for example by electrostatic manipulation or by surface acoustic waves (SAWs). Therefore fluidic transport paths (hydrophilic areas) are defined lithographically in a hydrophobic environment on a piezoactive substrate like lithium niobate. The excitation of SAWs is realized by microlithographically prepared thin-film electrodes. SAW systems are of particular interest due to the fast manipulation and the very efficient mixing processes, which can be applied also for sample preparation. The small volumes of droplets make the evaporation of water a serious problem. This can be overcome to a certain extent by oil covering.

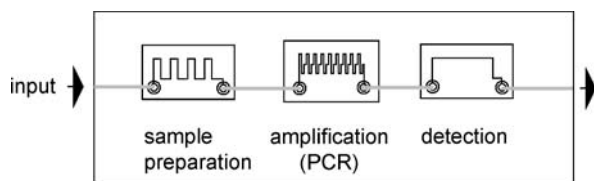
The evaporation of solvent can be avoided completely if closed microchannels are used for a micro-PCR with moving of samples between different temperature zones. The simplest variant consists of a flow-through PCR realized in a tube, which is conducted repeatedly through the three temperature zones. There are two principal geometries for the arrangement of tubes. In one variant the tube is coiled in one plane like a spiral (Fig. 6a). This geometry can be

easily realized. But, this arrangement suffers from increasing or decreasing residence times in all three temperature zones due to the changing diameter of the tube loops. The alternative is a three-dimensional, helical arrangement of the tube. In this geometry, the conditions at all temperature cycles can be kept equal (Fig. 6b).

A certain disadvantage of using tubes is the low heat conductivity of polymer tube materials and the comparatively high thicknesses of tube walls. This leads to low temperature gradients. As a result, the outside temperature must be adapted to the flow rate-dependent heat required inside the tube and the heat transfer conditions in the tube wall.

The principle of flow-through PCR can further be miniaturized and optimized with respect to heat transfer conditions by realizing flow-through chip thermocyclers. These are devices for flow-through PCR on silicon, glass or polymer chips. The use of silicon (high thermal conductivity) and glass (low heat conductivity) in one device allows convenient control of temperatures in the three process zones by choosing a suitable local heat capacity and heat exchange conditions. In analogy to chip chamber cyclers, etched holes in silicon support the thermal separation of temperature zones, and thin-film resistors are applied for heating and electroresistive temperature measurements.

DNA amplification can occur either in a continuous flow mode or in a serial flow mode. In the continuous flow mode, a template is mixed continuously with the PCR mix and transported through the flow thermocycler. This mode is well suited for preparing amplicates of a certain DNA sequence in constant quality. In contrast to continuous amplification, most PCR applications are found in analytics. Therefore, the enhancement of throughput of samples in one analytical system is the deciding parameter. In principle, series of template-containing samples can be applied in a certain clock pulse in a flow-through system. But, the subsequent injection of different samples in



PCR Lab-on-Chip Devices, Figure 7 Schematic of integration of PCR in a μ -TAS arrangement with sample pre-treatment and separation/detection unit

continuous flow leads to strong cross-talk effects due to the fluidic dispersion which is caused by the laminar flow behaviour in the microtubes and microchannels. This fluidic cross-talk is particularly critical due to the exponential character of the amplification process during PCR.

A crucial improvement of sample separation in serial processing can be achieved by the application of a segmented flow in the micro-PCR. The separation of aqueous fluid segments containing the PCR mix and the different templates by immiscible organic liquids converts the laminar flow to a plug flow. If the inner surface of the wall is hydrophobic, the chemical cross-talk between fluid segments can be suppressed completely. Thus, the segmentation of flow improves the transport behaviour as well as inhibiting effects of the wall. In addition, the plug flow guarantees constant residence times for all samples and all volume parts of one sample in all temperature regions and in all cycles. The segmented flow principle can be applied for both microtube PCR and on-chip PCR.

Serial PCR using flow segments allows comparatively high sample frequencies. The generation of segments can occur up to frequencies of some tens per second. Reconsidering a certain redundancy (several segments of the same sample), it is possible to process in the order of 10^4 samples per hour. This high throughput power of serial flow micro-PCR makes this technique interesting for screening procedures with well-standardized protocols. The fixed definition of temperature protocols is necessary because the length of capillaries or tubes in the different temperature zones is fixed. Changing of flow rate leads only to a proportional change in all temperature zones. The ratio of residence times is fixed by the design of the flow-through device. So, flow-through thermocyclers are much more efficient than chamber thermocyclers, but less flexible. Flow-through PCR is therefore preferably applied for well-defined analytical problems with high numbers of analogue samples.

A big additional advantage of flow-through processes in comparison with stationary processes consists in the possibility of implementation of process chains (Fig. 7). In this way, complete microanalytical systems can be designed.

They can include sample pre-treatment like cell separation, cell lysis or chemical separation. Reverse transcription processes can be directly connected on-chip with the DNA amplification by thermocycling (RT-PCR). The analytical chain can also include post-amplification steps like fluorescence labelling, separation by capillary electrophoresis or hybridization on DNA chips.

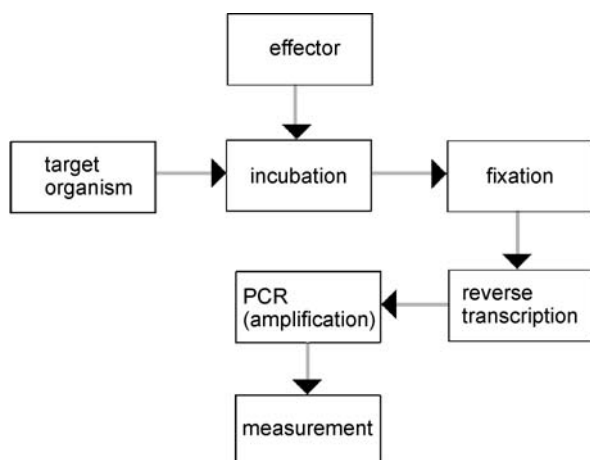
Chemical compatibility is a particularly important problem for Lab-on-Chip PCR devices. The taq-polymerase possesses a certain surface affinity due to their amphiphilic character. In addition, the low start concentrations of DNA for amplification are critical if the molecules are adsorbed at wall surfaces. The molecule-surface interactions are particularly critical due to the high surface-to-volume ratios in chip devices. The suppression of undesired adsorption demands inert materials. Polymers like polypropylene or polytetrafluoroethylene are much better suited than glass or silicon. The OH groups at the surface of native glass, silicon or silicon dioxide surfaces are mainly responsible for disturbing adsorption. The affinity of surfaces against the biomolecules can be suppressed by surface treatments. Therefore, alkylsilanization (by alkyltrichlorosilanes or alkyltrimethoxysilanes for example) is particularly suited. The surface properties change during the exposure of walls against cell suspensions and PCR solutions. Certain rinsing, cleaning and resilanization protocols are applied in order to achieve a high reproducibility in the case of reused Lab-on-Chip PCR systems.

Micro-flow-through PCR has been investigated in the last few years for several biological and medical applications. They relate to the identification of DNA in different research experiments, for genotyping of microorganisms, in particular the identification of pathogens, as well as for genotyping of tissue material, gene expression studies using a preceding reverse transcription step for conversion of messenger-RNA into DNA and for adaptation and development studies of insect embryos.

Future Directions of Research

Further research for Lab-on-Chip PCR devices is focused on four main topics: the improvement of the performance of the amplification process, the integration of new functions in PCR-based analytical chip devices, the integration of PCR in microautomated diagnostic systems and the extension of PCR-based nucleic acid microdiagnostics to new applications.

The improvement of heat transfer by optimized channel and wall geometries and the choice of best-suited materials contribute to the enhancement of process speed and to a better control of selectivity. Adapted selectivity can be achieved by realizing well-defined temperature proto-



PCR Lab-on-Chip Devices, Figure 8 Process scheme for use of Lab-on-Chip PCR for expression diagnostics in drug screening

cols for the primer annealing step. Therefore, steep cooling and heating ramps are of interest. Special material combinations and the adaptation of manufacturing technologies are under investigation for the improvement of speed and surface compatibility.

Sample throughput is particularly important for screening processes with large numbers of samples. Low-cost DNA diagnostic tools are necessary for biomedical screenings in order to identify genotype-determined health risks in the human population. They are also of interest for screenings for dangerous microbial genotypes. Despite biomedical, environmental and agricultural screenings, high sample throughput is also required for pharma screenings. Here, expression studies depending on drug concentrations and drug combinations demand low-cost solutions including fast RT-PCR (Fig. 8). Highly parallelized systems and in particular fast working serial PCR systems are of interest for such pharma screenings.

The specific fields of recent and future applications of rapid flow-through chip PCR demand the development of PCR-based chip devices with additional integrated functions for sample pre-treatment, on-line sensing, process control and real-time measurement, for referencing and reproducibility tests. Therefore, further efforts for integration of transducers as well as additional fluid manipulation functions are required.

Process speed, sample throughput as well as selectivity could be improved by ultra-miniaturization of Lab-on-Chip PCR. In nature, DNA amplification takes place in reaction volumes (cells) down to the sub-micrometre level. This corresponds to volumes in the sub-femtolitre range. In principle, the PCR process could be downscaled to this region. Recently, most PCR Lab-on-chip

devices are working with sample volumes in the sub- μl and in the nL range. By ultra-miniaturization, these volumes could be reduced by factors of between 10^{-3} and 10^{-7} . Therefore, special strategies of liquid handling and, in particular, microencapsulation techniques, vesicle techniques or ultra-miniaturized segmented flow techniques are necessary.

The very different possibilities of using information from nucleic acids for health care, drug development, environmental control, product and process safety, food control, control of trade and transport and so on demand cost-efficient special solutions with microsystems for DNA or RNA analysis. So, further research will address a wide spectrum of different special diagnostic solutions integrating Lab-on-Chip PCR. The questions of universality, reusability, safety and recyclability will become important, in future, like the questions of costs, operability, robustness and reliability.

Cross References

- ▶ Droplet-Based Lab-on-Chip Devices
- ▶ Lab-on-Chip Devices for Biodefense Applications
- ▶ Microfluidics for Bio- and Chemical Reactions
- ▶ Microsegmented Flow

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Peclet Number

Definition

Peclet number is a dimensionless number relating the rate of advection of a flow to its rate of thermal diffusion.

Cross References

- Polymer Synthesis within Microfluidic Reactor
- Microfluidics for Bio- and Chemical Reactions

Peltier Effect

Synonyms

Thermoelectric effect

Definition

The Peltier effect occurs when an electric current passes through two different connected materials (metals or semiconductors) and thereby generates a temperature difference across the materials.

Perfect Slip

- Boundary Slip of Liquids

Peristalsis

Definition

Peristalsis is transport in a flexible duct via traveling contraction waves. Peristalsis is the process by which material is moved through anatomical passages, for example of the digestive tract, by the contraction of smooth muscle in rhythmic waves. In biological systems, peristalsis moves liquids or solids. Micropumps based on the principle of peristalsis have been designed to move liquids and gases.

Cross References

- Linear Peristaltic Pumps
- Peristaltic Pumps
- Rotary Peristaltic Pumps

Peristaltic Pumps

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Synonyms

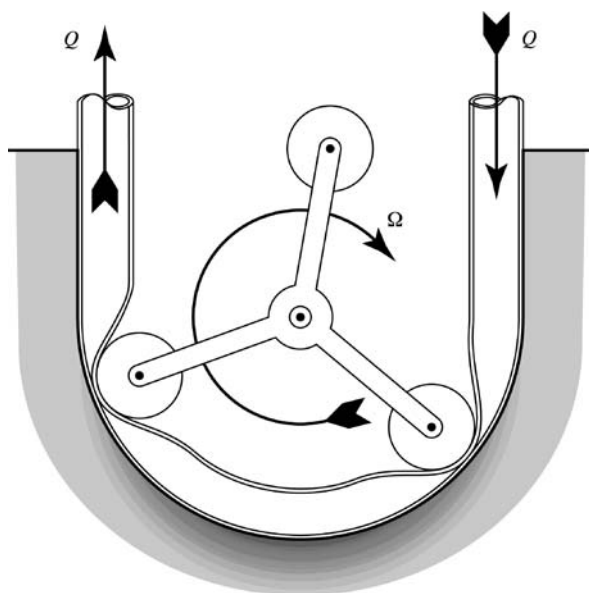
Flexible member pump; Membrane pump; Diaphragm pump; Valveless pumps

Definition

Peristaltic pumps are mechanical displacement pumps that induce flow in a fluid-filled, flexible-walled conduit through ►peristalsis – transport due to traveling contraction waves. While macroscale peristaltic pumps appear in a variety of configurations, micropumps based on this principle almost exclusively use the sequenced contraction and expansion of a small number of discrete actuators – typically three – placed along the fluid channel.

Overview

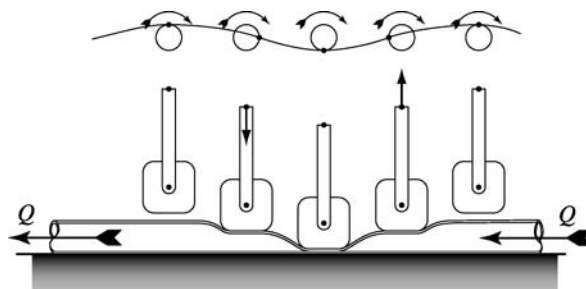
Peristaltic pumps emulate the biological process of peristalsis, in which material is moved through the esophagus or other anatomical passage by the contraction of smooth muscle in rhythmic waves. Figure 1 is a schematic of a typical ►rotary peristaltic pump, as implemented at the macroscale, in which a set of revolving contact elements creates a traveling compression wave in a section of a flexible tube. Figure 2 is a schematic of a macroscale implementation of a ►linear peristaltic pump, in which a number of translating piston actuators cyclically compress a flexible tube. In either case, a moving boundary displaces fluid and induces a flow, placing peristaltic pumps in the class of positive-displacement pumps. Increasing the tube diameter or the pumping cycle frequency increases the flow rate. A major attraction of macroscale peristaltic pumps is cleanliness. The fluid is completely isolated from the pump components since it never leaves the tube. Furthermore, it is a simple matter to change the tubing to avoid cross-contamination between fluids, and the tubing material may be tailored to ensure compatibility with a particular application. The pumping action is relatively gentle, making peristaltic pumps suitable for reactive liquids or cell suspensions. Also, the pumps may be self-



Peristaltic Pumps, Figure 1 A typical macroscale rotary peristaltic pump design. A set of rotating contact elements cyclically compresses a flexible tube. The elements may roll or slide along the tube; the number of contact elements may also vary. The fluid never leaves the tube, protecting the fluid from contamination by the pump elements, and the pump elements from corrosion or abrasion by aggressive fluids. The tube itself is easily replaced. Tube size and composition may be adjusted to accommodate different fluids, or to change the flow rate. Changing the rotation rate Ω changes the flow rate; changing the sign of Ω reverses the flow direction

priming due to the low-pressure region created behind the moving constriction, and the flow direction can be easily reversed. At the macroscale, use of rotary peristaltic pumps is widespread, while linear peristaltic pumps are found mainly in niche applications, such as intravenous drug delivery.

Cleanliness considerations do not translate to the microscale, nor does the notion of interchangeable components. Therefore, use of peristaltic micropumps must be justified for other reasons. Self-priming and bi-directionality remain appealing features, but perhaps the main benefit is simplicity of design. Any mechanism providing volume change in a chamber or channel is a candidate actuator for a peristaltic pump. For example, a typical active microvalve involves a membrane that can be forced against a seal to close, or pulled away from the seal to open [1, 12]. Two or more active microvalves can be arranged in sequence and actuated with the appropriate phasing to create a linear peristaltic micropump. The principles of peristaltic pumping apply to liquid or gas, and micropumps for both have been demonstrated. The ease with which an actuator can be transformed into a pump accounts for the wide variety of linear peristaltic micro-



Peristaltic Pumps, Figure 2 A typical macroscale linear peristaltic pump design. A set of translating actuators cyclically compresses a flexible tube. The number of actuators may vary. Features of macroscale rotating peristaltic pumps also apply to microscale linear peristaltic pumps. Linear peristaltic pumps are used in applications such as drug delivery, when accurate flow rate control is needed

pumps that have been reported, including an early micro-machined silicon and glass version by Smits (as reviewed in [3, 12], or see the original paper referenced therein).

In this article we refer to an actuator as *closed* when the associated chamber or channel volume is minimized, and as *open* when that volume is maximized. The volume in the closed position is called the **dead volume**, and the difference between the volume in the open position and the dead volume is the **stroke volume**. The **stroke volume** divided by the dead volume is the **compression ratio**. Compression ratio is particularly important when pumping gases, since the amount of gas ejected from the actuator is reduced by the compressibility of the fluid. In pure liquids this effect is much less, but compression ratio plays an important role in determining how severely bubbles impact a particular pump. In classifying micropumps, a distinction is sometimes made between **integrated actuators** – fabricated along with the other components of the microfluidic system – and **external actuators** – fabricated separately and attached later. It is common to see peristaltic micropumps of either kind.

While a linear pump can easily be built up from two or more basic actuation units, microscale rotary mechanisms face significant fabrication challenges. Accordingly, rotary peristaltic micropumps have seen very limited implementation. While several rotary mechanical micropumps are described in the literature (see the summaries in [1], [3] and [12], for example), to our knowledge there is only one example in the literature of a true rotary peristaltic micropump; Kim et al. (Sensors and Actuators A 128:43–51, 2006) use a silicone membrane for the deformable channel surface, and a rotating ferrofluid plug driven by an external magnet for actuation. Also notable is a miniature rotary peristaltic pump by Bar-Cohen and Chang (*Smart Structures and Materials 2000: Active Materials:*

Behaviors and Mechanics, Proceedings of SPIE 3992, 2000) that employs a traveling flexural wave in a piezoelectric ring. Reflecting their dominance in the literature, the subsequent discussion will consider only linear peristaltic micropumps, which for brevity are henceforth referred to simply as peristaltic micropumps. Most reported microscale implementations approximate continuous peristaltic action using two or more (most commonly three) discrete collapsible chambers. A notable exception, and one which does not fit well into the analysis scheme presented here, is the colloidal peristaltic pump presented in [15]. There a large number (approximately twelve) of 3 μm colloidal silica particles are oscillated sinusoidally in a 6 μm deep polydimethylsiloxane (PDMS) channel using external actuation by optical trapping with a scanned 532 nm laser. Flow rate on the order of 1 nl/h are induced in an aqueous solution by the resulting traveling wave. Peristaltic actuation through movable suspended particles is, to our knowledge, unique to [15]. The remainder of this paper focuses on the much more common case of discrete collapsible chambers.

Peristaltic pumps have found wide application in microfluidic systems based on PDMS, due to their ease of fabrication and versatility in operation. PDMS is a popular material for microfluidics for several reasons. Compared to traditional semiconductor industry fabrication schemes, both the materials and processing tools are inexpensive. PDMS is biocompatible, resistant to many chemicals, and transparent to wavelengths of light useful in chemical and biochemical analysis. Figure 3 shows a typical actuator design, similar to that first reported in [4], with the fluid channels formed between a thin PDMS layer and a glass, silicon, or PDMS substrate, and actuator channels formed at right angles to the fluid channels in a thicker PDMS layer, bonded to the planar side of the fluid layer. We refer to the area of the fluid channel under an actuator channel as a pump chamber, even though it is geometrically identical to the rest of the channel. The actuator channels are inflated from an external pressure source to close the pump chamber; when the pressure is removed the natural elasticity of the PDMS restores the channel to its original shape. As suggested by the figure, due to the low modulus of elasticity of PDMS, the pump chambers may be designed to collapse completely when closed, allowing them to act either in concert for pumping, or in isolation as pure valves.

In a peristaltic pump, the volume of fluid displaced by the individual actuators is rectified, that is, directed in the desired direction of flow, solely by the phasing of multiple actuators. Other types of displacement pumps rectify flow differently. A passive check-valve pump uses two passive, one-way check valves incorporating moving flaps or other

components. One such valve at the pump inlet permits fluid to enter the chamber as the pump chamber expands, but prevents it from leaving as the chamber contracts. Likewise, a passive check valve at the outlet prevents back-flow on the expansion stroke, but permits fluid to exit the chamber on the contraction stroke. Valveless rectification pumps are similar, but use flow diodes instead of mechanical check valves. Flow diodes are diffuser/nozzle structures or so-called Tesla valves that have no moving parts, but exhibit a preferential flow direction due to their geometry. More detail on passive check-valve or valveless rectifier pumps may be found in [1], [3] or [12]. Since these require only a single actuator, versus two or more for peristaltic pumps, they are natural design alternatives. Reasons to prefer a peristaltic design include ease of integration onto the microchip, with no special fabrication steps or external connections. For example, [13] presents an electrostatically-actuated peristaltic micropump suitable for incorporation into a micro-total analytical system (μTAS) with no additional special considerations. Peristaltic pumps avoid the more complex microfabrication typically required for passive check valves, and the susceptibility of such valves to fouling by particles or bubbles. Valveless rectifiers are simple to fabricate, and not prone to fouling, but may have a larger footprint than peristaltic designs. Finally, peristaltic pumps may be used to generate forward or reverse flow by modifying the actuator phasing, while bidirectional versions of the other two options are difficult to realize. Some guides to peristaltic pump design advise adding series passive check valves to aid in rectification if the application allows [1], though this reduces to some extent the operational and fabrication benefits that peristaltic designs offer over check-valve designs.

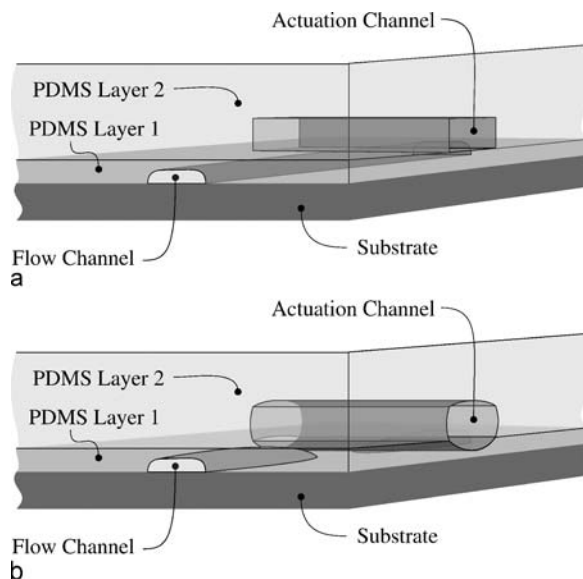
In many peristaltic micropump implementations identical actuators provide both stroke displacement and flow valving. However, these functions may also be divided between actuators of different design. One analysis reported in the literature treats a large stroke displacement actuator with a low flow resistance, flanked by active valves with small stroke displacement [5]. While similar in concept to a passive check-valve diaphragm pump, the valving actuators in this case are actively opened and closed by external commands, and flow direction is determined by the phasing of these commands with regard to actuation of the central diaphragm. Hence, this design may properly be classified as a peristaltic pump.

Due to a combination of surface tension and viscosity effects, the pressure drop per unit length along a channel with characteristic length l (radius or hydraulic diameter) scales unfavorably (as $1/l^3$ or $1/l^2$, depending on the analysis). Thus mechanical pumps must generate extremely high pressures to maintain flow rates as channel diameters

shrink. For this reason, it is widely accepted that mechanical pumping mechanisms, including peristaltic pumps, are not suitable for very small channels. Nguyen and Wereley survey the literature, and, using water as a working fluid, give $10\ \mu\text{l}/\text{min}$ as the flow rate below which nonmechanical pumping principles are preferred [1]. However, that boundary is somewhat arbitrary, as demonstrated by flow rate of $0.14\ \mu\text{l}/\text{min}$ reported in [4] for pneumatically-actuated peristaltic pumping of water in PDMS channels, the flow rate of $1.7\ \text{nl}/\text{min}$ for electrostatically-actuated peristaltic pumping of ethanol in parylene channels [13], and the flow rate of $1\ \text{nl}/\text{h}$ reported in [15] for optically-actuated peristaltic pumping by colloidal particles suspended in an aqueous solution.

Basic Methodology

A linear peristaltic micropump consists of two or more actuators arranged along a channel, and driven in a particular sequence. Important parameters that quantify pump performance are the maximum flow rate, denoted here by Q^* , the maximum pressure that can be generated between the pump inlet and outlet (called the **pump head**), denoted here by p^* , and the efficiency with which the pump converts external power into flow power, denoted here by η . Given a particular actuator, or set of available actuators, pump design consists of selecting the number of actuators, the actuation sequence, and the frequency of actuation. Important information about the individual actuators includes the stroke volume, V , the dead volume, V_0 , the flow resistance in open and closed configurations, R_o and R_c , respectively, the maximum pressure the actuator can generate, p_{act}^* , called the **closing pressure**, the minimum time required for the actuator to either open or close completely, T_{min} (the minimum time to open and the minimum time to close are typically different; here T_{min} would be the larger of the two) and the energy or power required to operate an actuator. Almost any mechanical transducer can be used as an actuator in a peristaltic micropump, making generalizations about pump performance difficult. For example, the response time of a thermopneumatic actuator may be dominated by heat transfer out of the actuator, while that of a piezoelectric actuator is primarily determined by material stiffness and membrane geometry. As another example, a thermal actuator may require continual power to remain in the closed position, while – once closed – an electrostatic or pneumatic actuator may be held closed with little or no additional power consumption. In this article we focus on those considerations that are common to all peristaltic pump designs. This rules out detailed physical analysis. However, some broad assumptions about actuator and system characteristics allow rough



Peristaltic Pumps, Figure 3 A typical pump chamber in a multilayer PDMS implementation (similar to those of [4]). The chamber is open by default, as shown in (a). Inflating the actuation channel with compressed air closes the chamber, as shown in (b). PDMS layer 1 defines the fluid channels (sometimes called the working channel), while PDMS layer 2 defines the actuation channels. In both cases casting or spinning PDMS on a negative mold creates the channel structures. The molds are defined photolithographically, with standard positive resists used for thin channels (on the order of a micrometer) and negative resists such as SU8 used for thicker channels (up to one millimeter). The patterned side of the fluid layer is bonded to a glass, silicon, or PDMS substrate, and the patterned side of the actuation layer is bonded in turn to the planar side of the fluid layer. For convenience, we refer to the area of the fluid channel under the actuation channel as the pump chamber, even though it is otherwise indistinguishable from the rest of the channel. Baking the fluid layer mold at high temperature after the pattern is developed causes the photoresist to *reflow*, yielding rounded corners in the fluid channels. These rounded corners reduce leakage when the chamber is closed

comparisons for pump design purposes. We begin by assuming ideal actuators, with infinite flow resistance in the closed position and zero dead volume. The first property allows easier accounting of the movement of fluid, the second allows issues related to fluid compressibility to be ignored.

The first step in this simplified analysis is estimating the volume of fluid transported in a complete pumping cycle. To do this, we apply the following rules:

1. When an actuator closes, the fluid within is completely expelled.
2. When an actuator opens, it completely fills with fluid.
3. If an actuator remains closed during a step in the sequence, no fluid may pass across it.
4. If, in a single step, some actuators open while others close, and there is a path for fluid to flow between them, then fluid is transferred from the latter to the former.

Step	Net In	C1 Vol	C2 Vol	C3 Vol	Net Out	Step	Net In	C1 Vol	C2 Vol	C3 Vol	Net Out
1	0	0	0	V	0	1	0	V	V	V	0
2	-V	V	0	0	+V	2	+V/2	0	V	V	+V/2
3	-V	0	V	0	+V	3	+V/2	0	0	V	+3V/2
1	-V	0	0	V	+V	4	+V/2	0	0	0	+5V/2

Peristaltic Pumps, Figure 4 A three-step, three-actuator pumping sequences, used in numerous pumps, including thermopneumatic pumps by Grosjean and Tai (see the reference in [1]) and Jeong, et al. [9]. The stroke volume is designated V . Chambers 1, 2, and 3 are abbreviated C1, C2, and C3. At every step two chambers are closed, reducing leakage. Note that Grosjean and Tai use a similar looking figure to define the pump sequence, but in their paper the raised level indicates actuation, which corresponds to *closing* the chamber

5. If an actuator opens or closes with no closed actuators between it and either the inlet or the outlet, the transfer of fluid is made equally from or to both sides. Once the fluid volume is computed, the flow rate flows from the actuation frequency. Maximum head is taken to be the maximum pressure exerted by an actuator (though this is not often known), and efficiency is estimated from the actuation sequence. Often, an estimate of flow rate is all that is desired. It must be noted that these estimates are very coarse, with for examples, errors of a factor of two reported, and attributed to unmodeled valve leakage [6].

Simplified Sequence Analysis

As an illustration of the simplified analysis, consider the three-step cycle shown in Fig. 4 for a pump consisting of three identical ideal actuators with stroke volume V . At the start of the cycle, no fluid has yet been transferred. In step 2, chamber C1 opens while chamber C3 closes. By rules 1 and 2, C1 must fill with volume V while C2 expels volume V . Since the intervening chamber C2 is closed, by rule 3 C1 must take volume V from the inlet, while C3 must expel volume V to the outlet. In step 3 C1 closes while C2 opens. By rule 4, since there is an open path between them, this transfers volume V from C1 to C2. Likewise the return to step 1, which completes the cycle, transfers volume V from C2 to C3. We conclude that, ideally, volume V is pumped per complete three-step cycle. At the final step the net fluid volume increase at the outlet must equal the net decrease at the input. This is a useful check that the analysis was done correctly.

5	-V/2	V	0	0	+5V/2
6	-3V/2	V	V	0	+5V/2
1	-2V	V	V	V	+2V

Peristaltic Pumps, Figure 5 A six-step, three-actuator pumping sequences proposed in the literature [6]. At step 1 all chambers are open, allowing backflow which will reduce the net flow rate

Maximum Flow Rate

If each step requires time T , the pumping rate is $V/3T$. The maximum flow rate Q^* then corresponds to the minimum cycle time, $T_{\min}$: $Q^* = V/3T_{\min}$.

Maximum Head

In step 2, C3 is emptied to the outlet. If the outlet pressure is greater than p_{act}^* , the maximum pressure the actuator can provide, the chamber will be unable to close. Thus, the maximum head can be approximated as the closing pressure, $p^* = p_{\text{act}}^*$.

Efficiency

Fluid power is the product of flow rate by head. In an ideal ► **positive displacement pump**, flow rate is independent of head. Therefore, the maximum fluid power is approximately $P^* = p^* Q = p_{\text{act}}^* V/3T$. One actuator closes at each step, and two actuators must be held closed at each step. If energy E_{act} is required to close an actuator, and power P_{act} is required to hold it closed, we see that each cycle requires total energy $3E_{\text{act}} + 6P_{\text{act}}T$ (we assume no energy is required to open an actuator). The total power consumed per cycle is $(E_{\text{act}}/T) + 2P_{\text{act}}$, and the maximum efficiency is $\eta = p_{\text{act}}^* V/(3E_{\text{act}} + 6P_{\text{act}}T)$.

Experiments suggest that a linear relation between flow rate and pump head may be a more appropriate model [5]. In that case, the maximum fluid power is given by $P^* = p^* Q^*/4$.

Design Comparison

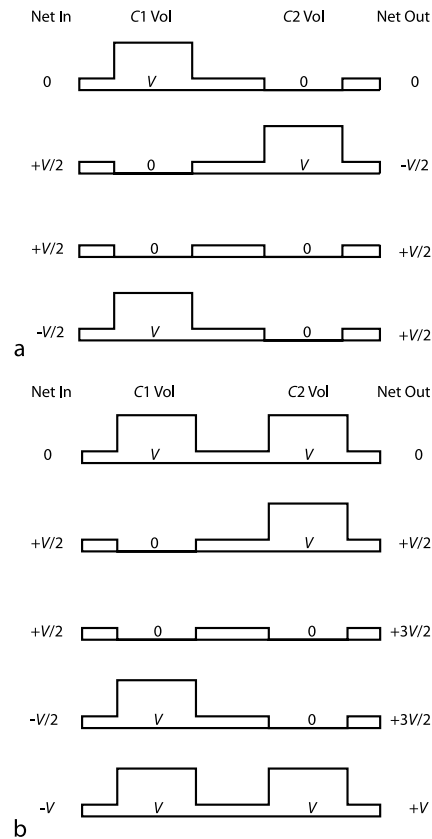
To illustrate the use of the simplified analysis in design, we compare the actuation sequence analyzed above to a six-step three-chamber sequence also reported in the literature [6], and shown in Fig. 5. The analysis is similar to the three-step sequence, except for steps 1 and 6, where an actuator closes with no closed actuators between it and either the inlet or the outlet. By rule 5, the expelled fluid is divided evenly between the inlet and the outlet. Two considerations make application of the simplified model more problematic in this case. The first is the all-actuators-open configuration in step 1. If there is significant head across the pump, this may produce an unmodeled backflow. The second is the assumption mentioned above, that the flow splits evenly in step 1 and 6. If the external system flow resistances as seen from the input and output are different, or if there is significant head, the split will *not* be even. These issues aside, we find that a volume $2V$ is pumped in each six-step cycle.

If each step requires time T , the pumping rate is $2V/6T$, or $V/3T$. The maximum flow rate is $Q^* = V/3T_{\min}$. This is the same as for the three-step sequence.

In steps 2, 3, and 4 chambers are emptied to the outlet. The maximum head can again be approximated as the closing pressure.

The maximum fluid power is approximately $P^* = p^*Q = p_{\text{act}}^*V/3T$. In each cycle, energy $3E_{\text{act}}$ is required for actuator closings, and energy $9P_{\text{act}}T$ is needed to hold actuators closed, for a total cycle power requirement $(E_{\text{act}}/2T) + (2P_{\text{act}}/3)$. The maximum efficiency is $\eta = 2p_{\text{act}}^*V/(3E_{\text{act}} + 9P_{\text{act}}T)$.

To the level of modeling accuracy applied, the two designs have the same maximum flow rate and maximum head. The six-step sequence is more power efficient, but there are unanswered questions about how well this design will perform, particularly if there is significant backpressure. If power usage is more important than maximum head, the six-step design may be preferable, but further investigation will be required. If efficiency is not important, the three-step sequence should be chosen for its higher resistance to backflow. An advantage of the six-step sequence is that only one chamber is actuated per step. This makes the sequence insensitive to asymmetries in opening versus closing actuators. For example, if the actuators close ten times faster than they open, fluid may not transfer smoothly between chambers in the sequence of Fig. 4, but instead may leak out of the chamber. Avoiding this risk by adding intermediate steps, as for example, in the five-step sequence used by Smits [2] will reduce the maximum flow rate, and make the sequence of Fig. 5 more attractive.



Peristaltic Pumps, Figure 6 Pumping sequences for two-actuator peristaltic pumps [7]. Sequence A has at least one actuator closed at all times. Sequence B has a higher maximum flow rate, but may have problems with backflow under high head conditions

Most reported peristaltic micropump designs use three actuators. A few use more, but typically not more than five. Since the actuators take up potentially valuable space, and require connections for power and switching, it is desirable in many applications to use the minimal possible number. Some authors have stated that three chambers are the fewest possible for peristaltic pumping. However, the sequences shown in Fig. 6 have been shown to provide peristaltic pumping using only two chambers [7].

Diagrams like Figs. 4–6 are easy to grasp visually, but a more compact description of pumping sequences is also useful. One method used in the literature (for example, in [8]) is to indicate an actuated (closed) chamber by a 1 and an un-actuated (open) chamber by a 0. Thus the sequence in Fig. 4 is written {110, 011, 101}, that of Fig. 5 becomes {000, 100, 110, 111, 011, 001}, and those of Fig. 6 A and B are {01, 10, 00} and {00, 10, 11, 01}, respectively. The five-step sequence used by Smits [2] is, following the convention of 1 for closed, 0 for open, {011, 001, 101, 100, 110}. (This sequence does involve an

actuator closing simultaneously with another opening, but they are separated by a closed chamber.) Some care is required in using this notation, since some mechanisms close when actuated (e. g. thermopneumatic), while others open (e. g. piezoelectric).

Dynamic Modeling

Leaks, backpressure, and actuator dynamics all influence the performance of peristaltic pumps. Leaks and backpressure effects will alter the distribution of fluid as actuators open and close. The dynamics of the actuators determines the maximum actuation rate, which in turn limits the maximum flow rate. These effects can be incorporated into lumped-parameter models for analysis and simulation. We do not pursue this further in this article, but refer the reader to the literature. One general approach is presented in [5]. Also relevant for further study are [6], [13] and [14], which present dynamic models for pneumatic and electrostatic pumps, respectively. All of these works are applied to liquid pumps. For gas pumps, or robustness to bubbles, compressibility becomes a factor. Some considerations of micropumps for compressible fluids may be found in [1]. Finite-element analysis of individual chambers can also be used to obtain detailed predictions of pump dynamic performance.

Key Research Findings

Fabrication Materials and Actuator Types

Peristaltic pumps have been demonstrated in all materials common for microfluidic applications, using a variety of physical principles. Space constraints prevent a comprehensive listing, but [1], [3], and [12] provide good overviews. Summarizing here just the main fabrication materials, the operating principle, and the working fluid, these include for silicon and/or glass devices, piezoelectric, thermopneumatic, and electrostatic actuators pumping water. For PDMS devices only pneumatic pumping of water is reported. For plastic or mixed-type devices, electrostatic, thermopneumatic, and pneumatic actuators are reported, pumping water and air. A novel traveling wave pump implemented using optically-actuated colloidal silica spheres is reported in [15]. Recently, Husband et al. (Microelectronic Engineering 73–74:858–863, 2004) report a piezoelectric silicon-glass device pumping water and air, Pan et al. (Proceedings of the 26th Annual International IEEE EMBS Conference, 2004) report a magnetically driven PDMS pump for water, and Jeong et al. report a thermopneumatic PDMS pump for water [9]. Electrostatic pumps implemented in parylene for integration with μ TAS devices are reported in [13].

Reducing Device Complexity

Peristaltic pumps already offer simpler fabrication than passive check-valve pumps. Even the simplest actuators, however, require at least two patterned layers, as, for example, in the PDMS device of Fig. 3. However peristaltic pumping was recently demonstrated in a device with a single patterned layer of PDMS on an un-patterned glass or PDMS substrate [10]. The single patterned layer incorporates both pneumatic actuation lines and the water-filled working channels. Rather than press on the fluid channel from above, the actuation lines in this device squeeze the working channel from the sides. While these actuators provide enough stroke and increased resistance for pumping, cell sorting, and mixing, in their closed configuration they are still quite leaky, and do not completely seal the channel. However, the elimination of the second layer, along with the associated alignment and bonding tasks, should yield a significant reduction in device cost.

Reducing Number of Off-Chip Connections

Each actuator in a peristaltic pump is an active element, requiring switching and power. In applications such as Lab-on-a-Chip, the number of required off-chip connections can become very large. As mentioned above, one means for doing this is by using two actuators instead of three [7]. The proliferation of pumps and valves is a particular problem in pneumatically actuated pumps that require external valving. An innovative approach to ameliorating this problem is described in [11] for pneumatically actuated pumps fabricated in PDMS. Here, a single pneumatic actuation channel has a serpentine shape, crossing over the straight channel containing the working fluid at multiple intersections. The phasing of the actuation is provided by the natural delay associated with the propagating inflation zone in the serpentine channel. In this way the equivalent of a three-, five-, or seven-actuator peristaltic pump is realized using a single actuator line.

Multifunction Components

The individual actuators in a peristaltic pump can also serve as individual valves. Additional flexibility in integrating peristaltic pumps into large microfluidic systems can be gained from providing components with multiple functions. An example of this is the rotary mixer described in [8], which can serve as an effective dynamic micromixer, or as a peristaltic pump, depending on the actuation pattern of the individual valves.

Future Directions for Research

Progress in peristaltic pump design has paralleled the advancement of microfluidics as a field. As new materials and applications have been developed, peristaltic pump designs have soon followed. This trend seems likely to continue.

Peristaltic pumps are commonly used in PDMS-based systems. As those systems have attracted increasing amounts of research effort, due to their biocompatibility and demonstrated success in manipulation of cells, genetic material, and proteins, new innovations in PDMS-based pumps have followed. Progress has been made in reducing the need for external connections, but improvement would be welcome in the area of fully integrated pumps of reasonable cost and low complexity.

Cross References

- Electromagnetic Valves
- Electrostatic Valves
- Magnetic Pumps
- Membrane Actuation for Micropumps
- Microactuators
- Microfluidic Mixing
- Piezoelectric Valves
- Piezo/PZT in Microfluidics
- Pneumatic Valves
- Soft Photolithography
- Thermomechanical Valves
- Thermopneumatic Valves
- Water Management in Micro DMFCs

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Permeabilization of Cells

- On-Chip Cell Lysis

Perovskite

Synonyms

Calcium titanium oxide; Orthorhombic crystal

Definition

Defines calcium titanium oxide in particular and generally all crystalline materials that exhibit a crystal structure similar to it, as with the crystalline structure of PZT.

Phase-Change Heat Transfer in Microchannels

- Condensation in Microchannels

Phase Field Method

- Interface Capturing Schemes for Free-Surface Flows

Phase Separation Kinetics

- Demixing

Phase Shift

- Frequency Response

pH Control Device

- pH-Sensitive Hydrogel

Phenol Extraction

- Lab-on-a-Chip Devices for Sample Extractions

Phenotypic Assay

- Cell Assays

Phenotypic Screening in Microfluidic Devices/on-Chip

- Cell Assays in Microfluidics

Phosphorescence

Definition

Phosphorescence is the emission of light from triplet-excited states, whereby one set of electron spins is unpaired. This means that electrons in the excited orbital have the same spin as the ground state electrons. Therefore, transition to ground state is not possible and emission rates are slow (in the region of $10^3 - 10^1 \text{ s}^{-1}$). Phosphorescent life-times have durations in the millisecond range. Following exposure to light, phosphorescent substances glow for several minutes as the excited phosphors return to the ground state very slowly.

Cross References

- Fluorescent Labeling

Photo-Activated Nonintrusive Tracking of Molecular Motion

- Visualization Based on Molecular Tagging Methods

Photobleached Fluorescence

- Visualization Based on Molecular Tagging Methods

Photobleaching

Synonyms

Photochemical decomposition; Photochemical degradation

Definition

Photobleaching is the decrease of the emitted fluorescence intensity by the excited fluorophores as a result of the decay of dye molecules due to their prolonged exposure to high-intensity incident light.

Cross References

- Fluorescent Thermometry
- Fluorescence Labeling

Photochemical Decomposition

- Photobleaching

Photochemical Degradation

- Photobleaching

Photofluidic Devices

- Optofluidics – Techniques for Fabrication and Integration

Photofluidics

- Optofluidics – Applications
- Optofluidics – Fluidics Enabling Optics
- Optofluidics: Optics Enabling Fluidics

Photolithography

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Synonyms

Optical lithography; Microlithography; Nanolithography

Definition

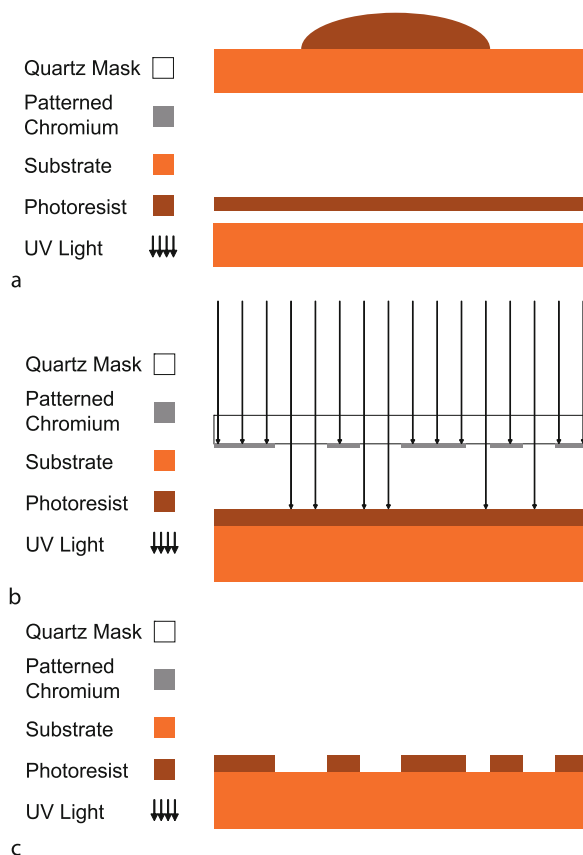
Photolithography is a process mainly used but not limited to manufacturing integrated circuits, and involves the photographic transfer of a desired pattern to a surface for further processing (e. g. etching).

Overview

Photolithography is the most popular method among various forms of lithography when transferring patterns from a mask (reticle) onto a surface of a substrate. Photolithography is widely used in the integrated circuit (IC) industry. However, it has other potential important applications such as developing methods for controlling the position of cells on a substrate or micropatterning of cells that has many bioengineering applications including the design of biosensors that can be used for drug and toxicity sensing devices and environmental monitoring devices, tissue engineering applications, and basic biology studies. Photolithography is widely used in the semiconductor industry to perform a series of successive patterning on a given substrate to achieve an electronically functioning product. Although, this patterning method is limited to planar surfaces, it can provide accurate patterning feature sizes at the submicrometer level.

Basic Methodology

Photolithography is comprised of sequential process steps that are schematically summarized in Fig. 1. The process begins by choosing or preparing a substrate for the photolithography process. The ideal substrate should be dehydrated before any application of photoresist. This dehydration is usually done by a baking step called *dehydration bake* to remove adsorbed water from the substrate, usually a silicon wafer. Thereafter, the dehydrated substrate is spin coated with a photoresist adhesion promoter using a spinner. Usually, hexamethyldisilazane (HMDS) is used for this step, which is called priming. Then, the primed substrate is spin coated placing excess amount



Photolithography, Figure 1 (a) The top image shows the deposition of photoresist on a dehydrated and then primed wafer. The image below shows the coated wafer with uniform layer of photoresist. (b) The photoresist-coated wafer is exposed to UV light with an intact mask. (c) The pattern is developed by rinsing the substrate in a developing solution or using a spray developer. This process will remove the unwanted photoresist and leaves the pattern on the surface of the substrate.

of a photoresist, a photosensitive material. The photoresist will be spread uniformly on the surface of the substrate by centrifugal force. This process is one of the most important parts of the process, as the quality of the pattern transfer depends on effective spin coating of photoresist material. The most important requirements of spin coating are having a known uniform thickness of the photoresist material throughout the substrate, and this should be optimized for different substrate–photoresist combinations to determine the appropriate amount of photoresist and spinning velocity to use during the spin coating step. The thickness of photoresist may be varied according to the spinning speed and the viscosity of the photoresist-containing fluid. Usually, photoresist vendors provide information on how thickness may vary according to the spinning velocity of a given photoresist. The photoresist-coated substrates are subjected to a prebaking

step or soft baking step before being exposed to UV light with an intact stencil called a *photomask*. This prebaking step is usually done at 80 – 110 °C for 5 to 10 min depending on the thickness of the photoresist layer. This step will enhance the adhesion of the photoresist to the substrate by removal of the solvent in the photoresist and any built in stresses.

The exposure can then be conducted using an aligner. An aligner is comprised of a light source including any reflecting and/or refracting optics needed for collecting, collimating, filtering, and focusing the light onto the mask with the desired image. The most common aligners are contact printers and projection printers. In the contact printer approach the mask is held against the photoresist; in the projection printing method a set of lenses (projector/objective) is placed between the mask and the wafer to refocus light onto the photoresist. After exposure, the pattern is developed by rinsing the substrate in a developing solution or using a spray developer. This process will remove the unwanted photoresist and leaves the pattern on the surface of the substrate. This is also a very important step as it is very temperature sensitive, and will affect control and maintenance of line widths of the pattern, resist contrast, and resist profiles. There are two types of pattern transfer: positive and negative. In the positive form, the photoresist pattern on the substrate is the positive image of the mask pattern; in the negative form, the image of the photoresist pattern is the inverse image of the photomask pattern.

After developing, the substrate is subjected to a hard baking process to harden the photoresist. The hardening of the photoresist is done as the patterned substrates may be used for further processing such as plasma etching or ion implantation. Once all the processing is complete, any remaining/developed photoresist can be removed using a commercially available organic photoresist stripping solvent. This substrate can be used for further patterning by starting another photolithography step.

Future Directions for Research

The future directions for research in photolithography will be in improving minimum feature size. The variables that involve improving minimum feature size include photoresist, mask, and exposing wavelength. Thus, one can think that the future improvement in photolithography will depend on improving resist types, and designing better mask-refining technology and better laser technology.

Cross References

- Cell Patterning on Chip
- Photo Patterned

- Photoresist SU-8
- Photoresist Reflow

References

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Photoluminescence

- Fluorescent Thermometry

Photomask

Definition

A plate of opaque material that has been patterned with transparent regions.

Cross References

- Sputtering for Film Deposition

Photometer

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Synonyms

Photometry; Illuminance; Irradiance; Scattered light intensity; Absorption; Spectroscopy; Spectrophotometer

Definition

The instrument used to measure illuminance or irradiance of light. It is the general term covering instruments for detecting scattered light intensity, absorption, fluorescence applied in the field of photometry. Spectroscopic techniques using photometer can probe the change of energy, frequency, wavelength and intensity of light caused by the interaction with matter. Most photometers are based on a photoresistor, photodiode, or photomultiplier tubes for photon observation. Either exhibits a change in electrical properties when exposed to light, which can be detected with a suitable electronic circuit.

Photometer, Table 1 Symbols, SI units and abbreviations used in photometer

Quantity	Symbol	SI unit	Abbreviation
Luminous energy	Q_V	lumen second	lms
Luminous flux	F	lumen (= cdsr)	lm
Luminous intensity	I_V	candela (= lm/sr)	cd
Luminance	L_V	candela per square metre	cd/m ²
Illuminance	E_V	lux (= lm/m ²)	lx
Luminous emittance	M_V	lux (= lm/m ²)	lx
Luminous efficacy		lumen per watt	lm/W
Radiant energy	Q	joule	J
Radiant flux	Φ	watt	W
Radiant intensity	I	watt per steradian	Wsr ⁻¹
Radiance	L	watt per steradian per square metre	Wsr ⁻¹ m ⁻²
Irradiance	E	watt per square metre	Wm ⁻²
Radiant exitance / Radiant emittance	M	watt per square metre	Wm ⁻²
Spectral radiance	L_λ or L_ν	watt per steradian per metre ³ or watt per steradian per square metre per hertz	Wsr ⁻¹ m ⁻³ or Wsr ⁻¹ m ⁻² Hz ⁻¹
Spectral irradiance	E_λ or E_ν	watt per metre ³ or watt per square metre per hertz	Wm ⁻³ or Wm ⁻² Hz ⁻¹
Radiant energy	Q	joule	J

Photometer, Table 2 Comparison of two photodetectors: photodiodes and photomultipliers

Advantages of photodiodes	Advantages of photomultipliers
• Excellent linearity of output current as a function of incident light • Spectral response from 190 nm to 1100 nm (silicon), longer wavelengths with other semiconductor materials • Low noise • Ruggedized to mechanical stress • Low cost • Compact and light weight • Long lifetime • High quantum efficiency • No high voltage required	• Large area • Flexible internal gain • Higher overall sensitivity • Photon counting is possible

P

Overview

The interaction of electromagnetic radiation with matter is essentially a quantum phenomenon and is dependent on the properties of the radiation and on the appropriate structural parts of the material involved. As the origin of the radiation is due to energy changes within the matter itself, an understanding of the properties of electromagnetic radiation and its interaction with matter leads to a recognition of the variety of types of spectrum, and consequently spectroscopic techniques, and their application to the solution of biological problems. Photometry is the

science of measurement of light, historically, in terms of its perceived brightness to the human eye. It is distinct from radiometry, which is the science of measurement of light in terms of absolute power; rather, in photometry, the radiant power at each wavelength is weighted by the luminosity function that models human brightness sensitivity. Thus many different units of measure are used for photometric measurements and sometimes that can't be converted from radiometry. For example, in photometric quantities, every wavelength is weighted according to how visible it is, while radiometric quantities use unweighted absolute power. Because the eye responds much more strongly

to green light than to red, a green source will have higher luminous flux than a red source with the same radiant flux would. Light outside the visible spectrum does not contribute to photometric quantities at all, so for example a 1000 W space heater may put out a great deal of radiant flux (1000 W, in fact), but as a light source it puts out very few lumens (because most of the energy is in the infrared, leaving only a dim red glow in the visible). Watts (radiant flux) and lumens (luminous flux) illustrates the distinction between radiometric and photometric units. The watt is a unit of power. As power is not a measure of the amount of light output, power is only a rough guide to light output. In contrast, the lumen is the photometric unit of light output. The lumen is defined as amount of light given into one steradian by a point source of one candela strength; while the candela, a base SI unit, is defined as the luminous intensity of a source of monochromatic radiation, of frequency 540 THz, and a radiant intensity of $1/683 \text{ Wsr}^{-1}$. (540 THz corresponds to about 555 nm, the wavelength, in the green, to which the human eye is most sensitive. The number 1/683 was chosen to make the candela about equal to the standard candle, the unit which it superseded). Combining these definitions, we see that $1/683 \text{ W}$ of 555 nm green light provides one lumen. The definition tells us that 1 W of pure green 555 nm light is equal to 683 lm. It should be noted that it does not say anything about other wavelengths. Because lumens are photometric units, their relationship to watts depends on the wavelength according to how visible the wavelength is. Infrared and ultraviolet radiation, for example, are invisible and do not count. That means one watt of infrared radiation is worth zero lumens.

In photometry, illuminance is the total luminous flux incident on a surface, per unit area. It is a measure of the intensity of the incident light, wavelength-weighted by the luminosity function to correlate with human brightness perception. Similarly, luminous emittance is the luminous flux per unit area emitted from a surface. Luminous emittance is also known as luminous exitance. In SI derived units, these are both measured in lux (lx) or lumens per square metre (cdsr m^{-2}). In the CGS system, the unit of illuminance is the phot. One phot is equal to 10,000 lx. Illuminance was formerly often called brightness, but this leads to confusion with other uses of the word.

Irradiance is a radiometry term for the power of electromagnetic radiation at a surface, per unit area, which is used when the electromagnetic radiation is incident on the surface. The SI unit for this quantity is watts per square meter (W/m^2), which is sometimes called **intensity**. This quantity characterizes the total amount of radiation present, at all frequencies. It is also common to consider each frequency in the spectrum separately. When this is done for

radiation incident on a surface, it is called spectral irradiance, and has SI units W/m^3 , or commonly $\text{Wm}^{-2}\text{nm}^{-1}$.

Basic Methodology

Photometric Measurements

Photometric measurement is based on photodetectors, devices that produce an electric signal when exposed to light. Simple applications of this technology include switching luminaries on and off based on ambient light conditions, and light meters, used to measure the total amount of light incident on a point. More complex forms of photometric measurement are used for detecting scattered light intensity, absorption, fluorescence.

Light Scattering

Light scattering is one of the two major physical processes that contribute to the visible appearance of most objects, the other being absorption. Major forms of elastic light scattering (involving negligible energy transfer) are Rayleigh scattering and Mie scattering. Inelastic light scattering effects include Brillouin scattering, Raman scattering. Light scattering can also give color to some objects, usually shades of blue, but resonant light scattering in nanoparticles can produce different highly saturated and vibrant hues, especially when surface plasmon resonance is involved. At values of the ratio of particle diameter to wavelength more than about 10, the laws of geometric optics are mostly sufficient to describe the interaction of light with the particle, and at this point the interaction is not usually described as scattering. For modeling of scattering in cases where the Rayleigh and Mie models do not apply such as irregularly shaped particles, there are many numerical methods that can be used. The most common are finite-element methods which solve Maxwell's equations to find the distribution of the scattered electromagnetic field.

Rayleigh Scattering

Rayleigh scattering is a process in which light is scattered by a small spherical volume of variant refractive index, such as a particle, bubble, droplet, or even a density fluctuation. This effect was first modeled successfully by Lord Rayleigh, from whom it gets its name. In order for Rayleigh's model to apply, the sphere must be much smaller in diameter than the wavelength (λ) of the scattered wave; typically the upper limit is taken to be about 1/10 the wavelength. In this size regime, the exact shape of the scattering center is usually not very significant and can often be treated as a sphere of equivalent volume. The inherent scattering that radiation undergoes

passing through matter is due to microscopic density fluctuations as the molecules in matter move around, which are normally small enough in scale for Rayleigh's model to apply. This scattering mechanism is the primary cause of the blue color of the Earth's sky on a clear day, as the shorter blue wavelengths of sunlight passing overhead are more strongly scattered than the longer red wavelengths according to Rayleigh's famous $1/\lambda^4$ relation. Along with absorption, such scattering is a major cause of the attenuation of radiation by the atmosphere. The degree of scattering varies as a function of the ratio of the particle diameter to the wavelength of the radiation, along with many other factors including polarization, angle, and coherence.

Mie Scattering

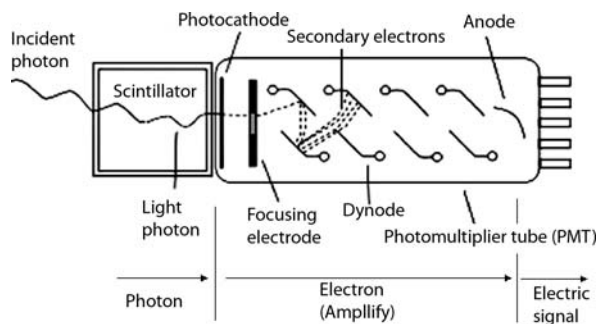
For larger diameters, the problem of electromagnetic scattering by spheres was first solved by Gustav Mie, and scattering by spheres larger than the Rayleigh range is therefore usually known as Mie scattering. In the Mie regime, the shape of the scattering center becomes much more significant and the theory only applies well to spheres and, with some modification, spheroids and ellipsoids. Closed-form solutions for scattering by certain other simple shapes exist, but no general closed-form solution is known for arbitrary shapes.

Raman Scattering

Raman scattering is the inelastic scattering of a photon. When light is scattered from an atom or molecule, most photons are elastically scattered (Rayleigh scattering). The scattered photons have the same energy (frequency) and, therefore, wavelength, as the incident photons. However, a small fraction of scattered light (approximately 1 in 1000 photons) is scattered from excitations with optical frequencies different from, and usually lower than, the frequency of the incident photons. Raman scattering can occur with a change in vibrational, rotational or electronic energy of a molecule. Raman spectroscopy employs the Raman effect for material analysis. The frequency of light scattered from a molecule is changed based on the structural characteristics of the molecular bonds. A monochromatic light source (laser) is required for sufficient illumination, and a spectrogram of the scattered light then shows the deviations caused by state changes in the molecule.

Doppler Shift

Electromagnetic radiation scattered by moving scattering centers does undergo a Doppler shift, which can be detected and used to measure the velocity of the scattering center/s in forms of techniques such as radar. Both Mie



Photometer, Figure 1 Photomultipliers are constructed from a glass vacuum tube which houses a photocathode, several dynodes, and an anode. Incident photons strike the photocathode material which is present as a thin deposit on the entry window of the device, with electrons being produced as a consequence of the photoelectric effect. These electrons are directed by the focusing electrode towards the electron multiplier, where electrons are multiplied by the process of secondary emission

and Rayleigh scattering are considered elastic scattering processes, in which the energy (and thus wavelength and frequency) of the light is not substantially changed. However, this shift involves a slight change in energy.

Absorption (Electromagnetic Radiation)

Absorption is the process by which the energy of a photon is taken up by another entity, for example, by an atom whose valence electrons make transition between two electronic energy levels. The photon is destroyed in the process. The absorbed energy may be re-emitted as radiant energy or transformed into heat energy. The absorption of light during wave propagation is often called attenuation. The absorbance of an object quantifies how much light is absorbed by it. This may be related to other properties of the object through the Beer-Lambert law. For most substances, the amount of absorption varies with the wavelength of the light, leading to the appearance of color in pigments that absorb some wavelengths but not others. For example, an object that absorbs blue, green and yellow light will appear red when viewed under white light. More precise measurements at many wavelengths allow the identification of a substance via absorption spectroscopy.

Fluorescent Measurement

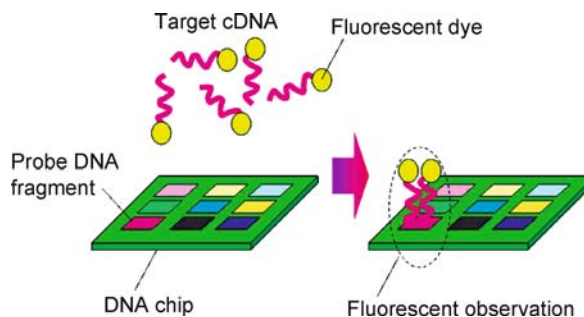
Fluorescent measurement uses the luminescence of fluorescence that is mostly found as an optical phenomenon in cold bodies, in which the molecular absorption of a photon triggers the emission of another photon with a longer wavelength. The energy difference between the absorbed and emitted photons ends up as molecular vibrations or

heat. Usually the absorbed photon is in the ultraviolet range, and the emitted light is in the visible range, but this depends on the absorbance curve and Stokes shift of the particular fluorophore. The fluorescence quantum yield gives the efficiency of the fluorescence process. It is defined as the ratio of the number of photons emitted to the number of photons absorbed. The maximum fluorescence quantum yield is 1.0 (100%); every photon absorbed results in a photon emitted. Compounds with quantum yields of 0.10 are still considered quite fluorescent. The fluorescence lifetime is an instance of exponential decay. Thus, it is similar to a first-order chemical reaction in which the first-order rate constant is the sum of all of the rates (a parallel kinetic model). Thus, the lifetime is related to the facility of the relaxation pathway. If the rate of spontaneous emission, or any of the other rates are fast the lifetime is short (for commonly used fluorescent compounds typical excited state decay times for fluorescent compounds that emit photons with energies from the UV to near infrared are within the range of 0.5 to 20 ns). The fluorescence lifetime is an important parameter for practical applications of fluorescence such as fluorescence resonance energy transfer. There are several rules that deal with fluorescence.

Photodetectors

Photoresistor

A photoresistor is an electronic component whose resistance decreases with increasing incident light intensity. It can also be referred to as a light-dependent resistor (LDR), photoconductor, or photocell. A photoresistor is made of a high-resistance semiconductor. If light falling on the device is of high enough frequency, photons absorbed by the semiconductor give bound electrons enough energy to jump into the conduction band. The resulting free electron (and its hole partner) conduct electricity, thereby lowering resistance. A photoelectric device can be either intrinsic or extrinsic. An intrinsic semiconductor has its own charge carriers and is not an efficient semiconductor, e. g. silicon. In intrinsic devices, the only available electrons are in the valence band, and hence the photon must have enough energy to excite the electron across the entire bandgap. Extrinsic devices have impurities added, which have a ground state energy closer to the conduction band – since the electrons don't have as far to jump, lower energy photons (i. e. longer wavelengths and lower frequencies) are sufficient to trigger the device. If a sample of silicon has some of its atoms replaced by phosphorus atoms (impurities), there will be extra electrons available for conduction. This is an example of an extrinsic semiconductor.



Photometer, Figure 2 DNA microarrays. The probes are complementary fragment oligonucleotides, of target DNA or mRNA are spotted onto the microarray surface. This type of array is typically hybridized with target cDNAs that are labeled with fluorophores. The samples are amplified by PCR and hybridized to the microarray, and then scanned in a microarray scanner to visualize fluorescence of the fluorophores on the chip. Absolute levels of gene expression cannot be determined, but relative differences in expression among different spots can be estimated when we used the two-color array, each color represents two different samples

Photodiode

A photodiode is a semiconductor diode that functions as a photodetector. Photodiodes are packaged with either a window or optical fibre connection, in order to let in the light to the sensitive part of the device. They may also be used without a window to detect vacuum UV or X-rays. A phototransistor is in essence nothing more than a bipolar transistor that is encased in a transparent case so that light can reach the *base-collector junction*. The phototransistor works like a photodiode, but with a much higher responsivity for light, because the electrons that are generated by photons in the base-collector junction are injected into the base, and this current is then amplified by the transistor operation. However, a phototransistor has a slower response time than a photodiode. A photodiode is usually a p-n junction structure. When a photon of sufficient energy strikes the diode, it excites an electron thereby creating a mobile electron and a positively charged electron hole. If the absorption occurs in the junction's depletion region, or one diffusion length away from it, these carriers are swept from the junction by the built-in field of the depletion region, producing a photocurrent. Photodiodes can be used under either zero bias (*photovoltaic mode*) or reverse bias (*photoconductive mode*). In zero bias, light falling on the diode causes a current across the device, leading to forward bias which in turn induces *dark current* in the opposite direction to the photocurrent. This is called the photovoltaic effect, and is the basis for solar cells – in fact, a solar cell is just a large number of big photodiodes. Reverse bias induces only little current (known as saturation or back current) along its direction. But a more important effect of reverse bias is widening of the depletion layer

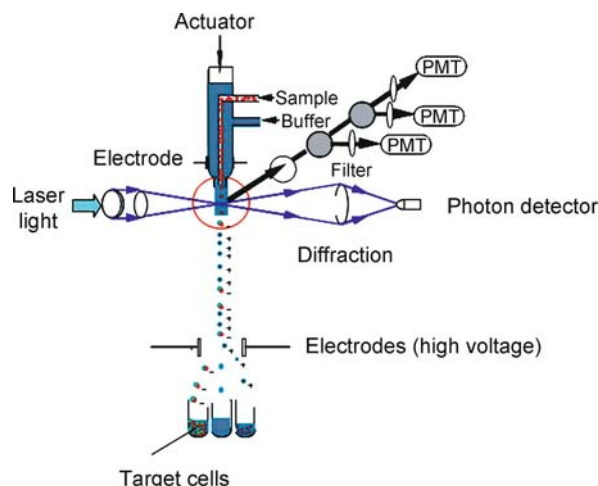
(therefore expanding the reaction volume) and strengthening the photocurrent. Circuits based on this effect are more sensitive to light than ones based on the photovoltaic effect and also tend to have lower capacitance, which improves the speed of their time response. On the other hand, the photovoltaic mode tends to exhibit less electronic noise.

Photomultiplier Tubes (Photomultipliers, PMT)

Photomultiplier tubes are extremely sensitive detectors of light in the ultraviolet, visible and near infrared. These detectors multiply the signal produced by incident light by as much as 10^8 , from which single photons can be resolved. Avalanche photodiodes have replaced photomultipliers in some applications, but photomultipliers are still used in most cases. The electron multiplier consists of a number of electrodes, called dynodes. Each dynode is held at a more positive voltage than the previous one. The electrons leave the photocathode, having the energy of the incoming photon (minus the work function of the photocathode). As they move towards the first dynode they are accelerated by the electric field and arrive with much greater energy. On striking the first dynode, more low energy electrons are emitted and these, in turn, are accelerated toward the second dynode. The geometry of the dynode chain is such that a cascade occurs with an ever-increasing number of electrons being produced at each stage. Finally the anode is reached where the accumulation of charge results in a sharp current pulse indicating the arrival of a photon at the photocathode.

Avalanche Photodiodes (APDs)

Avalanche photodiodes (APDs) are photodetectors that can be regarded as the semiconductor analog to photomultipliers. By applying a high reverse bias voltage (typically 100-200 V in **silicon**), APDs show an internal current gain effect (around 100) due to impact ionization (avalanche effect). However, some silicon APDs employ alternative doping and beveling techniques compared to traditional APDs that allow greater voltage to be applied (> 1500 V) before breakdown is reached and hence a greater operating gain (> 1000). In general, the higher the reverse voltage the higher the gain. Since APD gain varies strongly with the applied reverse bias and temperature, it is necessary to control the reverse voltage in order to keep a stable gain. Avalanche photodiodes therefore are more sensitive compared to other semiconductor photodiodes. If very high gain is needed (10^5 to 10^6), certain APDs can be operated with a reverse voltage above the APD's breakdown voltage. In this case, the APD needs to have its signal current limited and quickly diminished. Active and passive current quenching techniques have been used for this pur-



Photometer, Figure 3 Fluorescence-activated cell-sorting (FACS). The cell suspension is entrained in the center of a narrow, rapidly flowing stream of buffer liquid. The flow is arranged so that there is a large separation between cells relative to their diameter. A vibrating mechanism of actuator causes the stream of cells to break into individual droplets. The system is adjusted so that there is a low probability of more than one cell being in a microdroplet. Just before the stream breaks into microdroplets the flow passes through a fluorescence measuring station where the fluorescent character of interest of each cell is measured. An electrical charging ring is placed just at the point where the stream breaks into droplets. A charge is placed on the ring based on the immediately prior fluorescence intensity measurement and the opposite charge is trapped on the droplet as it breaks from the stream. The charged droplets then fall through an electrostatic deflection system that diverts droplets into containers based upon their charge. In some systems the charge is applied directly to the stream and the droplet breaking off retains charge of the same sign as the stream. The stream is then returned to neutral after the droplet breaks off

pose. APDs that operate in this high-gain regime are in Geiger mode. This mode is particularly useful for single photon detection provided that the dark count event rate is sufficiently low.

Key Research Findings

Biochemical and Medical Measurements

There is a wide range of applications for photometer in biochemistry and medicine. Especially, in micro/nano fluidics field, fluorescent chemical groups can be attached with small amount of biological molecules by a chemical reaction, and the fluorescence of the attached tag enables very sensitive detection of the molecule by fluorescent photodetector.

DNA Microarray

A DNA microarray (*gene chip*, DNA chip, or gene array) is a collection of microscopic DNA spots, commonly repre-

senting single genes, arrayed on a solid surface by covalent attachment to chemically suitable matrices. DNA arrays are different from other types of microarray only in that they either measure DNA or use DNA as part of its detection system. Qualitative or quantitative measurements with DNA microarrays utilize the selective nature of DNA-DNA or DNA-RNA hybridization under high-stringency conditions and fluorophore-based photodetection. DNA arrays are commonly used for expression profiling, i.e., monitoring expression levels of thousands of genes simultaneously, or for comparative genomic hybridization. Applications of these arrays include mRNA or gene expression profiling, comparative genomic hybridization (Array CGH), SNP detection arrays, and Chromatin immunoprecipitation (chIP) studies.

Fluorescent-Activated Cell Sorting (FACS)

Fluorescence-activated cell-sorting (FACS) is a specialised type of flow cytometry. It provides a method for sorting a heterogenous mixture of biological cells into two or more containers, one cell at a time, based upon the specific light scattering and fluorescent characteristics of each cell within microdroplet. It is a useful scientific instrument as it provides fast, objective and quantitative recording of fluorescent signals from individual cells as well as physical separation of cells of particular interest. The acronym FACS is trademarked and owned by Becton Dickinson although it is used within the scientific community as a general term.

Future Directions for Research

Photometer is important for identifying the materials in micro- and nano-scale flow, especially when the channel size is small and the flow speed is high. Although photometer itself does not have spatial resolution, the development of nanofluidics for single molecule detection should be desirable to develop new experimental methods and applications for single molecule based applications.

Cross References

- Bioprinting on Chip
- Infrared Imaging and Mapping for Biosensors
- Biosensors Using Laser
- Cell Sorting
- DNA Microarray
- Flow Cytometer Lab-on-Chip Devices
- Fluorescence
- Microfluidic Optical Devices
- SNP on Chip

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Photometry

- Photometer

Photo Patterned

Synonyms

Lithography; Optical lithography

Definition

Photo patterned is a process used in IC and microfabrication to selectively remove parts of a thin film (photoresist). It uses light to transfer a geometric pattern from a photomask to a light-sensitive chemical on the substrate. A series of chemical treatments then engraves the exposure pattern into the material underneath the photoresist.

Cross References

- Bulk Micromachining
- Photolithography

Photopatterned Materials

- Stationary Phases in Microchannels

Photoresist Reflow

Definition

Photoresist reflow refers to a unique method to generate the photoresist patterns with hemispheric cross-section using heating effect of photoresist for 3-D microstructure fabrications. The 3-D structures are formed by heating up the photoresist to a certain high temperature after the photoresist is patterned by the conventional lithography. When the temperature is higher than the glass transition temperature of the photoresist, the photoresist begins to melt

and reflow. Due to the surface tension of the liquid resist and the surface property of the substrate, the resist always tends to form a convex shape on top of the substrate. This is an effective way to form convex lens, cylinder type channels or round tops of the structures as the replica molds for MEMS applications.

Cross References

- Fabrication of 3-D Microfluidic Structures
- Photolithography
- Soft Photolithography
- Photoresist SU-8
- Photoresist Reflow
- Thick Resist Lithography

Photoresist SU-8

Definition

The photoresist SU-8 is an epoxy-based photosensitive polymer that loses its resistance to chemical etching when exposed to ultra-violet radiation. It is used especially in the transference of a structure pattern to a substrate. It offers properties such as high mechanical strength, good adhesion on many different substrate materials and biocompatibility. The fabrication of SU-8 structures is a low-cost process and ultra-violet lithography semiconductor compatible.

Cross References

- Lab-on-a-Chip Device for Chemical Analysis
- Microfabrication Techniques
- Photolithography
- Photoresist SU-8
- Photoresist Reflow
- Wafer Bonding

Photoresponsive Hydrogels

- Photosensitive Hydrogels

Photosensitive Hydrogels

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Synonyms

Photoresponsive hydrogels; Photoswitchable hydrogels

Definition

A hydrogel is defined as a network of hydrophilic polymer chains that is able to swell and retain large amounts of water. The crosslinks can be produced by chemical reactions to form covalent bonds, or physical interactions, such as hydrogen bonding, van der Waals force and polymer entanglement. Hydrogels can absorb over 99% water in their three-dimensional swollen network. Hydrogels possess a degree of flexibility very similar to natural tissue, due to their significant water content. Responsive hydrogels can reversibly change volume in response to a slight change of external stimulus, including pH, temperature, electric field, ionic strength, salt type, solvent, external stress or light. Photoresponsive is defined as a light-induced reversible change of chemical structures and physical properties.

Chemical and Physical Principles

Definition of Photochromism

Photochromism is the process of light-induced reversible color changes in a medium or more generally of their absorption spectra [1]. Photochromic compounds can exist in two different states, A and B, whose relative concentration depends on the wavelength of the incident light. The thermodynamically stable form A is transformed by irradiation into form B. The back reaction can occur either photochemically or thermally:



Photochromic molecules, or light-sensitive chromophores, are chemical species having photochromic properties. Examples of photochromic molecules and photochromic processes are summarized in Table 1.

Chemical Processes Involved In Photochromism

Pericyclic reactions Cycloadditions are found in spiropyrans, spirooxazines, chromenes, hexa-1, 3, 5-triene, diheteroarylethenes, cyclohexa-1, 3-diene and spirodi-hydroindolizines systems.

Cis-trans isomerizations Cis-trans isomerizations occur in stilbenes, azobenzene, azines, thioindigoids, as well as some biological receptors.

Intramolecular hydrogen transfer Intramolecular hydrogen transfer is found in anils, benzylpyridines, tri-

azoles, oxazoles, metal dithizonates and perimidine-spirohexadienones.

Intramolecular group transfers Intramolecular group transfers are found in polycyclic quinones (periaryloxyparaquinones).

Dissociation processes Heterolytic bond cleavages occur in triarylmethanes and related systems.

When incorporating a small amount of photochromic molecules into a hydrogel matrix, either through chemical coupling or physical impregnation, a molecular level conformational change often leads to a macromolecular structural change and then macroscopic change of shape due to swelling and contraction in the hydrogel. Such a photore-sponse has many advantages over other stimulation methods because:

- light is readily available;
- the amount of energy conversion can be carefully controlled by the wavelength, intensity of the exciting light, choice of chromophores and chromophore–polymer interactions;
- use of lasers offers high spatial resolution (up to nanometers), large areas and volumes, and fast speed; and
- it produces no waste during the photochemical process.

Key Research Findings

Hydrogels, which are crosslinked polymer networks surrounded with aqueous solution, are *intelligent* soft materials. In response to external stimuli, such as pH, temperature, light, electric potential, chemical agents and biological agents, polymer chains and water molecules will associate or dissociate, resulting in swelling and shrinking respectively, with a change of volume up to several hundred percent and a change of shape of the polymer.

In 1986, Irie and Kungwachakun reported a synthesis of photosensitive gels by incorporating photosensitive molecules (1–4 mol%), such as leucocyanide and leucohydroxide, into the gel network [2]. These gels underwent volume changes upon irradiation; however, no phase transition was observed. In 1990, a photoinduced phase transition was first reported in thermoresponsive *N*-isopropylacrylamide (PNIPAAm) gels, by copolymerizing light-sensitive bis(4-(dimethylamino)phenyl)(4-vinylphenyl)methyl leucocyanide into the gel network [3]. In the absence of light the gels are neutral and undergo a sharp, yet continuous volume change. Upon irradiation by ultraviolet light, the neutral triphenylmethane leuco derivatives are dissociated into triphenylmethyl cations, creating internal osmotic pressure that induces swelling of the gels, showing a discontinuous volume phase transition. This transition process is slow since it depends on

the photochemical ionization and subsequent recombination of ions.

Later, a much faster process (response time of the order of milliseconds) was reported. This was achieved by impregnating light-absorbing chromophores, trisodium salt of copper chlorophyllin, into PNIPAAm gels and illuminating with visible light [4, 5]. In this case, the phase transition was induced by local heating of polymer networks. According to the Flory–Huggins mean field theory, at equilibrium with zero osmotic pressure, the gel temperature, T , increases monotonically with the network density, ϕ :

$$T = T_{\text{gel}}(\phi) \quad (2)$$

When illuminated by light, the chromophore absorbs the light and subsequently dissipates heat locally, inducing an increase in the local temperature. The temperature increment is proportional to the incident light intensity and the concentration of chromophore, and therefore also to the network density ϕ . Thus

$$T = T_0 + \alpha I_0 \phi \quad (3)$$

where T_0 is the ambient temperature, I_0 is the illuminating light intensity and α is a constant related to the heat capacity of the gel. Therefore

$$T_0 = T_{\text{gel}}(\phi) - \alpha I_0 \phi \quad (4)$$

The term $\alpha I_0 \phi$ describes the ability of the system to undergo a local volume phase transition upon illumination of light. When the gel is in thermal equilibrium at a fixed gel temperature T , ϕ increase discontinuously with the increase of light intensity, I_0 , beyond a threshold value. When I_0 is decreased, ϕ decreases discontinuously and the gel does not exhibit hysteresis as a function of I_0 .

Because the increase of local temperature is the result of a balance between absorption of light and heat dissipation, the swelling and shrinking process of the gel in response to light is much faster compared to the response to ionization and ion migration, which requires collective diffusion of the gel network.

When introducing a small amount of ionic charge either through addition of salt or copolymerization of charged monomers in the light-sensitive gels mentioned above, the phase transition temperature was found also to be a function of pH. In ionic gels, the phase transitions are governed by the extent of hydrophobic interactions or the degree of hydrogen bonding induced between the polymer chains. The temperature and/or visible light intensity determine the extent of hydrophobic interactions, whereas pH affects the degree of hydrogen bonding. In the absence of external

Photosensitive Hydrogels, Table 1 Examples of photochromic molecules and photochromic processes

Photochromic molecules	Photochromic reactions
Spiropyrans	closed form $\xrightleftharpoons[hv \text{ (Vis)}]{hv \text{ (UV)}} \Delta$ open form
Spirooxazines	closed form $\xrightleftharpoons[hv \text{ (Vis)}]{hv \text{ (UV)}} \Delta$ open form
Acridizinium cations	2 $\xrightleftharpoons[hv \text{ (254nm)}]{hv \text{ (>300 nm)}}$ [Acridizinium cation]
Fulgides and fulgimides	open form $\xrightleftharpoons[hv_2]{hv_1} \Delta$ closed form X = O (fulgides) X = NR (fulgimides)
Azobenzene and derivatives	trans $\xrightleftharpoons[hv \text{ (Vis)}]{hv \text{ (UV)}} \Delta$ cis
Anils and related compounds (hydrogen transfer)	Anil $\xrightleftharpoons[\Delta]{hv}$ [Zwitterion]
Polycyclic quinones (periaryloxyquinones)	closed form $\xrightleftharpoons[hv \text{ (Vis)}]{hv \text{ (UV)}} \Delta$ open form
Triphenylmethane leuco derivatives	leuco form $\xrightleftharpoons[\Delta]{hv \text{ (UV)}} \Delta$ cationic form + X⁻

stimulus, the ionic gel exhibits a large hysteresis between the collapsed and swollen states, which is indicative of the existence of two free energy minima. It would, therefore, be possible to control the phase behavior by manipulation of other external factors in addition to visible light.

According to the principle of kinetics of gels, the time needed for swelling and shrinking of a gel is dependent on the diffusion of water in and out of the hydrogel, which is proportional to the square of the characteristic linear dimension of a gel, l^2 . It is expected that response time will be decreased to microseconds and nanoseconds if we can shrink the gel size to the submicrometer scale. This also means we need to perturb the environment extremely fast. Using a laser T -jump, a switching time of the order of ~ 100 ns was observed in individual PNIPAAm nanogels [6].

Another common approach to prepare light-sensitive hydrogels is to covalently couple chromophores either in the polymer side chains or in the main chains [7, 8]. The chromophores undergo crosslinking through photodimerization or *trans*–*cis* isomerization, which brings about a change of the dipole moment or structure, the dissociation of the chromophore into ions, or change of hydrophilicity of the polymer, thus varying the shape and volume phase transition of the hydrogel network.

Photocrosslinking linear PNIPAAm with photodimerizable chromophores, such as stilbene, stilbazolium and acridizinium cations, has been studied [8]. The reversible photodimerization reactions allow for dynamic tuning of the crosslinking density of the gel network, which in turn influences the phase transition temperature and swelling ratio. The introduction of the chromophores could affect the hydrophobic nature of the polymer. Compared to stilbene and stilbazolium, the positive charge in acridizinium cations compensates the hydrophobicity of aromatic groups, resulting in a smaller decrease of the critical transition temperature of PNIPAAm, -2.7°C vs. -11.0°C in the stilbazolium photopolymer, and a higher photoreversibility. A complete reversion to monomer will be impossible if the monomer absorption is higher than that of dimers at the irradiation wavelength.

A conformational change of a hydrogel can be induced by photoisomerization of a chromophore group, such as azobenzene and spiropyran. For azobenzene and derivatives, the thermodynamically more stable *trans* form is converted into the metastable *cis* isomer by UV light irradiation, and the *cis* isomer can return to the *trans* form by visible light irradiation or thermally in the dark. A spiropyran group is known to undergo photoisomerization from its neutral, *closed* spiro form (colorless and nonpolar) to the *open* merocyanine form (highly polar and zwitterionic) under UV light irradiation, which isomerizes back

to the spiro form under visible light irradiation or thermally in the dark [8]. The formation of zwitterionic merocyanines can be further exploited to bind with charged molecules to manipulate the phase transition of hydrogels. Kinetics study suggests that photoisomerization is dependent on solvent, polymer composition and the free volume distribution around the chromophore. A bulky substituent in the polymer chains seems to slow down the *trans*–*cis* isomerization but increases the thermal recovery rate (the *cis*–*trans* isomerization). Typically, it requires hundreds of seconds up to minutes of photoirradiation to induce reversible swelling and shrinkage of hydrogels using this approach.

Examples of Application

Because of their interesting properties, such as swellability in water, hydrophilicity, biocompatibility and lack of toxicity, hydrogels have been widely used in biological and biomedical applications. In response to an environmental stimulus, responsive hydrogels undergo various types of volume phase transition, which converts chemical potential energy into mechanical force. Thus, responsive hydrogels have been used in microfluidic devices such as actuators, pumps and valves, artificial muscles, controlled delivery, sensors and separation. Responsiveness to light and the potentially fast response time provide an additional opportunity to design microfluidic devices that can be triggered without disturbing the system. One could design a submicrometer optical switch on a gel surface, and control an on/off nature by focusing the light in a local area of the gel, generating features analogous to *memory* or *display*. When photoresponsive polymers are grafted onto a porous membrane, reversible UV or visible light irradiation induces a change of the polymer solubility in water, leading to photoregulating permeability and mass transport through the membrane [8].

Cross References

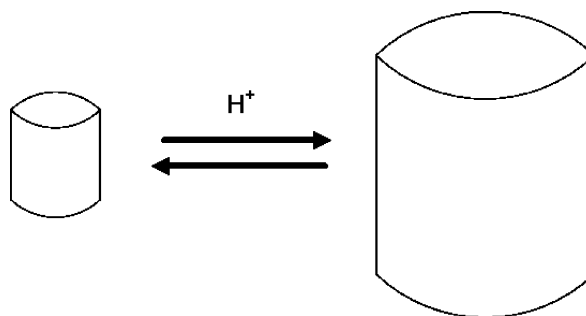
- pH-sensitive Hydrogel
- Photolithography
- Photoresist SU-8
- Photoresist Reflow
- Thermosensitive Hydrogel
- Soft Photolithography

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pH-Sensitive Hydrogel, Figure 1 Schematic representation of the reproducible swelling that occurs in a pH-sensitive hydrogel

Photoswitchable Hydrogels

► Photosensitive Hydrogels

Photothermal Effect

► Thermal Lensing, Detection

pH-Sensitive Hydrogel

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Synonyms

pH control device; pH-sensitive valve; pH sensor; Valve

Definition

A pH-sensitive hydrogel is a gel structure that is sensitive to changes in pH. Most commonly, the hydrogel will swell or contract in response to a change in the local chemical environment. Hydrogels can be made using in situ polymerization techniques, which makes it very useful in incorporating them into microfluidic systems. These pH-sensitive hydrogels have applications in forming valves that are sensitive to a change in pH or in systems that can release a compound when the pH is changed, and in pH sensors.

Overview

A pH-sensitive hydrogel consists of a gel structure which varies with a change in the pH of the system. The basis of a pH-sensitive hydrogel is a structure that swells or contracts depending on the pH of the solution (Fig. 1). The use of a hydrogel structure allows these devices to be easily fabricated within a microfluidic system. Placing these devices in a microfluidic device enables flow control to be accomplished by changing the pH of the solution that is being pumped through the system. Alternatively, a series of these hydrogels make it possible to sort solutions according to pH by placing two different hydrogels sensitive to either basic or acidic solutions into a microfluidic system, or by using a hydrogel to regulate the pH of solutions flowing out of a mixing hydrogel.

Basic Methodology

Hydrogels are polymer structures that contain a high water content. Because they contain a large amount of water, their physical configuration can change in response to the local chemical environment [1]. In particular, hydrogels undergo a change in volume in response to chemical or physical stimuli. Hydrogels that change their structure in response to temperature have been widely studied [2]. Hydrogels can also change in response to chemical conditions. The chemical environment that is often able to induce this type of volume change in a hydrogel is the pH of the system. In this type of a system the swelling of the hydrogel is produced from an ion-driven effect. How fast the gel expands is therefore dependent on the diffusion of ions into and out of the gel. Furthermore, swelling of the gel is an equilibration effect, and will be limited by how fast a steady state can be reached within the gel.

The ability to change in response to pH has been applied to two general applications. The first is as a valve. Because the volume of the hydrogel changes, a valve can be fabricated in which a hydrogel expands to plug a channel under

one set of conditions and contracts to allow flow under another set of conditions. The other application that hydrogels have been applied to is as pH detectors. This approach involves allowing a hydrogel to swell with a change in pH and to detect and measure the swelling of the hydrogel.

One of the great challenges in microfluidics is in preparing functional devices within a microfluidic system. Often, it is necessary to fabricate a functional device within a microfluidic system. Hydrogels are very useful in microfluidic systems because they can be either fabricated in bulk or they can be fabricated in situ. The in situ fabrication approach is particularly important in microfluidics because it greatly simplifies the fabrication of complex devices compared to more traditional methods of fabrication, which are usually based on semiconductor fabrication technology. The in situ synthesis of a hydrogel is usually performed using monomers that are somewhat hydrophilic, such as acrylic acid (AA), or 2-hydroxyethyl methacrylate (HEMA). A hydrophilic monomer allows water to penetrate the hydrogel, and therefore enables the polymer to swell and contract. The crosslinking monomer (such as ethylene glycol dimethacrylate, DGDMA) allows the polymer to retain its cohesiveness with swelling. The amount of crosslinker used affects many parameters in the resulting hydrogel such as how much it can swell, how much water can permeate the gel and the pore structure of the polymer. The actual polymerization step requires an initiator. The use of a photoinitiated monomer enables the hydrogel to be patterned into a specific portion of the microfluidic channel. The fabrication methods for placing a hydrogel into a microfluidic channel are critical in enabling their use in these applications.

Key Research Findings

There are two major areas that pH-sensitive hydrogels are used for in microfluidics. The first of these applications is as a valve. A hydrogel has the ability to change volume. When they expand, the hydrogel has the effect of closing off a channel. When the hydrogel contracts, the channel is opened. An early example of this approach involved introducing a pH-sensitive hydrogel into a microfluidic system in order to regulate the pH in the system [3]. This study utilized a circular chamber with a pH-sensitive hydrogel and included inlet channels that had solutions at a pH of 2 and 12. The outlet of the system was regulated at a pH of 7 by the action of the hydrogel. This study found that the geometry of the hydrogel has a strong effect on the regulation of the outlet pH. When a circular hydrogel is used the outlet pH was observed to oscillate; however, the use of a star-shaped hydrogel greatly reduced the oscillation. Another study fabricated two pH-sensitive hydrogels

in a “V” shape in order to mimic a biomimetic valve such as those found in the veins [4]. This valve arrangement has the effect that flow through the valve is enabled in the forward direction, but is restricted in the reverse direction, effectively acting as a check valve. Furthermore, flow regulation is enhanced because the pressure drop across the valve is constant across a wide range of flow rates (0.4–1.8 ml/min). This ability to have the same pressure drop across the valve is due to the elasticity of the hydrogel; it enables the hydrogel to act as a pressure regulator in the microfluidic system, while the pH sensitivity enables the valve to be activated and deactivated depending on the solution being used. In another study a pH-sensitive hydrogel sphere was used as the actuating element in a microfluidic valve [5]. This approach involved placing the pH-sensitive hydrogel microsphere in a chamber. The swelling microsphere contacts a thick-centered membrane. This contact causes the membrane to close the valve. This approach allows the valve to be activated without the hydrogel being present in the main microfluidic channel, and the activation of the valve can be performed independently of the solvent in the microfluidic channel. This is because the hydrogel microsphere is in a channel separate from the main channel, and the two channels are isolated from one another via the membrane. The primary drawback is that this approach requires a more complex fabrication process to incorporate the valve into the microfluidic system.

In addition to pH-sensitive hydrogels being used as valves in microfluidic devices, they have also been used as a method to determine the pH of a solution in a microfluidic device. The pH of the solution in a microfluidic system can be very important in performing functions on a solution, such as a protein digestion or a synthetic step. However, detection of the pH in a microfluidic channel can be very difficult due to the small volumes involved in these channels. One approach involved incorporating a pH-sensitive dye within a hydrogel [6]. The synthesis of the hydrogel with the pH-sensitive dye is relatively simple because the hydrogel is simply synthesized with the dye incorporated within the polymer gel structure. The pH of the system can then be determined by the color of the dye. Using UV-visible spectroscopy enables very specific determination of the dye state. The primary limitation on the time required to determine the pH is the diffusion of within the hydrogel. Another approach that has been demonstrated to measure the pH in a microfluidic channel involves a hydrogel actuating a piezoresistive device [7]. This approach involves fabrication of a pH-sensitive hydrogel in contact with a bending plate. This plate then actuates a piezoresistor, the electrical signal of which is then read to give an indication of the pH in the microfluidic device. While

the piezoresistor makes fabrication of these systems more complex, it has the advantage of enabling rapid and precise electronic indication of the pH of the system.

A key parameter that has been studied for pH-sensitive hydrogels is their mechanical properties [8]. This is very important because the macromolecular properties will determine how the gel performs in the system. For instance, the mechanical properties in a valve will determine how well it can seal a channel, and can also affect how fast the valve can close. A wide variety of experimental techniques have been developed to characterize the mechanical properties of microgels, particularly the tensile strength [9]. In these studies it was found that the mechanical properties are dependent on the extent of equilibrium swelling as well as the polymerization conditions used to produce the gel. The mechanical properties can be controlled by adjusting the amount of crosslinker that is used in the hydrogel composition. The ability to control the mechanical properties of a pH-sensitive hydrogel will be critical in applying these components into specific microfluidic applications.

Future Directions for Research

While there have been a number of devices demonstrated for pH-sensitive hydrogels, there is a great deal of development work that can still be done. In particular pH-sensitive hydrogels need to be demonstrated in more robust environments, and their reliability and longevity need to be explored. The use of these systems in more robust situations will enable them to be explored in more complex microfluidic systems. In particular, devices will need to be made that can incorporate multiple valves in order to enable complex flow paths to be constructed. While this may be accomplished utilizing pH-sensitive hydrogels, it may be advantageous to incorporate valves in the system that are actuated via different approaches. This may further enable complex systems to be fabricated. The synthesis of pH-sensitive hydrogels will also need to become more refined to enable them to be placed into these more complex systems. Part of this development will include refining the synthetic methods to produce hydrogels with the desired mechanical, physical and chemical properties. In developing hydrogels with specific chemical properties there are two possible approaches. The first is to incorporate specific chemical functionalities into the polymerization mixture of which the hydrogel is made. The other approach is to coat or react a specific chemical functionality on the surface [10]. The ability to control the chemistry will enable these devices to be used with a wide variety of solutions and may improve their versatility.

Cross References

- Capillary Force Valves
- Electrochemical Valves
- Electromagnetic Valves
- Microfluidic Sample Manipulation
- Photosensitive Hydrogels
- Sol-Gel Technique
- Thermosensitive Hydrogel

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P

pH-Sensitive Valve

- pH-Sensitive Hydrogel

pH Sensor

- pH-Sensitive Hydrogel

Physical-Chemical Etching

- Anisotropic Etching

Physical Domain

Synonyms

Physical region

Definition

This refers to the actual spatial domain under consideration. In computational methods, this is replaced by a computational domain using a simplified geometry and boundary conditions.

Cross References

- Meshless Methods
- Volume and Finite Difference Method for Modeling and Simulation

Physical Etching

- Sputter Etching

Physical Region

- Physical Domain

Physical Vapor Deposition

- Sputtering for Film Deposition

Picoliter Flow Calibration

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Synonyms

pL; Flow rate

Definition

Flow rate is defined as the volumes or masses of fluids flowing through a special area in unit time. The standard unit of flow rate is m^3/s . Picoliter (pL) is a unit of volume, $1 \text{ pL} = 10^{-15} \text{ m}^3$ and transfers to other units as: $1 \text{ pL} = 10^{-3} \text{ nL} = 10^{-6} \text{ }\mu\text{L} = 10^{-9} \text{ mL} = 10^{-12} \text{ L}$. An

example describes the volume of a picoliter: a droplet with a diameter of $20 \text{ }\mu\text{m}$ has a volume of about 4.2 pL .

Overview

Classic flow rate measurement tools are mass flowmeter, electromagnetic flowmeter, rotameter and turbine flowmeter, their measurement ranges being normally larger than $10^{-7} \text{ m}^3/\text{s}$ (0.1 mL/s). But in microscale flows, the range of flow rate is usually $10^{-7} - 10^{-12} \text{ m}^3/\text{s}$, in other words, $10^2 \text{ }\mu\text{L/s} - 1 \text{ nL/s}$. In nanoscale flows, the flow rate is from 10^{-15} to $10^{-18} \text{ m}^3/\text{s}$ ($1 \text{ pL/s} - 1 \text{ fL/s}$). So the measurement and calibration methods will be different for micro- or nanoscale flows, compared to classic macroscale flows.

Colin [1] has reviewed some measurement methods in microscale flow, including the weighing method, thermal method, wave reflection method, acoustic vibration method, and so on. Here two common methods, the weighing and displacement methods, are introduced. In the weighing method [2], the total mass M of fluids flowing out within a time Δt are measured, and the flow rate is calculated as:

$$Q_g = \frac{M}{\Delta t} \quad (1)$$

The uncertainty of this method is directly proportion to the accuracies of balance and stopwatch. If a balance with an accuracy of within $\pm 0.001 \text{ g}$ is used to measure a kind of flow whose range is larger than $1 \text{ }\mu\text{L/s}$, the measurement accuracy is about $\pm 1\%$.

The displacement method [1, 3] is another kind of indirect method, which calculates flow rate by measuring the displacement L of a fluid in a capillary tube with a diameter D and interval Δt :

$$Q_m = \frac{\pi D^2 L}{4 \Delta t} \quad (2)$$

Because flow rate Q_m is proportion to the second power of diameter D , the uncertainty of flow rate is mainly related to the accuracy of D . The measuring precision of the diameter of capillary tubes using a toolmaker's microscope is $\pm 0.01 \text{ mm}$, so the uncertainty can reach $\pm 2\%$, for a flow rate range of $1 - 100 \text{ nL/s}$.

For saving samples, reducing cross-contamination and operating fluids with smaller volume in microfluidic devices, designers hope that the dimensions of channels or tubes are as small as possible. When the characteristic dimension of channels decreases to the nanoscale, picoliter-scale flow will occur. De Heij et al. [4] designed a nozzle array of $50 \text{ }\mu\text{m}$ diameter, which can produce

100–700 pL discrete droplets. Ahmed and Jones [5] controlled dielectrophoretic (DEP) force, capillary force and inertial force by an electrophoresis method, and produced discrete droplets on the substrate of a channel, whose volumes were smaller than 10 pL. He et al. [6] drove water flow through a nozzle of diameter 1.7 μm by electric pulse, and formed 8–14 pL water droplets in an oil environment making use of Rayleigh instability. Ellson [7] used focused acoustic energy to spray droplets from the surface of fluids, whose volumes are about 0.1 pL–1 μL . Karnik et al. [8] controlled protein to transmit in nano- and microchannels under the effect of an electrostatic field, the dimension of the channel being 1–100 nm, equivalent to the Debye length. The height of the channel cross-section was just 30 nm, so this device can be used to control picoliter-scale flow of protein solution.

From these literature reports we can see that pL flows have been achieved and applied in sampling devices of biology chips, in micro- or nanochannels and two-phase flows, so the measurement and calibration of pL volume fluid flow have great importance attached to them.

Basic Methodology

The methods that are usually used to calibrate ultra small-volume fluids are the static weighing method, comparing method, flow velocity measurement method, and so on. The study of picoliter flows has just begun, so the perfect calibration methods have not been formed yet. This section will review partial methods to produce pL flow, their measurement accuracy and the relevant calibration methods. A fuller coverage is not possible because of space limitations.

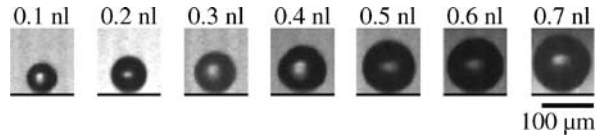
Visual Inspection Measurement

As shown in the literatures, pL flows always present as droplet form. So it is important to pay attention to measuring the volume of droplets in pL flow measurement. The common methods are calculating volume by the size of the microvial or microchannel in which droplets are formed, or measuring droplet diameters from their images and calculating volume by the ball volume formula:

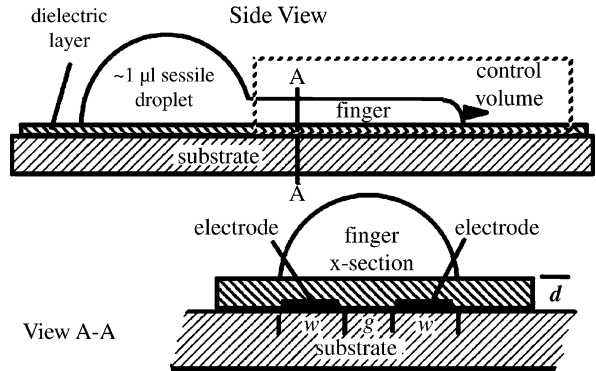
$$V_d = \frac{4}{3}\pi R^3 = \frac{1}{6}\pi D^3 \quad (3)$$

Here, the radius R or diameter D are measured from images captured by a microscope, with a calibrated microruler.

De Heij et al. [4] designed a nozzle array to produce continuous droplets. The diameter of the nozzle is 50 μm .



Picoliter Flow Calibration, Figure 1 The droplet volume measured with stroboscope camera [4]

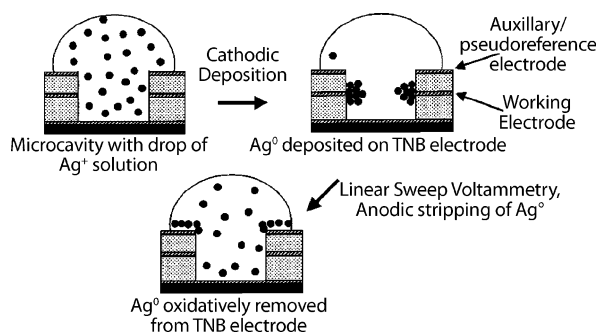


Picoliter Flow Calibration, Figure 2 Side view of a DEP-driven finger emerging from a sessile droplet using parallel co-planar electrode (top) and the cross-sectional (view A-A) [5]

A digital stroboscope (Visit Video Stroboskop MOCRON-RT) was used to record the spraying process of droplets, and then the image processing software was used to calculate the diameter (Fig. 1). The accuracy of image measurement is $\pm 5\%$. They have measured droplets from 0.1 to 0.7 nL, and the measurement accuracy is ± 0.1 pL.

Ellson et al. [7] used the method of focused acoustic energy to eject a pL droplet from a nanoliter-scale well, the droplet diameter being also measured by a digital stroboscope. In their experiment, 20 pL droplets were ejected from 20 nL liquids. The accuracy of liquid volume measurement is determined by the accuracy of image measurement. They used a precisely calibrated micrometer to measure the image size, the accuracy being estimated as about $\pm 5 \mu\text{m}$ based on the image resolution. The minimum volume of experimental droplets is 20 pL, and the measurement accuracy is ± 0.1 pL. Thousands of droplets were produced continuously, and the whole fluid volume divided by the number of droplets calibrated the average volume.

Ahmed et al. [5] proposed a method to produce discrete pL droplets on substrate by an electrophoresis method. A finger will be formed when water is driven by electrophoresis force above an electrode (Fig. 2). The key parameter dominating capillary instability is the Bond number, which is expressed by $Bo = \rho g R^2 \gamma$, which presents the relation between hydrostatic pressure $\rho g R$ and Laplace pressure



Picoliter Flow Calibration, Figure 3 Schematic diagram of the electrode and the procedure for small volume measurement using anodic stripping voltammetry [9]

γ/R . If the electrical potential is taken out for $Bo < 10^{-3}$, the capillary force will dominate, and the finger will be broken to form droplets. The front part of the finger is assumed to form a hemispherical shape droplet, whose diameter is about $D = 2w + g$, where w is the width of the electrode and g is the interspace between the electrodes. The volume of the hemispherical droplet can be calculated by reference to formula (3).

Ahmed et al. used a camera to record the droplet forming process at 500 frames/s, 6 continuous frames being used to calculate the extending length of the finger, and compared to the estimated value of droplet diameter. In the experiment, the average volume of 30 droplets is 40 pL. If the size of the microelectrodes decreases further, 10 pL volume droplets would be obtained. The minimum measurable volume is of the order of 10 pL, and the accuracy is about ± 5 pL, but the volume measured by this method has not been calibrated.

In the methods discussed above, fluid volumes were measured by visual inspection, so other methods are needed for fluids which cannot be inspected directly or which flow in opaque channels.

Electrical Measurement

Coulometric Method

Vandaveer et al. [9] proposed anodic stripping voltammetry to measure fluids with ultra-small volume. The advantages of this method are avoiding optical detecting devices, fast and easy and suitable for irregular shapes of fluid flows.

A microvial with electrodes in the wall is used in this method. The volume of fluids can be calculated by exhaustive reduction of current under the effect of continuous changed voltage (Fig. 3). Recessed microdisk (RMD) electrode, tubular nanoband (TNB) electrode and reference electrodes were installed on the bottom, sidewall and

top of the microvial, respectively. Soluble metal species were added into the sample fluids, and then they were deposited on the electrodes on the sidewall driven by voltage. After that, the deposited metal species were removed under a voltage of -0.6 to 1 V. Droplets volumes can be measured indirectly by the quantity of electricity integrated from the relation between removed current and voltage of reference electrodes, the formula being expressed as [9]:

$$V_s = \frac{Q}{nFC^*} \quad (4)$$

where Q is the total Coulomb value integrated from the anodic removed curve, n is the moles of electrons transferred per mole of metal ions, F is the Faraday constant ($F = 96,485$ C/mol) and C^* is the initial concentration of metallic ions in solution.

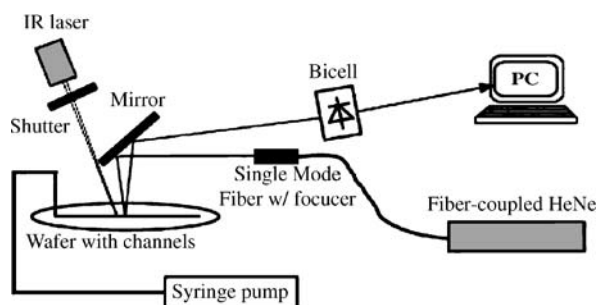
Electrodes are installed on the wall of the microvial, and as a result the minimum size of the microvial is $8 \mu\text{m}$ in height and $13 \mu\text{m}$ in diameter; in other words, 1 pL measurable volume. The accuracy of this method is $1.49 \text{ nL} \pm 20 \text{ pL}$. It is easier for smaller volume fluids to dissipate and deposit, so this method is better for measuring fluid volumes below 1 nL. Also, because the method is coulometric, calibration curves are not required [9].

Impedance Method

Hjelt et al. [10] proposed an integrated, noninvasive method to measure sub-nanoliter volume droplets. Microelectrodes were integrated on the bottom of a reactor and covered with a layer of SiN. According to the impedance characteristic between electrodes, the size of droplet can be obtained. The volumes of the reactor fabricated in the experiment are 60–40 pL. The relation between liquid volumes and electrodes voltages was calibrated, and the accuracy of voltage measurement was $\pm 5\%$, and the resolution of volume measurement could be up to 1 pL. To calibrate the measurement results of the impedance method, the authors used a fluorescence microscope to inspect the liquid volume inside the reactor, and they concluded an accuracy could be achieved of $\pm 5\%$. The minimum measuring volume in the experiment is 30 pL, and the measurement accuracy is ± 1.5 pL.

Optical Measurement

Markov et al. [11] measured the velocity of the flow in an electrophoresis chip by backscatter interferometry (Fig. 4). An infrared laser with a shutter was used to heat the fluid periodically in an upstream position, and a bicell was used as a position-sensitive device in the detecting section. The



Picoliter Flow Calibration, Figure 4 Diagram of backscatter interferometry for fluid flow measurement [11]

changes of refractive index of fluids as a function of temperature were converted to electrical signals, and these signals had the same frequency as that of the heating laser, but with different phase. The phase difference between the shutter signal and the bicell signal was used to calculate the time interval Δt during which fluids flowed from the heating section to detecting section. Then the velocity can be calculated. The authors also calibrated the relations between the flow rate and the phase difference.

Experiments were carried out in a $200\text{ }\mu\text{m}$ channel, with a measurement accuracy of $0.833 - 1.66\text{ nL/s} \pm 0.127\text{ nL/s}$, but the velocity measurement results were not calibrated.

Key Research Findings

Several methods to obtain and measure pL fluid volumes have been reviewed: visual inspection measurement, electrical measurement and optical measurement. Visual inspection measurement captures a droplet image by microscope or stroboscope, and measures the size of the droplet in the image to calculate its volume. The minimum volume of droplets can reach 10 pL , with a measurement accuracy of $\pm 0.1\text{ pL}$. Coulometric and impedance methods were introduced as electrical measurements, which measure the quantity of electricity or impedance of fluids to calculate the droplet volume. The minimum volume of droplets measurable with these methods is 30 pL , with $\pm 1\text{ pL}$ accuracy. Finally, optical measurement, especially the backscatter interferometric method, was introduced. This method measures the difference of light phase to calculate the velocity of a fluid, and then calculates the volume. Flow rates from 0.833 to 1.66 nL/s were measured in experiments and the accuracy was $\pm 0.127\text{ nL/s}$.

Obviously, visual inspection measurement is the easiest method with the highest accuracy to date. The impedance method also uses it for calibration [10].

For calibration, if the cumulative volume of liquid is larger than $1\text{ }\mu\text{L}$, the volume weighing method is suggested.

But for calibration of single droplet volume, the visual inspection method is simpler.

Future Directions for Research

In principle, calibration methods of ultra small-volume droplets include static weighing method, comparing to standard volume and calibration by high-accuracy velocity measurement. Traditional optical devices, microscopes, stroboscopes, etc., are still the main measurement method for pL fluid volume calibration, and the accuracy is about 0.1 pL . Novel measurement techniques are needed for calibration of sub-picoliter fluid volume.

Cross References

- Fluid Metering
- Droplet Dispensing
- Nanoscale Flow Visualization

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Piezoelectric

Definition

Coupling phenomenon between mechanical strain and stress, and the electric field strength and displacement in

a material. Requires a polarization that remains within a material in the absence of the application of an external field, in turn requiring a non-enantiomorphic polar symmetry in the composition of the material.

Cross References

- [Non-Centrosymmetric](#)
- [Ferroelectric](#)
- [Acoustics Based Biosensors](#)

Piezoelectric Drop-on-Demand

- [Piezoelectric Microdispenser](#)

Piezoelectric Ink Jet

- [Piezoelectric Microdispenser](#)

Piezoelectric Materials for Microfluidics

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Synonyms

Lead zirconate titanate; PZT; Zinc oxide; Quartz; AlN; Electroactive materials; Electromechanically coupled materials; Artificial muscle materials; Electrets

Definition

Piezoelectricity means *squeeze* or *pressure* electricity, from the Greek *pieze* which means to press or squeeze. The casual definition of piezoelectric materials is that they develop an electric charge differential along an axis of ► [piezoelectric](#) polarization if placed under appropriate mechanical strain, called the ► [direct piezoelectric effect](#), and deform if an electric field is applied along the same axis of polarization, the converse piezoelectric effect. The electric field to mechanical strain coupling occurs when the positive and negative charge of the individual ions within the material do not displace symmetrically upon the application of an external strain. Strictly speaking, piezoelectric materials exhibit electric polarization upon the application of a stress and vice versa. Often the casual and strict definitions are the same, though confusion may arise with the casual definition. All piezoelectric materials

are ► [anisotropic](#), and there are useful examples of piezoelectric materials that are single crystals, polycrystalline ceramics, and polymers. Many of the known piezoelectric materials have at one time or another been used as sensors or actuators in microfluidics, for which their rapid response and large force transmission have been beneficial, though researchers have had to learn to overcome small strains, thermal losses, aging, and difficulties in fabrication, especially in process compatibility.

Overview

Piezoelectric materials offer the ability to directly transform (transduce) from electrical to mechanical energy and vice versa, convenient for microtechnology in sensing and actuation; a material is said to be piezoelectric if an applied stress generates an electric field within, and vice versa. They may be used to provide static and dynamic deformation at frequencies up to several gigahertz, depending on the motion to be induced, the scale of the device, and the material used. Compared to other methods of power transduction, including electrostatic, electrostrictive, magnetostrictive, and electromagnetic means, piezoelectric transduction typically provides large forces at small strain rates (< 1% strain typically) for actuation, and relatively large voltages and small currents for sensing. Piezoelectricity is usually treated as a linear interaction between mechanical and electrical phenomena, though this assumption is generally invalid for high-power applications or in materials with either large losses or large deformations (polymers). The ► [hysteresis](#) in such conditions is the deviation from linear transduction. This is different than the typically weak ► [electrostrictive](#) [2] behaviour of many insulating materials which exhibit a strain proportional to the square of the applied field, though some materials, like ► [relaxor ferroelectrics](#) are exceptions. Single crystal materials are usually naturally polarized, implying they may be used without additional treatment, while polycrystalline and polymeric materials must be polarized with either an applied electric field or an applied strain prior to use.

Recognition of the piezoelectric effect has been known since ancient times [6]. Theophrastus noted in 314 BC the attraction of small pieces of straw and wood to tourmaline after heating, but recognition of the presence of an electrostatic field causing the attraction had to wait for the nineteenth century, along with a name provided by Brewster in 1824: ► [pyroelectricity](#). All pyroelectric materials are piezoelectric; Jacques and Pierre Curie studied piezoelectric materials in the 1880's and were able to describe many of the characteristics of the phenomena used in applications today. Once synthetic quartz and high-performance

materials like barium titanate, lead zirconate titanate and subsequent variations became available in the 1950's and 60's, a wide variety of applications beyond sonar became feasible.

Many of the piezoelectric materials used for applications are single crystals or polycrystalline matrices of single-crystal grains. Overall, crystalline media may be organized into thirty-two crystal classes or point groups based on their symmetry; the names of some of these groups are commonly known; triclinic, orthorhombic, cubic, and isotropic are especially well-known. ► **Hermann–Mauguin notation** [1] is often used to precisely describe the properties of these materials. For example, triclinic materials are represented with either a 1 or $\bar{1}$, depending on whether or not the material has ► **inversion symmetry**, respectively. Of these thirty-two crystal classes, twenty-one possess no centre of symmetry; they are said to be ► **non-centrosymmetric** or to lack inversion symmetry, and all but one of these classes are piezoelectric. The exception is class 432, which has other symmetries which negate the non-centrosymmetric nature of the crystal structure. The remaining crystal classes which are piezoelectric are 1, 2, m, 222, mm2, 23, 3, 3m, 32, 4, $\bar{4}$, 422, 4mm, $\bar{4}2m$, $\bar{4}3m$, 6, $\bar{6}$, 622, 6mm, and $\bar{6}2m$.

Out of these twenty piezoelectric classes, ten also lack symmetric characteristics along a particular axis, and are therefore able to retain a natural ionic charge separation in the crystalline structure along that axis—a ► **spontaneous polarization**—that forms an electric dipole which remains even without an externally applied electric field. This is in contrast to all insulating or dielectric materials which polarize to form such dipoles only with an externally applied field. These crystal groups are *pyroelectric*, including 1, 2, m, mm2, 3, 3m, 4, 4mm, 6, and 6mm. A crystal belonging to one of these crystal groups that has a polarization that may be reversed through the application of an external electric field is said to be ► **ferroelectric**. A distinct characteristic of ferroelectric materials, in the strictest definition, is the presence of hysteresis. All ferroelectric materials are both piezoelectric and pyroelectric. Lead zirconium titanate (PZT or $\text{Pb}_x\text{Zr}_{(1-x)}\text{O}_3$) is one of the most important ferroelectric materials, class 6mm, though PZT is only found in polycrystalline form as a ► **perovskite** ABO_3 .

Most piezoelectric materials tend to be quite capacitive. The pyroelectric nature of all polarized piezoelectric materials can cause the development of significant charging across the material due to thermal expansion coupled to the direct piezoelectric effect causing a charge displacement from the thermal strain; voltages of 1 kV are not unheard of in PZT after a temperature change of 50 °C, and caution is warranted in any application. However,

a piezoelectric material will lose its spontaneous polarization if its temperature is raised above the ► **Curie temperature**, near the temperature of maximum dielectric constant, the ► **Curie–Weiss temperature**. Since the mechanical and electrical behavior are coupled, a change in boundary conditions in one domain will change the behavior of the material in the other domain. By leaving a polarized piezoelectric element open-circuited, the mechanical stiffness of the material will be higher, called stress stiffening. Similarly, mechanically clamping a piezoelectric element will tend to increase the impedance.

With polycrystalline and amorphous *piezoelectric* materials, a spontaneous polarization can rarely be obtained as a consequence of the natural ordering within the material during fabrication. Some form of applied strain or electric field is necessary to polarize the material before use. A common sensor material, polydivinylfluoride (PVDF), is stretched to polarize it, while standard PZT must have a large ($> 10 \text{ kV/cm}$) electric field applied to it to order the electric Weiss domains, or groups of dipoles, in the general direction of the applied field. It is a common misconception that polarization of these materials implies the dipole polarization is rotated to be aligned with the electric field or that the atomic structure itself is realigned within the material. In reality, the atomic structure is essentially fixed, preventing such rotation or realignment. Once polarized, the material can be said to be a member of the pseudocrystal group $\infty \text{ mm}$, and is often represented as a 6 mm material. Generally, the domains are randomly oriented beforehand.

At times, particularly with amorphous materials, the term *electret* is used instead of *piezoelectric* to describe a material; the term was introduced as an electric counterpart to the magnet, defined as all materials that are able to retain at least a quasi-permanent electric polarization within, and perhaps trap charge on the surface (or within in a porous material). Polytetrafluoroethylene (PTFE) is a powerful electret material. The critical difference among the terms is that electrets retain polarization in thermodynamic nonequilibrium, pyroelectric materials retain polarization in equilibrium, ferroelectrics permit the equilibrium polarization to be reversed along the polarization axis, and piezoelectric materials may or may not have any polarization at all.

Basic Methodology

Modeling Linear Piezoelectricity

Equations for structural dynamics within the solid material, along with the standard equations for electrostatics, are sufficient to model most piezoelectric materials and their applications. The full electromagnetic equations are

T	S	E	D
<i>Mechanical Stress</i>	<i>Mechanical Strain</i>	<i>Electric Field Strength</i>	<i>Electric Field Displacement or Flux Density</i>
g	h	e	d
$= \frac{\text{field}}{\text{appl. stress}} = \frac{\text{strain}}{\text{applied charge electrode area}}$	$= \frac{\text{field}}{\text{appl. strain}} = \frac{\text{stress}}{\text{applied charge electrode area}}$	$= \frac{\text{stress}}{\text{applied field}} = \frac{\text{short circuit charge}}{\text{electrode area applied strain}}$	$= \frac{\text{strain}}{\text{applied field}} = \frac{\text{short circuit charge}}{\text{electrode area applied stress}}$
<i>Piezoelectric Field to Stress Coupling</i>	<i>Piezoelectric Field to Strain Coupling</i>	<i>Piezoelectric Stress to Field Coupling</i>	<i>Piezoelectric Strain to Field Coupling</i>
s^E	c^E	β^T	ϵ^T
<i>Compliance</i>	<i>Stiffness</i>	<i>Inverse Permittivity</i>	<i>Permittivity</i>
s^D	c^D	β^S	ϵ^S
<i>Compliance</i>	<i>Stiffness</i>	<i>Inverse Permittivity</i>	<i>Permittivity</i>
k_i	K^S	K^T	
$i = \begin{cases} P & \text{Planar} \\ 33 \text{ or } t & \text{Thickness} \\ 15 & \text{Shear} \\ 31 & \text{Transverse} \end{cases}$		$= \frac{\epsilon^T}{\epsilon^0} = \frac{\epsilon^T}{8.85 \times 10^{-12} \text{ F/m}}$	
<i>Electromechanical Coupling Factor</i>	<i>Relative Dielectric Constant</i>	<i>Relative Dielectric Constant</i>	

Piezoelectric Materials for Microfluidics, Figure 1 Variables used in linear piezoelectric material analysis

unnecessary since the mechanical motion propagates at $\sim 10^3$ m/s, several orders of magnitude less than the electric field, permitting the so-called *quasistatic* assumption. For the same reason, magnetic fields are typically weak and may be safely ignored. Still, equations to describe the electrical and mechanical behaviour are needed in addition to the constitutive equations for the material.

Using Auld's notation [1], the full tensor form of the linear constitutive equations with the strain and charge displacement as dependent on the stress and applied field may be

written as

$$\begin{aligned} \mathbf{T} &= \mathbf{c}^E : \mathbf{S} - \tilde{\mathbf{e}} \cdot \mathbf{E} \\ \mathbf{D} &= \mathbf{e} : \mathbf{S} + \tilde{\mathbf{\epsilon}}^S \cdot \mathbf{E} \end{aligned} \quad (1)$$

where the terms are defined with units in Figs. 1 and 2 (using contracted notation to be described later). The tilde ($\sim$) is a transpose while the single and double dots represent an inner product across one and two dimensions

Mechanical	S_{11}^D — Compliance measured with electrodes open circuited Related to stress in 1 dir. (strain for stiffness, c) Related to strain in 1 dir. (stress for stiffness, c) T_1 — Indicates stress is in 1 dir.	C_{36}^E — Stiffness measured with electrodes short circuited Related to strain in 3 dir. (stress for compliance, s) Related to shear stress around 3 dir. (strain for compliance, s) S_6 — Indicates strain about 3 dir.		
	d_{31} — Electrodes to support this field are perpendicular to the 3 dir. Piezoelectrically induced strain or the applied stress is in the 1 direction g_{31} — Electrodes to detect this field are perpendicular to the 3 dir. The applied stress or the piezoelectrically induced strain is in the 1 direction	h_{15} — Electrodes to detect this field are perpendicular to the 1 dir. The applied strain or the piezoelectrically induced stress is in shear about the 2 direction e_{15} — Electrodes to support this field are perpendicular to the 1 dir. Piezoelectrically induced stress or the applied strain is in shear around the 2 dir.		
Electric	ϵ_{11}^S — Permittivity measured with material fixed (clamped) Charge displacement in 1 dir. Electric field in 1 dir. E_1 — Indicates field is in 1 dir.	β_{33}^T — Inverse permittivity measured with material free Electric field in 3 dir. Charge displacement in 3 dir. D_3 — Indicates charge displacement is in 3 dir.		
Units	Variable	Name	Derived Units (MKS)	Fundamental Units
	c	Stiffness	Pa	kg / m • s ²
	s	Compliance	1 / Pa	m • s ² / kg
	T	Stress	Pa	kg / m • s ²
	S	Strain	—	—
	d	Piezo. Strain Coefficient	m / V or C / N	C • s ² / kg • m
	g	Piezo. Voltage Coefficient	m • V / N or N • m / C	m ² / C
	e	Piezo. Stress Coefficient	C / m ²	C / m ²
	h	Piezo. Stiffness Coefficient	V / m	kg • m / C • s ²
	ϵ	Permittivity	F / m	C ² • s ² / kg • m ³
	β	Inverse Permittivity	m / F	kg • m ³ / C ² • s ²
	D	Electric Charge Displacement	F • V / m	C / m ²
	E	Electric Field Strength	V / m	kg • m / C • s ²
	K	Relative Dielectric Constant (ϵ/ϵ_0)	—	($\epsilon_0 = 8.85 \times 10^{-5}$ F / m)
	k	Electromechanical Coupling Factor	—	—

Piezoelectric Materials for Microfluidics, Figure 2 Description of subscripts used in piezoelectric analysis

of the adjacent tensors, respectively. In piezoelectric materials analysis, the contravariant basis vectors are typically Cartesian, therefore orthonormal and so the covariant basis is not needed. Upon choosing a coordinate system, an indicial notation may be adopted [14],

$$T_{ij} = c_{ijkl}^E S_{kl} + e_{ijk} E_k$$

$$D_i = e_{ijk} S_{jk} + \epsilon_{ik}^S E_k, \quad (2)$$

where summation is assumed across repeated subscripts in products of terms, i.e., k and l in $c_{ijkl}^E S_{kl}$; all of the sub-

scripts range from 1 to 3. The strain S_{ij} is given by a strain-displacement relationship $S_{ij} = \frac{1}{2} (u_{i,j} + u_{j,i})$ where the comma in the subscript represents a spatial derivative ($u_{i,j} = \partial u_i / \partial x_j$). The momentum equation is $T_{ij,i} + f_i = \rho \ddot{u}_j$ where $\ddot{u} = \partial^2 u_j / \partial t^2$ and f_i represents a body force. The time-dependent electric field, E_i , and charge displacement, D_i , may be related through the polarization and permeability equation $D_i = \epsilon_0 E_i + P_i$ where P_i is the polarization. A scalar potential φ may be defined to represent the electric field, $E_i = \varphi_{,i}$, while the charge displacement is spatially constant along a particular direction, $D_{i,i} = 0$.

A piezoelectric Poynting energy flux akin to those defined for nonpiezoelectric acoustic waves and electromagnetic waves is given by $-(T_{ij}\dot{u}_j - \varphi\dot{D}_j)$.

In attempting to use these equations to solve problems, a complete and consistent set of electromechanical boundary conditions are needed. Tiersten [13] provides such a derivation, though solution of anything beyond simple problems requires computation. Fortunately, the mechanical and electrical boundary conditions are usually separable; $u_i = u_i^*$ or $n_i T_{ij} = n_i T_{ij}^*$, and $n_i D_i = n_i D_i^*$ or $\varphi = \varphi^*$. The starred terms represent constants, time-varying values, or values defined within another media adjacent to the piezoelectric material. The outward vector normal to the boundary surface is given by n_i .

Even in the most anisotropic of materials substantial symmetry exists in Eq. (2), permitting contraction of the subscripts [4] and giving a simpler form for the constitutive equation;

$$\begin{aligned} T_p &= c_{pq}^E S_q + e_{mp} E_m \\ D_n &= e_{nr} S_r + \varepsilon_{nm}^S E_m, \end{aligned} \quad (3)$$

where p, q , and r range from 1 to 6, while m and n range from 1 to 3. There is *not* a one-to-one correspondence between the two representations in Eqs. (2) and (3). For example, the strain S_{ij} is $S_p = 2^{|\text{sgn}(i-j)|}$. For this reason and the confusion that can arise in defining the momentum and strain-displacement equations, one must use care in solving problems with the contracted notation. The value of p in terms of i, j varies notoriously between authors, but the IEEE standard [4] is the most commonly accepted version; here $p = \frac{1}{2}(i+j)(1 - |\text{sgn}(i-j)|) + |\text{sgn}(i-j)|(9-i-j)$ for $i, j = 1, 2, 3$; here the vertical bars $|\cdot|$ represent an absolute value while $\text{sgn}(\cdot)$ is the *signum*, zero when x is zero and the sign of x otherwise.

Most solutions found by solving the complete set of piezoelectric material equations are very simple approximations. Computation, typically using finite element analysis software, is often necessary for practical cases. The ability to handle piezoelectric materials in commercially available finite element analysis software varies, though ANSYS (ANSYS Inc., Canonsburg, PA, USA) and COMSOL (COMSOL Inc., Burlington, MA, USA) are two well-known packages with decent implementations. However, a one-dimensional model of the piezoelectric material behavior offers many of the salient features without the complexity.

Piezoelectricity in One Dimension: Transducer Equations

If we assume a longitudinal plane wave propagates along the z -axis—assumed to be a principal axis of the

material—the only non-zero displacement will also be along the z - or 3-axis, $u = u(z, t)$ and the strain will be $S_{33} = \partial u / \partial z$. Further, assuming the applied electric field is only along the 3-axis, $E_3 = E$ while $E_1 = E_2 = 0$, we may find the resulting constitutive equations from substitution into Eq. (2), closely following the approach used by Dieulesaint and Royer [3]:

$$\begin{aligned} T_{ij} &= c_{ij33}^E S_{33} + e_{3ij} E_3 \\ D_i &= e_{i33} S_{33} + \varepsilon_{i3}^S E_3 \end{aligned}, \quad (4)$$

where i must be equal to j . If we set $i = j = 1$, the motion would be transverse to the applied field (setting i and j to 2 would give the same result), while if $i = j = 3$, the motion is parallel. Taking the latter option with Eq. (4),

$$\begin{aligned} T_{33} &= c_{3333}^E S_{33} + e_{333} E_3 & T &= c^E S + e E \\ D_3 &= e_{333} S_{33} + \varepsilon_{33}^S E_3 & D &= e S + \varepsilon^S E \end{aligned}, \quad (5)$$

where we discard the subscripts here and subsequently. The electric field, E , may be eliminated from Eq. (5):

$$T = \left(c^E + \frac{e^2}{\varepsilon^S} \right) \frac{\partial u}{\partial z} - \frac{e}{\varepsilon^S} D, \quad (6)$$

giving

$$\frac{\partial T}{\partial t} = \left(c^E + \frac{e^2}{\varepsilon^S} \right) \frac{\partial^2 u}{\partial z \partial t} - \frac{e}{\varepsilon^S} \frac{\partial D}{\partial t} \quad (7)$$

if we take a time derivative. The quantity $\left(c^E + \frac{e^2}{\varepsilon^S} \right) \equiv c^D$; this relation is also true for the tensor form, and indicates the relationship between the constant-field stiffness c^E and the constant field displacement c^D . There are relations between all of the coefficients as described later. The current passing through the piezoelectric element is given by $I = \int \frac{\partial D}{\partial t} dA$ where A is the cross-sectional area perpendicular to the z -axis. Since the wave is planar, $\frac{\partial D}{\partial t}$ is a constant across the cross-sectional area A , so $\frac{\partial D}{\partial t} = I/A$, and Eq. (7) simplifies to

$$\frac{\partial T}{\partial t} = c^D \frac{\partial^2 u}{\partial z \partial t} - \frac{eI}{\varepsilon^S A}. \quad (8)$$

Differentiating Eq. (8) with respect to z ,

$$\frac{\partial^2 T}{\partial t \partial z} = c^D \frac{\partial^3 u}{\partial z^2 \partial t} = c^D \frac{\partial^2 v}{\partial z^2}, \quad (9)$$

noting $I = I(t)$ alone, and $v = \partial u / \partial t$. From the momentum equation, if the body force $f = 0$,

$$\frac{\partial T}{\partial z} = \rho \frac{\partial^2 u}{\partial t^2} \implies \frac{\partial^2 T}{\partial z \partial t} = c^D \frac{\partial^2 v}{\partial z^2} = \rho \frac{\partial^2 v}{\partial t^2}, \quad (10)$$

The latter part of Eq. (10), $c^D \partial^2 v / \partial z^2 = \rho \partial^2 v / \partial t^2$, may be solved as a wave equation:

$$v(x, t) = e^{J\omega t} (V_1 e^{-Jkz} + V_2 e^{Jkz}). \quad (11)$$

where the circular frequency-wavenumber ratio is related to the material properties by $\omega/k = \sqrt{c^D/\rho}$ and $J = \sqrt{-1}$; the first and second parts of the solution in Eq. (11) are waves traveling in the positive and negative z -directions, respectively.

An acoustic impedance may be defined as $Z = T/v = \sqrt{c^D \rho}$ —the latter term found by substituting Eq. (11) into Eq. (10)—and used to describe the stress wave propagation by substitution of Eq. (11) into Eq. (8) and integrating,

$$T = e^{J\omega t} \left[\frac{kc^D}{\omega} (-V_1 e^{-Jkz} + V_2 e^{Jkz}) + J \frac{eI_0}{\omega \varepsilon^S A} \right]. \quad (12)$$

Notice that the wave is *electroelastic*, with a propagating electric field wave in addition to the elastic behavior, and that $I = I_0 e^{J\omega t}$.

The relationship between the mechanical and electrical behavior may now be determined using the stress and particle velocity fields for a piezoelectric element of homogeneous composition, finite thickness t , and constant cross-sectional area A as shown in Fig. 3. Note the polarization of the material indicated by the arrow adjacent to P . The stress T , when expressed on the faces of the piezoelectric element normal to the z -axis, becomes the force delivered by the element:

$$\begin{aligned} F_1 &= AT|_{z=z_1} = e^{J\omega t} \cdot \left[\frac{Akc^D}{\omega} (-V_1 e^{-Jkz_1} + V_2 e^{Jkz_1}) + J \frac{eI_0}{\omega \varepsilon^S A} \right], \\ F_2 &= AT|_{z=z_2} = e^{J\omega t} \cdot \left[\frac{Akc^D}{\omega} (-V_1 e^{-Jkz_2} + V_2 e^{Jkz_2}) + J \frac{eI_0}{\omega \varepsilon^S A} \right]. \end{aligned} \quad (13)$$

The velocities on the faces $z = z_1$ and $z = z_2$ are given by

$$\begin{aligned} v_1 &= -v|_{z=z_1} \\ &= -e^{J\omega t} (V_1 e^{-Jkz_1} + V_2 e^{Jkz_1}), \\ v_2 &= v|_{z=z_2} \\ &= e^{J\omega t} (V_1 e^{-Jkz_2} + V_2 e^{Jkz_2}) \end{aligned} \quad (14)$$

Using Eq. (14), it is possible to eliminate V_1 and V_2 from the force Eq. (13);

$$\begin{aligned} V_1 e^{J\omega t} &= -\frac{v_1 e^{Jkz_2} + v_2 e^{Jkz_1}}{2J \sin kd}, \\ V_2 e^{J\omega t} &= \frac{v_1 e^{-Jkz_2} + v_2 e^{-Jkz_1}}{2J \sin kd}, \end{aligned} \quad (15)$$

where $d = z_2 - z_1$, giving

$$\begin{aligned} F_1 &= \frac{kc^D A}{J\omega} \left[\frac{v_1}{\tan kd} + \frac{v_2}{\sin kd} \right] + J \frac{eI_0}{\omega \varepsilon^S} \\ F_2 &= \frac{kc^D A}{J\omega} \left[\frac{v_1}{\sin kd} + \frac{v_2}{\tan kd} \right] + J \frac{eI_0}{\omega \varepsilon^S} \end{aligned} \quad (16)$$

making use of several trigonometric identities.

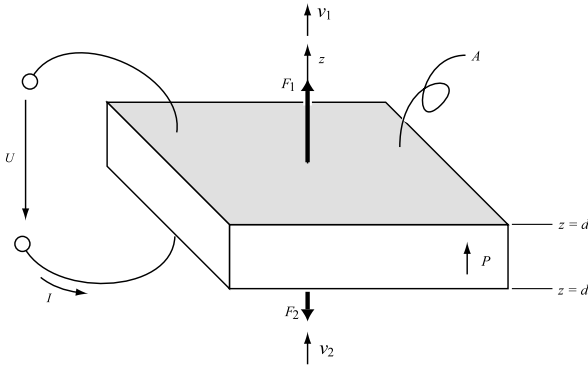
From the field displacement equation in Eq. (5), the applied voltage V is given by

$$\begin{aligned} V &= \int_{z_1}^{z_2} E dz \\ &= \int_{z_1}^{z_2} \left(-\frac{e}{\varepsilon^S} S + \frac{D}{\varepsilon^S} \right) dz \\ &= -\frac{e}{\varepsilon^S} u|_{z=z_1}^{z_2} + \frac{1}{\varepsilon^S} \int_{z_1}^{z_2} D dz \\ &= \frac{e}{\varepsilon^S} (u|_{z=z_1} - u|_{z=z_2}) + \frac{I_0 d}{JA\omega \varepsilon^S} \end{aligned} \quad (17)$$

since $D = (1/A) \int I_0 dA = I_0 / (J\omega A)$, implying $\int_{z_1}^{z_2} D dz = I_0 d / (J\omega A)$. Since $u = -j\omega v$ and the *static* or *shunt capacitance* is $C_0 = \varepsilon^S A / d$,

$$V = -J \left[\frac{\omega e}{\varepsilon^S} (v_1 - v_2) + \frac{I_0}{\omega C_0} \right], \quad (18)$$

forming the third of three equations which may be combined to give a matrix relationship between the input voltage and current and the output forces and velocities on the



Piezoelectric Materials for Microfluidics, Figure 3 Schematic of a piezoelectric element to illustrate the process of analysis

faces of the piezoelectric element:

$$\begin{bmatrix} F_1 \\ F_2 \\ V \end{bmatrix} = -J \begin{bmatrix} \frac{\zeta}{\tan kd} & \frac{\zeta}{\sin kd} & -\frac{e}{\omega \epsilon^S} \\ \frac{\zeta}{\sin kd} & \frac{\zeta}{\tan kd} & -\frac{e}{\omega \epsilon^S} \\ \frac{\omega e}{\epsilon^S} & -\frac{\omega e}{\epsilon^S} & \frac{1}{\omega C_0} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ I \end{bmatrix} \quad (19)$$

if $\zeta \equiv \frac{kc^D A}{\omega}$ as the mechanical impedance, with units of mass/time. Notice that

- The output forces F_1 and F_2 are linearly proportional to the input current I and piezoelectric stress coefficient e , and inversely proportional to the permittivity ϵ^S and circular frequency ω .
- The change in velocity across the element $v_1 - v_2$ is linearly proportional to the input voltage V and permittivity ϵ^S , and inversely proportional to the circular frequency ω and piezoelectric stress coefficient e .

Approximating Transducer Behavior with Equivalent Circuit Equations

From the expression in Eq. (19) most forms of equivalent circuit models of piezoelectric elements may be found. The Van Dyke circuit [14] is the simplest, using discrete electrical components combined to approximate the piezoelectric element's behavior. It is used to represent the electrical impedance about one resonance of a freely suspended piezoelectric element, using a shunt capacitor in parallel with an inductor, resistor, and capacitor placed in series to represent the *motional* or resonance behavior of the element. A multivibrator Van Dyke model may be formed by adding additional motional legs, each representing another resonance. Since it lacks any explicit treatment of the output force and velocity, it is not especially useful beyond electrical characterization (see

Smits' method below). Other circuit models incorporate the mechanical behavior, and the most prevalent of these are the *Mason* [14] and *Krimholtz–Leedom–Matthaei* [7] (KLM) circuits.

The Mason equivalent circuit may be derived directly from Eq. (19). It is sometimes called a *transmission-line* circuit model since the transcendental terms in the matrix appear in the same way when modeling power transmission lines. Most importantly, the circuit represents more than one resonance with these transcendental terms. Consider first an element that does not have piezoelectricity, implying the piezoelectric stress coefficient $e = 0$. The force-velocity relationships in the nonpiezoelectric element would then be

$$F_i = \frac{v_1 + v_2}{J \sin kd} \zeta + (J \zeta \tan kd/2) v_i \quad (20)$$

for $i = 1, 2$. This relates the force and velocity on one face of the nonpiezoelectric element to the other face. For an element that is piezoelectric, $e \neq 0$, and we can use the third equation in Eq. (19) to determine the actual force due to the applied voltage by substitution in the top two equations.

Piezoelectric Coefficient Relations

Typically, a piezoelectric material's properties are measured through a series of experiments on a set of samples, and these measurements give results easily substituted into the matrix representations of the material. Unfortunately, very few of the components within each of the matrices are defined with the measured data, but the remaining data may be calculated using relations between the coefficients. At times, not enough data are provided to do this; for whatever reason, the data set is incomplete. This problem is especially common with piezoelectric material manufacturers. Further complicating the problem, companies sometimes provide redundant data that are inconsistent—calculated results from some of the provided data does not match the rest of the data that are provided.

The simplest relationship between the electrical property tensors is the inverse relationship between the permittivity, ϵ , and the inverse permittivity, β . The equations are

$$\beta^S = \epsilon^{S-1} \quad \text{and} \quad \beta^T = \epsilon^{T-1}. \quad (21)$$

Other relations transform the matrix representation of the permittivity or inverse permittivity from one set of boundary conditions (either fixed stress or fixed strain) to the other; for the permittivity,

$$\epsilon^S = \epsilon^T - \mathbf{d} \mathbf{c}^E \tilde{\mathbf{d}} = \epsilon^T - \epsilon^S \mathbf{e}^E \tilde{\mathbf{e}}, \quad (22)$$

and for the inverse permittivity,

$$\beta^S = \beta^T + \mathbf{h} \mathbf{s}^D \tilde{\mathbf{h}} = \beta^T + \mathbf{g} \mathbf{c}^D \tilde{\mathbf{g}} \quad (23)$$

Again, there is a simple inverse relationship between the mechanical coefficient matrices, namely, the stiffness, $\mathbf{c}$ and the compliance, $\mathbf{s}$;

$$\mathbf{c}^E = \mathbf{s}^{E-1} \quad \text{and} \quad \mathbf{c}^D = \mathbf{s}^{D-1}. \quad (24)$$

The boundary conditions on the mechanical property matrices may be changed with the following transformations:

$$\mathbf{c}^D = \mathbf{c}^E + \tilde{\mathbf{e}} \beta^S \mathbf{e} = \mathbf{c}^E + \tilde{\mathbf{h}} \epsilon^S \mathbf{h}, \quad (25)$$

for the stiffness, and

$$\mathbf{s}^D = \mathbf{s}^E + \tilde{\mathbf{g}} \epsilon^T \mathbf{g} = \mathbf{s}^E + \tilde{\mathbf{d}} \beta^T \mathbf{d}, \quad (26)$$

for the compliance.

Unfortunately, there are no inverse relations between the piezoelectric coefficients. But, there are a wide collection of equations that describe one piezoelectric quantity in terms of the others. The following is a complete list of these equations:

$$\mathbf{d} = \epsilon^T \mathbf{g} = \mathbf{e} \mathbf{s}^E, \quad (27)$$

$$\mathbf{g} = \beta^T \mathbf{d} = \mathbf{h} \mathbf{s}^D, \quad (28)$$

$$\mathbf{h} = \beta^S \mathbf{e} = \mathbf{g} \mathbf{c}^D, \quad (29)$$

and

$$\mathbf{e} = \epsilon^S \mathbf{h} = \mathbf{d} \mathbf{c}^E. \quad (30)$$

Through substitution between these equations, many more relations between the variables can be expressed. Written differently, the relations are clearer. In the following equation, the dots within the parentheses represent where the variable at the top of each column would be placed to form the equation for the variable on the left side of each row:

	$\mathbf{d}$	$\mathbf{g}$	$\mathbf{h}$	$\mathbf{e}$
$\mathbf{d} =$	$(\cdot)$	$\epsilon^T (\cdot)$	$\epsilon^T (\cdot) \mathbf{s}^E$	$(\cdot) \mathbf{s}^E$
$\mathbf{g} =$	$\beta^T (\cdot)$	$(\cdot)$	$(\cdot) \mathbf{s}^D$	$\beta^S (\cdot) \mathbf{s}^D$
$\mathbf{h} =$	$\beta^S (\cdot) \mathbf{c}^E$	$(\cdot) \mathbf{c}^D$	$(\cdot)$	$\beta^S (\cdot)$
$\mathbf{e} =$	$(\cdot) \mathbf{c}^E$	$\epsilon^S (\cdot) \mathbf{c}^D$	$\epsilon^S (\cdot)$	$(\cdot)$

(31)

Experiment and Use

Physically measuring the coefficients of a piezoelectric material can be pursued in several different ways. Two of the most popular are the IEEE method [14] and Smits' technique [11]. Smits' technique has the advantage in that it requires fewer samples to be made by eliminating the radial resonator shape, especially important when considering the time necessary to make, machine, electrode, pole, and test piezoelectric material. Further, Smits' method uses exact equations in the calculation of the material constants, makes possible the calculation of loss data at the same time, and the electric and mechanical constants are measured at the same frequency unlike the IEEE method.

The technique is iterative, but the iteration process is strongly convergent. To determine all of the coefficients of a given piezoelectric material, four different geometries must be used. Each of these geometries encourage a particular vibration mode shape which reduce the number of material coefficients that are participating in the motion to a reasonable number. Named after the desired mode shape, the *thickness/shear* resonator, the *length expansion* resonator with perpendicular field, the *thickness expansion* plate resonator, and the *length expansion* resonator with parallel field provide different material constants when using Smits' technique on each of them. Convergence criteria and proof are provided in Smits' paper [11], but he claims that convergence to the correct values is usually achieved in eight iterations.

Manufacturing Methods

There are numerous ways to manufacture piezoelectric materials, roughly divided into crystal-growth methods for the single crystal media and a collection of techniques for polycrystalline piezoelectric media. The most obvious and troublesome difference in fabricating piezoelectric materials is the presence of lead in lead-based polycrystalline piezoelectrics like PZT; the lead is present in solid solution and tends to leach from the ceramic during sintering unless a source of lead is provided to maintain equilibrium. For all piezoelectric materials that exhibit remnant polarization, charging effects can be a problem during processing, enough to cause arcing and device failure or substrate cracking.

Among the crystal-growth techniques, the Czochralski (CZ) crystal growth technique is most common, particularly for lithium niobate, lithium tantalate, barium gallium orthophosphate (BGO), langasite. There are other techniques, and this is an active area of research. The development of single-crystal high-strain relaxor ferroelectrics like lead manganese niobate-lead titanate (PMN-PT) [8]

using the high-temperature flux technique is a notable example of another approach.

For ceramic materials, there are many approaches available to a researcher for fabrication, including the simple pellet press, tape casting, screen printing, sputter deposition (especially PZT and ZnO), sol-gel techniques for thin films, pulsed-laser deposition, and the hygrothermal technique [10] for deposition of quality PZT onto titanium.

Key Research Findings

During World War II, the discovery of PZT was a fortuitous and remarkable improvement on the state of the art in piezoelectric materials engineering. Since then, modest improvements in performance of piezoelectric materials have been made, with the discovery of single-crystal high-strain PMN-PT materials [8] an important milestone. In the past few years, however, methods for studying and manipulating the structure of ceramics on the scale of the unit cell and lower—the grain structure, the ferroelectric domains, and the structure of the unit cells themselves—have appeared [5]. These techniques are especially well-suited for improving the quality of piezoelectric materials; a recent potassium-sodium niobate/lithium tantalate/lithium antimonate ceramic has a piezoelectric performance equivalent to PZT while eliminating toxic lead [12]. Further, given the need to fabricate ►thin-film devices with features measured in nanometers in conjunction with micro- and nanofluidics, an understanding of the nature of polarization and ferroelectric domain growth in materials compatible with fabrication at these scales is vital. The applications for this technology span many disciplines: nonvolatile memory for computer technology, sensors and sensors for medical and biological diagnostics and actuators and microwave devices for consumer applications [9].

Future Directions for Research

The development of characterization and fabrication methods approaching the unit-cell size of known piezoelectric materials has reenergized the area of piezoelectric materials research in the past few years. By tailoring the structure of piezoelectric materials to maximize the polarizability, strength, and maximum strain, major advances are just beginning to appear. Work on applying the replete knowledge of bulk piezoelectric materials to thin-film materials research remains incomplete, from fabrication of PMN-PT to lead-free ceramics at small scales.

Looking beyond ceramic materials to electrets, polymer and elastomeric piezoelectric materials, and so-called electroactive materials, a wide-open field of research in high-strain piezoelectric materials is appearing. Using the same technology to manipulate the chemical and physi-

cal structures of complex ionic or nanotube-imbibed polymers, strains of well over 50 % have been obtained in this new class of material, many examples of which are biocompatible. Improvements in reliability, tolerance of extreme ambient conditions, and modeling are important areas that remain to be considered.

Cross References

► Piezoelectric Microdispenser

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Piezoelectric Microdispenser

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Synonyms

Piezoelectric ink jet; Piezoelectric drop-on-demand; Droplet generator; Piezoelectric micro/nanoliter droplet dispenser

Definition

A fluid-filled chamber that has one or more exit orifices (nozzles) through which the fluid is driven by piezoelectric material causing either a rapid increase in pressure within the fluid or an acceleration of the orifice with respect to the fluid body. Additional fluid is provided through an inlet orifice by external pressure or through a suction mechanism designed into the microdispenser. The fluid, once forced through the nozzles, forms one or more droplets through an instability mechanism – main and perhaps ► [satellite droplets](#) – that propagate away from the nozzle, usually at a speed several times the diameter of the droplet per second at speeds up to 10 m/s. The droplets characteristically have a diameter of the same order as the nozzle diameter, usually 1 to 100 μm , and have a volume of several picoliters to nanoliters as they form over a period of about 100 μs . To deliver the fluidic pressure wave required to form the droplets, a pulse or tailored electric drive signal must be provided to the piezoelectric material at a voltage of 5 to 200 V.

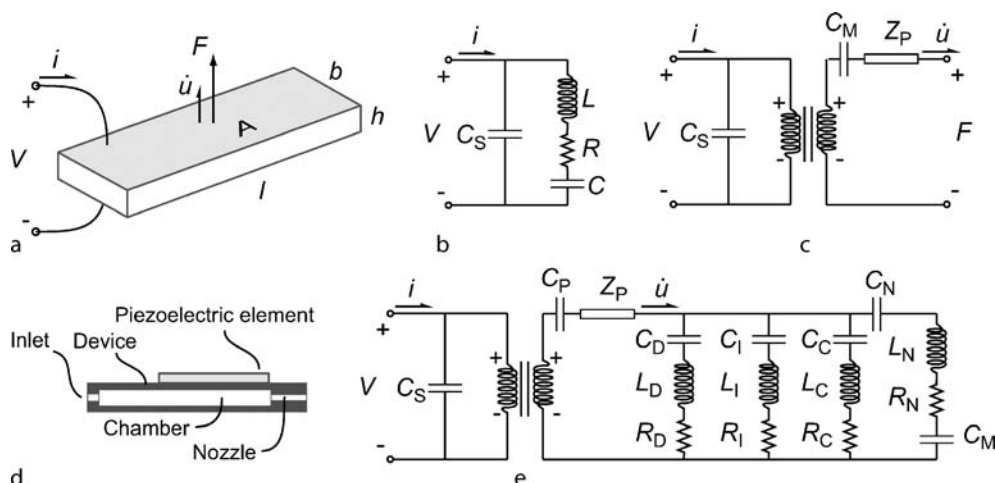
Overview

The ► [piezoelectric microdispenser](#), closely related to ► [piezoelectric ink-jet printing](#), a specific application of the microdispenser traditionally for consumer printing, encompasses three separate operating schemes depending on the intended application. A single ► [droplet on demand](#) may be delivered by a dispenser through either impulse excitation of the piezoelectric element or continuous vibration of the nozzle with short-term contact onto an application surface. Continuous droplet generation may be obtained through formation of a long-term Rayleigh instability of the fluid surface at the nozzle. ► [Fluid jetting](#) is another method to form fibers or droplets, the latter through subsequent breakup of the jet after it has left the microdispenser's exit nozzle due to Rayleigh or Kelvin–Helmholtz instabilities. Another less common method, ► [free-surface atomization](#), makes use of sessile droplets or relatively large fluid volumes placed on the surface of an ultrasonically vibrating structure which induces breakup through an as-yet poorly understood combination of Rayleigh instability and ► [cavitation](#) mechanisms. Piezoelectric microdispensers have existed since the 1950's, though their use increased dramatically with a need for inexpensive printing from the early 1980's, an

application shared with thermoelectric (bubble) microdispensers. Generally, piezoelectric microdispensers offer superior flexibility, simplicity, and scalability compared to other known droplet-forming technologies. However, the use of piezoelectric materials causes problems, particularly in the use of lead zirconate titanate (PZT) and its higher-performance doped variants, which are not generally compatible with standard micromachining processes. As a consequence, scaling down dispensers that make use of PZT is difficult. Though other piezoelectric materials have been used over the years, including quartz, aluminum nitride, and zinc oxide, PZT is favored for its large ► [electromechanical coupling](#) and relatively large ► [piezoelectric](#) strain coefficients. The generation of a sufficient electric field in the piezoelectric materials, particularly for pulsed drop-on-demand dispensers, can require high voltages not directly compatible with integrated-circuit voltages (5 or 12 V). The placement, diameter, and configuration of the inlet and outlet orifices is critical to obtain suitable droplet formation from input strain pulses along the boundary of the fluid chamber, and microdispenser design is a challenge met through a combination of approximate analytical methods, computational modeling, and experimental development, using many versions of a design to discover the one most suitable for the application. Over the years, different operating schemes have been considered for piezoelectric microdispensers, and current technology [1] is essentially a refinement of the spectrum of configurations available over 25 years ago [2].

Regardless, a majority of the systems operate through direct squeezing of the fluid volume present within the dispenser, and as a consequence the volume of fluid ejected from the dispenser at the nozzle is a linear function of the applied voltage, that is, once the stability of the fluid ► [meniscus](#) present across the nozzle is overcome. This is in contrast to the generation of jets and droplets via ► [acoustic streaming](#), which would result in an ejected fluid volume proportional to the square of the applied voltage. Applying low voltage signals to the piezoelectric element will result in a deformation of the meniscus, and as the applied voltage is increased, the deformation grows until the meniscus becomes unstable.

If the capillary wavelength is approximately the same size or larger than the nozzle diameter, the formed droplet diameter will be related to the nozzle diameter, and will increase with an increase in the applied voltage. At a certain threshold voltage, more than one droplet will form from the extension and breakaway of the meniscus. Typically, each of the extra droplets – a ► [satellite drop](#) – is smaller in size than the primary droplet, but as the voltage continues to increase, these droplets tend to increase in number and volume. There may be yet another transi-



Piezoelectric Microdispenser, Figure 1 Equivalent circuit modeling of (a) planar, thickness-polarized, thickness-vibrating piezoelectric transducers using (b) a simple lumped-element Van Dyke model and (c) a more complex transmission-line model. A model of a piezoelectric microdispenser (d) may be formed by connecting the output terminals to the equivalent circuit model of the remaining components (e)

tion to chaotic droplet and stream generation as the voltage continues to increase, and cavitation within the chamber or at the nozzle may also occur.

However, if the capillary wavelength is much less than the diameter of the nozzle, the droplet diameter will instead be related to the capillary wavelength: the droplets form from crest instabilities on the free surface of the fluid. The droplet dimensions for this arrangement follow similar trends as the voltage is increased. Using this technique, it is possible to generate droplets less than $1\ \mu\text{m}$ in diameter through the use of frequencies beyond 10 MHz generated from ► [surface acoustic wave](#) devices.

This review considers the variety of device designs for piezoelectric microdispensers, methods to model their overall behavior, in addition to the characteristics of atomization mechanisms available for dispensing droplets using piezoelectric materials.

Basic Methodology

Device modeling

Equivalent circuit models have been used to represent piezoelectric material behavior for many years, but are not without limitations. The most common representation for a piezoelectric material, as recommended by the IRE/IEEE standards on piezoelectricity, is the Van Dyke model as described in detail in ► [piezoelectric materials in microfluidics](#), with a capacitor, resistor, and inductor in series representing a single resonance, or motional component of the piezoelectric element, all set in parallel with a second capacitor, representing the ► [shunt capacitance](#). Additional resonances may be included by placing additional

capacitor-resistor-inductor sets in parallel, but therein lies some of the limitations of equivalent circuit modeling: accurate determination of the equivalent circuit components requires either a physical prototype or an accurate computational model to provide the data. Further, assuming the loss mechanisms are isolated to a single resistance in the motional component per resonance may lead to error in predicting both the system response and the resonance frequencies. Computational models using finite-element analysis or the like require similar physical data, and so a physical prototype becomes absolutely necessary in the prediction of microdispenser behavior.

Additional matters to consider in using piezoelectric materials that require a physical prototype include the aging of the piezoelectric material and the temperature change during operation. Over time, the resonance frequency and static capacitance will change in an exponential fashion; most of the aging will occur in the first two months, but the subsequent aging can be important in circuit and resonant chamber design. As an electric field is applied to a piezoelectric element, the element will heat due to electromechanical losses in the material. The heating can be significant for soft piezoelectric materials, and can affect the resonance frequencies and capacitance for both hard and soft piezoelectric media.

Equivalent circuit modeling is a useful and commonly applied analysis approach for designing piezoelectric microdispenser systems. The method has limited accuracy, and the ► [lumped analysis](#) approach requires the wavelength of the input signal to be larger than any characteristic length within the device. In a microdispenser with an internal cavity, for example, the ► [Hemholtz resonance](#)

of the orifice, nozzle, and cavity structure should be far higher than the frequency of excitation of the device used to perform the dispensing. The time required to transmit a change in the applied electric field on the piezoelectric element to a change in the meniscus shape at the nozzle is therefore negligible. Using ► **transmission-line** models of the piezoelectric element, fluid chamber, and other components in the microdispenser permit analysis without the wavelength limitation [3], and provide the ability to analyze the approximate interaction between multiple resonances in the complete system.

In this situation the coupling coefficient of the piezoelectric material is especially important. Though there are many specific forms of the coupling coefficient, an effective coupling coefficient may be used for a transducer operating near a particular resonance:

$$k_{\text{eff}} = \sqrt{\frac{\text{mechanical energy stored}}{\text{electrical energy input}}} = \sqrt{1 - \frac{f_r^2}{f_a^2}}$$

where f_r is the resonance frequency and f_a is the antiresonance frequency (► **resonance and antiresonance frequency**) associated with the vibration mode that dominates the motion induced by the applied voltage signal, if one such mode can be assumed to exist. The mechanical energy stored in the piezoelectric device may then be used to perform work on the fluid meniscus, and in this way the amount of energy applied to form a droplet for a microdispenser as a fraction of the input electrical energy may be estimated once one knows the resonance and antiresonance frequencies. The resonance and antiresonance frequencies are typically near the frequencies of minimum and maximum impedance, respectively, for a specific resonance.

Choosing a thickness-polarized, thickness-vibrating piezoelectric element as an example, we can define the applied voltage V , current i , dimensions b , h , and l , and the output force and velocity F and $\dot{u}$; the cross-sectional area bounded by b and l can be defined by A ; here it is assumed this area is electroded on the top and bottom faces of the element and that b and l are much greater than h . Treating it as a collection of discrete circuit elements as shown in Fig. 1b, the Van Dyke circuit, allows the analysis of one resonance within the isolated element. Most piezoelectric materials are capacitive insulators, and the ► **shunt capacitance** $C_S = bl/\beta_{33}^S h$ is the constant capacitance present across the element. The additional branch in the circuit represents the specific resonance being analyzed, the motional branch with inductance L , resistance R , and capacitance C . Many impedance analyzers provide this circuit as a means to model the electrical behavior of the piezoelectric element. The coupling coefficient for this

specific arrangement is $k_T = \sqrt{h_{33}^S \epsilon_{33}^S / c_{33}^S}$. Unfortunately, the model only accounts for one resonance and provides no information on the output force F and velocity $\dot{u}$. By treating the element as a transmission-line impedance Z_P as shown in Fig. 1c, one can incorporate an arbitrary number of resonances, where $C_M = -h_{33}^S bl / \beta_{33}^S h$ and $Z_P = \frac{k}{j\omega} \frac{\omega h / c}{\tan(\omega h / c)}$ where $c = \sqrt{c_{33}^D / \rho}$ is the longitudinal wave velocity for an open-circuit element. The resonances appear when $(\tan \omega h / c)$ goes to infinity. The quantities h_{33} , β_{33} , ϵ_{33} , and c_{33} represent the piezoelectric field to strain coupling, permittivity, inverse permittivity, and material stiffness along the thickness direction in this model. The S superscript on these quantities represents constant strain (clamped) values, and the D represents constant electric field displacement (open-circuit) values. By connecting the output of the piezoelectric element, or port, to a microfluidics structure (see Fig. 1d) with nozzle, chamber, inlet and compliant structure, a microdispenser can be formed. A simple representation of this situation can be modeled as a collection of RLC resistance-inductance-capacitance branches in Fig. 1e connected to the output of the piezoelectric element represented by the open terminals on the right in Fig. 1c. Note the inclusion of RLC sets for the device, inlet, device, and nozzle, each treating a specific resonance of the component. Transmission-line impedances could be used for these components as well to handle multiple resonances. There is an additional capacitor for the meniscus to take into account the formation of capillary waves; the charging of the capacitor is tied to the appearance of the capillary waves and can suggest whether a droplet will be formed as a consequence of an input signal V . The values of the resistance, inductance and capacitance may be related back to the dimensions and configuration of the microdispenser as shown by Berger and Recktenwald [3].

Generation

Modeling of piezoelectric microdispensers provides useful information on the potential for further optimization and insights into the operating mechanism underlying their behavior. However, the reliable formation of stable, monodisperse droplets from a device generally requires a tedious process of trial-and-error design of the driving scheme, fluid, nozzle, chamber, and inlet. Many factors affect the behavior of microdispensers, only a few of which can be incorporated into a mathematical model of the overall device at present. Most of the difficulty is associated with the behavior of the fluid at the exit nozzle. Generally, if the fluid to be dispensed has a low viscosity of about 1–50 cS, is slightly hydrophobic, and does

not contain particles greater than 10% of the diameter of the nozzle, droplet generation becomes a function of the design of the microdispenser alone [4].

The generation of droplets can be divided into three schemes depending on the form of the fluid leaving the orifice. Individual droplets may be formed on demand through discrete signals applied to the microdispenser, droplets may be formed continuously, with a short transition to and from steady droplet formation upon the application and removal of the driving signal from the microdispenser, or jets of fluid can be generated from a microdispenser, which may later break up into droplets as the jet interacts with the environment. A comparison of a characteristic length of the fluid leaving the microdispenser with the nozzle meniscus size provides another way to categorize the generation mechanism of a device. If the characteristic length – droplet diameter, for example – and the meniscus diameter are similar, the characteristic length is typically controlled by the meniscus size. If, however, the characteristic length is significantly smaller than the meniscus size, the characteristic length is controlled instead by the instability mechanism that generated it.

Droplets from Nozzles

Piezoelectric microdispensers that generate droplets of a size defined by the nozzle diameter function either through the interaction of a pressure wave induced in the fluid with the meniscus located at the nozzle, or a rapid acceleration of the meniscus. Either way, a column of fluid is pushed from the nozzle into the surrounding atmosphere. If the column reaches a sufficient length before being drawn back into the nozzle after passage of the pressure wave (πd_{noz} as predicted by Rayleigh [5], where d_{noz} is the nozzle diameter), the free surface will become unstable and pinch off, forming at least one droplet; the droplet diameter he predicted as a consequence of this process, $d_{\text{Rayleigh}} = 1.89d_{\text{noz}}$, represents an upper bound for the actual droplet diameter. This instability mechanism is not the free-surface Rayleigh instability as commonly understood and described later.

Lee and Lal [6] summarize the work of Lord Rayleigh, Weber, Sterling and many others quite well, and consultation of references in their work provides an introduction to the literature. Rayleigh's result represents the upper bound since it ignores the drag on the column and droplet from the surrounding atmosphere, and subsequent research considered these factors in the analysis. Through the inclusion of a drag model, the droplet size prediction can be made to be more accurate. Consider the column of fluid pushed from the nozzle, and treat the velocity of the column as U , the fluid's density, surface tension, and dynamic viscosity

as ρ , σ , and μ , respectively, and the ambient atmosphere's density as $\hat{\rho}$. The ► [Weber number](#) of the ambient atmosphere and ► [Ohnesorge number](#) of the fluid may then be defined as

$$\hat{We} = \frac{\hat{\rho} U^2 d_{\text{noz}}}{\sigma} \quad (1)$$

$$\text{and } Z = \frac{\mu}{\sqrt{\rho \sigma d_{\text{noz}}}}. \quad (2)$$

Typically, the Weber and Ohnesorge numbers are dimensionless measures of the relative strength of the inertia to the surface tension in a fluid, and the viscosity to the surface tension of a fluid, respectively. Here the Weber number is modified (with a hat, $\hat{We}$) to consider the inertia of the surrounding atmosphere in comparison to the surface tension of the fluid extending from the nozzle.

As the fluid column extends from the nozzle, the instability of the free surface is exhibited across several length scales, represented by a collection of solutions of Sterling's equation of motion describing the instability. One of these solutions grows in amplitude more rapidly than the others, which caused Sterling to assume that it would come to dominate the droplet formation process. The selected solution describes the maximum column length that can be obtained without actually causing sufficient surface instability to form a droplet. It further describes the diameter of the droplet formed as a consequence of the instability once the fluid column surpasses that maximum length. The wavelength of the instability as it forms a droplet may be written as $\lambda = \pi d_{\text{noz}} \hat{\vartheta}$, where

$$\hat{\vartheta} = \sqrt{2(1 + 3Z)} e^{-0.035 \hat{We}} \quad (3)$$

represents the reciprocal non-dimensional wavenumber of the most rapidly growing disturbance. For water being ejected from a 10 μm diameter nozzle at $U = 1 \text{ m/s}$, the modified Weber number is $\hat{We} = 1.81 \times 10^{-4}$, the Ohnesorge number is $Z = 0.0373$, and so $\hat{\vartheta} = 1.49$ and the wavelength of the instability is $\lambda = 46.9 \mu\text{m}$. The diameter of the droplet may be taken as one-half the instability wavelength, $d = 23.4 \mu\text{m}$, because one-half of the wavelength forms the droplet while the other half forms the neck region that separates the droplet from the remaining portion of the fluid column. The fluid's density, surface tension, and dynamic viscosity were treated as $\rho = 1002 \text{ kg/m}^3$, $\sigma = 0.072 \text{ N/m}$, and $\mu = 1.002 \times 10^{-3} \text{ kg/(m-s)}$, respectively, and the ambient atmosphere's density was set at $\hat{\rho} = 1.3 \text{ kg/m}^3$.

The formation of additional droplets – ► [satellite droplets](#) – in addition to the desired or primary droplet for each

extension of the fluid column induced by an applied pressure wave or acceleration of the nozzle is a troublesome problem with microdispensers. Dong, et al., in a thorough work [7], examined the formation of the primary and satellite droplets and suggested a criteria for avoiding the formation of satellite droplets. As the primary droplet forms from the liquid column extending from the nozzle, a fluid thread may form from the neck. In some cases, this fluid thread may become quite long in comparison to the nozzle diameter, the ratio of which could be defined as l_b/d_{noz} . If the fluid thread length is larger than a limiting value l_b^* , the thread will break up into satellite droplets. There is the possibility that the satellite droplets will merge with the primary droplet if the relative velocities are in the correct direction and large enough to induce recombination. In any case, the limiting value of the fluid thread length may be written as

$$l_b^* = d_{\text{noz}} \left(\frac{(C_2 - C_1)a}{2\alpha_{\text{max}}^*} + 1 \right), \quad (4)$$

where

$$\alpha_{\text{max}}^* = \sqrt{\frac{2\vartheta^2 - 2 + 9Z^2}{4\vartheta^4}} \quad (5)$$

is a term related solely to the nozzle and fluid characteristics; $\vartheta = \sqrt{2(1 + 3Z)}$ here, an approximation similar to Dong, et al., in line with Sterling's analysis given by Eq. (3) with the omission of the Weber-related air drag term. The nondimensional constants C_1 and C_2 are related to the pinch-off time of the liquid thread from the nozzle and primary droplet, respectively, while a represents another nondimensional constant that relates the speed of the wave disturbance along the periphery of the fluid column to the speed of the retreating fluid column back towards the nozzle after the formation of the droplet and fluid thread stretching between the column and droplet. Dong, et al., claim the product $(C_2 - C_1)a$ is between 0.9 and 1.1 for their experiments with water and glycerin compounds ejected into air. For the water/air system considered in this section, the limiting fluid thread length is $l_b^* = 22.8 \mu\text{m}$ if $(C_2 - C_1)a$ is 0.9. Experimental visualization would be necessary to determine the actual fluid thread length. Dong, et al., suggest that the most effective method to eliminate satellite droplet formation is an increase in the fluid viscosity which would increase the Ohnesorge number and reduce α_{max}^* .

Most piezoelectric microdispensers use the single-droplet nozzle ejection mechanism to form droplets. Through careful design, droplets may be rapidly ejected one after another from the meniscus at frequencies of up to 10 kHz,

though there are limitations from refill speed, cavitation, bubble ingestion from the nozzle into the fluid chamber, and satellite droplet formation. Generally the piezoelectric material is not a limitation in droplet delivery speed unless soft PZT is used, which has a larger electromechanical transduction loss and is susceptible to heating at operating frequencies above 1 kHz.

Jetting from nozzles in piezoelectric microdispensers is also possible, usually through a combination of constant applied pressure on the fluid with a lesser modulation of the pressure using the piezoelectric element. The modulation introduces an instability in the jet traveling out of the orifice which is amplified to cause the breakup of the jet into droplets through the interaction of the jet with the ambient environment; this instability mechanism is known as the **Kelvin–Helmholtz instability** and requires the jet to be moving at several meters per second in air to cause droplet formation [8]. Secondary breakup of the droplets into finer droplets may then occur via the Rayleigh–Taylor mechanism described later, though the conditions under which this can happen do not typically occur on the scales of most piezoelectric microdispensers.

Droplets from Free Surfaces

It is also possible to dispense microdroplets from free surfaces, either formed on a sessile droplet or along the meniscus formed at a nozzle. In the latter case, the critical difference with the previous section is that the primary droplet formed from the nozzle is much smaller than the diameter of the nozzle, eliminating the direct influence of the nozzle's size on the size of the droplet. The study of the formation of droplets from free surfaces has a long and controversial history, which has been suggested to be due to either capillary wave breakup, cavitation, cavitation-induced capillary waves which subsequently break up to form droplets, or even cavitation-screened capillary wave breakup, where a cavitation layer beneath the surface acts to suppress atomization. It is likely that over the wide range of operating frequencies and configurations available each of these physical mechanisms is possible, but for microdispensing fluids, capillary wave breakup appears to be the most likely mechanism [9]. Even so, the complexity of the analysis and nonlinear behavior [10] should not be underestimated in considering the use of free surfaces.

In 1950 Taylor studied the formation of sinusoidal waves on the free surface of a fluid in air undergoing acceleration perpendicular to the free surface and in the direction of the force of gravity, and almost a century prior Rayleigh considered the formation of similar waves if the denser fluid was placed above the free surface; both illus-

trate a similar mechanism that has come to be described by the Rayleigh–Taylor instability. Curiously, the instability will appear between the interfaces of other phases of media, including even solid–solid interfaces, and the form of the equations that describe the instability will remain essentially the same [11]. The formation of droplets from capillary waves may be described by the Rayleigh–Taylor instability.

Loosely following Piriz, et al. [11], if $\xi = \xi(x, t)$ represents the transverse displacement of the inviscid fluid surface where x is the position coordinate along the fluid interface¹, $\ddot{\xi} = A_T \kappa g \xi$ is the equation of motion of the surface if we ignore surface tension and let the ▶ **Atwood number** be $A_T = \rho_2 - \rho_1 / \rho_2 + \rho_1$, $\rho_2 > \rho_1$, where ρ_2 is the density of the fluid above the interface, and g and κ be the gravitational acceleration (9.8065 m/s^2) and the perturbation wave number, respectively. This can be integrated to give the exponential growth of the interface amplitude at a rate of $\sqrt{A_T \kappa g}$. The wavenumber is related here to the frequency of the applied oscillation through $\kappa = f/c$, where f is the frequency of the applied oscillation and c is the bulk sound speed within the fluid carrying the acoustic energy to the interface. If we place the dense fluid beneath the interface so that now ρ_2 is the density of the fluid below the interface, and incorporate surface tension, dynamic viscosity, and the acceleration of the fluid interface $\ddot{\psi}(t)$ in the direction of the gravity force g , the equation of the transverse displacement of the surface $\xi(t)$ becomes

$$\ddot{\xi} + [A_T \kappa (g - \ddot{\psi}(t)) + \sigma / \rho_1] \xi + 2 \frac{\mu_1 + \mu_2}{\rho_1 + \rho_2} \rho_2 \kappa^2 \dot{\xi} = 0. \quad (6)$$

Lamb recognized in 1932 that the surface tension acts to stabilize disturbances with wavenumbers greater than the cutoff value of

$$\kappa_C = \sqrt{\frac{g(\rho_1 + \rho_2)}{\sigma}}.$$

Above this value, oscillations induced on the free surface are stable, while below this value, it is possible to induce instability on the surface and form droplets. Viscosity cannot prevent the instability, but it does increase the acceleration $\ddot{\psi}(t)$ necessary to cause the instability to appear. In the equation of motion (6), acceleration due to gravity, g , also acts to stabilize the interface. There are different ways to destabilize the surface, the simplest of which is to $\ddot{\psi}(t)$ to a constant value above $g + \sigma / (\rho_1 A_T \kappa)$, forming a gravity wave. Other ways include harmonic excita-

tion with $\ddot{\psi}(t) = \Psi e^{i2\pi Ft}$ where $i = \sqrt{-1}$, Ψ is the amplitude, and F is the frequency of excitation. Lang [9] found the diameter of the droplets formed from the free surface closely follows Kelvin's equation for the capillary wavelength,

$$\lambda = \sqrt[3]{\frac{2\pi\sigma}{\rho f^2}}, \quad (7)$$

where f represents the frequency of the most easily excited capillary waves on the free surface. Based on experimental results, it has been shown that the frequency of the capillary waves is one-half the excitation frequency F if it is assumed to be harmonic, and Lang showed that the droplet diameter is 0.34 times the capillary wavelength based on experimental results across a wide range of frequencies:

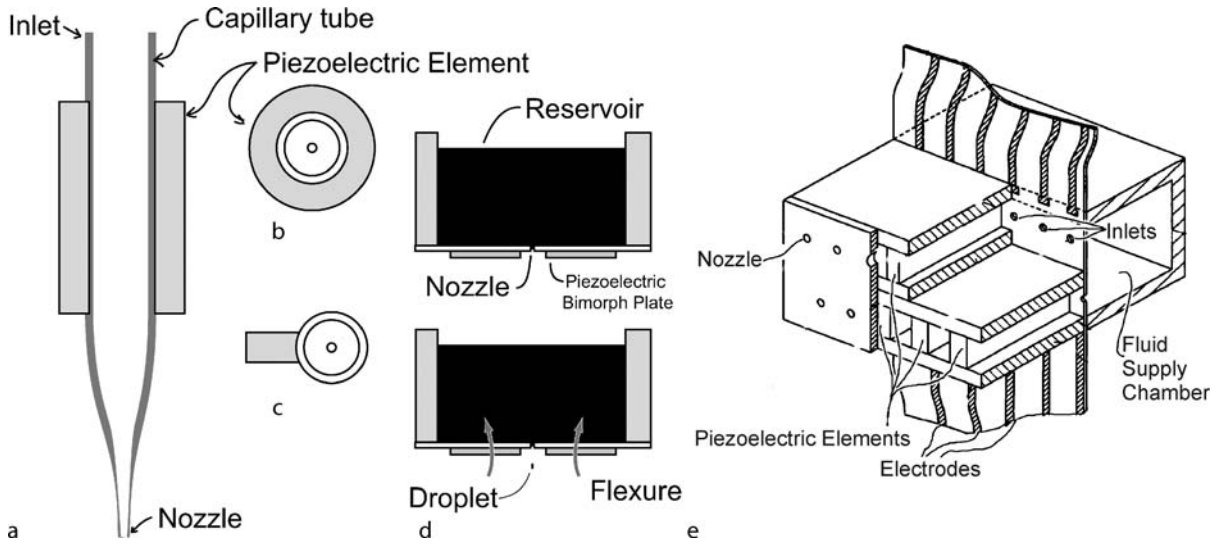
$$d = 0.34 \sqrt[3]{\frac{8\pi\sigma}{\rho f^2}}.$$

Similar to the droplet breakup model in the previous section, many different vibration modes may be excited on the surface by the acceleration $\ddot{\psi}(t)$, but the one represented by λ in equation is assumed to dominate in the generation of droplets. This assumption is valid at powers just sufficient to cause the generation of droplets, but as the amplitude Ψ is further increased, other modes may become large enough to form droplets, increasing the droplet size distribution and the complexity of analysis.

Impulsive excitation may also be used to form droplets from a free surface in this way. Continuous wave excitation can provide very monodisperse distributions of droplets with diameters much less than the characteristic size of the free surface [12], but impulsive excitation has also been used with success in commercial applications by Xerox [4]. A critical part of their success has been in finding ways to form a single droplet from the surface per excitation pulse; typical free surface droplet generators form n droplets where $n = (1/2)\pi(d_{\text{surf}}/\lambda)^2$, d_{surf} is the diameter of the free surface, and λ is the capillary wavelength. Impulsive excitation has the advantage of offering droplet-on-demand operation at the expense of requiring thorough analysis of the free surface to ensure reliable formation of individual droplets.

Jetting from the free surface may also be obtained, and droplets could be formed from these jets through secondary breakup mechanisms if the Ohsonorge number is less than about 0.1 and the Weber number is above 5. The conditions necessary to obtain free-surface jetting from piezoelectric microdispensers are fairly challenging; even though there are examples of high-frequency free-surface

¹A 2-D assumption is implicit here, as it is throughout most literature on instability calculations.



Piezoelectric Microdispenser, Figure 2 The capillary tube microdispenser, viewed from the side (a) and top (b), with a variation (c) by Lee and Lal [6]. Though almost unrecognizable after years of refinement and adoption of a parallel layout scheme, the Konica-Minolta device (d) is also a capillary tube microdispenser

jetting [13], they at present remain little more than curiosities.

Droplet Motion After Ejection

Once formed, most piezoelectrically-dispensed droplets have a drag force proportional to their velocity. In contrast to the viscous drag force on macroscopic objects,

$$F_{\text{Viscous Drag}} = \frac{1}{2} \rho C_d A v^2, \quad (8)$$

which scales with the square of the velocity of the object relative to the surrounding fluid, the drag force due to Stokes' law scales linearly with velocity:

$$F_{\text{Stokes}} = 6\pi\eta r v; \quad (9)$$

r , v , A , C_d , ρ , and η are the drop's radius, velocity, and cross-sectional area, the coefficient of drag, and the surrounding fluid's density and viscosity, respectively. The viscous drag scales inversely to the square of the diameter of the droplet, and so is negligible in air for droplets less than about 25 μm in diameter. The most common figure of merit used in choosing the appropriate drag force is the [Reynolds number](#).

For droplets in air less than about 5 μm in diameter, the actual drag force is even lower, since the surrounding air is not a continuum at these scales. Millikan's resistance factor (or [Cunningham's correction](#)) multiplies the terms on the right-hand side of equation (9) to form the corrected

drag force

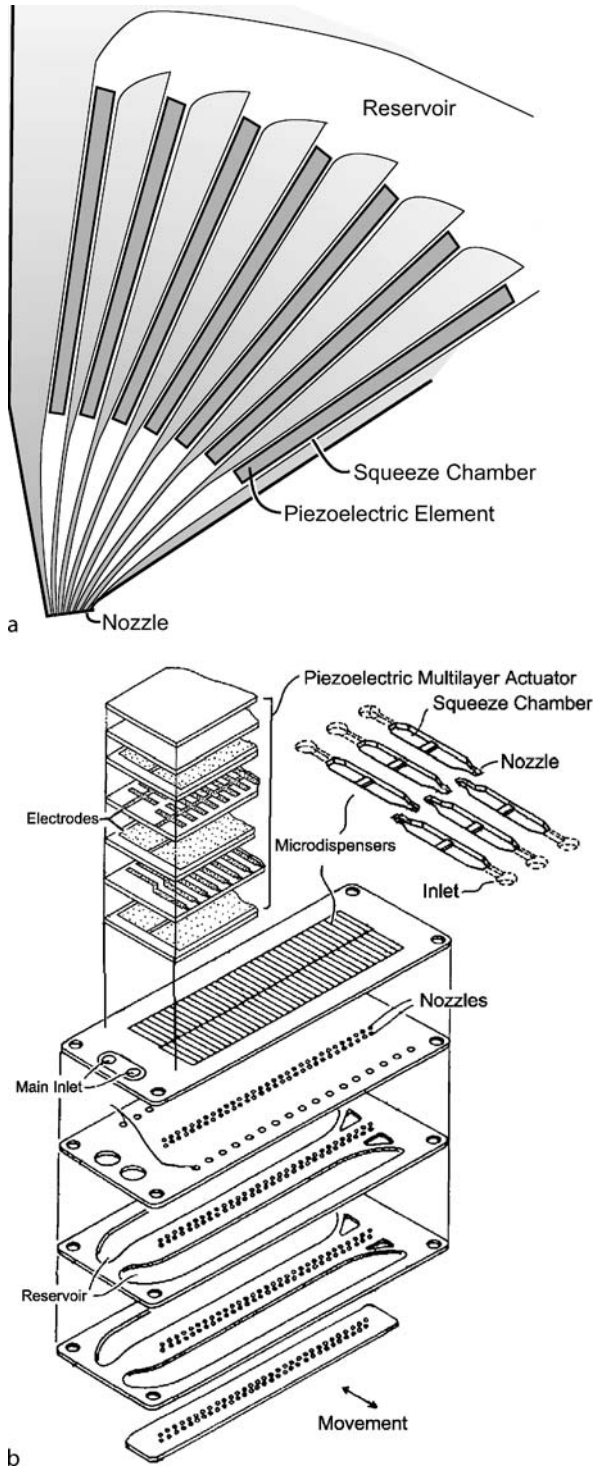
$$F'_{\text{Stokes}} = 6\pi\eta r v \left[1 + \left(\frac{0.130 \mu\text{m}}{d} \right) \times \left(1.252 + 0.399 e^{-1.100d/0.130\mu\text{m}} \right) \right] \quad (10)$$

A 10 μm diameter droplet moving at 1 m/s in air would have 14.1 pN, 1.72 nN, and 2.00 nN of force on it from viscous drag, uncorrected Stokes' drag, and Cunningham-corrected Stokes' drag, respectively. More importantly, the deceleration of a 10 μm diameter water droplet moving at 1 m/s would be 3820 m/s^2 due to the Cunningham-corrected Stokes drag calculated from

$$\dot{v} = F'_{\text{Stokes}} / \left(\frac{\pi}{6} \rho d^3 \right) \quad (11)$$

where the term in parentheses is the droplet mass. The deceleration decreases rapidly as the droplet slows; the velocity is an exponential function of time. Under gravity, the terminal velocity can be found by setting the drag force equal to the force of gravity, though this is far less than the ejection velocities for most dispenser systems; for example, a 10 μm droplet has a terminal velocity of only 3 mm/s.

In addition to the correction on the drag force from the noncontiguous surrounding fluid, there is also a randomizing effect on the droplets' motion due to the interaction with the environment: [Brownian motion](#). This motion can cause error in droplet placement or



Piezoelectric Microdispenser, Figure 3 The side-shooter design as conceived by Kyser and his associates at Silonics in the 1970's (a), and a modern version by Brother (b) using planar micromachining technology. Note the dramatic increase in nozzles and the more detailed cross-sectional design. The chamber cross-section in the Kyser design is rectangular

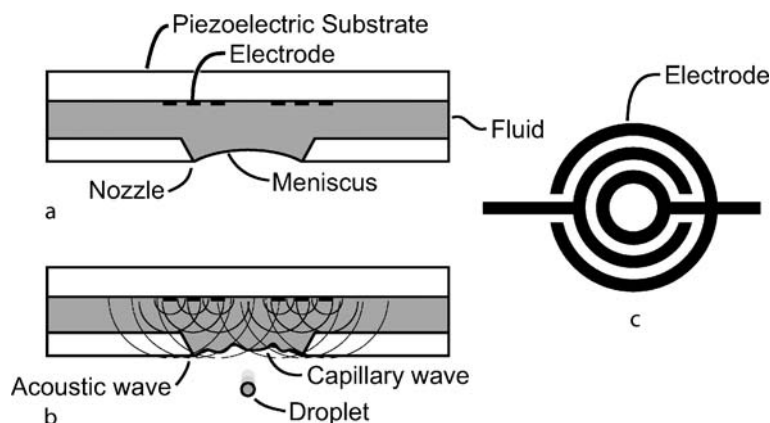
agglomeration of droplets in continuous droplet formation or atomization systems. The mean displacement of a droplet, δ , due to Brownian interaction over time t is a function of the temperature T , Boltzmann's constant $k = 1.38065 \times 10^{-23}$ J/K, and drag force, F_{Drag} , which is a choice between Eqs. (8) through (10) suited to the diameter of the droplet under consideration,

$$\delta = \sqrt{\frac{2kTt}{F_{\text{Drag}}}}. \quad (12)$$

For a $5 \mu\text{m}$ droplet, the average displacement after one second at 20°C is about $2.66 \mu\text{m}$, over half the droplet's radius, using Eq. (10) for the drag force.

Key Research Findings

The devices resulting from the combination of piezoelectric materials and fluidic microstructures to provide dispensing of small amounts of liquid fall into several distinct categories based on their construction. Most common of these is the capillary tube drop-on-demand system, shown in Fig. 2, first developed by Zoltan at Cle vite in the early 1970's, and commonly called the Gould or Bogy device. Many variations on this design exist due to the simplicity of the configuration and ease in forming the micropipette tips from which droplets may be dispensed; one can even manufacture their own capillary tube microdispenser for very little cost [4]. Fig. 2a illustrates the typical form of this device, though there have been many variations in the shape of the piezoelectric element used, as shown with the top views in Fig. 2b and c; the latter from Lee and Lal [6], and an interesting variation (see Fig. 2d) from Percin and Khuri-Yakub [14]. Through mass production and microfabrication techniques, the concept has been refined to the form shown in Fig. 2e by Konica-Minolta [15]. The device functions either by squeezing the fluid or accelerating the meniscus through actuation of the piezoelectric material; in most devices both mechanisms are in part responsible for the ejection of droplets. Note even in the Konica-Minolta device the piezoelectric material is a simple shape that may be constructed from machining bulk pieces of PZT. Another design that has been used for microdispensers is the so-called side-shooter design originally developed by Kyser and others in the late 1970's, as illustrated in Fig. 3 in its original configuration, and a modern configuration by Brother making use of planar micromachining technology. A relatively unique version of droplet generation from a free surface is focused acoustic beam ejection by Elrod, et al., at Xerox PARC in the early 1990's [4] illustrated in Fig. 4; it represents a refined version yet lower-frequency ver-



Piezoelectric Microdispenser, Figure 4 Focused acoustic beam ejection technology by Elrod and associates at Xerox PARC. Using a piezoelectric element with electrodes immersed in a thin layer of fluid below a nozzle (a), an electrical impulse will induce an acoustic wave that, when interacting with the meniscus, forms a capillary wave that ejects a single droplet (b). The form of the electrode (c) and the driving signal are two of the features of the design that must be carefully designed

sion of the surface acoustic wave atomizer by Shiokawa et al. [13].

Future Directions for Research

Recently, many new piezoelectric materials have appeared as a consequence of the ability to observe and tailor crystal, polycrystalline, and polymer structures on the nanoscale. Indeed the evolution of high-performance piezoelectric materials like PMN-PT, polypyrrole, IPMC, and other electroactive polymers promise to usher in a new generation of materials that offer advantages over PZT for dispensing applications. Meanwhile, compatibility problems in the fabrication of silicon, PZT, and other standard micromachining materials are being solved, offering the ability to apply PZT in ways not currently possible. In analysis, most microdispensing work assumes the working fluid is Newtonian, while many fluids that are useful for dispensing in the printing, biomedical and microfabrication industries are strongly non-Newtonian. A return to the fundamentals is in order to revisit the use of these fluids in forming droplets. Improved computational algorithms capable of dealing with the free surfaces and multiple physical phenomena of piezoelectric media, structural deformation, and nonlinear acoustic wave propagation will be central to this effort. As the excitation frequencies and power per unit volume are increased, the incorporation of cavitation models will also be beneficial.

Cross References

- Acoustic Streaming
- Biosample Preparation Lab-on-a-Chip Devices

- Cavitation in Microdomains
- Control of Microfluidics
- Droplet Based Lab-on-Chip Devices
- Droplet Dispensing
- Membrane Actuation for Micropumps
- Microactuators
- Microneedles – Applications & Devices
- Supersonic Micro-Nozzles
- Piezo/PZT in Microfluidics
- Ultrasonic Pumps

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Piezoelectric Micro/Nanoliter Droplet Dispenser

► [Piezoelectric Microdispenser](#)

Piezoelectric Pumps

► [Piezoelectric Valves](#)

Piezoelectric Valves

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Synonyms

Microvalve; Piezoelectric pumps; No-moving-part valves

Definition

Structures that regulate the flow of fluids using piezoelectric materials, either composed of active regulation through deflection of a structure to block or allow passage of the fluid using electrically-driven piezoelectric elements, or as a piezoelectric pump in combination with mechanical or passive diffuser valves.

Overview

Active piezoelectric valves offer a unique combination of large closing forces – hundreds to thousands of new-

tons – against fluid loads and small displacements measured in the tens of microns. Such valves may be designed to close or open upon the application of an electric field to the piezoelectric material, and offer flow rates from nearly arbitrarily low amounts to tens of liters per minute for gases, water, and similar fluids [1]. Given the limited strain, 0.1% or less, available from typical high-performance piezoelectric ceramic materials, a majority of the research and development effort has been in amplifying this strain to permit efficient valving action. Flap or cantilever valves seated across an orifice actuated as a piezoelectric ► [bimorph](#) or ► [unimorph](#) or ball or grooved structures seated against an orifice and moved using a large-displacement ► [linear multilayer piezoelectric actuator](#) is typical of this approach.

Passive piezoelectric valves function in an entirely different manner, though they also make use of piezoelectric bimorph or unimorph structures. Forming one side of a small fluid chamber, or ► [Helmholtz cavity](#), the piezoelectric element can excite chamber resonances when driven at appropriate frequencies via an oscillatory electric field, usually from 10 Hz to 100 kHz depending on the device dimensions and fluid. With two or more orifices connecting the chamber to the outside, fluid may be passed into and out of the chamber; by using orifices with different shapes the flow direction may be regulated without requiring a mechanical blockage of the flow [2] or moving parts. Though these configurations are effective as pumps [3], they also can serve as valves for applications where some leakage is tolerable, or where ball or flap valves may be used as passive mechanical restrictions on the orifices to augment the sealing action and improve performance.

Cross References

- [Piezoelectric Microdispenser](#)
- [Piezoelectric Materials for Microfluidics](#)

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pL

► [Picoliter Flow Calibration](#)

Planar Laser-Induced Fluorescence

- Laser-Induced Fluorescence (LIF)

Plasma Bonding

- Plasma in Microchannel

Plasma Chromatography

- Miniaturized Ion Mobility Spectrometry

Plasma Etching

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Synonyms

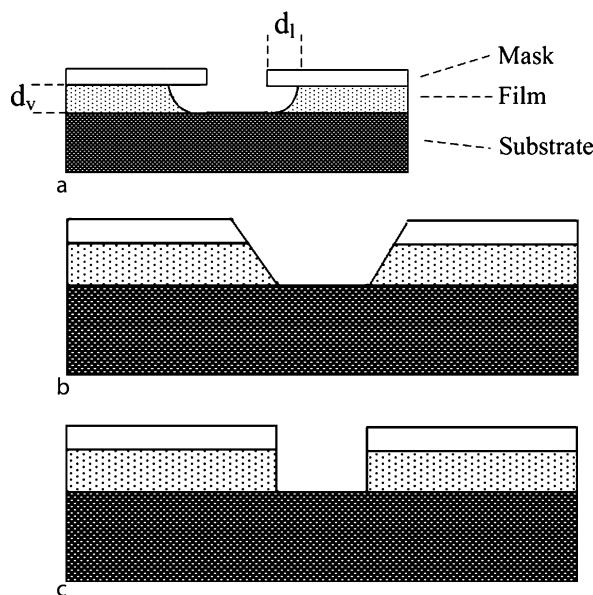
Advanced silicon etching (ASE); DRIE (Deep Reactive Ion Etching); High aspect ratio Si etching

Definition

Plasma etching is the removal process of materials performed using a plasma, which involves the physical, chemical and mixed etching mechanisms. Ion-enhanced ► [chemical etching](#), i. e., a reactive radical dominated etching process with ion assistance, is used in microelectronics and MEMS fabrication. ICP (inductively coupled plasma) etching is the etching process using a high-density plasma generated by the inductive coupling of RF power through an antenna, which is widely used in Si micromachining for high aspect ratio and deep microstructure fabrication. DRIE (deep ► [reactive ion etching](#)) is another term for ► [ICP etching](#) due to the deep or high aspect ratio microstructure fabricated by RIE principles, which is also called high aspect ratio Si etching or Advanced Silicon Etching (ASE™, a trademark for this technology of the company Surface Technology Systems, STS).

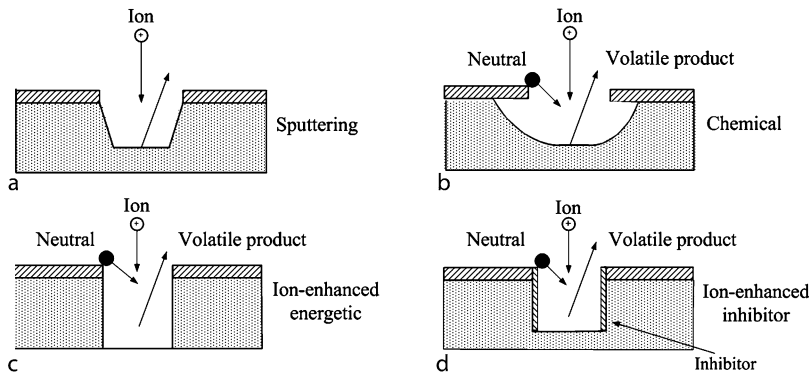
Overview

Etching is used in microelectronic processing and MEMS for pattern transfer during fabrication of devices. The pattern transferred is defined using lithographic techniques by UV exposure on a photoresist through a mask. The



Plasma Etching, Figure 1 Three types of plasma etching of a film on a substrate under a mask: (a) isotropic etching, (b) sloped wall via etch and (c) vertical wall (anisotropic etching)

photoresist material often loses adhesion in solutions during wet etching processes. Therefore, the pattern dimensions can vary and there can be a lack of control of line width. The process also etches laterally at approximately the vertical rate during wet chemical etching downward into the film and loses the etching directionality. So, wet etching in many semiconductor processes has been replaced by plasma-assisted etching [1, 2]. The driving force of plasma-based etching technology in semiconductor manufacturing is primarily due to the achievement of ► [etch directionality](#) and the ability for true pattern transfer from lithographically defined photoresist patterns into underlying layers. The isotropy, undercutting and adhesion problems encountered in wet ► [chemical etching](#) can be overcome by anisotropic dry plasma etching; other advantages are its cleanliness and compatibility with automation and vacuum processing technologies. Plasma etching is performed using a discharge gas formed by reactive radicals, ions and electrons to create ion sputtering or reactive radical species that react with surfaces with or without ion assistance to form volatile compounds. The physically sputtered or chemically reacted products evaporate from the surface and proceed to etch the film or substrate. When plasma etching is used, adhesion is no longer a major problem, undercutting can be controlled by varying the plasma chemistry and parameters, and directional or anisotropic profiles can thus be achieved.



Plasma Etching, Figure 2 Four types of reaction mechanism during plasma etching: (a) sputter etching, a purely physical process, (b) chemical etching, or chemical volatilization, a purely chemical process, (c) ion-enhanced energetic, i. e., ion-enhanced chemical, etching, a combination of physical and chemical processes and (d) ion-enhanced inhibitor, i. e., inhibitor ion-enhanced chemical, etching, also a combination of physical and chemical processes but that involves an inhibitor that passivates the surfaces that are not exposed to ion bombardment. The last two types (c) and (d) are related to reactive ion etching (RIE)

The **etch directionality** in plasma etching is due to directed energy input to an etching reaction at a surface and can be accomplished by radical atoms, electrons, photons or ion bombardment of a surface exposed to a chemical etchant. It can be obtained by controlling the interaction between the plasma physical and chemical mechanisms to achieve the required etching profiles using a suitable reactor and adjusting the plasma parameters. Figure 1 shows three types of plasma-etched profiles of films on a substrate under a mask: isotropic etching, sloped wall via etch and vertical wall (anisotropic etching) [2]. Anisotropic etching occurs when it proceeds much faster in one direction as compared to other directions. The directionality, or anisotropy, is defined by the ratio of vertical to lateral etching rate. The degree of isotropic etching, IE, can be defined in Fig. 1a by equation (1) [2]:

$$IE = \frac{d_l}{d_v} = \frac{V_l}{V_v} \quad (1)$$

where d_l is the amount of undercutting due to the lateral etching in rate of V_l , and d_v is the amount of the etched thickness of film due to the vertical etching in rate of V_v . So, the degree of anisotropic etching, AE, is correspondingly given by

$$AE = 1 - IE = 1 - \frac{V_l}{V_v} \quad (2)$$

In addition to isotropic and/or anisotropic behavior, another important etching characteristic is the **etch selectivity**. Etch selectivity between two materials is defined as the ratio between their etching rates at identical plasma conditions. High selectivity is usually preferred in etching and is related to a high etching rate

ratio between chemically different materials or between the etched and the underlying layer. In addition, high selectivity is required between the substrate and etching mask or between different substrate materials.

Also, a plasma etching process should satisfy a number of requirements, including [2]:

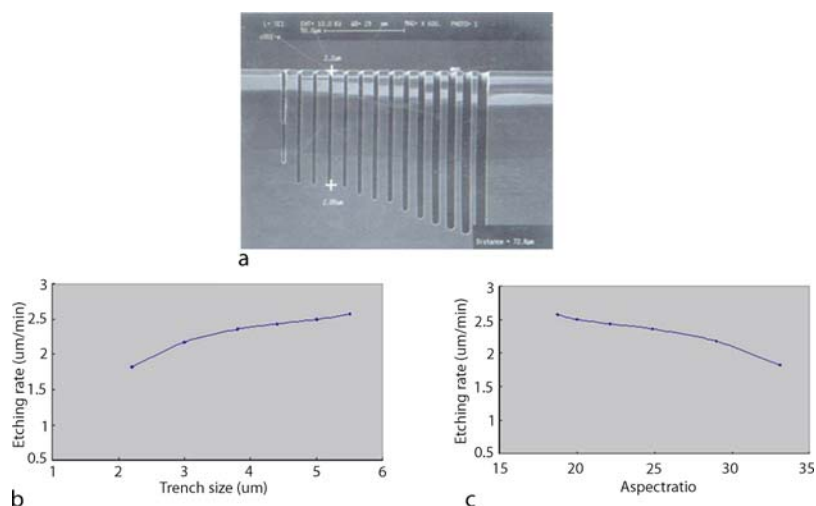
- high etching anisotropy for the desired etching profile
- high etching selectivity
- high etching rate
- good uniformity of etching across wafer surface
- minimal material damage
- smooth surface without residues after etching.

Since plasma etching is related to the reactant activated by input energy to active radicals, electrons and ions, it occurs through the following steps for physical or chemical processing [2]:

1. Active species are formed in the plasma from non-reactive gases.
2. The active species formed in the plasma adsorb on the surface of the substrate and react with the surface to form volatile products.
3. The volatile products with a high vapor pressure leave the surface at the substrate temperature and are pumped out with the effluents of the plasma.

The reaction during plasma etching can be categorized into four types of mechanism [2, 3] as illustrated in Fig. 2.

- (a) **Sputter etching**, a purely physical process caused by surface bombardment with high-energy ions, leading to anisotropic etching; also with poor selectivity, high surface damage, and low etching rates.
- (b) **Chemical etching**, or chemical volatilization, caused by neutral species generated in the plasma, which react with the etched layer to form volatile compounds.



Plasma Etching, Figure 3 High aspect ratio silicon trenches with nearly vertical sidewall profile obtained by inductively coupled plasma (ICP) etching: (a) SEM micrograph, (b) RIE lag and (c) aspect ratio dependent etching phenomena

Chemical etching is typically isotropic and is characterized by high etch rates, usually good [etch selectivity](#), and low substrate damage.

- (c) Ion-enhanced energetic, i.e., ion-enhanced chemical, etching, a combination of physical and chemical processes in which the chemical reactions at the surface of the processed sample are enhanced by ion bombardment, enabling one to achieve material selective etch anisotropy.
- (d) Ion-enhanced inhibitor, i.e., inhibitor ion enhanced, chemical etching, also a combination of physical and chemical processes but that involves an inhibitor that passivates the surfaces that are not exposed to ion bombardment. It is used in the sidewall passivation process for anisotropic etching.

The last two types, types (c) and (d), are related to [reactive ion etching](#) (RIE).

Plasma chemical etching is generally performed under low ion bombardment and chemical reactions with plasma species under conditions at which the physical effects are negligible and is usually an isotropic etching process. The pressure used in plasma chemical etching is relatively high, greater than 0.2 torr, often about 1 torr [2, 3]. If a parallel plate reactor is used, the substrates are placed on the ground electrode at a low potential relative to the plasma and the ion bombardment on the substrates is very weak. It is also performed in barrel reactors often with an etch tunnel to limit the discharge to a region outside the substrates. The substrates interact only with plasma species diffusing out of the plasma and are only chemically etched by atoms and radicals. RIE is generally per-

formed in a parallel plate reactor. The substrates are placed on an electrode, which is capacitively coupled to a generator, usually operated 13.56 MHz. The pressure used in RIE is intermediate in the range 0.01–0.2 torr. The problem is the increase of ion bombardment damage with increasing substrate bias as high RF power is required for a high etching rate. To solve the surface damage problem in RIE, a high-density low-pressure (HDLP) system has been developed with the substrates placed out of the high density plasma region. The plasma diffuses downstream to the substrates biased at low power for etching. The benefits of HDLP are high etching rate with low damage, high selectivity and good directionality control for high anisotropy. Electron cyclotron resonance (ECR) and ICP are the two popular excitation techniques for generating HDLP plasma. In MEMS, [ICP etching](#) is a very powerful tool for the fabrication of deep and high aspect ratio (HAR) Si microstructures. It will be discussed in detail in the following sections. The common applications of plasma/reactive ion etching in Si technology are listed in Table 1.

Many materials, e.g., single/poly-Si, SiN_x, SiO_x, glass, metals, silicides and polymers, can be etched by plasma-assisted etching to get anisotropic profiles by changing the etching chemistry. A large variety of fluorine-, chlorine-, bromine- and oxygen-based etching plasmas with a few gas additives, e.g., H₂, Ar and He, are used in the semiconductor industry as listed in Table 2 [1–3]. A process recipe involving the specific etching gas mixture and operation parameters is generally based on a large amount of empirical evidence obtained for a particular etching application rather than a real fundamental understanding of the rele-

vant plasma chemistry. In practice, etching is performed by selecting a typical etching recipe and adjusting using a certain basic understanding of plasma physics and chemistry for process improvement.

Basic Methodology

Anisotropic etching is desired in the fabrication of numerous devices, especially the high aspect ratio microstructure (HARM) for MEMS applications. Many phenomena become pronounced in HARM etching compared to the low aspect ratio features ($< 3:1$), e. g., ► **RIE lag**, ► **aspect ratio dependent etching** (ARDE), bowing (barreling) of sidewalls, faceting, profile shape dependence, microtrenching and notching (footing). These effects result from several mechanisms as follows [4]:

- angular dispersion of ions and neutrals as etchants due to collisions within the plasma sheath;
- ion and etchant transport within the feature onto the sidewalls;
- depletion of the reactant as microscopic loading under conditions of high probability at the surface;
- charging of feature sidewalls, as local charging of substrate surface or microstructure sidewalls deflects ions and prevents them from reaching the bottom surface to be etched;
- etching yield dependency upon the angle of incidence;
- deposition of material produced in the discharge within the feature; and
- redeposition of material released by ion-induced etching and reaction of the neutrals on the sidewalls to form passivation layers.

In order to obtain HARM with plasma-assisted etching, the significant issue is to reduce the lateral etching rate by controlling the ratio of ion flux to radical flux and sidewall passivation through the precursor gases and process parameters. Ion-assisted reactions at the bottom surface lead to highly anisotropic etching with the assistance of sidewall passivation or reduction of lateral reactive radical reactions. The key factors for controlling the micro-

scopic etch uniformity in HAR etching are the selection of reactant chemistry, process conditions and the transport behavior of reactant to the surface and product away from the surface. Kassing and Rangelow [4] reported two main etching strategies for reducing the adverse effects listed above: controlled formation of sidewall-inhibitor films, and ► **reactive ion etching** at low substrate temperatures. Also, Laermer and Schilp [5] used alternating etching and passivation steps in the Si etching process, known as Bosch patented technology, which has become the main technique in the fabrication of Si HARM devices. Based on the Bosch technology, Surface Technology Systems (STS) continues promoting the ASE process in equipment design and process enhancement [6], which has been widely used in many universities, research institutes and industries. Figures 3a–c show an SEM micrograph, feature-dependent ► **RIE lag** and ARDE phenomena of HAR Si trenches with nearly vertical sidewall profile, respectively [7]. The structure was obtained using the ASE process with an STS Multiplex ICP system. The feature sizes of trenches from the fourth to the ninth trenches from the left side are between 2.2 μm and 5.5 μm . The etch rates measured range from 1.82 $\mu\text{m}/\text{min}$ to 2.58 $\mu\text{m}/\text{min}$ while the aspect ratios obtained are from 33.1 to 18.7. The larger the trench size, the higher the etching rate, and the lower the aspect ratio.

In the following, the STS Multiplex ICP system and ASE process, etching performance and chemistry, end point detection and experiment will be introduced for anisotropic plasma etching.

STS Multiplex ICP Etching System and ASE Process

Figure 4 shows schematically the STS Multiplex ► **ICP etching** system. The source plasma is generated by an inductively coupled coil supplied by a 1–3 kW 13.56 MHz RF generator. Another 13.56 MHz generator is used to power the platen electrode which allows independent control of the bias potential of the wafer relative to the source plasma. The wafer temperature is maintained at typically less than 80 °C through temperature-controlled, pressurized helium supplied to the back of the wafer. Wafers may be either mechanically or electrostatically clamped to the platen electrode. Typical base pressures are in the 10^{-7} torr range. The system is equipped with an automatic pressure control valve which can operate in one of two modes. In fixed automatic pressure control (APC) mode, the APC position remains constant and the chamber pressure is a function of gas flows and RF powers. In automatic APC mode, the APC valve adjusts to maintain the chamber pressure at a constant value. The tool has been developed to produce excellent

Plasma Etching, Table 1 Common applications of plasma/► **reactive ion etching** in silicon technology

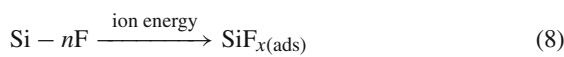
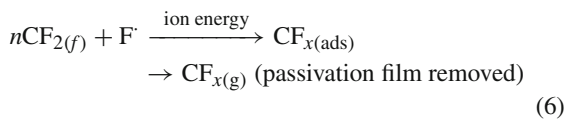
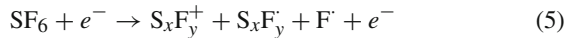
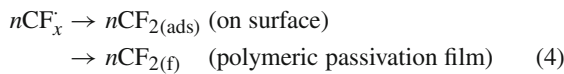
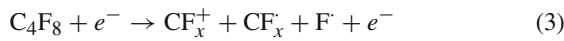
Application	Etch step
Device formation	Polysilicon/polycide gate; sidewall; storage node; emitter contact
Isolation	Deep trench; shallow trench or recessed oxide
Interconnection	Contacts; metal; interlevel vias
Other	Multilayer resist definition; resist strip; backside strip; planarization

Plasma Etching, Table 2 Materials and reactive gases used for plasma/reactive ion etching in Si technology

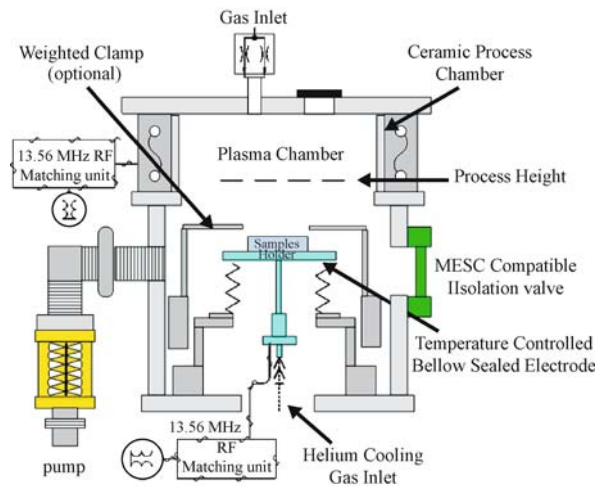
Material	Reactive chemistry	Remarks
Si	CF ₄ /O ₂ , SF ₆ , NF ₃ Cl ₂ , BCl ₃ , CCl ₄ HBr, CF ₃ Br	Anisotropic; difficult Directional; good SiO ₂ selectivity Directional
Oxide	CF ₄ /H ₂ , CHF ₃ /C ₂ F ₆ , CHF ₃ /CO ₂	Etching selectivity to Si
Nitride	CF ₄ , CHF ₃ , SF ₆ , NF ₃	Characteristics intermediate to Si, SiO ₂
TiSi ₂	CCl ₂ F ₂ , CCl ₄	Control of oxygen impurities
WSi ₂	CF ₄ /O ₂ , SF ₆	
W	CF ₄ /O ₂ , SF ₆	
Al	Cl ₂ , BCl ₃ , CCl ₄ , SiCl ₄	Removal of native oxide
Al(Cu)	Cl ₂ , BCl ₃ , CCl ₄ , SiCl ₄	Cu removal, removal of native oxide
Polymers	O ₂ , O ₂ /CF ₄	

repeatability from cycle to cycle for the switched ASE process.

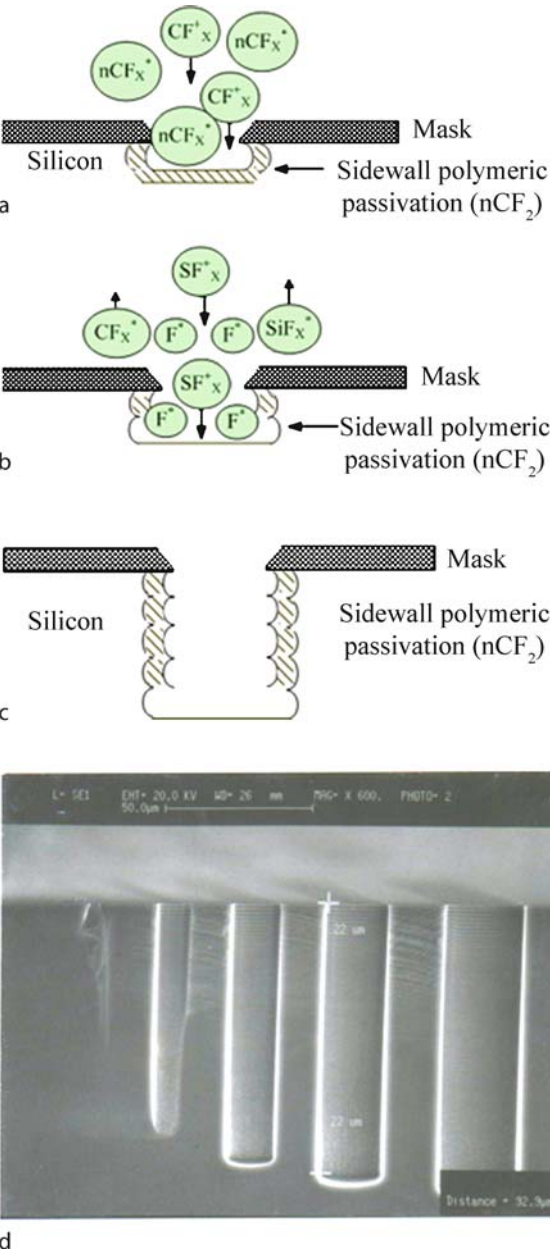
The ASE process uses sequentially alternating etching and passivation steps based on Bosch technology for anisotropic Si etching with feature sizes ranging from sub-micrometer to several millimeters. SF₆ and C₄F₈ are the primary gases used for etching and passivation purposes, respectively. O₂ is generally added to SF₆ in the etching step for prolonging the turbo pump life with reduction of sulfide coating on turbo blades [6] and for reducing the bowing profile at high bias with the enhancement of side-wall passivation [8]. The STS ASE procedure is shown schematically in Figures 5a–c: (a) passivation step after one cycle, (b) the etching step after one cycle, i. e., the second etching, and (c) profile after four cycles. Fig. 5d shows a cross-sectional SEM micrograph of the etched holes with scalloped surface due to the cycling ASE process. The etching and passivation mechanism can be explained with the following Eqs.:



The ionization and dissociation of C₄F₈ plasma deposits a nCF₂ or (C_xF_y)_n polymeric passivating film on the side-

**Plasma Etching, Figure 4** Schematic diagram of the STS Multiplex ICP etching system

walls and the base of the features (Eqs. (3) and (4)). This polymer, nCF_{2(f)}, is preferentially removed from the base of features by the directional ion bombardment energy during the following etching cycle to expose Si for spontaneous etching (Eqs. (5) and (6)). The sidewalls remain protected by the polymer which receives no ion bombardment and subsequently no **chemical etching** either. SF₆ gas supplies fluorine radicals (Eq. (5)) for spontaneous isotropic etching of exposed Si (Eqs. (7)–(9)). Si is mainly etched by the fluorine radicals (F[•]) and assisted by ion bombardment. The ion bombardment only serves to remove polymer from the base of the features. Then the passivation and etching steps are repeated for Si etching. The balance between etching and passivation determines the final process results and can be controlled by a wide variation of process parameters, e. g., the cycle time, flow rate, coil power, platen power and pressure in the etch-



Plasma Etching, Figure 5 Schematic diagram of the STS ASE procedure: (a) passivation step after one cycle, (b) the etching step after one cycle, i.e., the second etching, (c) profile after four cycles and (d) cross-sectional SEM micrograph of the etched holes with scalloped surface

ing and passivation step, as well as the substrate temperature and the distance from the source plasma. Table 3 lists the typical characteristics of Si etching using the STS ASE process [3, 6].

Figures 6a–d show cross-sectional SEM micrographs of etched profiles in different geometry [7]: (a) high aspect ratio microstructure of holes with the same nominal diam-

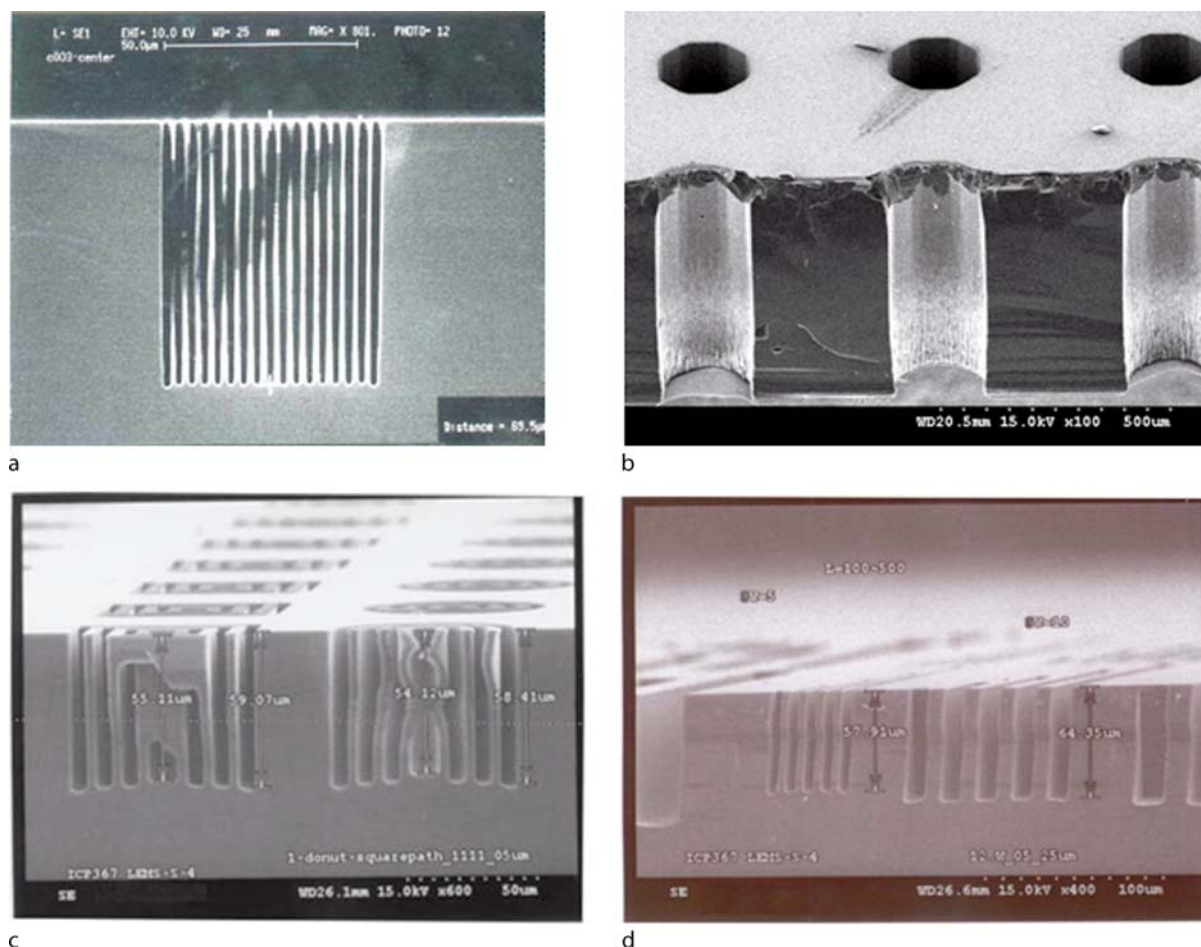
Plasma Etching, Table 3 Typical characteristics of Si etching using the STS ASE process [3, 6]

SF ₆ flow	30 – 150 sccm
C ₄ F ₈ flow	20 – 100 sccm
Etch cycle	5 – 15 s
Deposition cycle	5 – 12 s
Pressure	0.25 – 10 Pa
Temperature	20 – 80 °C
Etching rate	1.5 – 4 μm/min
Aspect ratio	> 20 : 1
Etch depth capability	10 – 800 μm (/through wafer)
Sidewall angle	90° ± 2°
Selectivity to photoresist	50 – 100
Selectivity to SiO ₂	100 – 200

eter of 1.5 μm and nearly the same etching depth of 69.5 μm, (b) through-wafer etching of 525 μm thick Si holes with a diameter of 200 μm, (c) rectangular and circular donut trenches with constant width or radius of 5 μm and (d) trenches with feature length from 100 to 500 μm at constant width of 5 – 10 μm, respectively. Compared to Fig. 3 with the larger feature width for the deeper etching depth, the larger continuous feature area also leads to the deeper etching depth. The rectangular donut has a larger area than the circular donut, which leads to the higher etching rate or depth. In addition, the etching depth of trenches is nearly the same as the feature width is fixed. But the etching depth varies as the feature width is changed. The feature width of pattern geometry is a dominating factor affecting the ►RIE lag in ►ICP etching while secondary factors are feature area, shape as well as length-to-width ratio. Etching rates of rectangular trenches are sensitive to width while the ring trenches are sensitive to both width and area. Process parameters can be adjusted to control RIE lag magnitude.

Etching Performance and Chemistry

The performance of anisotropy etching can be evaluated from the etching rate, selectivity, profile (directionality), loading effect, ►RIE lag, ►aspect ratio dependent etching (ARDE) and uniformity. Besides the design of equipment configuration, it is related to the process condition, pattern geometry and reactant chemistry. In general, there is a compromise between the etching rate, selectivity and profile. The higher the etching rate, the better the throughput. Etching rate can be enhanced by high plasma power, substrate bias, process pressure and gas flow and a reduced total etching area. But the high ion energy from high bias,

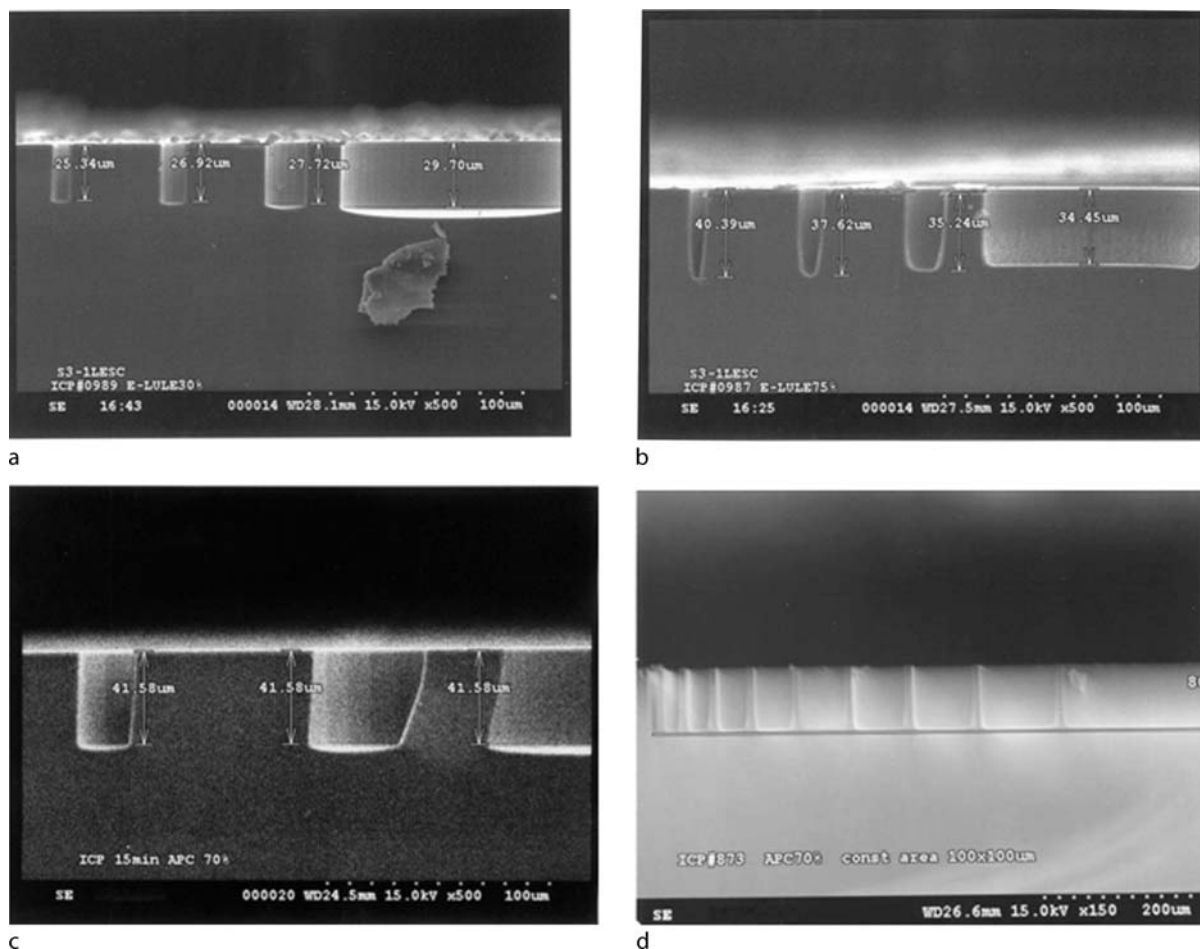


Plasma Etching, Figure 6 Cross-sectional SEM micrographs of: (a) high aspect ratio microstructure of holes with the same nominal diameter of $1.5\ \mu\text{m}$ and nearly the same etching depth of $69.5\ \mu\text{m}$, (b) through-wafer etching of $525\ \mu\text{m}$ thick Si holes with a diameter of $200\ \mu\text{m}$, (c) rectangular and circular donut trenches with constant width or radius of $5\ \mu\text{m}$ and (d) trenches with feature length from 100 to $500\ \mu\text{m}$ at constant width of 5 – $10\ \mu\text{m}$

which is increased with RIE power or independently controlled in ICP, leads to low selectivity and surface damage. Good selectivity, profile control and uniformity are more easily obtained at lower etching rate and shallower depth. In HARM with high etching rate, **RIE lag** and ARDE will become pronounced, thus reducing the uniformity. The larger the feature size, the higher the etching rate while the higher the aspect ratio, the lower the etching rate. Most isotropic etchants, e. g., SF_6 gases, exhibit a loading effect which can be reduced to some extent by using large-volume reactors and be eliminated as etchant loss, i. e., surface reaction instead of diffusion mechanism, dominates the etching behavior [1, 2]. The high ion energy for HAR features will result in bowing of the profile due to the scattering of ions at high pressure, which can be eliminated by enhanced sidewall passivation with the addition of O_2 in the etching gas [8]. In the ASE process [6], the primary dominant factors for etching rate enhancement are increase

of etching step time, decrease of passivation step time and decrease of passivation gas flow. Increasing plasma power, etching step pressure and platen power also increase the etching rate. In order to create a vertical sidewall angle, the process parameters may be adjusted by increasing passivation gas flow, substrate bias, substrate temperature and etching gas flow. The reverse adjustment is made to obtain sloped or tapered sidewall profile for a wafer-level package. To reduce the scallop size for a smooth profile, it can be controlled by decreasing the etching step time, increasing the deposition step time and decreasing the etching gas flow. The patented *Parameter Ramping* has been developed to continually adjust the key parameters in ASE to meet the requirements for good profile in HARM etching [6].

As listed in Table 2, a large variety of fluorine-, chlorine-, bromine- and oxygen-based etching plasmas are used in the Si semiconductor industry. Etching of Si can be accom-



Plasma Etching, Figure 7 Cross-sectional SEM micrographs of etched donut trenches with constant area at pressures: (a) APC 30% and (b) APC 75% for $\pi \times 52.5^2 \mu\text{m}^2$, (c) APC 70% for $\pi \times 105^2 \mu\text{m}^2$ and (d) APC 70% for the etched rectangular trenches with constant area of $100 \times 100 \mu\text{m}^2$, in which the width varies from 10 to 100 μm while length varies from 1000 to 100 μm

polished using F-, Cl- and Br-based chemistries and the etch products include volatile SiF_4 and SiCl_4 . The etching of Si can be carried out in a CF_4 or SiF_6 plasma with negligible ion bombardment of the Si substrate [1]. The correspondence of Si etch rate and atomic fluorine emission is as a function of the RF power through the intensity variation of the 704 nm atomic fluorine-related optical emission. Reactive F atoms are directly involved with Si etching. The gas phase density influences the etching rate in simple CF_4 or SiF_6 plasmas and a large undercut occurs in F-based plasma etching due to the isotropic chemical etch. The directionality in SiF_6 plasma etching can be improved by cooling the substrate to a temperature of -110 to -120°C , as cryogenic etching enables one to get near vertical profiles [1]. The spontaneous [chemical etching](#) reactions are suppressed at *cryogenic* temperatures. The etching at the bottom of the feature dominated by ion-assisted etching reactions is not affected at

low substrate temperature while the vertical etching rate increases with decreasing temperature, attributed to condensation of etchant for high density. The drawbacks of Si etching at cryogenic temperature are the cracking of resist mask, the increase of surface contamination (leading to black Si) and the increase of equipment complexity. A hard mask, e.g., oxide or metal, is needed in cryogenic etching. Anisotropic Si etching can also be obtained by Cl- and Br-based plasmas at room temperature. Si and oxide passivation are not spontaneously etched by chlorine and bromine atoms for directionality improvement. The size of the halogen relative to the Si atom plays an important role in the spontaneous etching. The spontaneous etching of Si exposed to large Cl atoms is absent compared to small F atoms for isotropic profiles. Although Cl- and Br-based plasma chemistries may achieve highly anisotropic structures, the shallow depth ($< 20 \mu\text{m}$) obtained limits the potential for MEMS applications. Modified F-based

plasmas for Si etching by sequentially switching etching and passivation gases, i. e., Bosch technology, demonstrate anisotropic etching for high aspect ratio microstructure and through-wafer at room temperature. It has become the main method in Si etching as described in the previous sections.

End Point Detection (EPD)

The most direct need for plasma diagnostic techniques results from the determination of the etch end point for a given process. In addition, plasma diagnostic techniques are used for process monitoring and provide information on the types of species present in a plasma etching, the concentration and the energy content. Laser interferometry (or reflectance) and optical emission spectroscopy (OES) are two commonly used techniques for EPD and require only an appropriate optical window attached to the chamber. They are easily implemented to obtain information about etching plasmas [1].

Laser Interferometry (Reflectance)

This technique measures the laser light reflected from the surface of a wafer being etched. For transparent films, e. g., SiO₂, an oscillation is observed for the reflected laser light intensity due to interference of the reflected light from surfaces of both the film and the substrate. The spacing between adjacent maxima (or minima) is $\delta d = \lambda/2n$, where λ is the wavelength of the laser light and n is the refractive index of the transparent layer. Etch rates can be examined in real time. For nontransparent films, e. g., metals, one measures the change in reflectivity before and after complete removal of the metallic film. For patterned wafers without an etch stop layer, we determine the depth of the etched pattern in situ through the used pattern as a diffraction grating. There are two principal drawbacks of these techniques. Firstly, they usually require a special test site especially as the features being etched, e. g., contact holes into a SiO₂ film, are too small for measurements. Secondly, the obtained EPD information is only for one specific area on the wafer, and not suitable for batch processing.

Optical Emission Spectroscopy (OES)

OES is the most widely used techniques for etch end point detection. The change in emission from a characteristic species is observed as etching of a film is completed. Either the decrease in emission of a suitable etch product species or the increase in etchant species at the end of the etching process is monitored. By measuring the wavelengths and intensities of the emitted spectral lines,

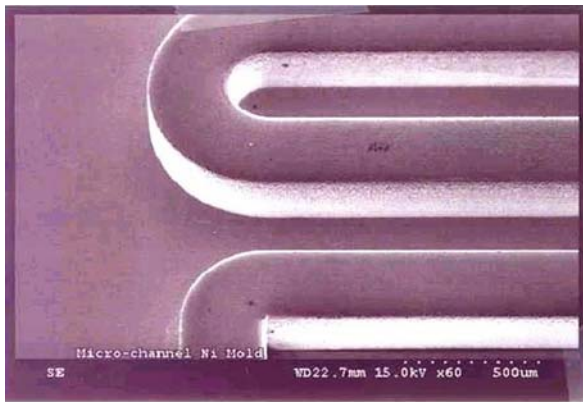
one can identify the end points both in research laboratories and in manufacturing for production control. Some important electronic materials and common optical emission lines of etching gases used for EPD are listed in Table 4. The OES technique is limited to the monitoring if light-emitting species and the emission intensities are not always directly related to the concentration of the species in the plasma. The sensitivity of this technique depends on how much etchant is consumed or how much film material is etched per unit. If the etch rate is too slow or the size of the etched pattern too small, e. g., contact hole etching into SiO₂ layers, this technique may not be suitable for EPD.

Experimental

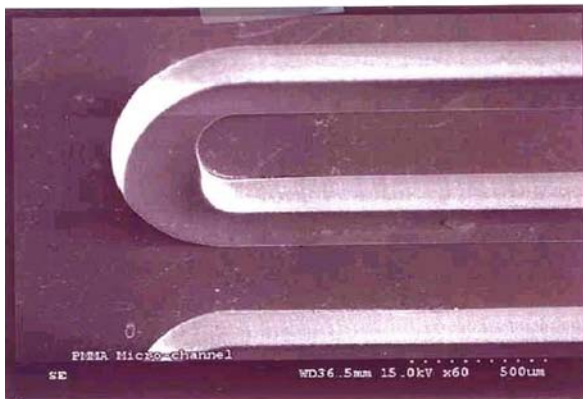
Si wafers are initially cleaned in a solution of H₂SO₄ : H₂O₂ at 3 : 1 ratio. Thick resist or thermal oxide SiO₂ are used as ICP etch mask of Si. Standard lithography and RIE are used to transfer the photo mask pattern to thick resist or oxide mask. Si deep etching is performed using the ASE process in an STS Multiplex ICP system. The etching and passivation gases are (SF₆ + O₂) and C₄F₈, respectively, and are switched during the process. Three major kinds of patterns of constant length, constant width and constant area with different feature sizes of 2 to 100 μ m are divided to understand the dominating factor of geometry for [▶ RIE lag](#) in [▶ ICP etching](#). Process pressure is adjusted by auto pressure control (APC) from 30% to 75% to understand the process pressure effect on RIE lag. The plasma source is generated by a 600–800 W 13.56 MHz RF generator

Plasma Etching, Table 4 Common optical emission lines of etching gases used for EPD [1]

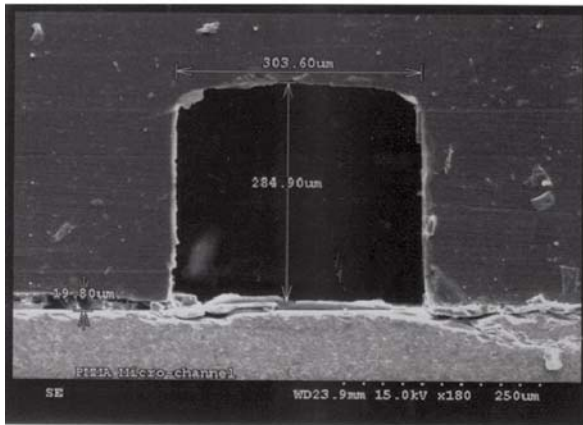
Material	Etchant gas	Emitting species	Wavelength (nm)
Si	CF ₄ /O ₂ , SF ₆	F (etchant)	704
	CF ₄ /O ₂ , SF ₆	SiF (product)	440, 777
	Cl ₂ , CCl ₄	SiCl (product)	287
SiO ₂	CHF ₃	CO (product)	484
Si ₃ N ₄	CF ₄ /O ₂	N ₂ (product)	337
	CF ₄ /O ₂	CN (product)	387
	CF ₄ /O ₂	N (product)	674
W	CF ₄ /O ₂	F (product)	704
Al	Cl ₂ , BCl ₃ , CCl ₄	Al (product)	391, 394, 396
	Cl ₂ , BCl ₃ , CCl ₄	AlCl (product)	261
Resist	O ₂	O (etchant)	777, 843
	O ₂	CO (product)	484
	O ₂	OH (product)	309
	O ₂	H (product)	656



a



b



c

Plasma Etching, Figure 8 SEM images of (a) an electroplated Ni mold, (b) hot embossed PMMA replica and (c) PMMA adhesive bonding in ICP LIGA

and biased by a 10–15 W 13.56 MHz RF generator at the platen during the etching step. The wafer is electrostatically clamped and cooled by backside helium flow. Scanning electron microscopy (SEM) is used to examine the etching result after the ASE process.

Key Research Findings

Fabrication of Deep/HAR Si Microstructure with High Etching Rate and Good Uniformity to Large Diameter Wafers (Up to 200 mm)

For the requirements of full-scale commercial production, Si DRIE for MEMS applications and advanced packaging are needed for high etching rate with excellent uniformity. The wafer size up to 200 mm is expected and important for high throughput and yield for a cost-effective end product. In mid-2005, STS [6] introduced a new generation of Si etch system, known as Pegasus, for the above purposes. Pegasus employs a revolutionary new de-coupled ICP source design by enhancing both the throughput and the yield to reduce the cost per die.

In conventional plasma processing with the *center-high* spatial profile of both ions and neutral radicals, the density profile variation of the ions will usually be more extreme than for the radicals, leading to large uniformity deviation. Two new de-coupled plasma sources are patented technologies of STS for ramping and SOI modules. They offer the following benefits:

- To reduce the volume of the plasma source for more efficient breakdown of the precursor gas due to the higher power density delivered from the RF power supply.
- To adjust the geometry and volume of the plasma source region to match the silicon's loading characteristics for the ions and neutral radicals diffusing down to the wafer with the desired profiles to obtain high etch uniformity.
- To alter the balance between numbers of ions and neutral radicals reaching the wafer to control selectivity to mask and/or ion damage to feature profiles.

With the appropriate magnitude of parameter values during an etching and precise control of notching at the surface of Si to the insulator, a high etch rate with good uniformity can be achieved. For example, Si etch rates of more than 30 $\mu\text{m}/\text{min}$ can be achieved at a more realistic exposed area of around 10% and exceeding 50 $\mu\text{m}/\text{min}$ for a 1% exposed area have been observed using the Pegasus source with the ASE process. Yield is optimized by improving absolute etch depth uniformity and by reducing trench tilting artifacts observed with conventional ICP plasma sources. The very high etch rate uniformity and feature profile control is relevant for up to 200 mm wafers.

Inverse RIE Lag and Lag Elimination

A ► **RIE lag** phenomenon occurs in many patterns and becomes more severe at smaller feature width and higher etching depth. In general, the smaller the feature size, the

Examples of Applications

Many examples of applications have been reported in the STS Innovations magazine issues 1–3 since April 2006 [6]. They include MEMS inertial and pressure sensors, Si springs for luxury wristwatch components, optical switches and resonators for communication, microphones for portable electronic equipment, advanced packages (system-in-package), and inkjet channels and nozzles. For example, Figs. 9a and b show SEM micrographs of a plane-view nozzle array with circuit layout and cross-section of one nozzle, respectively, for high-resolution inkjet head application. It combines the fabrication of backside wet anisotropic etching and front-side dry DRIE etching. The nominal nozzle diameter is 10 μm with vertical profile as well as scallop rings due to the ASE switching process. The emerging nanoscale etching has been applied in photonic crystals, magnetic nanostructures (magnetoresistive random access memory, MRAM), high-Q MEMS resonators, sub-micrometer pillar array or HAR structures made of Si for bioanalysis and lab-on-a-chip, microchannels and inkwells for dip pen nanolithography, and Si or quartz stamps for nanoimprint lithography.

Future Directions for Research

ICP is an important technology for micro- and nanoscale patterning processes. High etch rate, high aspect ratio and anisotropic, good profile control, high uniformity and RIE lag elimination are important issues in the development of MEMS and nanotechnology devices. In MEMS fabrication, it is steadily moving from the development and pilot stages to full-scale production. In nanofabrication, it is promising to enable nanometer–micrometer–millimeter multiscale structures in device design and application. In combining isotropic and anisotropic etching, different promising HARM or three-dimensional structures may be realized for the application of Si-MEMS or SOI MEMS. ICP technology will be required in many applications including MEMS inertial and pressure sensors, optical switches and resonators for communication, inkjet channels and nozzles for high-resolution printheads, microphones for portable electronic equipment, advanced packages, and so on.

Cross References

- Anisotropic Etching
- Dry Chemical Etch
- ICP Etching
- Isotropic Etching
- Laser Ablation
- Laser Based Micromachining

- LIGA Like Techniques
- Plasma in Microchannel
- Reactive Ion Etching (RIE)
- Silicon Micromachining

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Plasma in Microchannel

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Synonyms

Microchannel glow discharge; Microchannel surface modification; Microchannel surface sterilization; Plasma bonding

Definition

Gas plasma, a fourth state of matter, is a neutral composition of ions, electrons, radicals, and photons that is synthesized by applying energy to a gas at atmospheric pressure or within a vacuum. Gas plasma can be used to enhance the functionality of microdevices for various biological and chemical applications. The effects of plasma in microchannels are regulated by parameters such as the type of gas, power intensity, chamber pressure and duration of the treatment.

Overview

Microchannels that range in diameter from tens to hundreds of microns have emerged as potentially powerful tools for a variety of biomedical applications. They can be used to minimize sample volume and reduce costs, as well as to increase throughput and analysis sensitivity. Since microfluidic devices operate at small length scales, their functionality is greatly dependent on their surface properties, which vary depending on the type of material that is used to fabricate the channel and the subsequent surface modifications.

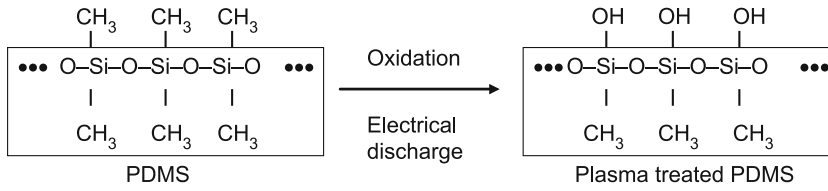
Microchannels have traditionally been fabricated from commonly used materials from the semiconductor industry, such as silicon. Silicon offers many advantages for microchannel fabrication, such as standardized fabrication procedures and easy integration with electronics. Despite these merits, silicon is opaque and brittle and cannot be used for fabricating transparent microfluidic devices that require light based sensors. To overcome these difficulties, glass microfluidic channels have also been developed, however, the fabrication of these devices, similar to those of silicon microchannels is time consuming, expensive and requires the use of a clean room facility. As a result, polymeric microchannels have recently emerged as potentially valuable alternatives. Polymers that have previously been investigated for microchannels include: polycarbonate (PC), polystyrene (PS), poly(dimethylsiloxane) (PDMS) and poly(methylmethacrylate) (PMMA). One of the main advantages of using polymeric materials to make microchannels is that they can be fabricated from soft lithographic techniques such as micromolding. Furthermore, the ability to tailor the mechanical and optical properties of polymers can be used to fabricate microchannels with a wide range of properties. Despite the great advance-

ments in the materials used to fabricate microchannels, the surface properties of these channels often require additional modifications based on the specific needs of the device. Thus, approaches to activate microchannel surfaces have been of great interest in fully realizing the potential of these techniques.

During the 1920s and 1930s, Dr. Irving Langmuir proposed the term *plasma* while he was studying the physical properties of gases and the balance of charges between ions and electrons. Commonly considered as the fourth state of matter, basic gas phase plasma contains a neutrally charged mixture of negative and positive ions, photons, electrons and radicals that are produced by applying energy to a gas at low pressure (from 10^{-6} to 10^{-4} bar). In this process, the temperature of plasma electrons is extremely high (on the order of 10^4 K), however, the plasma gas remains close to ambient temperature. This is because plasma components found at higher temperatures (such as the sun) are made of equilibrium (thermal) plasmas while low-pressure gas plasmas are non-equilibrium (non-thermal) type. Plasma is formed by the exposure of free electrons to an external energy source that increases the kinetic energy, resulting in the formation of radicals and ions. These excited molecules then form photoemissions, ions and free radicals due to a loss of excess energy from the plasma. The energy released from these chemical species is capable of breaking covalent bonds and altering the surface chemistry of many materials.

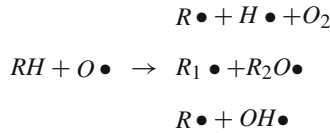
The bombardment of a surface with reactive chemical species can be used to drastically change its surface properties and create functional surfaces within microchannels. Traditionally, corona discharge and oxygen plasma treatment have been used. Corona discharge is the formation of sparks at the tip of a wire created by electromagnetic fields that result in the ionization of nearby molecules at low temperatures and pressures. Corona discharge can be used to oxidize a surface near the tip of the wire. Although widely used in various industrial processes, a negative aspect of corona discharge is its tendency to modify the surface non-uniformly. To overcome this limitation, surface plasma treatment can be used. Plasma treatment of microchannel surfaces is particularly attractive, as it does not require the tedious placement of a wire inside microscale channels. In addition, plasma treatment of a surface is faster and more reproducible compared to corona discharge.

The reaction between the plasma and a surface is highly complex and comprised of numerous and simultaneous chemical reactions [1]. Upon the initial application of energy, the gas molecules, such as oxygen, are converted to various unstable intermediates that can react with the surface. Some of the initiation mechanisms between plasma



Plasma in Microchannel, Figure 1 Schematic of the oxygen plasma treatment of a PDMS substrate

(which may include oxygen radicals) with a surface are shown below:



These radical species then induce chain scission, radical transfer, oxidation and disproportionation to generate functional surface groups such as hydroxyl and carboxyl moieties (Fig. 1). For some materials, such as PDMS, these activated surfaces can be bonded to fabricate channels, as shown in Fig. 2. Plasma treatment of microchannels can also be used to control surface properties by incorporating reactive molecules or by increasing the hydrophilicity of the surfaces that eases the flow of fluids and minimizes the formation of bubbles.

In addition to adding surface functional groups, plasma can be used to remove organic contaminants and sterilize surfaces. In this process, plasma radicals break the covalent bonds and oxidize the surface contaminants, which are then cleared by the vacuum. Here, we will outline the basic methodology of gas plasma modification in microchannels, including theory, experimental methods, pertinent process parameters and characterization methods. Additionally, key research findings will be discussed.

Basic Methodology

In this section we will outline the basic methodology of the plasma treatment process. First, the theory behind plasma treatment will be presented. Following this, experimental protocol outlining the treatment process, pertinent parameters and their relevance will be discussed. Finally, surface analysis methods for plasma treated surfaces will be described, such as X-ray photoelectron spectroscopy (XPS), atomic force microscopy (AFM), and contact angle measuring tools.

Theory

To establish an analytical framework of plasma formation and initiation, thermodynamic models have been devel-

oped. For example, for low pressures and high frequencies that generate plasma gases at ambient temperatures the power required to generate plasma can be determined from molecular collision analysis. In this model, the reactive species within the plasma process are created from inelastic collisions between electrons and ground-state atoms [2], where electrons have energy, u , given by

$$u = \frac{(mw^2)}{2}$$

where w is the electron velocity and m is the electron mass. The excitation rate coefficient, C_j , of a particular species is then given by

$$C_j = \left(\frac{2}{m}\right)^{\frac{1}{2}} \int_0^{\infty} \sigma_j(u) F_0(u) u du$$

where $\sigma_j(u)$ is the particular process cross-section and F_0 is the electron energy distribution function. Nearly all processes at this level that are important possess energy threshold, u_j , where

$$\sigma_j(u) \equiv 0, \quad u \leq u_j$$

There is also a dependence of the number density $\dot{n}_j$ of species j produced per second in the plasma from ground state molecules (of number density $N \text{ cm}^{-3}$) on the electron density n , where

$$\dot{n}_j = C_j N n.$$

The applied electromagnetic field (where frequency $f = \omega/2\pi$) and the plasma has a power balance that is expressed by

$$P_a = \xi n V,$$

where the power absorbed is P_a , the volume of plasma is V and the average power absorbed per electron is ξ . ξ also refers to the power necessary to sustain an electron-ion pair in the plasma.

Experimental

To plasma treat a surface, samples are initially placed within a plasma vacuum chamber. Gas plasma is then generated using various energy sources such as alternating current (AC), radio frequencies (RF) or microwaves (MW) (Fig. 3). For example, in apparatuses that use RF, two electrodes apply energy in the form of magnetic and electric fields to a gas at low pressure (~ 266 – 800 mbar) inside the vacuum. The application of appropriate amount of energy to the gas discharge results in the formation of a neutral gas plasma. During this process, a characteristic glow is emitted from the apparatus indicating the plasma state of the gas.

Parameters

There are a number of parameters that affect the surface modification during the plasma treatment process. These parameters can be separated into internal process parameters, which are dependent on the plasma, and external process parameters, which are external to the plasma but within the system. Examples of internal parameters include the type and the degree of dissociation of the gas and the residence time of the active species in the vacuum chamber. Examples of the external parameters include, but are not limited to, the substrate material, amount of time that the material is exposed to the plasma, gas pressure, energy frequency, power and substrate temperature. It is important to note that the internal parameters of the plasma are indirectly controlled by the external process parameters. Figure 4 depicts the relations among several parameters. This entry will discuss how the type of gas, exposure time, gas pressure and power affect the plasma initiated surface modifications and the resulting bonding strengths of substrates.

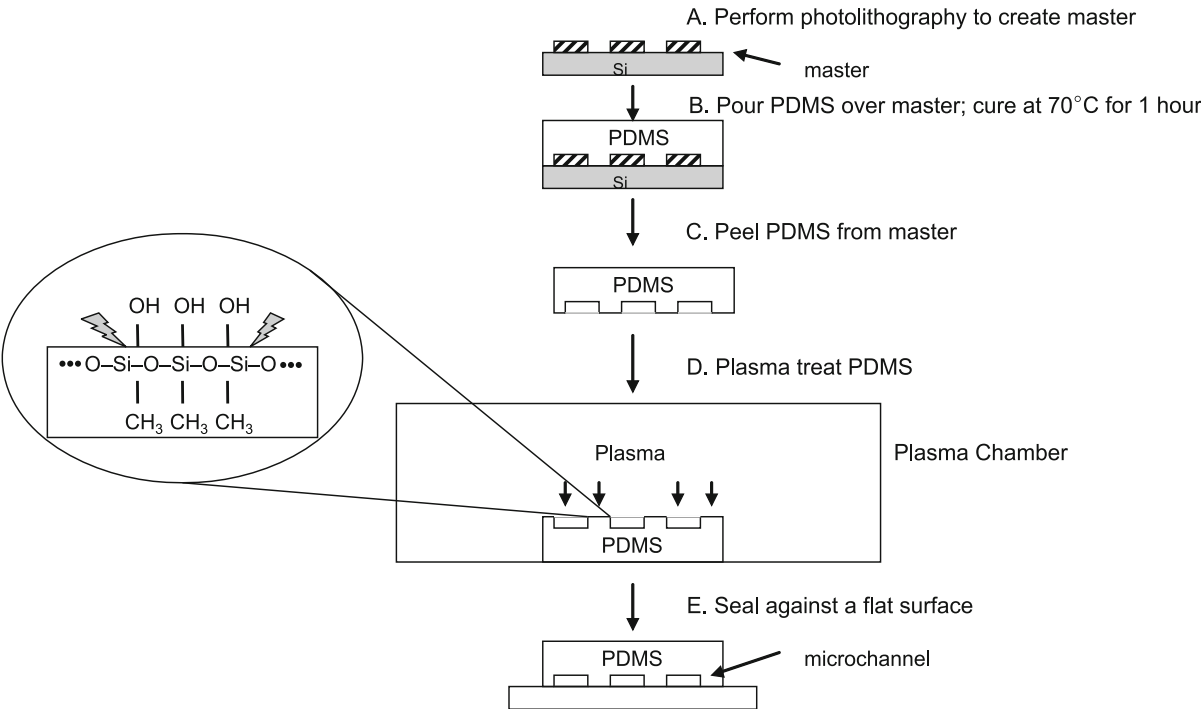
The type of gas is a key parameter in any plasma surface modification process. Plasma treatments using inert gases, such as argon, have been shown to increase the hydrophobicity of oxygen-containing polymers without adding detectable chemical species to the surface. These properties are a result of the degradation and rearrangement of molecules on the surface of the polymer. Degradation is caused by an increase in the kinetic energy in the plasma treatment chamber when inert plasmas are present. This increased energy breaks molecular bonds on the surface, which result in the degradation of the impurities on the surface of the substrate. Rearrangement of chemical species occurs as a result of cracks that are introduced onto the surface of the substrate. These cracks enable molecules from beneath to rearrange with molecules present on the surface. This is indicated by a decrease in the oxygen/carbon ratio within the treated

surface, possibly resulting from loss of CO or CO₂. In contrast, plasma treatment by using reactive gases results in the direct conjugation of the reactive molecule on the surface. For example, in the case of oxygen plasmas, hydroxyl and carboxyl functional groups are formed on the surface and can react with each other between two treated surfaces through a dehydration reaction to produce covalent linkages that result in bonding of two substrates.

The exposure time of the plasma to the surface can greatly affect its surface properties. For example, the bonding of PDMS to PDMS for the fabrication of microchannels has been shown to be time dependent (Table 1). If the substrate is not adequately treated with plasma then there will not be enough functional groups present to achieve bonding. If the substrate exposure exceeds an optimal time. Then there will be excessive oxidation of the PDMS surface and the formation, which leads to the formation of a brittle layer of silica. Thus, prolonged plasma treatment results in polymer reorganization on the substrate surface that causes decreased bonding strength due to induced hydrophobic properties.

It has also been determined that the internal pressure of the treatment chamber has a significant impact on the surface properties of treated substrates (Table 1). When the internal pressure is high, the mean free distance between molecular collisions within the plasma is significantly reduced. For example, for PDMS substrates the upper pressure region is characterized as greater than approximately 0.1 mbar. The molecules within this condensed plasma increasingly interact with the polymer surface molecules, resulting in increased bonding strength. Conversely, if the pressure within the chamber is too low (i. e. less than approximately 0.1 mbar) there is reduced interaction between the plasma and the surface and the bond strength is not optimized.

Finally, power variation has also been shown to affect the surface properties of the substrate (Table 1). In the case of PDMS at low power settings, plasma electron speeds are reduced and there is less interaction between the radicals in the plasma and the substrate. This minimizes the formation of functional groups on the surface. Conversely, extreme voltages result in reduced bonding strengths because of the formation of Si–O–Si groups on the PDMS surface. The dissociation energy of these specific bonds is larger than the dissociation energy of the Si–C bonds in the backbone of the polymer, therefore causing substrate degradation and less functional group formation for PDMS–PDMS bonding. At optimal power levels the number of functional groups present on the surface of the substrate is high enough to encourage bonding but low enough to not affect the siloxane backbone of the PDMS.



Plasma in Microchannel, Figure 2 Schematic of the process used to fabricate microchannels through a plasma induced bonding process

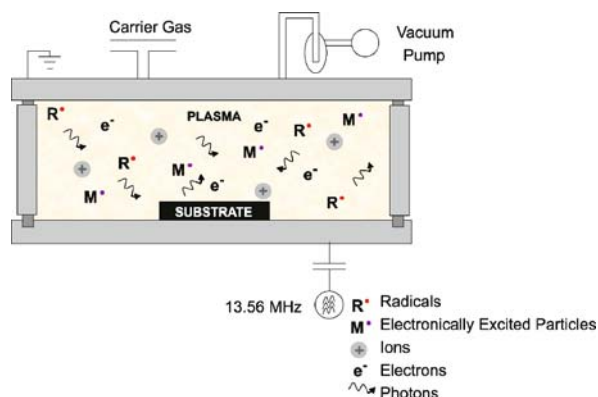
Plasma in Microchannel, Table 1 Comparison of the bonding results with RIE process parameters for bonding of plasma treated surfaces. Bonding 1 metric based on permanently bonded area after immediate contact (no bonding = 0 ~ whole area bonded = 10). Bonding 2 metric based on permanently bonded area when 3 days have passed after immediate contact. Adapted from [11]

#	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	
Power (W)	70				200				300				400				500					
Pressure (mTorr)	65		120		180		65		120		180		65		120		180		120		160	
Time (sec)	10	40	10	40	10	40	10	40	10	40	10	40	10	40	10	40	10	40	10	40	20	
Bonding 1 (max=10)	10	6	10	9	10	9	10	6	10	7	10	3	4	0	3	0	3	0	2	0	0	
Bonding 2 (max=10)	10	10	10	10	10	10	10	8	10	8	10	8	4	0	3	0	3	0	2	0	0	

Evaluation of Plasma Treated Surfaces

Surface chemical composition, structure and functional properties can be characterized by various techniques, such as XPS, AFM, and contact angle measurements. XPS measures the short range of photoelectrons leaving the surface. The sensitivity (~1 – 10 nm) is semi-quantitative and detects chemical bonding information at the surface. Composition depth profiles, known as XPS spectra, can be generated from this method. Conversely, AFM utilizes a nanoscale probe to characterize surfaces and produce 3D surface profiles. AFM can determine the surface roughness

and elastic modulus through phase imaging and nanoiden-tations, respectively. In addition to analyzing the surface of substrates, cross-sections can be imaged by using AFM to picture the stiffer surface layer and determine thick-ness. Furthermore, contact angle analysis can also be used to study surface wetting behavior. Contact angle measur-ing devices measure the angle of a liquid–vapor interface with a solid surface to assess surface hydrophilicity. Water droplets on hydrophilic substrates spread out and have relatively low contact angle measurements. Conversely, they maintain a more spherical shape on hydrophobic sub-strates, thereby producing higher contact angle measure-



Plasma in Microchannel, Figure 3 Schematic of a radio frequency plasma treatment chamber

ments. Common contact angle analysis methods include drop–bubble, reflective light and interference microscopy.

Key Research Findings

Various research findings on the effects of plasma treatment on microchannels have reported improved surface characteristics within microfluidic channels. Researchers have also been able to adapt plasma treatment technologies to develop novel applications in the field of engineering. Furthermore, applications of plasma in microchannels include the development of on-chip light sources and the fabrication of micropatterns and gradients for cell adhesion studies.

Hydrophobic Recovery and Bonding Enhancement Techniques

As previously mentioned, surface properties of a substrate can be tailored by applying various plasma treatments. The stability of plasma treatment is dependant on the material properties. For instance, for PDMS, oxygen plasma treated surfaces regain their hydrophobicity over time. This may be due to

- surface silanol groups reorienting into the bulk;
- movement of free PDMS chains from the bulk phase to the surface through cracks in the Si–O–Si structure;
- silanol groups at the surface condensing;
- volatile oxygen or other species escaping into the atmosphere;
- changes in the surface roughness [4].

Conversely, other explanations suggest hydrophobic recovery can be described in two distinct phases: adsorption of atmospheric contaminants and surface reorganization. Within the first few hours, the initial phase of adsorption of atmospheric contaminants begins. These atmospheric species react with the surface moieties that were

introduced during treatment and reduce their functionality. In the case of PDMS, surface reorganization begins when siloxane groups from the backbone of the polymer migrate through cracks created during surface modification and reorganize to yield the original properties of the polymer. Similarly to hydrophilicity, bonding characteristics are also dependent on time (Table 2). It has been shown that the adsorption of contaminants and surface reorganization also has an adverse effect on the bonding properties of PDMS.

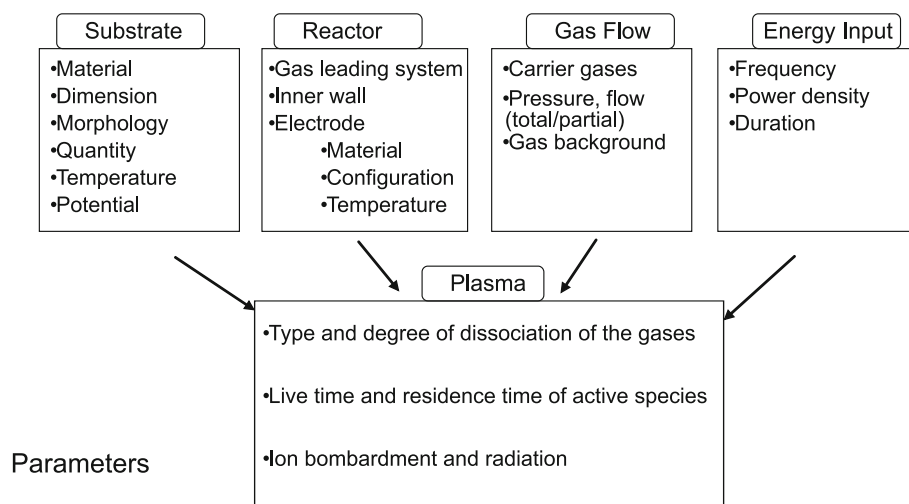
To increase the stability of a surface treated substrate a variety of approaches have been developed. For example, different plasma gases can be applied to reduce the amount of surface reorganization. Specifically, nitrogen treated surfaces maintain their hydrophilicity longer than oxygen plasma treated surfaces. Also, to maintain stable hydrophilic PDMS it is possible to fabricate C-F bonds or react poly(HEMA) bonds on PDMS microchannel surfaces [6].

Enhancing Microchannel Functionality

Plasma treatment of microchannels can be useful for improving the functionality of microdevices. For example, previous studies have shown that PDMS microchannels can be made hydrophilic by the addition of silane molecules with polar head groups [7]. In this process (3-mercaptopropyl)trimethoxysilane (3-MPS) was absorbed to PDMS to increase the hydrophilic properties of microchannels. Additionally, plasma polymerization has been used to induce in the long-term hydrophilic surface modification by covalently bonding a polymer layer to the surface. Barbier et al. [8] describe a method based on plasma polymerization modification with acrylic

Plasma in Microchannel, Table 2 Aging of activated PDMS surfaces. Bonding metric is based on permanently bonded area after aging the activated surfaces (no bonding = 0 ~ whole area bonded = 10). Adapted from [11]

#	1	2	3	4	5	6	7	8	9
Power (W)	70								
Pressure (mTorr)	75								
Etch Time (sec)	10								
Aging (min)	2	3	5	7	10	15	20	25	30
Bonding 1 (max=10)	10	10	10	10	10	10	6	7	8



Plasma in Microchannel, Figure 4 External parameters that influence the quality of surface plasma treatment

acid coatings. First, argon plasma pretreatment was used to activate trace oxygen molecules in the chamber, which partially oxidize the top layer of the substrate. This step crosslinked the surface to reduce ablation of silicon during deposition as well as limits the movement of siloxane chains for hydrophobic recovery. Secondly, plasma with acrylic acid was deposited onto the pretreated PDMS. Finally, helium plasma treatment was used to crosslink the deposited polymer film to reinforce the cohesion. This process provides a homogeneous stable hydrophilic polymer layer for robust applications within microfluidics and nanofluidics.

Applications

Applications of plasma in microchannels include the development of chromatography detectors and on-chip light sources. Evju et al. [9] describe the creation of microdischarges or microplasmas by direct current within microfluidic channels on polystyrene. The three different gases tested included helium (He), water-saturated helium ($\text{H}_2\text{O}/\text{He}$), and hexafluoroethane ($\text{C}_2\text{F}_6/\text{He}$). These gases were released through a microchannel and ignited at atmospheric pressure. The treatments were completed in several seconds to reduce heat formation. XPS confirmed that oxygen and fluorine on the surfaces contributed to hydrophilic and hydrophobic regions, respectively.

Complex biomolecular micropatterns and microgradients have also been synthesized with the use of plasma treated surfaces, with possible future applications in tissue engineering, biosensors, and immunoassays. Various method-

ologies, such as microcontact printing and micromolding, have been used to create micropatterns where wettability was controlled through selective plasma treatment. These micropatterns are made up of many ligands and cell types for spatial control and cellular organization. This organization can induce cell–cell signaling and recreate the hierarchy lost in standard in vitro methods to produce more tissue-like structures. These advances are critical in understanding biology and enabling improved tissue culture and tissue engineering techniques.

An application of plasma treated surfaces has also been shown in microcontact printing, where elastomeric stamps are *inked* with chemical species and ligands, which can then be patterned onto different surfaces. Langowski et al. [10] have shown that by applying oxygen plasma to the original *stamp* there is a reduction in low-molecular-weight silicon containing fragments transferred to PMMA substrates. The effect of the treatment increased ligand bioactivity, cell function and proliferation due to reduced contamination from the initial untreated substrate. This finding allows for proper modification for applications desiring the removal of low-molecular-weight silicon fragments, especially for biological applications.

Microgradient textures within microchannels have also been synthesized by utilizing plasma treatments. By having gradients of ligands, microchannels enable visualization of the effects on cell behavior, such as differentiation, migration and adhesion. Because of the quick and inexpensive nature of soft lithographic techniques and plasma treatments, biological applications can be done more cost effectively and in a faster manner.

Future Directions for Research

Plasma treatment of microfluidic surfaces can be used to improve microdevice functionality, to build devices through bonding processes and to activate and sterilize surfaces. Future directions for this technology include improved modeling of techniques with respect to parameter and surface plasma interaction. The development of refined experimental methods, theoretical models, and experimental studies is required to have more control over plasma treatment within microchannels.

Cross References

- Plasma Treatment Physics
- Plasma Etching
- AFM
- Bioprinting on a Chip
- Cell Adhesion and Detachment
- Control of Microfluidics
- Wetting and Spreading

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Plasma Oxidation

Definition

The treatment of a surface with a plasma containing oxygen generally performed using a vacuum chamber and an RF discharge.

Cross References

- Sputtering for Film Deposition

Plasma Treatment of Nanoparticles for Nanofluids

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Synonyms

Diamond nanoparticles; Nanofluids; Thermal conductivity

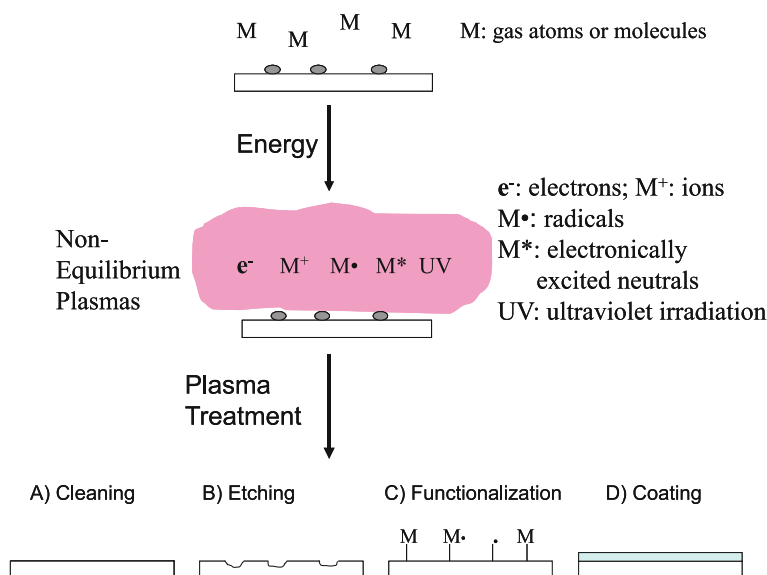
Definition

Plasma treatment refers to the surface modification processes of materials using non-equilibrium gas plasmas. Non-equilibrium plasmas with a low degree of ionization, so-called cold plasmas or low-temperature plasmas, are mainly composed of electrons, ions, free radicals, and electronically excited atomic and molecular species. These highly reactive plasma species interact non-thermally with material surfaces, and can react with and bond to various substrate surfaces, or combine together to form an ultra thin layer of plasma coating, and consequently alter the surface chemistry and surface properties. The plasma treated nanoparticles with desired surface functionalities can strongly interact with liquid molecules and thus better disperse into the base fluid to form stable suspension.

Overview

Plasma Treatment

Non-equilibrium plasmas, i. e., low-temperature plasmas, which can be easily created by electrical discharges under reduced pressures (e. g., 10 mtorr to 10 torr), are composed of electronically excited atomic, molecular, ionic and free radical species. Depending on the plasma chemistry or gas



Plasma Treatment of Nanoparticles for Nanofluids, Figure 1 Plasma treatment effects on materials surfaces

composition, these highly reactive plasma species react with, clean, and etch surface materials, bond to various substrates, or combine to form a nanoscale thin layer of plasma coating, and consequently alter the surface characteristics. Figure 1 schematically illustrates the plasma-surface interaction results when a material substrate is exposed to low-temperature plasmas. Such plasma-surface interactions are limited to only the material surface without affecting bulk properties of the substrate materials. Since the gas phase is close to room temperature, low-temperature plasmas are suitable and very efficient in surface modification of various materials such as metals, polymers, and ceramics. Plasma treatments with and/or without deposition are currently being widely used in surface modification of various materials for improving adhesion, wettability, printability, corrosion resistance, surface etching or cleaning, and biocompatibility etc. [1–6].

Plasma treatment for material surface modification can be achieved using either simple gas plasmas without deposition or organic vapor plasmas to deposit an ultra-thin layer of plasma coatings on the surfaces. Plasma treatment possesses many unique advantageous features in surface modification of materials. Among these are:

1. plasma treatment processes are usually reliable, reproducible, non-line-of-sight, and applicable to different sample geometries;
2. plasma deposition creates an ultra-thin (in nano-scale) film coating, which are conformal, chemically inert and robust, corrosion-resistant, and strongly bonded to various substrates including both organic and inorganic materials;

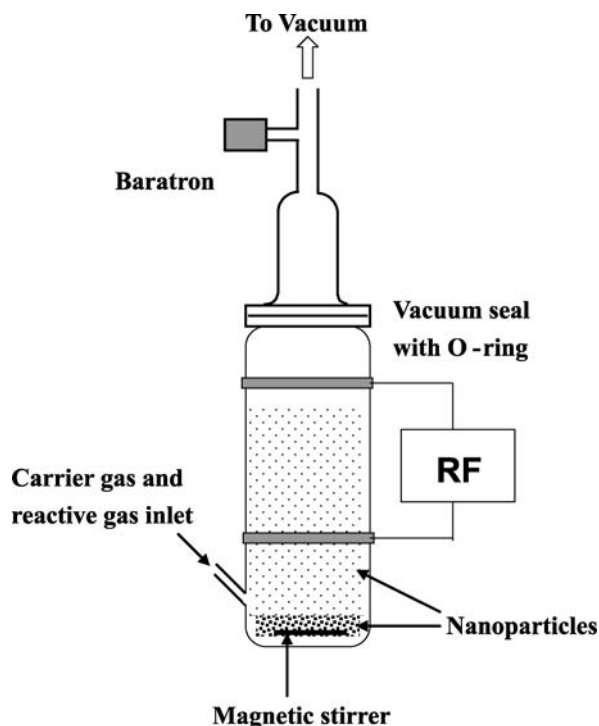
3. plasma treatment is relatively inexpensive, easily operated, dry and rapid;

4. plasma treatment can tailor the surface with desired functionalities and properties through simply adjusting the plasma chemistry.

These advantageous features have been demonstrated in the continuous operation of plasma deposition and plasma etching processes in semiconductor manufacturing industry.

Recently, low-temperature plasma treatment has been explored for surface modification of various nanosized materials [7–13]. Surface functionalization of carbon nanotubes (CNTs) using plasma treatment was recently attempted by several research groups. Oxygen plasmas generated at both radiofrequency (RF) and microwave (MW) were used to treat CNTs [7, 8]. It was found that acidic groups were implemented to the CNT surfaces with improved surface wettability. Several other studies [9, 10, 13] reported the surface modification of both CNTs and nanoparticles using plasma treatment. With a simple fluidized-bed plasma apparatus, plasma coatings of 2 nm from pyrrole and 15 nm from acrylic acid were able to be uniformly deposited on 10–150 nm Al_2O_3 and 50 nm ZnO nanoparticle surfaces respectively [11]. It was also reported that uniform plasma coating with 1–3 nm thickness was achieved on carbon nanotubes with inner diameter of 20 nm [12].

Plasma treatment of nanoparticles for nanofluids can be obtained by plasma treatment using low-temperature plasmas. The diamond nanoparticle surfaces, for example, can be modified by the low-temperature plasmas of methane

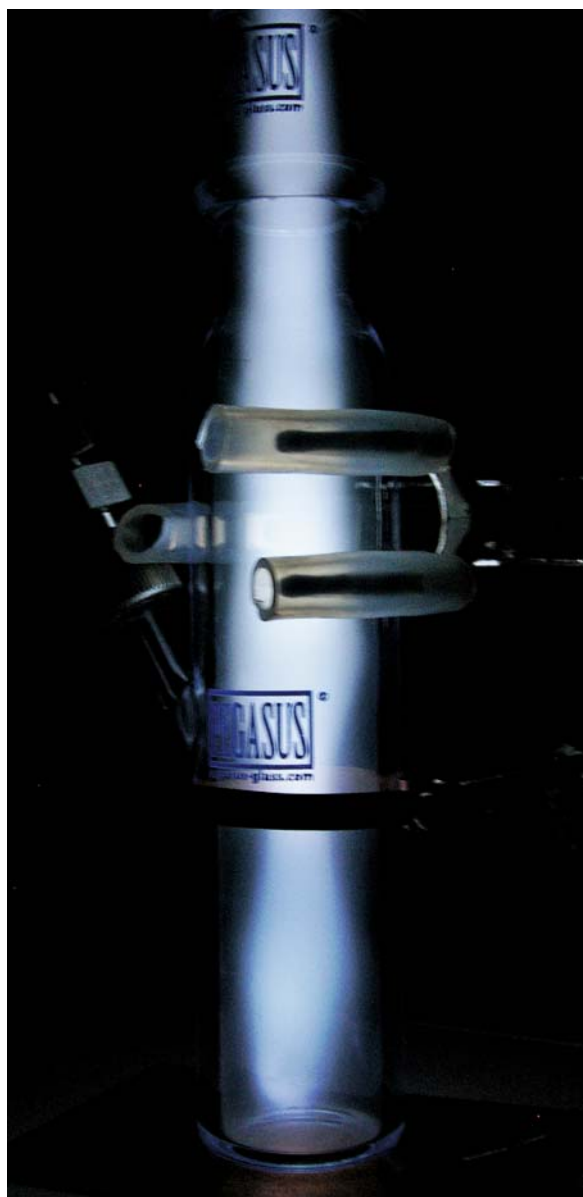


Plasma Treatment of Nanoparticles for Nanofluids, Figure 2 Schematic diagram of the magnetically assisted fluidized bed plasma reactor system

and oxygen mixtures, which can improve their dispersion characteristics including dispersibility and suspension stability in water for nanofluid. Plasma treatment of diamond nanoparticles with selected plasma chemistry imparted various functional groups on the particle surfaces and tailored the surface energy to the required levels. Plasma coated nanoparticles with desired surface functionalities, e.g., polar and/or charged groups, can strongly interact with water molecules and thus better disperse into water to form stable suspension through mechanism of *electrostatic repulsion* or *steric repulsion*, or combination of both.

Nanofluids

Nanofluids, a concept developed by Choi at Argonne National Laboratory [14], is a new class of heat transfer fluids that are composed of suspended nano-sized metallic or nonmetallic solid particles in liquid media. Nanofluids when properly prepared display significantly enhanced heat transfer capacities relative to those of traditional heat transfer fluids. From the practical application viewpoint, the most critical concerns are the agglomeration and poor suspension stability of the metallic or non-metallic nanoparticles in a given fluid medium.



Plasma Treatment of Nanoparticles for Nanofluids, Figure 3 Pictorial view of the plasma reactor system

For most fine powders including nanoparticles, strong interparticle attractions tend to stick the particles into agglomerates, which make the independent dispersion of the powders difficult and, more importantly, deteriorates the final property including the thermal conductivity, viscosity, suspension stability, reliability and reproducibility. There is a great difference of density between metallic particles and fluids. As a result, the stability of the suspension of metallic particles in liquid media is usually poor, especially with the relatively high-density and large-size parti-

cles. The agglomerates of particles have a great tendency to precipitate and thus further worsen the suspension stability. The particle precipitation will not only decrease the heat transfer capacity of the nanofluids but also adversely limits the application of nanofluids in practical devices. The consequence of the agglomerate precipitation may cause clog issues of the nanofluids-based devices in which the fluids need to flow through mini and/or microchannels.

One approach to achieve the homogeneous dispersion and stable suspension of fine particles is using various dispersing agents (or surfactants) to stabilize the particle suspension via a steric and/or electrostatic repulsion [15]. It is well known that, however, the synthesis of stable dispersion of nanosized inorganic materials in liquid requires amounts of stabilizers several orders of magnitude higher than the available space on the surface of nanoparticles can accommodate [16]. The presence of excess amount of stabilizers could significantly alter the fluids characteristics, such as viscosity and thermal conductivity, and could be a big concern in the practical application of the fluids. In contrast, plasma treatment of nanoparticles provides a new promising approach in producing well-dispersed and stably suspended heat transfer nanofluids via fabrication of surface-functionalized nanoparticles without using dispersing agents (or surfactants).

Basic Methodology

Plasma Treatment

A magnetically-assisted fluidized bed plasma reactor, which is schematically shown in Fig. 2 and pictorially shown in Fig. 3, can be used to treat diamond nanoparticles (average size of 50 nm). The reactor shown in Fig. 3 is a long cylinder with an inside diameter of 2.54 cm and a length of 15.24 cm. The plasma reactor is pumped down to a base pressure less than 3 mtorr. Then, methane and oxygen mixture at various ratios is used as the plasma fuel and introduced into the reactor to a preset pressure for plasma treatment. Methane and oxygen gases both have a purity of 99.997%. The flow rates of the methane and oxygen were controlled and adjusted using needle valves. The gas pressure in the reactor was monitored using a Baratron (MKS instruments, Inc.). A radio-frequency plasma source (RFX-600, Advanced Energy Industries, Inc.) was used to ignite the plasmas for plasma treatment of the nanoparticles.

Surface Energy and Polar Interactions of Nanoparticles

The surface energy of diamond nanoparticles can be measured and calculated through Lifshitz–van der Waals/acid-

base (LW-AB) approach by surface contact angle measurements using three different liquids with known surface tensions, including HPLC grade water, acetone, and hexane. The surface contact angles of the untreated and plasma treated diamond nanoparticles were determined using sessile drop contact angle measurements with a VCA 2500XE system (Advanced Surface Technologies, Inc., Billerica, MD). Before each measurement, the diamond nanoparticles were sticking to a glass slide with a piece of double-layer tape, and then pressed and smoothened using another glass slide under ~ 1.0 kg weight. For surface contact angle measurement, a sessile droplet of $0.4 \mu\text{l}$ liquid was placed on the smoothened nanoparticle surfaces to obtain an optical image of the liquid droplet. The computer user manually traced the arc that was created by the droplet and a computer program was used to calculate the contact angles of liquid on the nanoparticle surfaces. Six measurements for each sample were performed and the average number was reported in this study. The experimental error for the static contact angle measurement was within ± 2.0 degree from the average value.

Plasma treatment of diamond nanoparticles can functionalize the nanoparticles' surfaces with polar groups that have improved interactions with water molecules and thus strong affinity to water medium. Lifshitz–van der Waals/acid-base (LW-AB) approach can be used to quantify the contributions of these plasma-induced polar groups to surface energy increase of the plasma treated nanoparticles. As described by Van Oss et al. [17], the total surface energy of nanoparticle, which combined both Lifshitz–van der Waals and Lewis acid-base (LW-AB) interactions, can be expressed by following expression:

$$\gamma = \gamma^{\text{LW}} + \gamma^{\text{AB}} \quad (1)$$

In other words, the total potential energy of nanoparticle-liquid interactions is the sum of a Lifshitz–van der Waals interaction (γ^{LW}), which means the surface free energy (dipole–dipole and dipole-induced dipole interaction), and Lewis acid-base interaction (γ^{AB}). The Lewis acid-base interaction (γ^{AB}) for the component i is simply calculated by the geometric mean of electron acceptor (γ^+) and donor (γ^-):

$$\gamma^{\text{AB}} = 2\sqrt{\gamma^+\gamma^-} \quad (2)$$

In order to calculate the surface energy of nanoparticles from our experimentally measured surface contact angle

data, Young–Dupre equation can be used:

$$(1 + \cos \theta) \gamma_L = 2 \left(\sqrt{\gamma_S^{LW} \gamma_L^{LW}} + \sqrt{\gamma_S^+ \gamma_L^-} + \sqrt{\gamma_S^- \gamma_L^+} \right) \quad (3)$$

By measuring the contact angle (θ) of the diamond nanoparticles with three different liquids including at least two polar solvent with known γ_L^{LW} , γ_L^+ , and γ_L^- values, the γ_S^{LW} , γ_S^+ and γ_S^- of the untreated and plasma treated diamond nanoparticles were determined.

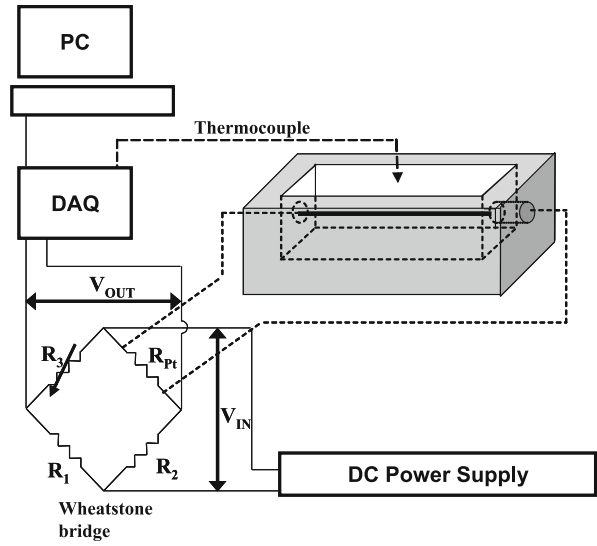
Fourier Transform Infrared Spectra (Ftir)

FTIR measurements can be used to identify and semi-quantitatively verify the surface functionalities/polar groups on diamond nanoparticles' surface induced by plasma treatment. Nicolet 6700 FT-IR Spectrometer (Thermo Scientific Inc.) was used to take Fourier Transform Infrared Spectra (FTIR) of the untreated and plasma treated diamond nanoparticles. Well-ground mixtures of nanoparticle and KBr powders (IR grade) were kept in an oven and baked at 200 °C for 48 h to get rid of moisture before performing the FTIR measurements.

Thermal Conductivity Measurement

The transient hot-wire method can be used to measure the thermal conductivity of nanofluids in this study [18]. As schematically shown in Fig. 4, the transient hot-wire measurement system contains a DC power supply, the wheatstone bridge, a data acquisition system, a nanofluid container, and a computer system to analyze the measurement data. The wheatstone bridge was constructed to determine the change of resistance. The input voltage of 5 V was supplied by an Agilent E3645A DC power supply and the output voltage was measured by a National Instruments SCXI-1000 DAQ. This data acquisition system was also used to measure the temperature of the liquid during the measurement. Each dataset was collected at a rate of 500 points per second. The relation between the output voltage and the resistance of the platinum wire is given by following Eq. (4):

$$R_{Pt} = R_2 \frac{\left(\frac{V_{OUT}}{V_{IN}} + \frac{R_3}{R_1 + R_3} \right)}{\left(1 - \frac{V_{OUT}}{V_{IN}} - \frac{R_3}{R_1 + R_3} \right)} \quad (4)$$



Plasma Treatment of Nanoparticles for Nanofluids, Figure 4 Schematics of the transient hot wire system for thermal conductivity measurement

Using the linear relation between resistance and temperature, Eq. (5) is derived.

$$T = \frac{1}{\alpha} \left(\frac{R_{Pt,T}}{R_{Pt,T=0}} - 1 \right) \quad (5)$$

where α is

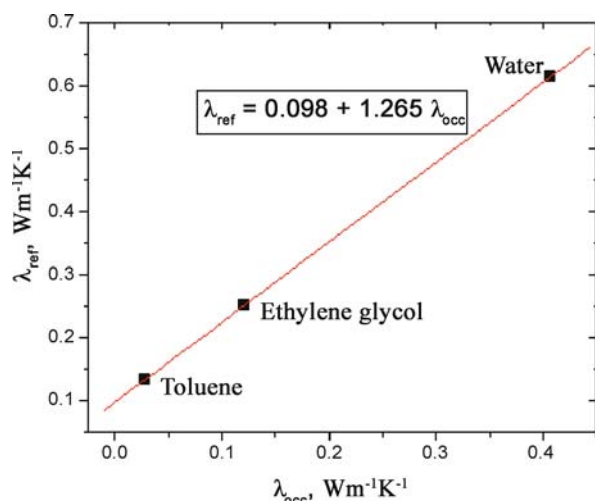
$$\alpha = \frac{\partial \frac{R_{Pt}}{R_{Pt,T=0}}}{\partial T} \quad (6)$$

Finally, thermal conductivity (k) can be found from the following liner heat source model:

$$k = \frac{q}{4\pi(T_2 - T_1)} \ln \left(\frac{t_2}{t_1} \right) \quad (7)$$

In order to eliminate the effect of convection of the nanofluid, time interval in Eq. (7) must be within the region where thermal conduction is dominant.

The rectangular nanofluid container was designed so that the wire was at least 10 cm long and would contain about 100 ml of fluid. The fluid container was constructed out of 1.49 cm thick Plexiglas. A 1.27 cm diameter hole was drilled through the container wall in each end. A 0.0254 mm diameter pure platinum wire was attached to 12 gage copper wire. The copper wire was crimped onto each end of the platinum wire at a spacing of 10 cm. The connections between platinum and copper wires were soldered to ensure a good contact and the two copper wires were inserted into the fluid container through two rubber stoppers. The copper portion that would be in contact with



Plasma Treatment of Nanoparticles for Nanofluids, Figure 5 Calibration curve for thermal conductivity measurements of nanofluids

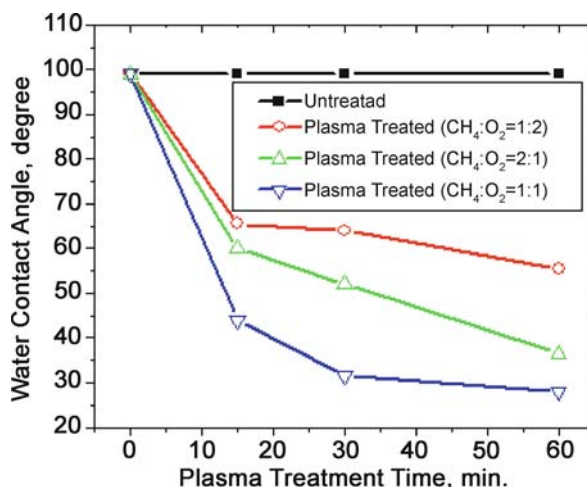
the test fluid was covered with Omegabond “100” epoxy to eliminate galvanic corrosion. A T type thermocouple was also placed in the fluid so that the temperature of the fluid could be monitored. The temperature measurements using this thermocouple gave reading with an error of $\pm 1^\circ\text{C}$. To relate the resistance of the wire to temperature, the resistance of the platinum wire over a range of temperatures should be known. Therefore, HPLC grade water of varying temperatures was placed into the container and the resistances of the platinum wire and the temperature of the water were measured.

The thermal conductivity measurement of nanofluids was conducted with the fluid without any vibration. Each test was conducted at ambient condition and the data were collected for 131 s. After repeated trials it was found that the variable resistor should be balanced slightly above the known resistance of the platinum wire. Once the bridge is properly balanced the variation of voltage with time was measured and recorded. The measured thermal conductivity data was calibrated using three kinds of fluids, which was HPLC grade water, ethylene glycol and toluene. Figure 5 shows the calibration curve obtained by recording the thermal conductivity of those liquids measured. The calibration curve gave a linear calibration equation of $\lambda_{\text{ref}} = a + b\lambda_{\text{acc}}$, with a and b were 0.098 and 1.265. All the measured thermal conductivity data of nanofluids was corrected using this calibration equation.

Key Research Findings

Surface Energy of the Plasma Treated Nanoparticles

Figure 6 shows that the water surface contact angle change of plasma treated diamond nanoparticles with plasma



Plasma Treatment of Nanoparticles for Nanofluids, Figure 6 Contact angle change of diamond nanoparticles with plasma treatment time. Plasma treatment conditions were: 60 mtorr pressure, 4 W RF power

treatment time under different plasma conditions. It can be seen that the untreated diamond nanoparticles had a water surface contact angle of 99° . In contrast, plasma treatment significantly reduced the water surface contact angle to less than 30° , which indicated very hydrophilic surfaces with increased surface energy. Using Lifshitz–van der Waals/acid-base (LW-AB) approach, Table 1 summarized the surface energy data analysis results contributed from Lifshitz–van der Waals interaction (γ^{LW}) and Lewis acid-base interaction (γ^{AB}). Evidenced from the data shown in Table 1, the significant surface energy increase of the plasma treated diamond nanoparticles resulted from the contributions from enhanced Lewis acid-base interaction (γ^{AB}), which includes γ_S^+ and γ_S^- , the surface energies contributed from electron acceptors and the electron donors, respectively. This indicates that increased amount of polar functionalities on plasma treated diamond nanoparticles.

Fourier Transform Infrared Spectra

FTIR measurements were used to determine the surface functionalities induced by plasma treatment. Figure 7 presents the FT-IR spectra of untreated, plasma-treated diamond nanoparticles in plasma gas mixtures of methane and oxygen with ratios of 1 : 1 and 1 : 2. The band at 1630 cm^{-1} is associated with the vibrations of $\text{C}=\text{C}$ bonds which is corresponding to the characteristic of diamond. A wide band with its maximum at 3400 cm^{-1} in the IR spectrum is due to the vibration of OH groups. The peak at 1725 cm^{-1} is attributed by the vibration of $\text{C}=\text{O}$ bonds in weak-acid carbonyl. The increased peak

Plasma Treatment of Nanoparticles for Nanofluids, Table 1 The contact angle (θ) and its surface energy of untreated and treated ($\text{CH}_4 : \text{O}_2 = 1 : 1$, 60 mtorr, 4 W RF power) diamond nanoparticles. Surface energy of each solvent is $\gamma_i^+ = 71.99$, $\gamma_i^- = 23.46$, $\gamma_i^{\text{LW}} = 17.89$. γ_S^+ and γ_S^- are the surface energies contributed from electron acceptors and the electron donors, respectively. γ^{LW} represents Lifshitz–van der Waals interaction and γ^{AB} represents Lewis acid–base interaction

Plasma gases	Plasma time	Contact angle ($\cos \theta$)			Surface energy (mN/m)				
		Water	Acetone	Hexane	γ_S^+	γ_S^-	γ_S^{LW}	γ_S^{AB}	γ_S^{Total}
Untreated		−0.156	0.044	0.019	12.82	6.39	4.64	18.10	22.74
$\text{CH}_4 : \text{O}_2 = 1 : 1$	15 min	0.729	0.820	0.175	53.80	19.43	6.17	64.66	70.83
$\text{CH}_4 : \text{O}_2 = 1 : 1$	30 min	0.853	0.886	0.559	61.79	20.86	10.87	71.80	82.67
$\text{CH}_4 : \text{O}_2 = 1 : 1$	60 min	0.884	0.927	0.643	63.88	21.78	12.07	74.60	86.67

Plasma Treatment of Nanoparticles for Nanofluids, Table 2 FTIR peak ratios for untreated and plasma treated diamond nanoparticles. Plasma treatment conditions were: 60 mtorr pressure, 4 W RF power, 60 min treatment time

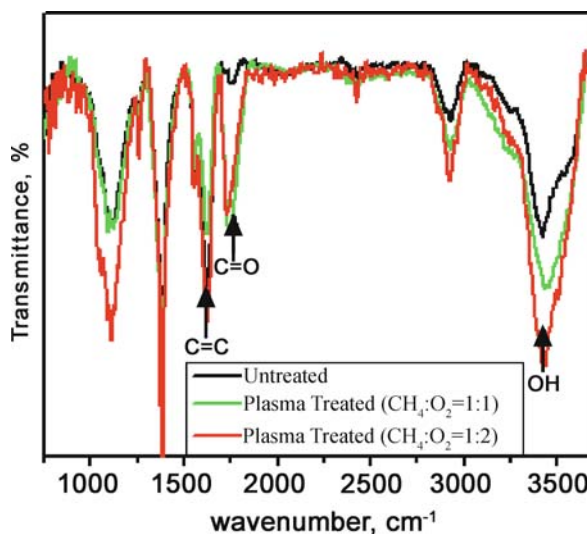
FTIR peaks	C = O/C = C	OH/C = C
(a) Untreated	0.07	2.37
(b) Plasma treated ($\text{CH}_4 : \text{O}_2 = 1 : 1$)	0.56	3.59
(c) Plasma treated ($\text{CH}_4 : \text{O}_2 = 1 : 2$)	0.82	4.01

intensity at 1725 and 3400 cm^{-1} indicates the incorporation of C = O and –OH polar functionalities on plasma treated nanoparticles' surfaces. Table 2 summarizes the peak area comparison of C = O and –OH relative to C = C peaks to semi-quantitatively determine the polar groups on plasma treated nanoparticle surfaces, because the amount of C = C group should be constant before and after the plasma treatment. From the Table 2, it is shown that, in comparison with the untreated control, the ratios of C = O/C = C and –OH/C = C groups for plasma treated diamond nanoparticles were significantly increased.

Thermal Conductivities of the Resulting Nanofluids

Effect of Oxygen/Methane (O_2/CH_4) Ratio in Plasma Gas Mixture

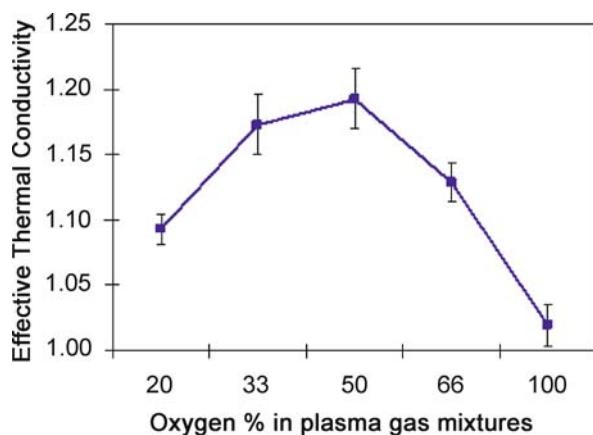
The untreated and plasma treated diamond nanoparticles were dispersed into HPLC grade water with the aid of mechanical agitation using a magnetic stirrer. We have reported that plasma treated diamond nanoparticles showed enhanced dispersibility and improved suspension stability in water as compared with the untreated controls [19]. After settled for one hour, the nanofluids thus prepared were characterized using the transient hot wire system shown in Fig. 4. Figure 8 shows the thermal conductivities of the nanofluids prepared using plasma treated nanoparticles with varying oxygen addition in plasma gas



Plasma Treatment of Nanoparticles for Nanofluids, Figure 7 FTIR spectra of untreated and plasma treated diamond nanoparticles. Plasma treatment conditions were: 60 mtorr pressure, 4 W RF power, 60 min treatment time

mixtures of methane and oxygen. When O_2/CH_4 ratio increased from 1 : 4 to 1 : 1 in the plasma gas mixture, the plasma treated nanoparticles gave increasing thermal conductivities of the resulting nanofluids with more oxygen addition. Further increasing the oxygen amount above 50% ($\text{O}_2/\text{CH}_4 = 1 : 1$), however, the thermal conductivity of the resulting nanofluids with the plasma treated nanoparticles started to decrease.

The thermal conductivity dependence of the resulting nanofluids on O_2/CH_4 ratios in plasma gas mixture shown in Fig. 8 is consistent with the water surface contact angle results shown in Fig. 6. 1 : 1 ratio of O_2/CH_4 gave the lowest water contact angles for the plasma treated diamond nanoparticles as compared with lower or higher O_2/CH_4 ratios. The lower water surface contact angle suggested higher amount of polar functionalities were introduced onto diamond nanoparticles' surfaces through

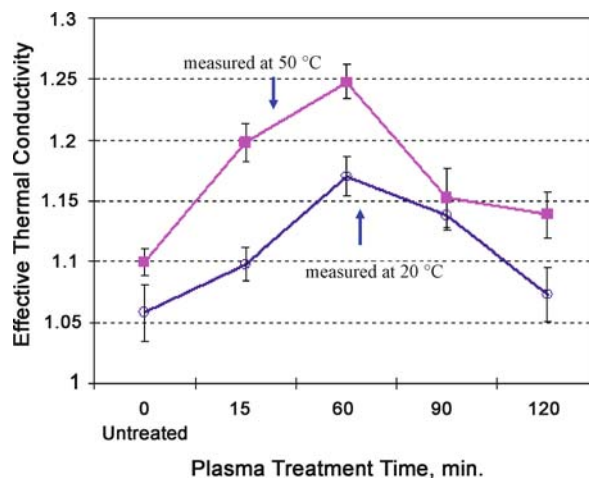


Plasma Treatment of Nanoparticles for Nanofluids, Figure 8 The effects of O_2/CH_4 ratio used for plasma treatment on thermal conductivity of the resulting nanofluids with addition of 0.15 Vol.% plasma treated diamond nanoparticles into water. The thermal conductivities were measured after 1.0 hour settling time. Plasma treatment conditions were: 60 mtorr pressure, 4 W RF power, 60 min treatment time

plasma treatment. As a result, a better dispersion of the nanoparticles in water medium was achieved and so was the thermal conductivity of the resulting nanofluids. It is believed that plasma treatment using higher amount of oxygen addition over 1 : 1 ratio of O_2/CH_4 could very likely result in a pronounced surface etching effect and etch the surface polar surface functionalities away.

Effect of Plasma Treatment Time

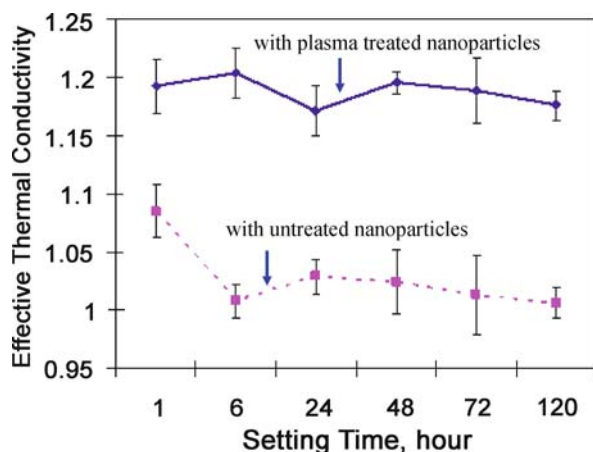
The water surface contact angle measurement and thermal conductivity measurement showed that plasma treatment of diamond nanoparticles with 1 : 1 ratio of O_2/CH_4 gave rise to the best surface wettability and the highest thermal conductivity of the resulting nanofluids. Using plasma gas mixture with a fixed O_2/CH_4 ratio of 1 : 1, therefore, effects of plasma treatment time on thermal conductivities of the resulting nanofluids were examined. In Fig. 9, the effective thermal conductivities of the resulting nanofluids at both 20 °C and 50 °C were plotted against the plasma treatment time for diamond nanoparticle treatment using plasma gas mixture of $O_2/CH_4 = 1 : 1$. With plasma treatment time varied from 15 to 120 min, plasma treatment of diamond nanoparticles for 60 min gave a nanofluids with the maximum thermal conductivity at both temperatures. The results in Fig. 9 are also consistent with the water surface contact angle measurement results shown in Fig. 6. The thermal conductivity decrease using plasma treatment time longer than 60 min could be due to the over treatment effects often observed in plasma treatment processes.



Plasma Treatment of Nanoparticles for Nanofluids, Figure 9 Plasma treatment time effects on thermal conductivity of the resulting nanofluids with addition of 0.15 Vol.% plasma treated diamond nanoparticles into water. The thermal conductivities were measured after 1.0 h settling time. Plasma treatment conditions were: 60 mtorr pressure, 4 W RF power, 1 : 1 ratio of $CH_4 : O_2$

Stability of the Resulting Nanofluids

We have recently reported that plasma treatment of diamond nanoparticles significantly improved their suspension stability in water [19]. After 45 day sediment test, the nanofluids using plasma treated diamond nanoparticles didn't give any phase separation, while the nanofluids using untreated controls showed clear phase separation of water. In this study, the settling time dependence of the nanofluid thermal conductivity was monitored and the data are shown in Fig. 10. It can be seen that the nanofluids using untreated nanoparticles showed a decreasing trend with settling time. After 120 h settling, the effective thermal conductivity reached to ~ 1.0 , i.e., the same as pure water. In contrast, the nanofluids prepared using plasma treated diamond nanoparticles showed very little change in the effective thermal conductivity with settling time. From Fig. 10, it was noted that, with the same amount nanoparticle addition of 0.15 Vol.%, nanofluids using plasma treated nanoparticles gave much higher thermal conductivity than that using untreated nanoparticles. It was usually observed that the thermal conductivity of nanofluids is proportional to the volume fraction of nanoparticles added in base liquid media [20]. Results shown in this study suggest that plasma treatment of diamond nanoparticles could well disperse the nanoparticles in water and thus increase the thermal conductivity of the resulting nanofluids. It is suspected that the enhanced interfacial interactions between nanoparticles – base water fluids due to introduction of polar surface functionalities by plasma treat-



Plasma Treatment of Nanoparticles for Nanofluids, Figure 10 Thermal conductivity change of the resulting nanofluids (with 0.15 Vol.% diamond nanoparticle addition in water) with settling time. Plasma treatment conditions were: 60 mtorr pressure, 4 W RF power, 60 min treatment time, 1 : 1 ratio of CH_4 : O_2

ment could be another important factor contributed to the thermal conductivity increase of the resulting nanofluids.

Future Directions for Research

Plasma treatment provides a new approach to functionalize nanoparticle surface for heat-transfer nanofluid preparation without using stabilizing agents. More experimental investigation is very necessary to optimize the plasma treatment process in term of plasma chemistry, plasma operating conditions, and proper plasma apparatus design. As pointed out in this paper, besides nanoparticle concentration, the enhanced interfacial interactions between nanoparticles – base fluids could be another important factor that contributed to the thermal conductivity increase observed in heat-transfer nanofluids. A detailed characterization of the nanofluids with the aid of theoretical simulations will help us to understand the fundamental mechanisms for the observed heat-transfer enhancement.

Cross References

- ▶ Plasma Surface Modification
- ▶ Surface Tension Measurements
- ▶ Surface Conductivity Measurement

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Plastic

- Polymer

Plasticity

- Viscosity

Platinum Resistance Thermometer

- Resistance Temperature Detectors

Plug Flow

- Taylor Flow in Microchannels

Pneumatic Valves

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Synonyms

Elastomeric valve

Definition

Mechanical structure designed to control the flow of fluids in one channel through changes in pressure in a second channel from which it is separated by a membrane or equivalent structure.

Overview

Pneumatic valves rely on the deformation of a membrane, or other thin structure, by pressure, with the end result of altering the flow of a fluid in a channel. The size of the pneumatic valves is usually comparable to the size of the channels they control and consequently these valves are easy to integrate in the overall design of the microfluidic chips. Although several designs have been proposed over time, using different materials, the most successful valves in terms of applications have been the valves using elastomeric materials. Elastomeric materials are easily deformable, and are already widely used for microfluidic

applications, making them ideal for developing pneumatic valves.

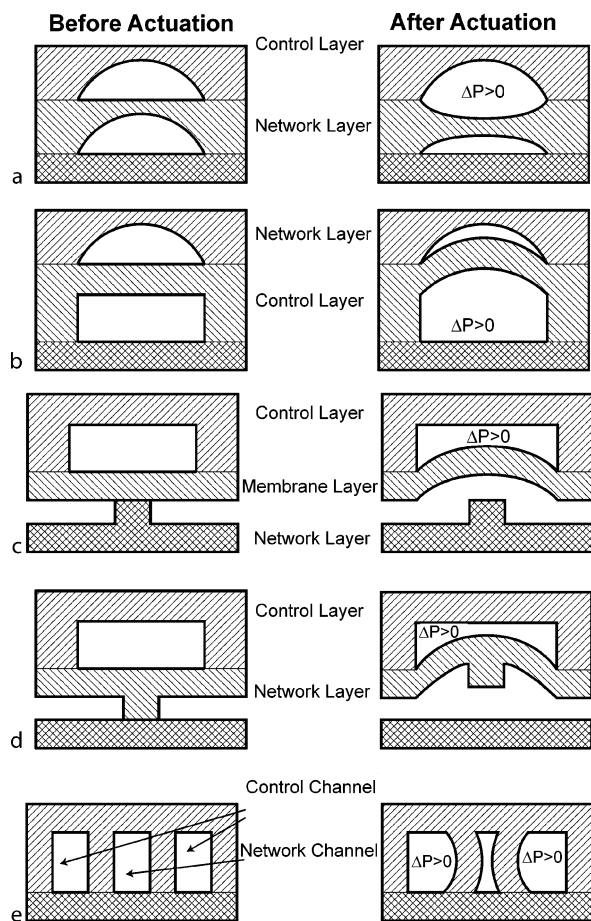
In the simplest design, the deformation of a circular membrane under uniform pressure on one side can be described by the following equation:

$$\Delta x = \Delta P \frac{3r^4}{16t^3} \cdot \frac{(1 - \nu^2)}{E}$$

where Δx is the deformation of the membrane at its center, ΔP is the pressure difference on the two sides of the membrane, E and ν are characteristics of the membrane material (the Young modulus and the Poisson's ratio, respectively), and r and t are dimensional characteristics of the membrane (the radius and the thickness of the circular membrane, respectively). Although for more complicated shapes of the membrane the dependence of deformation to actuation pressure has to be calculated numerically using specialized software, the formula for the circular membrane already suggests a very important impact from the valve dimensions on valve functionality.

Basic Methodology

Pneumatic valves are fabricated using standard microfabrication and assembly technologies. The membrane of the valve can be one uniform film of elastomeric material, or can have microchannels already imprinted on one side (see ► [microfabrication techniques](#)). The actuation chamber is usually in a separate layer, either elastomer or a rigid material. The layer including the controlled microchannel, the membrane, and the layer containing the actuation chamber are then bonded together using standard techniques (e.g. oxygen plasma for elastomer and glass layers see ► [microfluidic assembly](#) and microfabrication techniques). The decision for the precise design and integration of pneumatic devices into the manufacturing of microfluidic devices is usually based on several criteria, each of which may have different weight in different applications. The first criterion is usually the compatibility of the technologies for the valve with the rest of the manufacturing. Some valve require the use of positive resist and overflow at higher temperature, other require glass etching, and some can be manufactured using only negative epoxy resist (SU8). Most of the pneumatic valves require the precise aligning and bonding of two layers of elastomeric material. The second criterion is the nature of the fluid to be handled. While most of the elastomeric materials are compatible with water solutions, only few materials could handle solvents and more aggressive chemicals. The third criterion is the number and size of the valves on one device. Smaller valve size is possible when controlling single rounded channels, while larger valves could incorpo-



Pneumatic Valves, Figure 1 Schematics of pneumatic valves. (a) Pneumatic valve using the compression of a rounded channel in the network layer. (b) A bulging membrane can be used to control flow in a rounded channel. (c) Negative pressure can deform a membrane and open communication in a network layer. (d) A microstructured membrane could be used to control flow in a complex network layer. (e) In a single layer valve the control channels are placed on the side of the network layer

rate more features or control multiple channels at the same time. Finally, valves that are normally closed or normally open have been described, and the choice depends on the functionality and actuation pressure available for operating the valves.

Key Research Findings

Several approaches have been described for the practical implementation of the pneumatic valves in microfluidic systems. In general, the control and network channels (Fig. 1) are placed in different layers and the membrane structure performing the actuation is in between the layers. Both higher pressure (Fig. 1a,b,e) and vacuum (Fig. 1c,d) have been used to control the fluid in the network layer. In

the case of vacuum-actuated valves, the membrane could seal channels in the network layer in the absence of the actuation pressure, only by elasticity of the membrane pressing against the valve seat (Fig. 1c,d).

While smaller aspect ratios of the network layer usually lead to smaller dead volumes for the valves (Fig. 1), various designs that allow the control of higher aspect ratios has been described, especially for applications that require the transport of larger cells or particles (Fig. 1b-e). Pneumatic valves with rounded (Fig. 1a-c) as well as rectangular (Fig. 1d-e) channels in the network layer have been implemented. The control layer could be above or below the network layer (Fig. 1a-d), and at least in one implementation the control and network channels were implemented at the same level (Fig. 1e). Hierarchical arrangements of pneumatic valves in microfluidic devices, where some microscale valves are used to control the operation of other microscale valves, have been demonstrated as a way to reduce the number of control lines and increase the sophistication of actuation patterns.

For most of the designs, the membrane is relatively flat and feature-less and in some implementations it can be different material from the control and network layer materials (Fig. 1c). However, the addition of microstructures to the valve membrane (Fig. 1d) has been demonstrated to result in higher degree of complexity of the functionality that can be achieved using pneumatic actuation.

Future Directions for Research

There are two main challenges for the development of applications using pneumatic valves. The first one has to do with the choice of materials for the valve. Although elastomeric materials are very good for prototyping, scaling up and reducing the fabrication costs would require the use of polymeric materials. Either these new materials would have to have comparable mechanical properties with the current elastomers, or different designs would have to be imagined. The second challenge is related to the control system and pressure generation that is usually off chip. Integrating these on the chip or at least standardizing and miniaturizing these would be needed for compact, reliable miniaturized devices.

Cross References

- [Capillary Force Valves](#)
- [Control of Microfluidics](#)
- [Digital Microfluidics](#)
- [Electrochemical Valves](#)
- [Electromagnetic Valves](#)
- [Electrostatic Valves](#)
- [Fabrication of 3D Microfluidic Structures](#)

- Volume and Finite Difference Method for Modeling and Simulation
- Fluid Metering
- Lab-on-a-chip (General Philosophy)
- Lab-on-chip Devices for Particle and Cell Separation
- Membrane Actuation for Micropumps
- Microactuators
- Microfluidic Assembly
- Microfluidic Circuits
- Peristaltic pumps
- Piezoelectric Valves
- Thermopneumatic Valves
- Water Management in Micro DMFCs

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Poiseuille Coefficient

Definition

Poiseuille coefficient determines a gas flow induced by a pressure gradient.

Cross References

- Gas Flow in Nanochannels
- Micro- and Nanoscale Gas Dynamics

Poiseuille Single Phase Gas Flows

- Pressure-Driven Single Phase Gas Flows

Polyclonal

Definition

Polyclonal antibodies are antibodies that are derived from several different t B-cell lines. The selections of antibody molecules bind to different parts of the antigen, with different binding affinities. Within a polyclonal sample, some of the antibodies will be specific for the antigen with which the animal was immunised. The rest of the antibodies have been obtained from meetings with other foreign antigens that the animal has been exposed to through its life time.

Poly(dimethylsiloxane) (PDMS)

Definition

PDMS is popular polymer in the microfluidics community; often used for rapid fabrication of microchannel structures. PDMS is optically-transparent and is generally considered to be inert, non-toxic and non-flammable. PDMS is formed by mixing a *base* with a *curing agent* (typically mixed in a 10 : 1 ratio, but this can vary depending on the desired stiffness of the resultant polymer) and curing in a vacuum-oven for several hours. Cured PDMS will form a reversible-bond with various surfaces (glass microscope slides, for example). Irreversible sealing can be achieved by exposing the PDMS to oxygen plasma prior to bonding.

Cross References

- Temperature Effects on the Zeta Potential

Polydispersity

Synonyms

Polydispersity index; Mw/Mn

Definition

The distribution of molecular weights in a given polymer sample.

Cross References

- Polymer Synthesis within Microfluidic Reactor

Polydispersity Index

- Polydispersity

Polyelectrolyte

Definition

A species (for example, a colloidal particle or a flexible chain) that is partially ionizable when placed in solution. The polyelectrolyte dissociates into a macroion and counterions.

Cross References

- Electrokinetic Flow and Ion Transport in Nanochannels
- Electrophoresis

Polyelectrolyte Patterning Technique

- Surface-Charge Patterning Techniques

Polymer

Synonyms

Plastic; Organic material

Definition

A polymer is a material composed of large macromolecules. These macromolecules are formed by chains of hundreds or thousands of connected (polymerized) monomer molecules. The three main classes of polymers are thermoplastics, elastomers and thermosets. They differ in the degree of cross-linking of their macromolecules – from no cross-linking (thermoplastics) to moderate cross-linking (elastomers, rubbers) to high cross-linking (thermosets). Thermoplastics commonly used in microfluidics include materials like polymethylmethacrylate (PMMA), polycarbonate (PC), polyethylene terephthalate (PET) or cyclic olefin copolymers (COC). Silicones (like polydimethylsiloxane, PDMS) are a typical class of elastomers. Thermosets include photoresist materials like SU-8 and others.

Cross References

- Glass-Polymer-Bonding

Polymer Casting

- Micromolding

Polymer Electrolyte Membrane Fuel Cells

- Microstructured Hydrogen Fuel Cells

Polymerization Reaction in Microreactor

- Polymer Synthesis within Microfluidic Reactor

Polymer Monoliths

- Stationary Phases in Microchannels

Polymer Replication

- Micromolding (Injection and Compression Molding)

Polymer Solutions

- Non-Newtonian Fluids in Microfluidics

Polymer Synthesis within Microfluidic Reactor

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Synonyms

Synthesis of polymer molecules using microfluidics; Microfluidic reactor for polymer synthesis; Polymerization reaction in microreactor

Definition

Polymer synthesis requires strict control of reaction conditions such as temperature, reaction time, and mixing of reagents. Microfluidic reaction systems enable efficient mixing and rapid heat exchange. Moreover, a laminar stream in a microfluidic channel enables control of the resulting shape of a molecule. These features of microfluidic reaction systems are suitable for polymer synthesis.

Overview

Reactions involved in chemical polymerization should be highly controlled because polymer properties such as ►average molecular weight, molecular weight distribution, and size and shape of particles strongly affect the functions of polymer materials. When using polymers of less well known functions, the molecular weight or size and shape should be carefully selected, and the molecular weight or size distribution should be controlled to be as narrow as possible. Such requirements could be satisfied by utilizing microfluidic systems. Microfluidic reaction systems have several distinctive features. One of the unique features is the ability to control molecular weight distribution and simple control of the molecular weight of linear polymers. In addition, microfluidic systems enable efficient emulsification with controlled size and shape. Present-day investigations on polymer materials are directed towards the development of novel functional substances by using high-throughput and combinatorial synthesis methods. The microdevices developed for this purpose should be simple, controllable, and capable of continuous combinatorial synthesis process with high-throughput screening systems for polymer material libraries. Moreover, there is a requirement for accelerated multi-parameter optimization of reaction conditions. In this article, efforts towards the development of microreactors for linear polymer synthesis and preparation of complicated polymer particles are summarized.

Synthesis of Linear Polymers in Microfluidic Reactor

Microreaction systems involve microreaction apparatuses that enable high controllability of chemical reactions. Such controllability results from efficient heat transfer, mass transport, and/or a larger surface/interface area. Recent studies have shown the potential benefits of using microfluidic reactors for various chemical reactions. Reactions using micromixing devices give better results than batchwise reactions. This is because a micromixer enables rapid mixing and therefore yields excellent controllability of rapid reactions.

Application of microreaction technology for linear polymer syntheses was initially performed by several researchers, but the field is still in its infancy. Most of the work utilized a combination of micromixing and reaction parts.

The most popular polymerization method is ►radical polymerization. Wu et al. performed radical polymerization by using a chip-type microreactor [1]. The system is composed of a microstirrer mixer connected to a microchannel reaction part. The molecular mass of the polymer produced is governed by the flow rate or polymerization time. The monomer conversion agrees well with the bulk reaction kinetics reported in the literature. The reactor is convenient and inexpensive to manufacture, with a versatile design that can be reconfigured and prototyped in less than a day. The radical polymerization technique was also applied for ►block copolymerization of poly[(ethylene oxide)-*block*-(2-hydroxypropyl methacrylate)] [2]. A series of well-controlled polymerizations was carried out at different pumping rates or reaction times with a constant ratio of monomer to initiator. The stoichiometry of the reactants was also adjusted by varying relative flow rates to change the reactant concentrations. Another example of radical polymerization in a microreactor was described by Iwasaki and Yoshida, who reported a microreactor consisting of commercially available micromixers (T-shape, 500 μm internal diameter) connected with capillary tubes (500 μm internal diameter) [3]. Thermal decomposition of 2,2-azobis(isobutyronitrile) (AIBN) was examined first and subsequent polymerizations of butyl acrylate (BA), benzyl methacrylate (BMA), methyl methacrylate (MMA), vinyl benzoate (VBz), and styrene (St) were performed. The microreactor was quite effective in controlling the molecular weight distribution for highly exothermic reactions (BA, BMA, and MMA) but not so effective for less thermic polymerizations (VBz and St).

Cationic polymerization was performed in a microreaction system consisting of two micromixers connected by microcapillaries [4]. A method that used irreversible generation and accumulation of highly reactive cations in the absence of nucleophile was applied for polymerization. The molecular weight distribution was controlled by extremely fast micromixing, and the resulting polymer could be used for follow-up reactions.

The benefit of microreaction systems is not limited to cationic or radical polymerization. Amino acid polymerization using α -amino acid *N*-carboxyanhydride (NCA) yielded better molecular weight distribution when performed in a microreactor [5]. Also, the average molecular weight could be controlled by changing the flow rates.

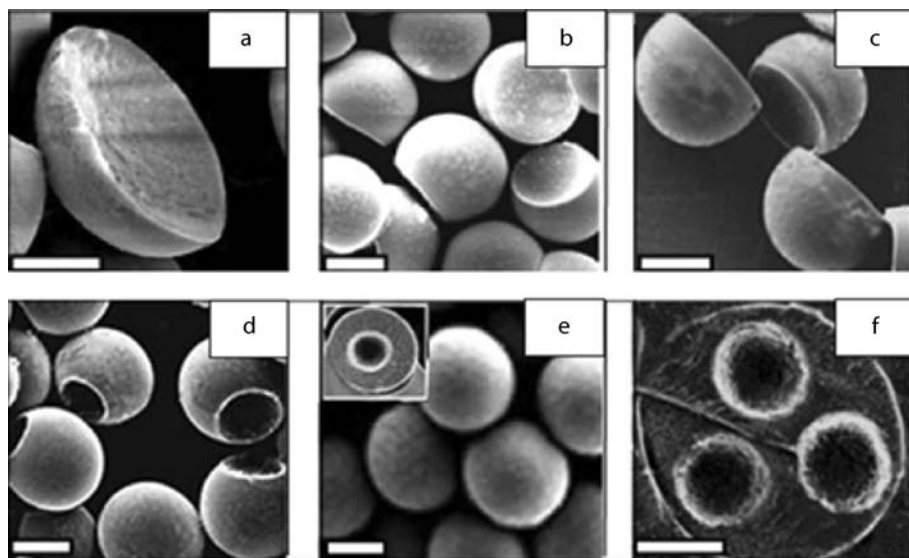
Synthesis of dendric molecules has also been performed. Chang and coworkers synthesized ethylenediamine-cored polyamidoamine (PAMAM) dendrimer using an interdigital micromixer [6]. Using this method, synthesis of generation two through generation five PAMAM dendrimer was confirmed by microchip capillary zone electrophoresis. Furthermore, the synthesized polyamide dendron was deposited onto a functionalized glass surface through the formation of amide bonds using a facile coupling procedure via a microreactor. The dendron deposition process involved two major steps: functionalization of the native glass surface using 3-aminopropyl triethoxysilane, and coupling of the synthesized polyamide dendron with the aminosilanized surface.

Compared with conventional synthesis methods, the microreactor provides better yield and selectivity, and a considerably faster synthesis rate. Nanostructured polyamide G1 dendron thin films could be deposited in minutes using an impinging flux from this microreactor. Not only have efforts been focused on the synthesis of linear or branched polymer molecules, but theoretical studies have also been performed. Serra et al. conducted numerical simulations of free radical polymerization in microfluidic devices [7]. Three microfluidic devices were modeled: two interdigital multilamination micromixers with a long and short focusing section, respectively, and a simple T-junction followed by a microtube reactor together considered as a bilamination micromixer with a large focusing section. The simulations showed that in spite of the heat released by the polymerization reaction, the thermal transfer in such microfluidic devices was high enough to ensure isothermal conditions. Moreover, for low radial [Péclet number](#), microfluidic devices with a large focusing section could achieve better control over the polymerization than a laboratory-scale reactor as the [polydispersity](#) index obtained was very close to the theoretical limiting value. Serra et al. [7] concluded that the reactive medium cannot be fully homogenized by the diffusion transport before leaving the system resulting in a high polydispersity index and a loss in the control of polymerization as the characteristic dimension of the microfluidic device increases, i. e. for high radial [Péclet number](#).

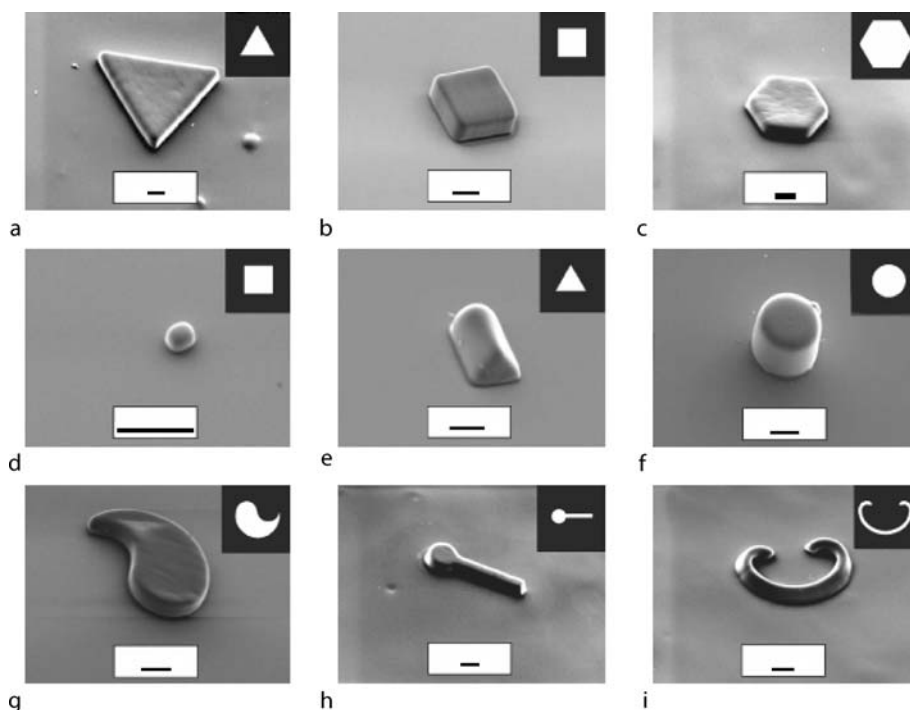
Complicated Polymer Particle Synthesis

There is an increasing demand for polymer particles by industry, particularly for painting formulations and drug delivery applications. A microfluidic reaction system provides superior controllability of fluid because the flow forms simple laminar streams. Thus, efforts have been exerted on polymer particle synthesis using microfluidics because of the unique capacity to generate microdroplets

with very narrow size distributions. Polymer particles by simple emulsification have been widely studied and well summarized in other entries. Here, we focus on polymer synthesis by complicated emulsification. Nie et al. reported a novel approach to continuous and scalable production of core-shell droplets and polymer capsules in microfluidic devices [8]. They used laminar flow of three immiscible liquids, aqueous SDS, monomer, and silicon oil, in a microfluidic system. The described method is also useful for the synthesis of polymer particles with nonspherical shapes. They used capillary instability-driven break-up of a liquid jet formed by two immiscible fluids. Precise control of emulsification of each liquid allowed production of highly monodispersed core-shell droplets with a predetermined diameter of cores and thickness of shells. They also achieved control over the number of cores per droplet and the location of cores in a droplet. The authors carried out fast throughput photopolymerization of the monomeric shells and obtained polymer particles with various shapes and morphologies, including spheres, truncated spheres, hemispheres, and single and multicore capsules. Photolithography was also applied for the synthesis of polymer particles. Dendukuri et al. reported a one-phase method using a T-junction microchannel to create solid disk- and plug-shaped particles (Fig. 1) [9]. Snap-off at the junction yielded plugs containing Norland optical adhesive (NOA) 60, a photopolymerizable resin. When an ultraviolet (UV) beam was shone on the plug in the narrow channel, a plug structure was captured, but if the plug was allowed to expand laterally and then polymerized, a disk was formed. Because of the small dimensions of the plugs and disks, the photopolymerization occurred in less than 1 s, using only 3 J cm^{-2} of energy. They also developed an evolutionally advanced microfluidic system by combining the advantages of microscope projection photolithography and microfluidics to continuously form morphologically complex or multifunctional particles down to the colloidal length scale (Fig. 2) [10]. Exploiting the inhibition of free radical polymerization of diacrylate monomers near polydimethylsiloxane (PDMS) surfaces, they were able to repeatedly pattern and flow rows of particles in less than 0.1 s, affording a throughput of near 100 particles per second using the simplest of device designs. Polymerization was also carried out across laminar, co-flowing streams to generate [Janus particles](#), which are chemically biphasic particles. They synthesized a Janus polymer by using co-flowing laminar streams of monomer, one of which contained a rhodamine-labeled crosslinker, and forming particles across this interface. The relative proportions of Janus particles could be easily tuned. This new high-throughput technique offers unprecedented control over particle size, shape, and anisotropy.

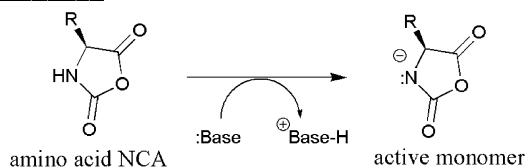


Polymer Synthesis within Microfluidic Reactor, Figure 1 (a)–(e) Scanning electron microscopy images of polymer microbeads obtained by polymerizing TPGDA in droplets obtained in regimes A, B, C, D, respectively, after removing a SO core. Inset: cross-section of the core-shell particle. (f) Cross-section of a polyTPGDA particle with three cores obtained by polymerizing core-shell droplets with three cores. The particle is embedded in epoxy glue. Scale bar is 40 μm . Reproduced from [9]

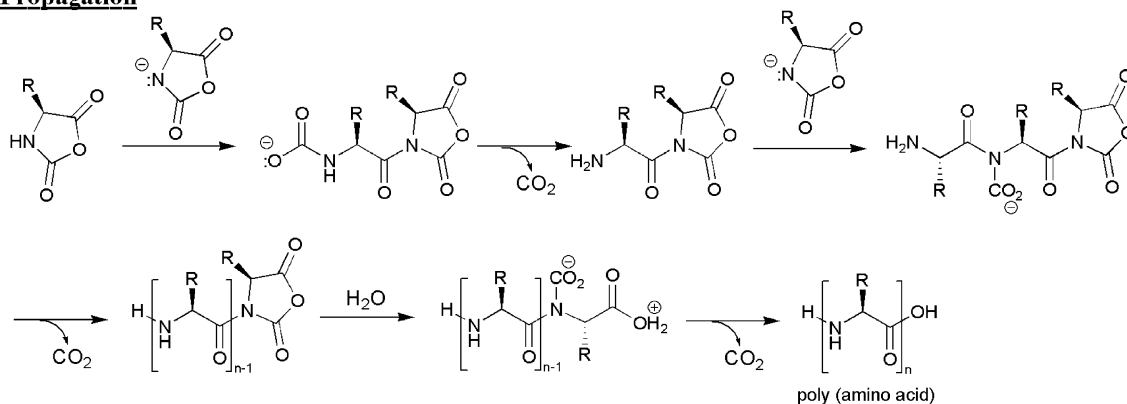


Polymer Synthesis within Microfluidic Reactor, Figure 2 Scanning electron microscope images of particles. Microparticles formed using a $\times 20$ objective (except (d), which was formed using a $\times 40$ objective) were washed before being observed. The scale bar in all of the figures is 10 μm . (a)–(c) Flat polygonal structures that were formed in a 20 μm high channel. (d) A colloidal cuboid that was formed in a 9.6 μm high channel. (e), (f) High-aspect-ratio structures with different cross-sections that were formed in a 38 μm high channel. (g)–(i) Curved particles that were all formed in a 20 μm high channel. The inset shows the transparency mask feature that was used to make the corresponding particle. Reproduced from [10]

Initiation



Propagation



Polymer Synthesis within Microfluidic Reactor, Figure 3 Mechanism of NCA polymerization initiated by a base such as tertiary amine. Reproduced from [5]

Basic Methodology

Basically, a microreaction system for polymerization reaction consists of a microfluidic device for mixing or droplet formation followed by microchannels as reaction parts. In this article, linear amino acid polymerization reaction is used as a basic example of polymer synthesis using microfluidic devices. Amino acid polymer was synthesized using the NCA method. This reaction is an anion ring-opening reaction which consists of two stages (Fig. 3): initiation by tertiary amine, and propagation of the polymer chain. The initiation and NCA anion attack are known to be relatively fast reactions. It is assumed that these steps are important in controlling polymerization of NCA. However, control of polymerization would be difficult in conventional batchwise systems due to the tendency to locally generate concentration gradients of the reagents. As a remedy, strict reaction control of a microreaction system can be employed to synthesize biopolymers.

Preparation of Microreaction Device

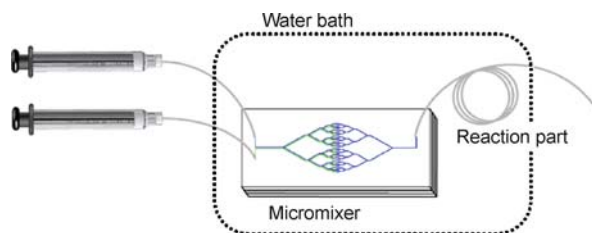
Generally, mixing is extremely important in chemical reactions. When using microreactors for chemical reactions, micromixers are often incorporated in the microreactor for effective mixing. Mixing of the monomer and base principally depends on diffusion through the interface of the two solutions. A multilayered micromixer increases

the interface area of two fluids and shortens the diffusion length, resulting in an increase in the mixing efficiency of reactants. Also, a characteristic feature of this mixer is that mixing efficiency increases with a decrease in flow rate. In order to gain long residence time required for polymerizations, low flow rate should be used in the microfluidic system; therefore, a multilayered micromixer was employed for NCA polymerizations in a microfluidic system.

Figure 4 shows the setup of a microreaction system. The micromixer was attached to a PTFE microtube (length = 500 cm) and served as a polymer-propagating region. The volume of fluid channel of the micromixer was 4.4 μL , and the total inner product volume of this microreactor corresponded to 250 μL . In the microreactor setup, monomer and initiator solutions were each charged into a 1 ml Hamilton gastight syringe. The solutions were supplied to the microreactor using a syringe pump with a parallel syringe holder, which enable identical pumping rates so that the final mixed solution would have a constant reagent concentration.

Polymerization Reaction

Polymerizations were performed in DMF and initiated by triethylamine. Both solutions of monomer and base were separately introduced into the microreactor (Fig. 4) using a syringe pump. The resulting mixture was heated at

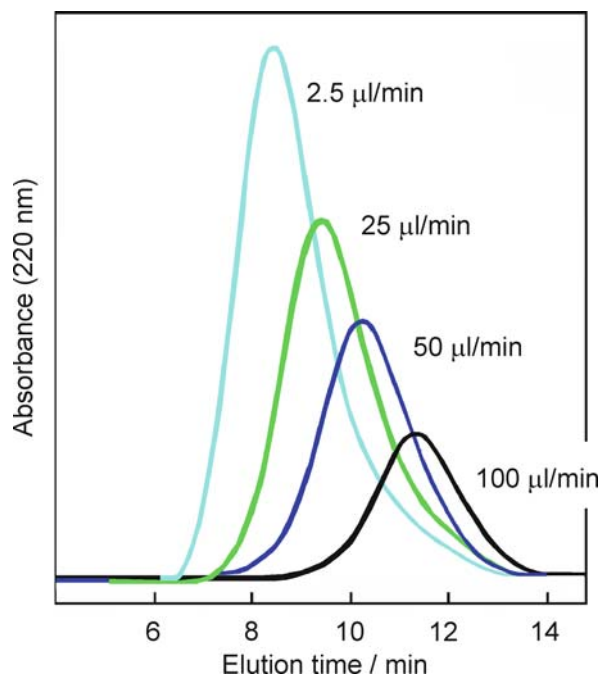


Polymer Synthesis within Microfluidic Reactor, Figure 4 Micro-reaction system used in this study. The system consists of a PDMS micromixer and PTFE microtubes. Two syringes containing reagents were coupled on the PDMS micromixer by PTFE microtubes (diameter = 250 μm , 10 cm). The PDMS micromixer was attached to a PTFE microtube (diameter = 250 μm , 500 cm) and served as a reaction region. The mixer and reaction region were kept at 30 $^{\circ}\text{C}$ in a water bath. Reproduced from [5]

100 $^{\circ}\text{C}$ for termination, then deprotection of the side-chain protective group was performed using HBr in AcOH. The sample was analyzed by gel filtration chromatography after dialysis against water to remove HBr. Figure 5 shows the gel filtration chromatography data for the poly(Lys) produced using the microreactor at different flow rates. The molecular weights of the respective samples regularly increased with a decrease in flow rates, that is, as a consequence of the increase in time required for polymerization to occur. It was found that the molecular mass of the polymer produced using the microreactor is simply governed by the flow (pumping) rate. Also, polymers produced by the microreaction showed narrower molecular weight distribution than that of macroscale reaction in homo- and copolymers (Fig. 6).

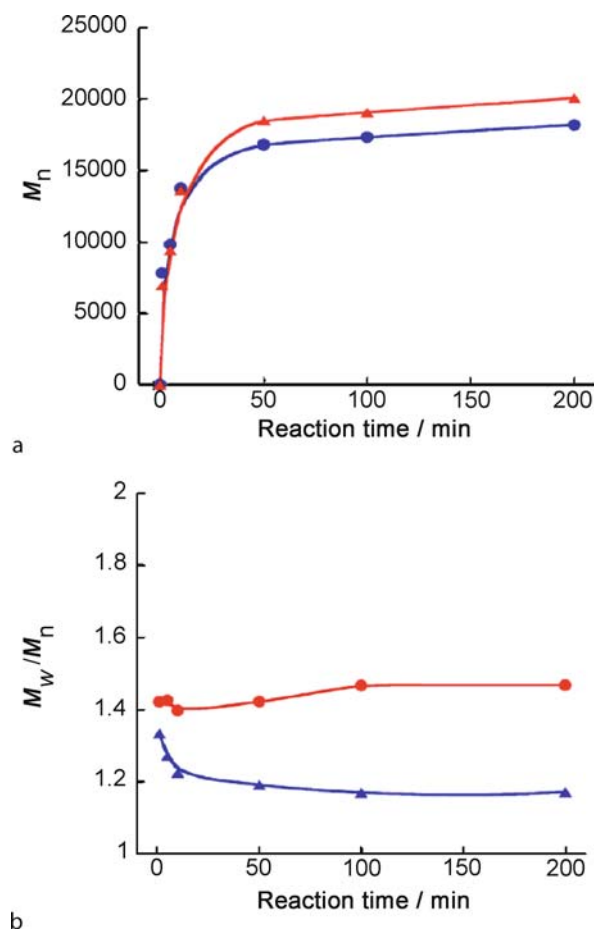
Key Research Findings

Many polymers with interesting functionalities have been developed and applied in various fields. At present, a rapid and combinatorial synthesis system with highly controllable polymerization reaction is much required. However, it is generally difficult to control polymerization reactions in conventional batchwise systems. Synthesis of polymer molecules using a microreactor enables strict control of chemical reactions which serves as a model of polymer synthesis. The performance of a microreaction system for polymerization was compared with that of a conventional batchwise system by analyzing the properties of the obtained polymers. In linear polymer products, the M_w/M_n values of the polymers synthesized by the batchwise system obviously increased with increasing concentration of monomer. In contrast, the M_w/M_n values in the microfluidic system remained low with high monomer concentrations [5]. It was demonstrated that the microreactor provided high controllability of polymeriza-



Polymer Synthesis within Microfluidic Reactor, Figure 5 Gel filtration chromatography analysis of poly(Lys) produced by the microreactor. The polymerizations were conducted with 0.1 M monomer and 0.01 M initiator for 1, 5, 10, and 100 min of reaction times corresponding to 250, 50, 25, and 2.5 $\mu\text{l/min}$ flow rates, respectively. Reproduced from [5]

tion even at concentrations that produced poor control of molecular weight distribution in the case of the batch reaction. The controllability of molecular weight distribution is governed by several factors. The first factor is the mixing efficiency of reactants. In the batchwise system, an increase in reagent concentration tends to decrease the mixing efficiency of reagents, and leads to generation of local concentration gradients of the reagents. In addition, the viscosity rise observed for highly concentrated monomer solutions will cause a decrease in the efficiency of mixing based on turbulent flow in the batchwise system. Consequently, it is assumed that such a decrease in mixing efficiency induces a broadening of the molecular weight distribution. In contrast, it is hypothesized that effective mixing of reactants by the attached micromixer in the microfluidic system enables high control of molecular weight distribution. Not only was linear polymer synthesis performed in the microfluidic reactor, but also synthesis of dendrimer molecules. Although still in their infancy, microfluidic systems open the use of microreactors for branched polymers. Overall, the usefulness of microreactors in controlling the reaction of linear and branched polymer synthesis has been demonstrated. For polymer particle synthesis, microfluidic reaction systems show several



Polymer Synthesis within Microfluidic Reactor, Figure 6 Comparisons of polymer properties in batchwise (circle) and microfluidic (triangle) systems at various monomer concentrations. (a) $\blacktriangleright$ number average molecular weight (M_n) and (b) polydispersity (M_w/M_n) of the obtained polymers against concentrations of NCA. Polymerizations were carried out for 200 min. Reproduced from [5]

advantages over the conventional reaction systems. These benefits arise from the small dimensions of microdevices which provide predictable thermal and mass transportation properties and unique control over the fluid structure. Enhanced transport characteristics of microreactors allow more precise synthesis of polymers compared to traditional batch reactors. Perhaps most importantly, extreme fluid controllability in a laminar flow system enables synthesis of particles with unique shapes not accessible by other methods. Core-shell type polymers and nonspherical polymers were prepared by controlling the fluidic behavior in microchannels. Also, solid, hollow, and multicore asymmetric and irregularly shaped polymer microparticles as well as membranes and fibers can all be produced with microfluidic devices. These particles are generally

monodispersed, and the sizes may be tuned from a few micrometers to hundreds of micrometers, depending on the specific device geometry and flow parameters. The design of fluidic systems therefore becomes quite important. Numerical simulation is useful in designing microreaction devices. Likewise, reproducibility of these particles is quite high when performing the reaction using the same microreaction apparatus because of the simple operation of the reactor and strict control of the reaction conditions provided by the system. The reaction does not require special techniques as in batchwise synthesis. This feature is quite advantageous for production of polymer particles by numbering-up. Utilizing the principle of shape control by fluidic behavior, the elegant technique was applied to control the shape of polymers. Microscope projection photolithographic synthesis of particles is a novel particle formation method [8, 9]. This method enables continuous-flow synthesis of particles with complicated shapes. Polymerization was also carried out across laminar, co-flowing streams to generate Janus particles containing different chemistries, whose relative proportions could be easily tuned. This new high-throughput technique offers unprecedented control over particle size, shape, and anisotropy.

To summarize, polymer synthesis in microfluidic reactors, although in its initial stage, holds a promising future. Many efforts are now in progress in many laboratories to develop generally applicable microreactors for polymer synthesis.

Future Directions for Research

Microreactors will represent an indispensable tool for synthesizing polymers with controlled properties. Microreaction in microfluidic systems will find its way in the shift from conventional batchwise reaction to microreaction strategy. In future, microfluidic devices will find broad applications particularly in linear polymer production aimed at obtaining polymers of desired average molecular weight with narrower molecular weight distribution. In addition, the prospect of making novel polymer particles with unique shapes can be realized by microreaction system via control of specific device geometry and flow parameters. The process of rapid prototyping and the emergence of simple, homemade devices that take only minutes or hours to construct have enabled many groups to explore this exciting field. More and more applications of microfluidic reaction systems for polymer synthesis are required to develop a generally applicable microreactor for polymerization reactions. In addition to simple reactions, an aspect of polymer synthesis that may offer great research opportunity is synthesis of novel particles with unique shapes, and thus the development of microscope

projection photolithography will be necessary. In addition, theoretical studies such as numerical simulations are also required to obtain a generally applicable microreactor. Matching of experimental achievements and theoretical studies may accelerate development of microreactors for polymerization reactions. Still, this field is in its initial stage and has much room for development.

Cross References

- Droplet Based Lab-on-Chip Devices
- Lab-on-a-Chip Devices for Chemical Analysis
- Microfluidics for Bio- and Chemical Reactions
- Droplet Microreactors

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Poor Man's LIGA

- LIGA-Like Techniques

Porous Electrodes

- Three-Dimensional Electrodes

Port Resonance

- Hemholtz Resonance

Positive Displacement Pump

Synonyms

Flexible member pump; Membrane pump; Diaphragm pump

Definition

In a *positive displacement pump* fluid is displaced by a moving boundary. Ideally, this displacement is independent of backpressure, up to the maximum pump head.

Cross References

- Compression Ratio
- Dead Volume
- Electromagnetic Valves
- Electrostatic Valves
- Magnetic Pumps
- Membrane Actuation for Micropumps
- Microactuators
- Peristaltic Pumps
- Piezoelectric Valves
- Pneumatic Valves
- Stroke Volume
- Thermomechanical Valves
- Thermopneumatic Valves

Positive Pressure or Suction (Vacuum) Pressure

Definition

Positive pressure or suction (vacuum) pressure is used to generate a pressure difference to the reservoir on chips and drives the sample solution into the microfluidic systems on chip.

Cross References

- Techniques for Transferring Samples to Chips

Potassium Hydroxide (KOH) Etching

- Anisotropic Etching

Powder Blasting

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Synonyms

Sand blasting; Micropowder blasting

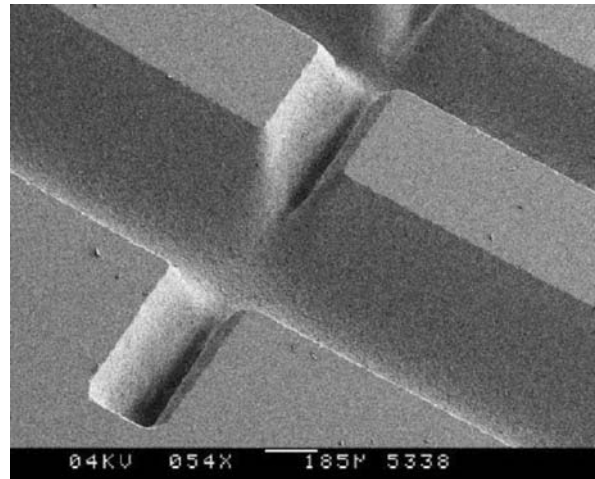
Definition

Powder blasting is a micromachining technique which uses a high-velocity air jet containing microparticles to impinge on and erode material from a surface.

Overview

Sand blasting is a traditional mechanical processing technique which can be found in a mechanical workshop. Sand blasting has been used commonly for removing rusty surface, dry paints or contaminants from a metal work piece. From the mid-1990s, sand blasting was used to machine large arrays of holes in glass panels for large-area flat panel display devices. Thousands of holes could be machined in the glass at once, which was highly efficient and low cost [1]. Sand blasting was introduced as a microstructuring technique in early 2000, with the name of powder blasting because much smaller particles were used [2]. The principle of power blasting micromachining is that when a powder particle indents the surface of a brittle material such as glass or silicon it generates cracks and a plastic zone around the impact point. There are two types of cracks: lateral and radial. The lateral cracks play a major role in the erosion process because they can reach the surface and remove upper material zone. The radial cracks have no significant contribution in the erosion process, but they weaken the material. The length of such cracks depends on a few parameters like the substrate's Young's modulus and hardness and the indentation volume. These parameters then depend on the kinetic energy of the particle and its shape. Powder blasting has the following features [3]:

- It is only suitable for machining brittle materials, such as glass, silicon and ceramics; it is not suitable for ductile materials, such as metals and polymers. Ductile materials can serve as masking material in powder blasting machining.
- The mask materials for powder blasting are required not only to be ductile but also thick enough. Because high-speed particles inevitably consume the masking

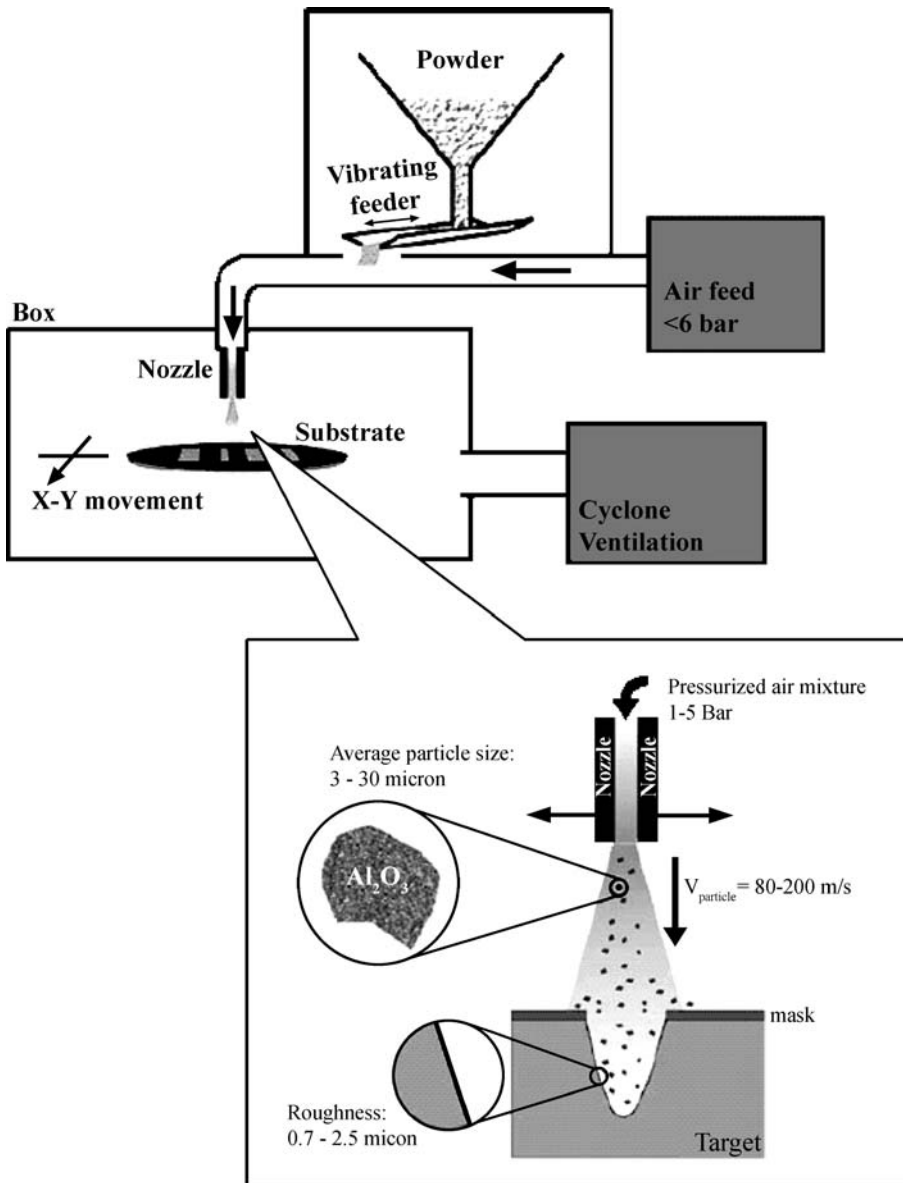


Powder Blasting, Figure 1 Typical etch profile of powder blasting in glass

layer, even though at a lower rate, when the masking layer reaches a certain thinness the particles with high kinetic energy can penetrate through the mask and damage the work piece surface. The best way of making the mask is to pattern it directly on the work piece surface. For example, a copper layer is deposited by electroplating on the substrate surface and then photolithographically patterned. The selectivity of powder blasting Pyrex glass to copper is 65:1 [4]. Some rubber-like polymers, which are also photon sensitive, can also be used as masking materials.

- Powder blasting is a pure mechanical processing technique. The particle beam from a nozzle is always of the highest speed in the centre, as a result of which the etched profile always has a sloped sidewall. Figure 1 shows typical structure profiles etched by powder blasting [4].
- The resolution of powder blasting is normally above 50 μm . This is mainly attributed to the particle sizes which are around 30 μm . The resolution may increase if smaller particles are used. However, the kinetic energy of particles also reduces for smaller sized particles.
- The roughness of an eroded surface produced by high-speed powder is around the micrometre scale, which is related to the size of the particles.

Powder blasting is not a high-tech machining method. It is for this reason that the technique is less expensive. Powder blasting is particularly useful for making patterns in glass. For micromachining of glass there are no other better techniques apart from wet etching by HF acid. It is well known that HF acid is a hazardous chemical and wet etching by HF is isotropic. Powder blasting is a much better suited



Powder Blasting, Figure 2 Schematic representation of a powder blasting system

method to process glass materials. The structures encountered in microfluidic and biochip systems are normally above a few tens of micrometres, which can be adequately resolved by powder blasting machining [5]. For drilling holes in glass chips, the conventional method is mechanical drilling with a diamond bit. Mechanical drilling can produce chippings on the edge and it becomes extremely difficult to drill thin glass. Powder blasting can machine a large number of holes in glass at once and the quality of the holes is much better than that produced by mechanical drilling.

Basic Methodology

A schematic representation of a powder blasting system is shown in Fig. 2. The microparticles used for powder blasting are not sands, but Al_2O_3 or SiC particles in the size range $20-30 \mu\text{m}$. These are injected from a vibrating feeder into a pressured air flow stream. The pressured particle jet flows through a nozzle and impinge at high velocity normal to the surface of a work piece. The velocity at which the particles impinge on a solid surface is in the range $80-200 \text{ m/s}$. These particles with high kinetic

energy can crack and erode the solid surface locally, with etching rate of a few tens to a few hundreds of micrometers per minute, and in extreme cases can be as high as 1 mm/min [2]. The etching rate is directly proportional to the air pressure and the gap distance between the nozzle and surface of the work piece. However, the etching rate becomes saturated when the air pressure is too high and the speed of the air stream approaches the speed of sound. An array of nozzles can be used to etch a large area of surface. The substrate to be blasted is normally mounted on a rotating table, while the nozzle array is fixed. By rotating the table, uniform erosion over a large area of substrate or a number of substrates can be achieved. While the powder jet impinges the substrate, a cyclone ventilation system collects the powders and reuses them. Although small particles can improve the surface roughness of powder blasting, particles smaller than 5 μm cannot be recycled, which is another reason for not using too small powder particles. After powder blasting and removal of the masking layer, the etched surface can be washed clean by water.

There are no commercial powder blasting systems available because there is not such a market. Only a few places worldwide have utilised this technique. There are companies that can make custom designed powder blasting systems. They only make to order. However, there are some companies, such as Micronit in the Netherlands, which provide powder blast micromachining services.

Key Research Findings

Much of the research in powder blasting has been focused on improving resolution and selectivity of masking materials. Currently the resolution of powder blasting is above 50 μm . This is mainly attributed to the particle sizes which are around 30 μm . Using smaller particles can improve resolution and give smaller pattern dimensions. However, there is a limit to the particle size that can be used for powder blasting. As mentioned earlier, there is the sound barrier which limits the velocity of microparticles not to exceed 335 m/s. There is also a threshold kinetic energy above which a particle can crack an eroding material. For powder blasting of soda lime glass with alumina (Al_2O_3) particles, the threshold energy is 25 nJ, from which the minimum size of alumina particles can be estimated to be 6 μm in order to acquire the threshold kinetic energy at the speed of sound [6]. For safety reasons 10 μm alumina particles were used with a velocity of 156 m/s, which helped to achieve a minimum feature size of 30 μm in soda lime glass. However, smaller particles resulted in much less efficiency. Experimental comparison demonstrated that a

10 μm particle is 50 times less efficient than a 30 μm particle in powder blasting. The etch rate is also limited by the mask feature size. The smaller the mask feature size compared to the particle size, the lower the etch rate.

Various photosensitive polymers, as well as metal films, have been used as masking materials. In general, depending on their erosion behaviour, materials can be divided into two classes: one class comprising all materials showing maximum erosion rate at normal impact of particles, and the other class of materials having a maximum erosion rate at a glancing impact angle. Glass belongs to the first class of materials, while metals belong to the second class. Due to their high erosion resistance at normal impact, metals therefore have been used as a masking material. Metallic contact masks can be made by laser ablation or electroplating using a photoresist as a mould. Recent development has shown that elastomers such as polydimethylsiloxane (PDMS) can be used as masking materials [6]. A PDMS mask can be made by casting into a patterned SU-8 mould. The PDMS mask has selectivity of 5000:1 to soda lime glass in powder blasting [7].

Future Directions for Research

Powder blasting as an alternative technique has been increasingly adopted for making microstructures in glass, ceramics or graphite, which are not easy to make by other techniques. Future research will be aimed at improving etching efficiency and resolution by trying out different masking materials or parameters of microparticles and air jet flow.

Cross References

► Plasma Etching

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Power MEMS

- Micro Energy Conversion Devices

Precipitate

Definition

A phase change which results in the formation of a collection of a material. For example, solid precipitate can be formed in a liquid phase solution as a result of a chemical reaction or a supersaturation condition in the solution.

Cross References

- Fabrication of Self-Assembled Catalytic Nanostructures

Precipitation

- Nucleation and Growth

Precursor Materials

Synonyms

Predecessor material

Definition

A substance that, following a reaction, becomes an intrinsic part of a product chemical. The material must be stable at room temperatures. During CVD process, the precursor material is vaporised. The precursor gas is transported to the surface where it undergoes a chemical reaction to produce the desired thin film.

Cross References

- Chemical Vapor Deposition for Film Deposition
- Plasma-Treatment Physics

Predecessor Material

- Precursor Materials

Preparatory Separation

- Biosample Preparation by Lab-on-a-Chip Devices

Pressure

Synonyms

Absolute pressure; Gage pressure; Differential pressure

Definition

The absolute pressure is referenced to total vacuum (no gas molecule present in a defined volume). The gage pressure is the difference between absolute pressure and atmospheric pressure. The differential pressure is the difference between a measured pressure and a reference pressure. Pressures are measured in Pascal (N/m^2) according to the international standards system (SI). Further commonly used units include bar, Torr (millimeters of mercury column), pounds per square inch and others.

Cross References

- Glass-Polymer-Bonding
- Wall Shear Stress

Pressure-Driven Flow

Definition

Flow driving principle used to create fluid flows through microfluidic channels. By applying pressure at the inlet the fluid is pumped through the channel. The fluid velocity near the walls approaches zero and a parabolic velocity profile is produced within the channel.

Pressure-Driven Single Phase Gas Flows

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Synonyms

Poiseuille single phase gas flows

Definition

A pressure-driven gas flow is generated by applying a pressure difference between the inlet and the outlet of a fluidic system, for example a channel. Due to very small hydraulic diameters, pressure-driven gas flows in microchannels are laminar. Gas microflows are distinct from gas macroflows by rarefaction effects which appear as soon as the mean free path of the molecules is no longer negligible compared with the hydraulic diameter of the microchannel. For such rarefied flows, the classic Poiseuille model is no longer valid, and other models should be used, according to the rarefaction level, which is quantified by the Knudsen number.

Overview

Rarefaction Effects and Gas Flow Regimes

Usually, the regime of gas flow in a macrochannel refers to viscous and compressibility effects, respectively quantified by the Reynolds number Re and the Mach number Ma . For low Reynolds numbers (typically for $Re < 2000$), the flow is laminar and it becomes turbulent for higher values of Re . For $Ma < 1$, the flow is subsonic and for $Ma > 1$, it is supersonic. Generally, if $Ma > 0.3$, compressibility effects should be taken into account.

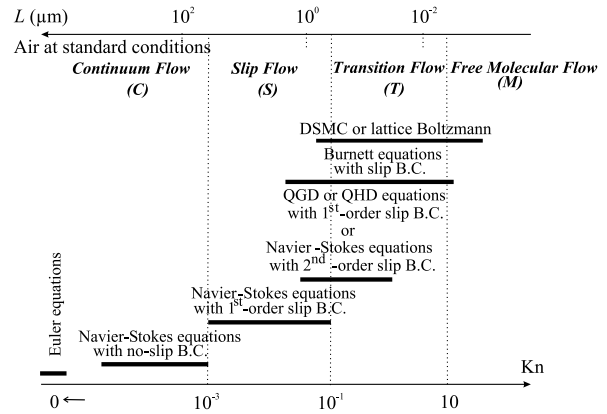
On the other hand, for gas microflows, the main micro-effect that results from shrinking down the devices size is rarefaction. Actually, there is a strong – although not complete – analogy between microflows and low pressure flows. Rarefaction, which implies a local thermodynamic disequilibrium, occurs as soon as the mean free path of the gas molecules is no longer negligible compared with a characteristic length of the system.

Thus, it is convenient to differentiate the flow regimes in function of the Knudsen number

$$Kn = \frac{\lambda}{L}, \quad (1)$$

ratio of the mean free path of the molecules λ over this characteristic length L , and the following classification is usually accepted:

- For $Kn < 10^{-3}$, the flow is a *continuum flow* (C) and it is accurately modeled by the compressible Navier–Stokes equations with classical no-slip boundary conditions.
- For $10^{-3} < Kn < 10^{-1}$, the flow is a *slip flow* (S) and the Navier–Stokes equations remain applicable, provided a velocity slip and a temperature jump are taken into account at the walls. These new boundary conditions point out that rarefaction effects become sensitive at the wall first.



Pressure-Driven Single Phase Gas Flows, Figure 1 Gas flow regimes and main models according to the Knudsen number

- For $10^{-1} < Kn < 10$, the flow is a *transition flow* (T) and the continuum approach of the Navier–Stokes equations is no longer valid. However, the intermolecular collisions are not yet negligible and should be taken into account.
- For $Kn > 10$, the flow is a *free molecular flow* (M) and the occurrence of intermolecular collisions is negligible compared with the one of collisions between the gas molecules and the walls.

The limits of these different regimes are only indicative and could vary from one case to another, partly because the choice of the characteristic length L is rarely unique. For flows in microchannels, L is generally the hydraulic diameter or the depth of the channel. In complex geometrical configurations, it could be preferable to define L from local gradients (for example of the density ρ : $L = 1/|\nabla \rho/\rho|$) rather than from simple geometrical considerations [1]; the Knudsen number based on this characteristic length is the so-called local rarefaction number. The upper axis of Fig. 1 locates these different regimes for air in standard conditions, as a function of a characteristic length, for example a microchannel hydraulic diameter. The lower axis shows the corresponding values of the Knudsen number.

Mean Free Path and Knudsen Number Calculation

Calculating the mean free path requires assumptions about intermolecular collisions. First, the gas is assumed to be a dilute gas, i.e. a gas for which the mean molecular diameter of the molecules is small compared with the mean molecular spacing. This condition is verified with a good precision for all usual gases under standard pressure and temperature conditions. It involves binary collisions between molecules and the mean free path may be expressed as $\lambda = c'/v = \sqrt{8RT/\pi}/v$, ratio of the mean

thermal velocity c' to the collision rate ν . The mean thermal velocity depends on the temperature T and on the specific gas constant R .

For classic collision models, based on an inverse power law (IPL) interaction scheme which only takes into account the repulsion part of the force between two molecules, the expression of ν leads to the general form of the mean free path:

$$\lambda = \frac{k_2 \mu}{\rho \sqrt{RT}}. \quad (2)$$

The mean free path is a function of the temperature T , the density ρ and the dynamic viscosity μ of the gas which itself depends on the temperature as

$$\mu(T) = \mu_0(T_0) \left(\frac{T}{T_0} \right)^\omega, \quad (3)$$

T_0 being a reference temperature for which the viscosity μ_0 is known.

The values of the dimensionless parameters k_2 and ω for the most classic collision models are given in Table 1. The Maxwell Molecules (MM) model assumes a linear relationship between viscosity and temperature, although for the Hard Sphere (HS) model, the viscosity is proportional to the square root of the temperature. These models could be roughly considered as limits for the real behavior of gases, and the Variable Hard Sphere (VHS) model proposed by Bird [2] is much more accurate. Another sophistication has been proposed by Koura and Matsumoto who developed the Variable Soft Sphere (VSS) model, which introduces via an adjustment parameter α limited corrections – less than 3% – to the evaluation of the mean free path deduced from the VHS model, assuming soft collisions between the molecules. For $\alpha = 1$, the VSS model reduces to the VHS model.

Experimentally deduced values of ω and α are available for different gases (Table 2).

Ideal Gas Assumption

For a dilute gas in thermodynamic equilibrium, the classic ideal gas equation of state

$$p = \rho RT = nk_B T \quad (4)$$

is valid. If the thermodynamic equilibrium is not verified, Eq. (4) remains valid, providing the thermodynamic temperature T is replaced with the translational kinetic temperature T_{tr} , which measures the mean translational kinetic energy of the molecules. Here, p is the pressure and $\rho = mn$ is the density, $k_B = R'/N = (M/N)(R'/M) = mR = 1.381 \times 10^{-23} \text{ J K}^{-1}$ is the Boltzmann constant,

Pressure-Driven Single Phase Gas Flows, Table 1 Mean free path and dynamic viscosity for classic IPL collision models

Model	$\mu \propto T^\omega$	$\lambda = k_2 \frac{\mu}{\rho \sqrt{RT}}$
	ω	k_2
Maxwell Molecules (MM)	1	$\sqrt{2/\pi} \approx 0.798$
Hard Spheres (HS)	$\frac{1}{2}$	$\frac{16}{5\sqrt{2\pi}} \approx 1.277$ (sometimes proposed as $\sqrt{\pi}/2 \approx 1.253$)
Variable Hard Spheres (VHS)	$\omega = \frac{\eta+3}{2(\eta-1)}$	$\frac{2(7-2\omega)(5-2\omega)}{15\sqrt{2\pi}}$
Variable Soft Spheres (VSS)	$\omega = \frac{\eta+3}{2(\eta-1)}$	$\frac{4\alpha(7-2\omega)(5-2\omega)}{5(\alpha+1)(\alpha+2)\sqrt{2\pi}}$

$R' = 8.315 \text{ J mol}^{-1} \text{ K}^{-1}$ being the universal gas constant, $N = 6.022 \times 10^{23} \text{ mol}^{-1}$ the Avogadro's number and M the molecular weight.

As a consequence, for an ideal gas and an IPL collision model, the Knudsen number can be written as

$$\text{Kn} = \frac{k_2 \mu(T) \sqrt{RT}}{pL}, \quad (5)$$

with the value of k_2 given in Table 1.

Modeling Gas Flow in Microchannels

It can be shown from Eq. (5) that for an ideal gas and an IPL collision model,

$$\text{Re} = k_2 \sqrt{\gamma} \text{Ma} / \text{Kn}, \quad (6)$$

since $\text{Re} = \rho u L / \mu$ and $\text{Ma} = u / \sqrt{\gamma R T}$, noting γ the ratio of the specific heats of the gas. As k_2 is of the order of unity (Table 1), the result is that for subsonic microflows, viscous effects are predominant and the flows are laminar. It can be seen from Fig. 1 that gas flows in micron size channels are typically relevant to the slip flow regime, at any rate for usual pressure and temperature conditions. For lower sizes, i. e. for Knudsen numbers higher than 10^{-1} , the slip flow regime could remain valid, provided that classical velocity slip and temperature jump boundary conditions are modified (taking into account higher order terms as explained below) and/or that Navier–Stokes equations are extended to more general sets of conservation equations, such as the Quasi-Gasodynamic (QGD), the Quasi-Hydrodynamic (QHD) or the Burnett equations [3]. However, for Knudsen numbers higher than unity, as it could be the case in nanochannels (► [gas flow in nanochannels](#)) or in low-pressure flows in microchannels,

Pressure-Driven Single Phase Gas Flows, Table 2 Molecular weight and dynamic viscosity under standard conditions ($p_0 = 101,325 \text{ Pa}$ and $T_0 = 273.15 \text{ K}$) estimated from a VHS or a VSS model (data from Bird [2]) with corresponding values of ω and α

Gas	$M \cdot 10^3$ (kg mol ⁻¹)	$\mu_0 \cdot 10^7$ (N s m ⁻²)	ω	α
sea level air	28.97	171.9	0.77	–
Ar	39.948	211.7	0.81	1.40
CH ₄	16.043	102.4	0.84	1.60
Cl ₂	70.905	123.3	1.01	–
CO	28.010	163.5	0.73	1.49
CO ₂	44.010	138.0	0.93	1.61
H ₂	2.0159	84.5	0.67	1.35
HCl	36.461	132.8	1.00	1.59
He	4.0026	186.5	0.66	1.26
Kr	83.80	232.8	0.80	1.32
N ₂	28.013	165.6	0.74	1.36
N ₂ O	44.013	135.1	0.94	–
Ne	20.180	297.5	0.66	1.31
NH ₃	17.031	92.3	1.10	–
NO	30.006	177.4	0.79	–
O ₂	31.999	191.9	0.77	1.40
SO ₂	64.065	116.4	1.05	–
Xe	131.29	210.7	0.85	1.44

the continuum approach is not longer valid, and molecular methods such as the Direct Simulation Monte Carlo – DSMC – (► [Monte Carlo method](#)) or the Lattice Boltzmann methods (► [Lattice Boltzmann method](#)) should be used.

Basic Methodology

For moderate pressure levels, and Knudsen numbers up to 0.5 or so, pressure-driven gas microflows can be modeled with the compressible Navier–Stokes equations, slip velocity and temperature jump boundary conditions, considering the gas as an ideal gas.

Compressible Navier–Stokes Equations

The compressible Navier–Stokes equations are the governing conservation laws for mass, momentum and energy. These laws are written assuming that the fluid is Newtonian and follows the Fourier law of diffusion, so that the stress tensor is a linear function of the velocity gradients and the heat flux vector is proportional to the temperature gradient. Adding Stokes’ hypothesis, which expresses that the changes of volume do not involve viscosity, the com-

pressible Navier–Stokes equations may be written as:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \quad (7)$$

$$\frac{\partial (\rho \mathbf{u})}{\partial t} + \nabla \cdot \left(\rho \mathbf{u} \otimes \mathbf{u} - \mu \left[\nabla \otimes \mathbf{u} + (\nabla \otimes \mathbf{u})^T - \frac{2}{3} (\nabla \cdot \mathbf{u}) \mathbf{I} \right] \right) + \nabla p = \mathbf{f} \quad (8)$$

$$\frac{\partial E}{\partial t} + \nabla \cdot \left((E + p) \mathbf{u} - \mu \left[\nabla \otimes \mathbf{u} + (\nabla \otimes \mathbf{u})^T - \frac{2}{3} (\nabla \cdot \mathbf{u}) \mathbf{I} \right] \cdot \mathbf{u} - k \nabla T \right) = \mathbf{f} \cdot \mathbf{u} \quad (9)$$

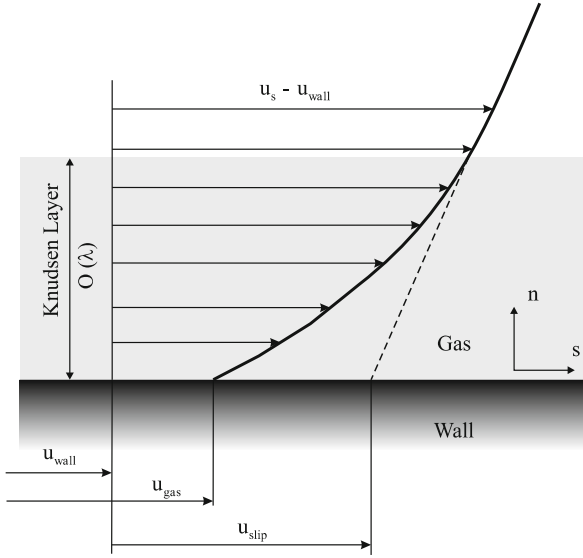
where $\mathbf{u}$ is the velocity vector, $\mathbf{I}$ the unit tensor, t the time and k the thermal conductivity. The total energy per unit volume $E = \rho(e + \mathbf{u} \cdot \mathbf{u}/2)$ is a function of the internal specific energy e that may be written $e = p/[\rho(\gamma - 1)]$ for an ideal gas.

The external volume forces $\mathbf{f}$ (gravitational forces, magnetic forces, etc.) are generally negligible in gas flow. It is even more the case in gas microflows, since the volume over surface ratio decreases with the characteristic length L and consequently, in microscale geometries, volume effects may be neglected when compared to surface effects.

Knudsen Layer

The Knudsen layer plays a fundamental role in the slip flow regime. This thin layer, between one and two molecular mean free paths in thickness, is a region of local non-equilibrium which is observed in any gas flow near a surface. In non rarefied flows, the Knudsen layer is too thin for having any significant influence, but in the slip flow regime, it should be taken into account.

Although the Navier–Stokes equations are not valid in the Knudsen layer, due to a nonlinear stress/strain-rate behavior in this small layer [4], their use with appropriate boundary velocity slip and temperature jump conditions proved to be accurate for predicting mass flow rates [5] (► [methods for flow rate measurements](#)) and velocity profiles out of the Knudsen layer. Classically, the real flow is not simulated within the Knudsen layer, but the influence of the Knudsen layer on the flow outside this non-equilibrium layer is taken into account, replacing the no-slip condition at the wall with a slip-flow condition. For that purpose, a fictitious slip velocity u_{slip} is introduced (Fig. 2). Real slip at the wall, $u_{\text{gas}} - u_{\text{wall}}$, is due to the fact that gas molecules very close to the wall have actually a mean



Pressure-Driven Single Phase Gas Flows, Figure 2 Gas flow regimes and main models according to the Knudsen number

tangential momentum partly due to the non zero momentum of molecules coming from the flow and to the momentum of the molecules reflected by the wall. These reflected molecules keep all their tangential momentum if the reflection is specular, loose all their tangential momentum if the reflection is diffuse or more generally keep only a part of it.

Velocity Slip in the Slip-Flow Regime

The classic slip-flow condition, deduced from a momentum balance at the wall, was first proposed by Maxwell and is generally presented in the form

$$u_{\text{slip}} = \frac{2 - \sigma}{\sigma} \lambda \left. \frac{\partial u_s}{\partial n} \right|_w + \frac{3}{4} \frac{\mu}{\rho T} \left. \frac{\partial T}{\partial s} \right|_w. \quad (10)$$

In this equation, σ is the tangential momentum accommodation coefficient, equal to unity for perfectly diffuse molecular reflection and zero for purely specular reflection. In Maxwell's model, u_{slip} overestimates the real velocity at the wall but leads to a rather good prediction of the velocity out of the Knudsen layer, as represented in Fig. 2. After non-dimensionalization with the characteristic length L , a reference velocity u_0 and a reference temperature T_0 , Eq. (10) is written as follows:

$$u_{\text{slip}}^* = \frac{2 - \sigma}{\sigma} \text{Kn} \left. \frac{\partial u_s^*}{\partial n^*} \right|_w + \frac{3}{4\gamma k_2^2} \frac{\text{Kn}^2 \text{Re}_0}{\text{Ma}_0^2} \left. \frac{\partial T^*}{\partial s^*} \right|_w, \quad (11)$$

with $u_s^* = u_s/u_0$, $T^* = T/T_0$, $n^* = n/L$ and $s^* = s/L$. Note that this non-dimensionalization is valid only if the ref-

erence lengths are the same in the s and n directions, which is generally not the case for a flow in a microchannel. In that case, it is preferable to keep the dimensional form of Eq. (10). The second term on the right-hand side, which is proportional to the square of the Knudsen number, involves the Reynolds number $\text{Re}_0 = \rho u_0 L / \mu$ and the Mach number $\text{Ma}_0 = u_0 / \sqrt{\gamma R T_0}$ defined at the reference conditions. This second term is associated with the thermal creep, or transpiration phenomenon. It shows that a flow can be induced by the sole presence of a tangential temperature gradient, without any pressure gradient. In that case, the gas experiences a slip velocity and moves from the colder toward the warmer region. This can also result in a pressure variation inside a microchannel only submitted to tangential temperature gradients. Given that we consider here pressure driven flows, we assume that there is no thermal creep and that the only cause of flow is a pressure gradient. As a consequence, the first-order slip flow boundary condition is written

$$u_{\text{slip}}^* = \beta_u \frac{2 - \sigma}{\sigma} \text{Kn} \left. \frac{\partial u_s^*}{\partial n^*} \right|_w = A_1 \text{Kn} \left. \frac{\partial u_s^*}{\partial n^*} \right|_w. \quad (12)$$

A corrective coefficient β_u has been calculated by different authors [6] who have shown that a better prediction of the flow out of the Knudsen layer would be obtained with this corrective coefficient slightly different from unity, unlike as initially proposed by Maxwell. Its value depends on the collision model: for example, $\beta_u = 1.11$ for a HS model [7]. Equation (12) is called first-order slip boundary condition, because it involves the Knudsen number ($\partial(\text{Kn})$) and the first derivative ($\partial u_s^* / \partial n^*$)_w.

For $\text{Kn} > 0.1$, experimental [5] or numerical studies with the DSMC method show significant deviations with models based on first-order boundary conditions. Therefore, since 1947, several authors have proposed second-order boundary conditions, hoping to extend the validity of the slip flow regime to higher Knudsen numbers. Second-order boundary conditions take more or less complicated forms, which are difficult to group together in a sole equation. Actually, according to the assumptions, the second-order terms ($\partial(\text{Kn}^2)$) may be dependent of σ [8, 9] or involve streamwise second derivatives ($\partial^2 u^* / \partial s^{*2}$)_w, in addition to spanwise second derivatives ($\partial^2 u^* / \partial n^{*2}$)_w [10]. For a developed flow between plane plates or in a circular microtube, the tangential second derivatives are zero and one may compare most of the second-order models that take the generic form

$$u_{\text{slip}}^* = A_1 \text{Kn} \left. \frac{\partial u_s^*}{\partial n^*} \right|_w + A_2 \text{Kn}^2 \left. \frac{\partial^2 u_s^*}{\partial n^{*2}} \right|_w. \quad (13)$$

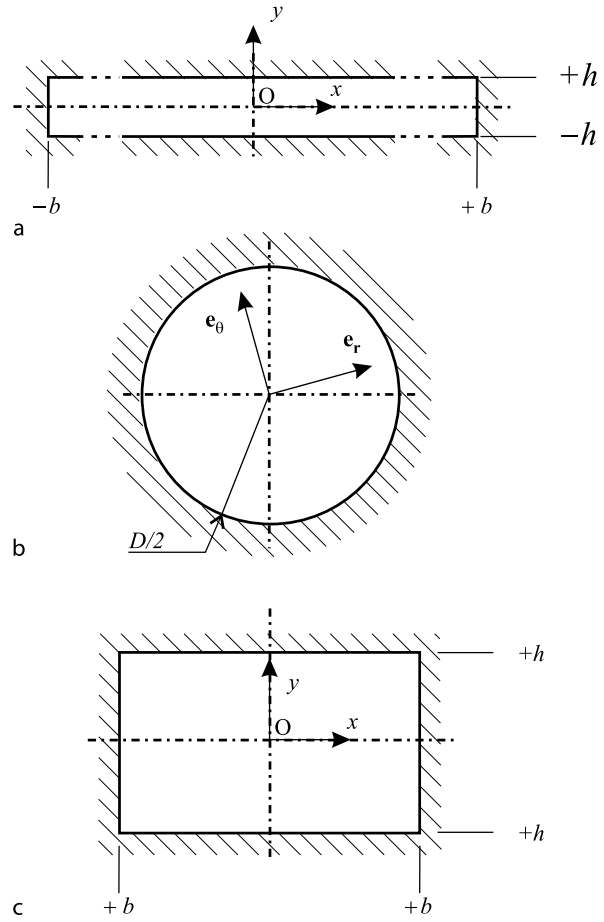
Pressure-Driven Single Phase Gas Flows, Table 3 Coefficients of some second-order boundary conditions proposed in the literature, for $\sigma = 1$

Author, Year	A_1	A_2
Maxwell, 1879	1	0
Schamberg, 1947	1	$-5\pi/12$
Chapman and Cowling, 1952	≈ 1	$\approx 1/2$
Deissler, 1964	1	$-9/8$
Cercignani, 1964	1.1466	-0.9756
Hsia and Domoto, 1983	1	$-1/2$
Mitsuya, 1993	1	$-2/9$
Karniadakis and Beskok, 2002	1	$1/2$
Hadjiconstantinou, 2003	1.11	-0.61

In the particular case of a fully diffuse reflection ($\sigma = 1$), some A_1 and A_2 coefficients proposed in the literature are compared in Table 3. We note significant differences, essentially for the second-order term. Recent experimental flow rate data and molecular numerical simulations show that the second order coefficient A_2 should be negative, which corresponds to a slip velocity higher than that predicted by a first-order slip boundary condition.

Pressure Driven Isothermal Flows in Microchannels

We consider here the flow of an ideal gas through different microchannels in the slip-flow regime. The flow is steady, isothermal and the volume forces are neglected. The flow is also assumed to be locally fully developed: the velocity profile in a section is the same as the one obtained for a fully developed incompressible flow, but the density is recalculated in each cross-section and depends on the pressure and the temperature via the equation of state (4). This assumption is no longer valid if the Mach number is greater than 0.3. The main kinds of sections of interest are the rectangular and the circular sections. Microchannels with rectangular cross-sections are easily etched in silicon wafers, for example by reactive ion etching (RIE) or by deep reactive ion etching (DRIE). If the aspect ratio of the section is small enough, the flow may be considered as a plane flow between parallel plates and the modeling is much simpler. Gas flows in circular microtubes are also frequently encountered, for example in chromatography applications. Fused silica microtubes are proposed by different suppliers with a variety of internal diameters, from one micrometer to hundreds of micrometers. Due to their low cost, they are an interesting solution as connecting lines for a number of microfluidics devices or prototypes.



Pressure-Driven Single Phase Gas Flows, Figure 3 Different microchannel sections; **a** plane channel limited by parallel plates ($b \gg h$), **b** circular duct, **c** rectangular channel

The streamwise coordinate is z and the cross-section is in the (x, y) plane. The different considered sections are represented in Fig. 3.

Two different Knudsen numbers are usually defined for microchannel flows:

$$\text{Kn} = \frac{\lambda}{D_H} = \frac{\lambda}{4A/P} \quad (14)$$

is defined from the hydraulic diameter D_H as a reference length. The section area is noted A and the wetted perimeter is noted P . On the other hand,

$$\text{Kn}' = \frac{\lambda}{L_{\min}} \quad (15)$$

is defined from the minimal representative length of the section.

Key Research Findings

Pressure Driven Flows Between Parallel Plates

In the case of a plane flow (Fig. 3a), $h/b \rightarrow 0$ and the velocity only depends on the transverse coordinate y . With the previous assumptions, the momentum Eq. (8) reduces to

$$\frac{d^2 u_z}{dy^2} = \frac{1}{\mu} \frac{dp}{dz}. \quad (16)$$

The first-order boundary condition (12) for a fixed wall and an isothermal flow is $u_s|_w = A_1 \lambda (\partial u_s / \partial n)_w$, i.e. for a flow between parallel plates:

$$u_z|_{y=h} = -A_1 \lambda \left. \frac{\partial u_z}{\partial y} \right|_{y=h}, \quad (17)$$

to be associated with the condition of symmetry (cf. Fig. 3)

$$\left. \frac{du_z}{dy} \right|_{y=0} = 0. \quad (18)$$

The solution of this set of equations is

$$u_z = \frac{1}{\mu} \frac{dp}{dz} \left(\frac{y^2 - h^2}{2} - A_1 \lambda h \right). \quad (19)$$

In a dimensionless form, the velocity distribution can be written as

$$u_z^* = 1 - y^{*2} + 8A_1 \text{Kn} = 1 - y^{*2} + 4A_1 \text{Kn}', \quad (20)$$

where $y^* = y/h$, $u_z^* = u_z/u_{z0}$ and $u_{z0} = u_z(y=0, \text{Kn}=0) = -(h^2/[2\mu])(dp/dz)$ represents the velocity at the center of the section when rarefaction is not taken into account.

The integration of the velocity distribution given by Eq. (20) over the cross-section leads to the mean velocity

$$\begin{aligned} \overline{u_z^*} &= \frac{1}{2} \int_{-1}^1 u_z^* dy^* = \frac{\overline{u_z}}{u_{z0}} \\ &= \frac{2}{3} + 8A_1 \text{Kn} = \frac{2}{3} + 4A_1 \text{Kn}'. \end{aligned} \quad (21)$$

Considering a long microchannel of length l with a pressure p_i at the inlet and a pressure p_o at the outlet and neglecting the entrance effects, we can deduce from Eq. (21) the mass flow rate $\dot{m} = \rho \overline{u_z} A = p \overline{u_z} A / (RT)$, which is independent of z . According to Eq. (5), the quantity $(\text{Kn}p)$ is constant for an isothermal flow and it can

be written as a function of the outlet conditions: $\text{Kn}p = \text{Kn}_o p_o$. Therefore, the integration of Eq. (21) yields

$$\int_{p_i}^{p_o} \left(\frac{2}{3} p + 8A_1 \text{Kn}_o p_o \right) dp = -\frac{2\mu \dot{m} RT l}{h^2 A}, \quad (22)$$

which leads to

$$\dot{m}_{\text{NS1,plan}} = \frac{2bh^3 p_o^2}{\mu RT l} \left[\frac{\Pi^2 - 1}{3} + 8A_1 \text{Kn}_o (\Pi - 1) \right], \quad (23)$$

where $\Pi = p_i/p_o$ is the ratio of the inlet over outlet pressures and Kn_o is the outlet Knudsen number. The subscript NS1 refers to the Navier–Stokes equations with first-order slip boundary conditions. In a dimensionless form that compares the actual mass flow rate $\dot{m}$ with the mass flow rate $\dot{m}_{ns}$ obtained from a no-slip hypothesis, Eq. (23) becomes:

$$\begin{aligned} \dot{m}_{\text{NS1,plan}}^* &= \frac{\dot{m}_{\text{NS1,plan}}}{\dot{m}_{ns,plan}} = 1 + 24A_1 \frac{\text{Kn}_o}{\Pi + 1} \\ &= 1 + 12A_1 \frac{\text{Kn}_o'}{\Pi + 1}. \end{aligned} \quad (24)$$

The pressure distribution along the z -axis can be found by expressing the conservation of mass flow rate through the channel, notably between any section defined by z and the outlet section:

$$\begin{aligned} \dot{m}_{\text{NS1,plan}} &= \frac{2bh^3 p_o^2}{\mu RT (l - z)} \\ &\times \left[\frac{(p/p_o)^2 - 1}{3} + 8A_1 \text{Kn}_o (p/p_o - 1) \right]. \end{aligned} \quad (25)$$

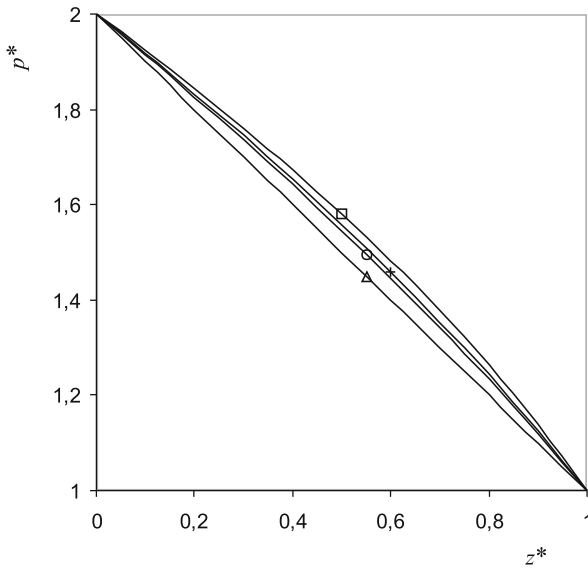
Equation (25) shows that the pressure $p^* = p/p_o$ is the solution of the polynomial

$$p^{*2} + \alpha_1 p^* + \alpha_2 + \alpha_3 z^* = 0, \quad (26)$$

where $z^* = z/l$ and $\alpha_1 = 24A_1 \text{Kn}_o$, $\alpha_2 = -\Pi(\Pi + \alpha_1)$, $\alpha_3 = (\Pi - 1)(\Pi + 1 + \alpha_1)$.

The solution of Eq. (26) is plotted in Fig. 4, for an inlet over outlet pressure ratio $\Pi = 2$. This figure shows that rarefaction and compressibility have opposite effects on the pressure distribution. Rarefaction effects (characterized by Kn) reduce the curvature in pressure distribution due to compressibility effects (characterized by Ma).

Therefore, the previous model can be improved to remain accurate for higher Knudsen numbers. For this purpose,



Pressure-Driven Single Phase Gas Flows, Figure 4 Pressure distribution along a plane microchannel, for $\Pi = 2$.

(Δ): incompressible flow; ($\square$): $\text{Kn}_0 = 0$; (+): $\text{Kn}_0 = 0.05$; ($\circ$): $\text{Kn}_0 = 0.1$

second-order – $\vartheta(\text{Kn}^2)$ – models, based on second-order boundary conditions, may be used.

It is easy to verify that the mass flow rate is consequently corrected as:

$$\begin{aligned} \dot{m}_{\text{NS2,plan}}^* &= \frac{\dot{m}_{\text{NS2,plan}}}{\dot{m}_{\text{ns,plan}}} \\ &= 1 + 24A_1 \frac{\text{Kn}_0}{\Pi + 1} - 96A_2 \text{Kn}_0^2 \frac{\ln \Pi}{\Pi^2 - 1} \\ &= 1 + 12A_1 \frac{\text{Kn}_0'}{\Pi + 1} - 24A_2 \text{Kn}_0'^2 \frac{\ln \Pi}{\Pi^2 - 1}. \end{aligned} \quad (27)$$

Pressure Driven Flows in Circular Microtubes

With the same general assumptions than in the previous section, the momentum Eq. (8) in polar coordinates reduces to

$$\frac{1}{r} \frac{d}{dr} \left(r \frac{du_z}{dr} \right) = \frac{1}{\mu} \frac{dp}{dz}. \quad (28)$$

The first-order boundary condition (12) for a non-moving wall and an isothermal flow is now (cf. Fig. 3b):

$$u_z|_{r=D/2} = -A_1 \lambda \left. \frac{\partial u_z}{\partial r} \right|_{r=D/2}, \quad (29)$$

which leads to the dimensionless velocity profile

$$u_z^* = 1 - r^{*2} + 4A_1 \text{Kn} \quad (30)$$

where the Knudsen number $\text{Kn} = \lambda/D_H = \lambda/D = \text{Kn}' = \lambda/L_{\min}$ is defined from a characteristic length $L = D$. Here, $r^* = 2r/D$ and $u_z^* = u_z/u_{z0}$ with $u_{z0} = -(D^2/[16\mu])(dp/dz)$. The integration of the velocity within a section and along the microtube leads to

$$\dot{m}_{\text{NS1,circ}} = \frac{\pi D^2 p_0^2}{16\mu R T l} \left[\frac{\Pi^2 - 1}{4} + 4A_1 \text{Kn}_0 (\Pi - 1) \right] \quad (31)$$

for a long microtube, neglecting entrance effects and

$$\dot{m}_{\text{NS1,circ}}^* = \frac{\dot{m}_{\text{NS1,circ}}}{\dot{m}_{\text{ns,circ}}} = 1 + 16A_1 \frac{\text{Kn}_0}{\Pi + 1}. \quad (32)$$

The pressure distribution $p^* = p/p_0$ along the microtube is the solution of the polynomial

$$p^{*2} + \beta_1 p^* + \beta_2 + \beta_3 z^* = 0, \quad (33)$$

where $\beta_1 = 16A_1 \text{Kn}_0$, $\beta_3 = (\Pi - 1)(\Pi + 1 + \beta_1)$ and $\beta_2 = -1 - \beta_1 - \beta_3$, which qualitatively leads to the same conclusions as for a plane flow.

The first-order solution can easily be extended to second-order boundary conditions, following the same reasoning as in the previous section. We find

$$\begin{aligned} \dot{m}_{\text{NS2,circ}}^* &= \frac{\dot{m}_{\text{NS2,circ}}}{\dot{m}_{\text{ns,circ}}} \\ &= 1 + 16A_1 \frac{\text{Kn}_0}{\Pi + 1} - 32A_2 \frac{\text{Kn}_0^2 \ln \Pi}{\Pi^2 - 1}. \end{aligned} \quad (34)$$

Pressure Driven Flows in Rectangular Microchannels

A rectangular section is characterized by its aspect ratio $a^* = h/b$, with $0 < a^* \leq 1$, assuming $h \leq b$ (Fig. 3c). Actually, calculation of the mass flow rate in a rectangular microchannel using the plane flow solution (24) or (27) is inaccurate as soon as the aspect ratio of the section a^* is higher than 10^{-2} , due to non-negligible friction on the side walls of equations $x = \pm b$. Typically, the overestimation $[(\dot{m}_{\text{plan}} - \dot{m}_{\text{rect}})/\dot{m}_{\text{rect}}]_{\text{NS2}}$ is about 0.5% for an aspect ratio $a^* = 0.01$, it rises to 5.3% for an aspect ratio $a^* = 0.1$ and reaches 112% for a square cross-section for which $a^* = 1$! As a consequence, a real 3D model is required. With the general assumptions of previous sections, the momentum equation takes the dimensionless form

$$a^{*2} \frac{d^2 u_z^*}{dx^{*2}} + \frac{d^2 u_z^*}{dy^{*2}} = -1, \quad (35)$$

with $x^* = x/b$, $y^* = y/h$ and $u_z^* = -\mu u_z/(h^2(dp/dz))$.

The first-order boundary conditions are

$$u_z^*|_{y^*=1} = -2A_1 \text{Kn}' \frac{\partial u_z^*}{\partial y^*} \Big|_{y^*=1}, \quad (36)$$

$$u_z^*|_{x^*=1} = -2a^*A_1 \text{Kn}' \frac{\partial u_z^*}{\partial x^*} \Big|_{x^*=1}, \quad (37)$$

with $\text{Kn}' = \lambda/(2h)$ and the conditions of symmetry are $(du_z^*/dy^*)_{y^*=0} = 0$ and $(du_z^*/dx^*)_{x^*=0} = 0$. The solution of Eqs. (35) to (37), compatible with the conditions of symmetry can be written as

$$u_z^* = - \sum_{i=1}^{\infty} \Psi_i(x^*) \cos(\delta_i y^*), \quad (38)$$

where the eigenvalues δ_i must verify

$$\delta_i \tan \delta_i = \frac{A_1}{2\text{Kn}'} \quad (39)$$

in order to be compatible with the boundary condition (36) without leading to a trivial velocity solution. The x^* dependent Ψ_i functions are found by expanding the right-hand side of Eq. (35), i. e. unity, in terms of the orthogonal functions $\cos(\delta_i y^*)$:

$$\begin{aligned} -1 &= - \sum_{i=1}^{\infty} \frac{2 \sin \delta_i}{\delta_i + \sin \delta_i \cos \delta_i} \cos(\delta_i y^*) \\ &= - \sum_{i=1}^{\infty} \Omega_i \cos(\delta_i y^*). \end{aligned} \quad (40)$$

Therefore, to verify the momentum Eq. (35), the functions Ψ_i must obey

$$\frac{d^2 \Psi_i}{dx^{*2}} - \left(\frac{\delta_i}{a^*} \right)^2 \Psi_i - \frac{\Omega_i}{a^{*2}} = 0. \quad (41)$$

The solution of Eq. (41), with the boundary and symmetry conditions and using Eq. (39), is

$$\Psi_i = - \frac{\Omega_i}{\alpha_i^2} \left(1 - \frac{\cosh\left(\frac{\alpha_i x^*}{a^*}\right)}{\cosh\left(\frac{\alpha_i}{a^*}\right) + 2A_1 \text{Kn}' \alpha_i \sinh\left(\frac{\alpha_i}{a^*}\right)} \right). \quad (42)$$

Consequently, the velocity distribution is given by Eq. (38), where Ψ_i and δ_i are obtained from Eqs. (40) and (42). This solution is given as a function of a Knudsen number $\text{Kn}' = \lambda/(2h)$ based on the channel depth, but it

can also be expressed as a function of the Knudsen number $\text{Kn} = \lambda/D_H$ based on the hydraulic diameter, using the relation $2\text{Kn} = \text{Kn}'(1 + a^*)$.

The solution of the problem with Deissler [10] second-order boundary conditions is detailed in [3]. These boundary conditions may be written in a dimensionless form as

$$\begin{aligned} u_z^*|_{y^*=1} &= -2A_1 \text{Kn}' \frac{\partial u_z^*}{\partial y^*} \Big|_{y^*=1} \\ &\quad - \frac{9}{4} \text{Kn}'^2 \left(\frac{\partial^2 u_z^*}{\partial y^{*2}} \Big|_{y^*=1} - 1 \right) \end{aligned} \quad (43)$$

and

$$\begin{aligned} u_z^*|_{x^*=1} &= -2A_1 a^* \text{Kn}' \frac{\partial u_z^*}{\partial x^*} \Big|_{x^*=1} \\ &\quad - \frac{9}{4} \text{Kn}'^2 \left(a^{*2} \frac{\partial^2 u_z^*}{\partial x^{*2}} \Big|_{x^*=1} - 1 \right). \end{aligned} \quad (44)$$

for a locally fully developed flow in a rectangular section. A semi-analytical solution for the mass flow is found as [3]:

$$\dot{m}_{\text{NS2,rect}} = - \frac{4ph^4}{a^*RT\mu} \frac{dp}{dz} \left[a_1 + a_2 \text{Kn}' + a_3 \text{Kn}'^2 \right]. \quad (45)$$

Values of a_1 , a_2 and a_3 coefficients are given in Table 4, for different aspect ratios and values of A_1 .

Temperature Jump and Heat Transfer

As for the velocity, it can be shown that the temperature experiences a jump at the wall, which can be written

$$\begin{aligned} T^* - T_{\text{wall}}^* &= \beta_T \frac{2 - \sigma_T}{\sigma_T} \frac{2\gamma}{\gamma + 1} \frac{\text{Kn}}{\text{Pr}} \frac{\partial T^*}{\partial n^*} \Big|_w \\ &= B_1 \frac{2\gamma}{\gamma + 1} \frac{\text{Kn}}{\text{Pr}} \frac{\partial T^*}{\partial n^*} \Big|_w, \end{aligned} \quad (46)$$

where Pr is the Prandtl number and σ_T is the energy accommodation coefficient; β_T is a corrective coefficient introduced by Cercignani to improve the original formula proposed by Smoluchowski. This temperature jump must be taken into account as well as the velocity slip if the flow is not isothermal and if heat transfer is considered ([►convective heat transfer in microchannels](#)). For example, for a hydrodynamically and thermally developed flow in a plane microchannel (Fig. 3a) with a lower adiabatic

Pressure-Driven Single Phase Gas Flows, Table 4 Values of a_1 , a_2 and a_3 coefficients for different values of A_1 and a^* , for Deissler second-order boundary conditions

a^*	A_1	3.0000	2.3333	1.8571	1.5000	1.2222	1.0000
0		$a_2 = 6.0000$	$a_2 = 4.6667$	$a_2 = 3.7143$	$a_2 = 3.0000$	$a_2 = 2.4444$	$a_2 = 2.0000$
	$a_1 = 0.33333$	$a_3 = 4.5000$	$a_3 = 4.5000$	$a_3 = 4.5000$	$a_3 = 4.5000$	$a_3 = 4.5000$	$a_3 = 4.5000$
0.01		$a_2 = 5.9694$	$a_2 = 4.6459$	$a_2 = 3.7007$	$a_2 = 2.9917$	$a_2 = 2.4405$	$a_2 = 1.9993$
	$a_1 = 0.33123$	$a_3 = 4.4528$	$a_3 = 4.4542$	$a_3 = 4.4548$	$a_3 = 4.4550$	$a_3 = 4.4549$	$a_3 = 4.4549$
0.025		$a_2 = 5.9228$	$a_2 = 4.6141$	$a_2 = 3.6797$	$a_2 = 2.9788$	$a_2 = 2.4337$	$a_2 = 1.9977$
	$a_1 = 0.32808$	$a_3 = 4.3840$	$a_3 = 4.3875$	$a_3 = 4.3887$	$a_3 = 4.3895$	$a_3 = 4.3896$	$a_3 = 4.3892$
0.05		$a_2 = 5.8420$	$a_2 = 4.5591$	$a_2 = 3.6423$	$a_2 = 2.9547$	$a_2 = 2.4201$	$a_2 = 1.9925$
	$a_1 = 0.32283$	$a_3 = 4.2761$	$a_3 = 4.2813$	$a_3 = 4.2843$	$a_3 = 4.2858$	$a_3 = 4.2858$	$a_3 = 4.2853$
0.1		$a_2 = 5.6774$	$a_2 = 4.4431$	$a_2 = 3.5613$	$a_2 = 2.9000$	$a_2 = 2.3860$	$a_2 = 1.9751$
	$a_1 = 0.31233$	$a_3 = 4.0697$	$a_3 = 4.0832$	$a_3 = 4.0906$	$a_3 = 4.0943$	$a_3 = 4.0952$	$a_3 = 4.0944$
0.25		$a_2 = 5.1332$	$a_2 = 4.0464$	$a_2 = 3.2675$	$a_2 = 2.6823$	$a_2 = 2.2267$	$a_2 = 1.8626$
	$a_1 = 0.28081$	$a_3 = 3.6073$	$a_3 = 3.6287$	$a_3 = 3.6449$	$a_3 = 3.6560$	$a_3 = 3.6626$	$a_3 = 3.6647$
0.5		$a_2 = 4.2443$	$a_2 = 3.3519$	$a_2 = 2.7120$	$a_2 = 2.2289$	$a_2 = 1.8487$	$a_2 = 1.5395$
	$a_1 = 0.22868$	$a_3 = 3.3489$	$a_3 = 3.3588$	$a_3 = 3.3708$	$a_3 = 3.3856$	$a_3 = 3.4048$	$a_3 = 3.4292$
1		$a_2 = 3.1207$	$a_2 = 2.4552$	$a_2 = 1.9782$	$a_2 = 1.6187$	$a_2 = 1.3364$	$a_2 = 1.1077$
	$a_1 = 0.14058$	$a_3 = 3.3104$	$a_3 = 3.3176$	$a_3 = 3.3287$	$a_3 = 3.3438$	$a_3 = 3.3644$	$a_3 = 3.3908$

wall and an upper wall with constant heat flux, the Nusselt number Nu can be calculated and is given by

$$Nu^{-1} = \frac{\zeta^*}{4} + \frac{26 + 147\xi^* + 210\xi^{*2}}{140(1 + 3\xi^*)^2}, \quad (47)$$

where $\xi^* = 4A_1Kn$ and $\zeta^* = (8\gamma)/(\gamma + 1)B_1(Kn/Pr)$ are the dimensionless slip and temperature jump coefficients. This Nusselt number reduces to the classic value $70/13$ when $Kn = 0$.

Future Directions for Research

Theoretical knowledge on pressure-driven gas microflows is currently well advanced, especially in the slip-flow regime. However, there is yet a need for accurate experimental data, both for steady or unsteady gas microflows, with or without heat transfer. For example, in order to definitely validate the choice of the best boundary conditions in the slip flow regime, we need to isolate the influence of the accommodation coefficients, as it remains an open issue. Relationships between their values, the nature of the substrate and the micro fabrication processes involved are currently not available. There are also few data about flows of gas mixtures in microchannels.

Cross References

- Micro-PIV-Based Diffusometry
- Convective Heat Transfer in Microchannels
- Gas Flow in Nanochannels
- Lattice Boltzmann Method (LBM)

- Methods for Flow Rate Measurement
- Micro and Nanoscale Gas Dynamics
- Molecular Dynamics Simulation Method
- Monte Carlo Method
- Single Phase Gaseous Flow

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Pressure-Driven Single Phase Liquid Flows

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Definition

The forced mass movement of single phase liquids through microchannels with a hydraulic diameter less than 1 mm induced by a pressure gradient imposed at the ends of the microchannel is called pressure-driven single phase liquid flow.

Overview

Investigation of the flow phenomena in pipes and channels is the most important subject of hydraulics. In the middle of the nineteenth century an exact solution of the hydrodynamic equations was derived for the flow of a fluid through straight tubes of circular cross-section by Hagen [1] and Poiseuille [2] independently. It is interesting to highlight that the original experimental work of Poiseuille was conducted by testing water through glass capillary tubes having an internal diameter down to 15 μm . This work can be considered the first experimental test of pressure-driven single phase liquid flows in microchannels.

Basic Methodology

The behavior of isothermal pressure-driven single phase fluid flows in microchannels can be studied by determining the velocity distribution in the fluid region with the aid of the mass conservation principle (*continuity equation*) and the equations of conservation of fluid momentum [3]:

$$\begin{aligned} -\frac{1}{\rho} \frac{D\rho}{Dt} &= \nabla \cdot \mathbf{v} \\ \rho \frac{D\mathbf{v}}{Dt} &= -\nabla p + \rho \mathbf{g} - \nabla \cdot \boldsymbol{\tau} \end{aligned} \quad (1)$$

in which $\mathbf{v}$ is the velocity vector, ρ is the fluid density, t is the time, p is the pressure, D/Dt is the total derivative and $\boldsymbol{\tau}$ is the viscous tensor.

In Table 1 the components of the viscous tensor for Newtonian fluids are reported for Cartesian, cylindrical and spherical coordinates.

In order to solve the continuity and momentum balance equations (Eq. (1)) the following *constitutive laws* have to

Pressure-Driven Single Phase Liquid Flows, Table 1 Components of the viscous tensor for Cartesian, cylindrical and spherical coordinates

Cartesian coordinates (x, y, z)	$\tau_{xx} = -\mu \left[2 \frac{\partial v_x}{\partial x} - \frac{2}{3} (\nabla \cdot \mathbf{v}) \right]$
	$\tau_{yy} = -\mu \left[2 \frac{\partial v_y}{\partial y} - \frac{2}{3} (\nabla \cdot \mathbf{v}) \right]$
	$\tau_{zz} = -\mu \left[2 \frac{\partial v_z}{\partial z} - \frac{2}{3} (\nabla \cdot \mathbf{v}) \right]$
	$\tau_{xy} = \tau_{yx} = -\mu \left[\frac{\partial v_x}{\partial y} + \frac{\partial v_y}{\partial x} \right]$
	$\tau_{yz} = \tau_{zy} = -\mu \left[\frac{\partial v_y}{\partial z} + \frac{\partial v_z}{\partial y} \right]$
	$\tau_{zx} = \tau_{xz} = -\mu \left[\frac{\partial v_z}{\partial x} + \frac{\partial v_x}{\partial z} \right]$
Cylindrical Coordinates (r, θ, z)	$\tau_{rr} = -\mu \left[2 \frac{\partial v_r}{\partial r} - \frac{2}{3} (\nabla \cdot \mathbf{v}) \right]$
	$\tau_{\theta\theta} = -\mu \left[2 \left(\frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r} \right) - \frac{2}{3} (\nabla \cdot \mathbf{v}) \right]$
	$\tau_{zz} = -\mu \left[2 \frac{\partial v_z}{\partial z} - \frac{2}{3} (\nabla \cdot \mathbf{v}) \right]$
	$\tau_{r\theta} = \tau_{\theta r} = -\mu \left[r \frac{\partial}{\partial y} \left(\frac{v_\theta}{r} \right) + \frac{1}{r} \frac{\partial v_r}{\partial \theta} \right]$
	$\tau_{\theta z} = \tau_{z\theta} = -\mu \left[\frac{\partial v_\theta}{\partial z} + \frac{1}{r} \frac{\partial v_z}{\partial \theta} \right]$
	$\tau_{zr} = \tau_{rz} = -\mu \left[\frac{\partial v_z}{\partial r} + \frac{\partial v_r}{\partial z} \right]$
Spherical coordinates (r, θ, ϕ)	$\tau_{rr} = -\mu \left[2 \frac{\partial v_r}{\partial r} - \frac{2}{3} (\nabla \cdot \mathbf{v}) \right]$
	$\tau_{\theta\theta} = -\mu \left[2 \left(\frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r} \right) - \frac{2}{3} (\nabla \cdot \mathbf{v}) \right]$
	$\tau_{\phi\phi} = -\mu \left[2 \left(\frac{1}{r \sin \theta} \frac{\partial v_\phi}{\partial \phi} + \frac{v_r}{r} + \frac{v_\theta \cot \theta}{r} \right) - \frac{2}{3} (\nabla \cdot \mathbf{v}) \right]$
	$\tau_{r\theta} = \tau_{\theta r} = -\mu \left[r \frac{\partial}{\partial y} \left(\frac{v_\theta}{r} \right) + \frac{1}{r} \frac{\partial v_r}{\partial \theta} \right]$
	$\tau_{\theta\phi} = \tau_{\phi\theta} = -\mu \left[\frac{\sin \theta}{r} \frac{\partial}{\partial \theta} \left(\frac{v_\phi}{\sin \theta} \right) + \frac{1}{r \sin \theta} \frac{\partial v_r}{\partial \phi} \right]$
	$\tau_{\phi r} = \tau_{r\phi} = -\mu \left[\frac{1}{r \sin \theta} \frac{\partial v_r}{\partial \phi} + r \frac{\partial}{\partial r} \left(\frac{v_\phi}{r} \right) \right]$

be taken into account:

$$\begin{cases} \rho = \rho(p, T) \\ \mu = \mu(p, T) \end{cases} \quad (2)$$

If the flow is isothermal, Eqs. (1) and (2) allow the calculation of the pressure field and of the velocity vector ($p, \mathbf{v}$) in the fluid region.

It is possible to demonstrate that, for Newtonian fluids with constant viscosity μ , the following equality is satisfied:

$$(\nabla \cdot \boldsymbol{\tau})_i = -\mu \left[\nabla^2 v_i + \frac{1}{3} \frac{\partial}{\partial i} (\nabla \cdot \mathbf{v}) \right] \quad (3)$$

where $i = x, y, z$ for Cartesian coordinates.

Combining Eq. (3) and the momentum equation one obtains

$$\rho \frac{D\mathbf{v}}{Dt} = -\nabla p + \rho \mathbf{g} + \mu \left[\nabla^2 \mathbf{v} + \frac{1}{3} \nabla (\nabla \cdot \mathbf{v}) \right] \quad (4)$$

This version of the momentum equation is known as the *Navier–Stokes equation*.

For an incompressible ($\rho = \text{const}$) and Newtonian fluid with constant viscosity ($\mu = \text{const}$) the continuity and the momentum balance equations can be written as follows:

$$\nabla \cdot \mathbf{v} = 0$$

$$\rho \frac{D\mathbf{v}}{Dt} = -\nabla p + \rho \mathbf{g} + \mu \nabla^2 \mathbf{v} \quad (5)$$

The momentum balance equation can be simplified in the case of very slow fluid motion. In this case the total derivative of the velocity and the acceleration due to the gravity can be neglected in the *Navier–Stokes equation*:

$$\mu \nabla^2 \mathbf{v} = \nabla p \quad (6)$$

Equation (6) applies to the analysis of fluid behavior in tribology and microchannel flows: this kind of flow is called *Stokes flow*. In this case, the velocities are free of macroscopic fluctuations at any point in the flow field and the flow is defined as *laminar*.

For a laminar flow through a straight microchannel the fluid particles move in definite paths called streamlines and there are no components of fluid velocity normal to the duct axis. The projection of Eq. (6) along the axial direction (z) gives

$$\mu \nabla^2 v_z = \frac{dp}{dz} \quad (7)$$

Equations (6) and (7) are valid only far from the inlet section of the microchannel (see ► [developing flows](#) and ► [entrance region](#)) where the velocity profile can be considered hydrodynamically fully developed. Laminar flow is designated as hydrodynamically fully developed when the fluid velocity distribution at a cross-section is of an invariant form independent of the axial distance z .

Laminar Regime

Let us consider a liquid flowing through a straight microchannel having an axially unchanging and uniform cross-section with an area equal to Ω and the perimeter equal to Γ . It is possible to define the *hydraulic diameter* of the microchannel as follows:

$$D_h = \frac{4\Omega}{\Gamma} \quad (8)$$

and the *mean fluid velocity* as the integrated average axial velocity over the flow area Ω :

$$W = \frac{1}{\Omega} \int_{\Omega} v_z d\Omega = \frac{\dot{m}}{\rho \Omega} \quad (9)$$

in which the link with measurable quantities, like the mass flow rate, is highlighted.

It is suitable to introduce the dimensionless quantities

$$\Omega^* = \frac{\Omega}{D_h^2}; \quad \Gamma^* = \frac{\Gamma}{D_h}; \quad \nabla^* = D_h \nabla; \\ V_z = \frac{v_z}{W}; \quad c_1 = -\frac{D_h^2}{\mu W} \frac{dp}{dz} \quad (10)$$

Consequently, the dimensionless momentum balance equation is readily obtained in the following form:

$$\nabla^{*2} V_z + c_1 = 0 \quad (11)$$

In order to characterize a liquid flow through microchannels under an imposed pressure gradient Eq. (11) can be used together with the appropriate boundary condition that, for Newtonian liquids, is the no-slip boundary condition at the walls:

$$V_z|_{\Gamma^*} = 0 \quad (12)$$

For a Newtonian liquid flow, this assumption is justified if the microchannel has a hydraulic diameter larger than $1 \mu\text{m}$. In fact, for liquids the typical mean free path λ of molecules under ambient conditions is $0.1 - 1 \text{ nm}$. Since the fluid velocity tends to evidence a slip at the microchannel walls for Knudsen numbers, defined as $\text{Kn} = \lambda/D_h$, larger than 0.001 , the no-slip boundary condition has to be abandoned only when the hydraulic diameter of the microchannel becomes less than $1 \mu\text{m}$.

On the contrary, for non-Newtonian liquids, like polymer solutions and surfactants, a slip velocity at the walls can be observed even for microchannels having a hydraulic diameter greater than $1 \mu\text{m}$.

The solution of Eqs. (11) and (12) yields the velocity distribution inside the microchannel in the laminar regime for fully developed flow. The analytical solution of this differential problem is available only for very simple geometries.

From the velocity distribution obtained by the solution of Eqs. (11) and (12) it is possible to calculate the value assumed by the main flow parameters which can be useful during the design stage of a microfluidic device:

- the ratio between the maximum and the mean value of the velocity (V_{max}):

$$V_{\max} = \frac{v_{z,\max}}{W} \quad (13)$$

- the asymptotic value of the *momentum flux correction factor* K_d defined as:

$$K_d = \frac{1}{\Omega^*} \int_{\Omega^*} V_z^2 d\Omega^* \quad (14)$$

- the asymptotic *kinetic energy correction factor* K_e defined as:

$$K_e = \frac{1}{\Omega^*} \int_{\Omega^*} V_z^3 d\Omega^* \quad (15)$$

- the asymptotic *incremental pressure drop number* $K(\infty)$:

$$K(\infty) = \frac{2}{\Omega^*} \int_{\Omega^*} \left[(V_z^3 - V_z^2) \right] d\Omega^* = 2(K_e - K_d) \quad (16)$$

- the *Poiseuille number* defined as the product of the Fanning friction factor for fully developed flow (f) and the Reynolds number ($Re = \rho W D_h / \mu$):

$$(fRe)_{fd} = -\frac{1}{2\Omega^*} \int_{\Gamma^*} \frac{\partial V_z}{\partial n^*} \Big|_{\Gamma^*} d\Gamma^* = \frac{c_1}{2} \quad (17)$$

where n^* is the normal direction to the walls.

For Newtonian laminar flows these parameters depend on the geometry of the microchannel cross-section only.

Circular Cross-Section

For a channel with a circular cross-section it is possible to demonstrate that the velocity distribution, written in dimensionless form, is

$$V_z(r^*) = 2(1 - 4r^{*2}) \quad (18)$$

in which r^* is the dimensionless radius defined as r/D_h with D_h being the inner diameter.

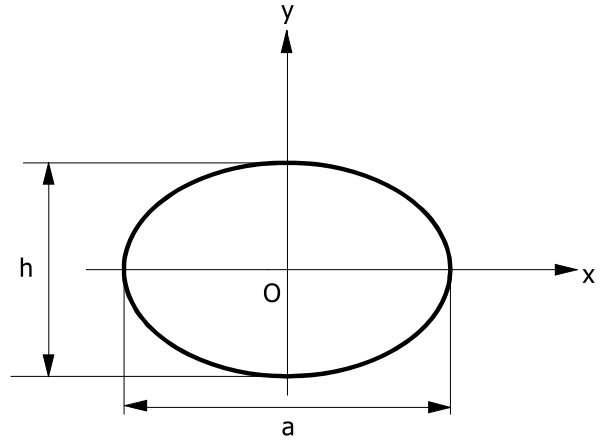
In Table 2 the values assumed by $V_{\max}$, K_d , K_e , $K(\infty)$ and fRe_{fd} are quoted for this geometry.

Elliptical Cross-Section

Microtubes with elliptical cross-section in which fully developed laminar flow occurs are geometrically characterized by the value of the aspect ratio β defined as ratio between the minor (h) and the major (a) axes (see Fig. 1).

Pressure-Driven Single Phase Liquid Flows, Table 2 Analytical laminar fully developed values of $V_{\max}$, K_d , K_e , $K(\infty)$ and of the Poiseuille number (fRe_{fd}) for circular microtubes

$V_{\max}$	K_d	K_e	$K(\infty)$	fRe_{fd}
2	4/3	2	4/3	16



Pressure-Driven Single Phase Liquid Flows, Figure 1 Schematic of the elliptical cross-section and of the Cartesian coordinate system

The fully developed laminar velocity profile is

$$V_z(x^*, y^*) = 8 \left(\frac{1}{4} - x^{*2} - y^{*2} \right) \quad (19)$$

in which x^* and y^* are dimensionless coordinates defined as x/a and y/h respectively and the Cartesian coordinate system has its origin at the center of the channel (see Fig. 1).

It is possible to demonstrate by using Eq. (19) that $V_{\max}$, K_d , K_e and $K(\infty)$ assume the same values as those for circular cross-section (see Table 2) for any β .

The fully developed Poiseuille number can be calculated analytically by means of the following relation:

$$fRe_{fd} = 2\pi^2 (1 + \beta^2) \left[\frac{1}{E(1 - \beta^2)} \right]^2 \quad (20)$$

where E denotes the complete elliptical integral of the second kind.

In Table 3 the value of the fully developed Poiseuille number is quoted as a function of the aspect ratio β of the elliptical cross-section.

Concentric Annular Cross-Section

The solution of Eq. (11) with the no-slip boundary condition (Eq. (12)) through an annular channel with an inner radius equal to r_i and an outer radius equal to r_o (see Fig. 2) can be written as follows:

$$V_z(r^*) = \frac{2}{[1 + \beta^2 - 2r_m^{*2}]} \left(1 - r^{*2} + 2r_m^{*2} \ln(r^*) \right) \quad (21)$$

in which β is the aspect ratio of the annuli defined as r_i/r_o , r^* is the dimensionless radius defined as r/r_o and r_m^* is

given by

$$r_m^* = \left[\frac{1 - \beta^2}{2 \ln(\frac{1}{\beta})} \right]^{\frac{1}{2}} \quad (22)$$

and designates the dimensionless radius corresponding to the maximum value of the velocity.

For this geometry a closed form solution for $V_{\max}$ and the Poiseuille number is known:

$$\begin{aligned} V_{\max} &= 2 \frac{(1 - r_m^{*2} + 2r_m^{*2} \ln(r_m^*))}{1 + \beta^2 - 2r_m^{*2}} \\ f_i Re_{fd} &= 16 \frac{(1 - \beta)}{\beta (1 + \beta^2 - 2r_m^{*2})} (r_m^{*2} - \beta^2) \\ f_o Re_{fd} &= 16 \frac{(1 - \beta)}{(1 + \beta^2 - 2r_m^{*2})} (1 - r_m^{*2}) \\ f Re_{fd} &= \frac{16}{(1 + \beta^2 - 2r_m^{*2})} (1 - r_m^{*2}) \end{aligned} \quad (23)$$

Pressure-Driven Single Phase Liquid Flows, Figure 2 Schematic of the annular cross-section

Pressure-Driven Single Phase Liquid Flows, Table 3 Analytical laminar fully developed values of $V_{\max}$, K_d , K_e , $K(\infty)$ and of the Poiseuille number (fRe_{fd}) for elliptical microtubes as a function of the aspect ratio β

β	fRe_{fd}	β	fRe_{fd}
1	16	0.3	17.896
0.9	16.022	0.25	18.24
0.8	16.098	0.2	18.602
0.75	16.161	1/6	18.847
0.7	16.244	1/7	19.02
2/3	16.311	1/8	19.146
0.6	16.479	1/9	19.241
0.5	16.823	1/10	19.314
0.4	17.294	1/16	19.536
1/3	17.681	1/20	19.598

where f_i and f_o designate the Fanning friction factors at the inner and outer walls respectively and fRe_{fd} is the perimeter average Poiseuille number:

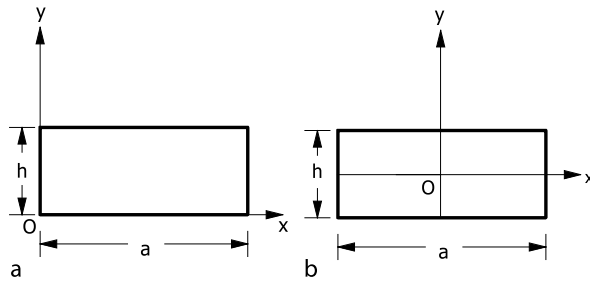
$$fRe_{fd} = \frac{\beta f_i Re + \beta f_o Re}{1 + \beta} \quad (24)$$

In Table 4 the value assumed by $V_{\max}$, K_d , K_e and fRe_{fd} are quoted for this geometry as a function of the aspect ratio β . It is evident that, the circular geometry can be considered as the limit case for $\beta = 0$ and the parallel plate geometry for $\beta = 1$.

Rectangular Cross-Section

For a rectangular channel with an aspect ratio $\beta = h/a$ (see Fig. 3a), for which a Cartesian system of coordinates x , y , z is assumed, with its origin in the left bottom corner of the inlet rectangular cross-section (y along the side with length h and z perpendicular to the cross-section), the analytical solution of Eqs. (11) and (12) gives the following velocity distribution:

$$\begin{aligned} V_z(x^*, y^*) &= \sum_{n=1, \text{odd}}^{\infty} \sum_{m=1, \text{odd}}^{\infty} v_{n,m} \sin\left(n\pi \frac{2\beta}{1 + \beta} x^*\right) \\ &\quad \times \sin\left(m\pi \frac{2}{1 + \beta} y^*\right) \end{aligned} \quad (25)$$



Pressure-Driven Single Phase Liquid Flows, Figure 3 Schematic of the rectangular cross-section with the Cartesian coordinate system used in (a) Eq. (25) and (b) Eq. (28)

Pressure-Driven Single Phase Liquid Flows, Table 4 Analytical laminar fully developed values of r_m^* , $V_{\max}$, K_d , K_e , and of the inner, outer and average Poiseuille number (fRe_{fd}) for annular microtubes as a function of the aspect ratio β

β	r_m^*	$V_{\max}$	K_d	K_e	$f_1 Re$	$f_0 Re$	fRe_{fd}
0.0001	0.233	1.767	1.298	1.863	9742.834	16.973	17.945
0.001	0.269	1.725	1.287	1.823	1352.776	17.337	18.671
0.01	0.329	1.661	1.266	1.753	219.426	18.034	20.028
0.05	0.408	1.598	1.243	1.675	74.459	18.923	21.567
0.1	0.464	1.567	1.230	1.634	50.886	19.489	22.343
0.15	0.508	1.549	1.222	1.610	42.027	19.904	22.790
0.2	0.546	1.537	1.217	1.594	37.264	20.253	23.088
0.3	0.615	1.522	1.210	1.573	32.166	20.850	23.461
0.4	0.677	1.513	1.206	1.561	29.437	21.375	23.678
0.5	0.736	1.508	1.204	1.555	27.719	21.859	23.813
0.6	0.791	1.504	1.202	1.549	26.529	22.318	23.897
0.75	0.872	1.501	1.200	1.546	25.295	22.971	23.967
0.8	0.898	1.501	1.200	1.544	24.978	23.182	23.980
0.9	0.950	1.500	1.200	1.543	24.440	23.596	23.996

in which the series coefficients $v_{n,m}$ are defined as

$$v_{n,m} = \frac{\pi^2}{4mn(\beta^2 n^2 + m^2)} \frac{1}{\sum_{i=1, \text{odd}}^{\infty} \sum_{j=1, \text{odd}}^{\infty} \frac{1}{i^2 j^2 (\beta^2 i^2 + j^2)}} \quad (26)$$

and x^* and y^* are the dimensionless coordinates defined as x/D_h and y/D_h respectively.

It is easy to demonstrate that for a rectangular cross-section the hydraulic diameter D_h , the aspect ratio β and the length of the sides (a, h) are correlated:

$$a = \frac{1+\beta}{2\beta} D_h; \quad h = \frac{1+\beta}{2} D_h \quad (27)$$

Since Eq. (25) involves considerable computational complexity, a simple approximation in the following form, validated for $\beta \leq 0.5$, can be suggested:

$$V_z(x^*, y^*) = \left(\frac{w+1}{w} \right) \left(\frac{k+1}{k} \right) \left[1 - \left(\frac{4}{1+\beta} y^* \right)^w \right] \times \left[1 - \left(\frac{4\beta}{1+\beta} x^* \right)^k \right] \quad (28)$$

but in this case the origin of the Cartesian system of coordinates x, y, z is assumed at the center of the cross-section ($a/2, h/2$) (see Fig. 3b).

The exponents w and k can be calculated by means of the following empirical correlations:

$$k = 1.7 + 0.5\beta^{-1.4}$$

$$w = \begin{cases} 2 & \text{for } \beta \leq \frac{1}{3} \\ 2 + 0.3 \left(\beta - \frac{1}{3} \right) & \text{for } \beta > \frac{1}{3} \end{cases} \quad (29)$$

In Table 5 the values assumed by $V_{\max}$, K_d , K_e , $K(\infty)$ and fRe_{fd} are quoted for this geometry as a function of the microchannel aspect ratio β . It is evident that, owing to the symmetry of the cross-section, for microchannels having β greater than 1, one can use $1/\beta$ in order to calculate the integral parameters from Table 5.

Parallel Plates

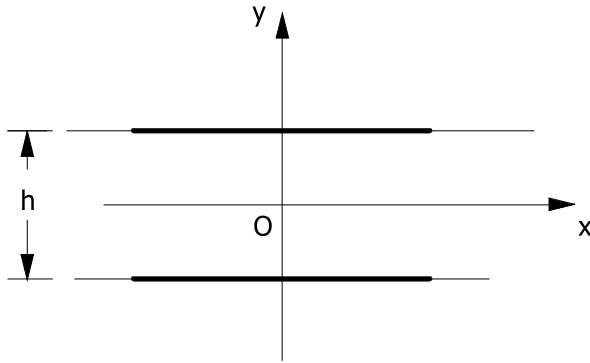
Very shallow rectangular microchannels with $h \ll a$ (or, for symmetry, in the case that $h \gg a$) are characterized by low values of the aspect ratio ($\beta = 0$); in this case the rectangular microchannel can be modeled as two parallel plates. It is possible to demonstrate that for parallel plates with a plate spacing equal to h , the fully developed laminar velocity profile can be written as follows:

$$V_z(y^*) = \frac{3}{2} \left[1 - 16(y^*)^2 \right] \quad (30)$$

in which y^* is the dimensionless coordinate defined as y/D_h with $D_h = 2h$, as obtained by Eq. (27) for $\beta = 0$. As depicted in Fig. 4, in this case the origin of the Cartesian coordinate system is assumed as placed at the center of the channel.

Pressure-Driven Single Phase Liquid Flows, Table 5 Laminar fully developed values of $V_{\max}$, K_d , K_e , $K(\infty)$ and of the Poiseuille number (fRe_{fd}) for rectangular microchannels as a function of the aspect ratio β

β	$V_{\max}$	K_d	K_e	$K(\infty)$	fRe
0	1.500	1.200	1.543	0.686	24.000
0.01	1.510	1.204	1.554	0.7	23.677
0.05	1.549	1.218	1.599	0.762	22.477
0.1	1.601	1.237	1.656	0.838	21.169
0.2	1.715	1.271	1.770	0.998	19.071
0.3	1.829	1.303	1.878	1.15	17.512
0.4	1.924	1.328	1.969	1.282	16.368
0.5	1.992	1.347	2.039	1.384	15.548
0.6	2.038	1.361	2.088	1.454	14.980
0.7	2.067	1.370	2.121	1.502	14.605
0.8	2.085	1.375	2.141	1.532	14.378
0.9	2.094	1.378	2.151	1.546	14.261
1	2.096	1.379	2.154	1.55	14.227



Pressure-Driven Single Phase Liquid Flows, Figure 4 Schematic of the parallel plate cross-section with its Cartesian coordinate system

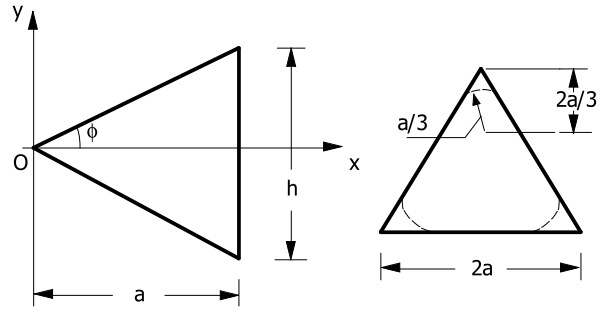
The values assumed by $V_{\max}$, K_d , K_e , $K(\infty)$ and fRe_{fd} are given in Table 5 for $\beta = 0$.

Triangular Channels

The fully developed laminar velocity profile for isosceles triangular ducts (see Fig. 5a) can be in general determined by solving numerically Eq. (11) with Eq. (12). Shah and London [4] reported the following approximate closed-form solution for the velocity profile for isosceles triangular channels:

$$V_z(x^*, y^*) = \frac{3(B+2)}{(B-2)} \left[4y^{*2} - x^{*2} \right] \left[x^{*B-2} - 1 \right] \quad (31)$$

$$B = \left[4 + \frac{5}{2} \left(\frac{1}{\tan^2 \phi} - 1 \right) \right]^{\frac{1}{2}}$$



Pressure-Driven Single Phase Liquid Flows, Figure 5 Schematic of (a) the isosceles triangular cross-section and (b) an equilateral triangular microchannel with rounded corners

where x^* and y^* are dimensionless coordinates defined as x/a and y/h (see Fig. 5a). For the isosceles triangular ducts the hydraulic diameter can be calculated by means of the following equation:

$$D_h = \frac{a \sin \phi}{1 + \sin \phi} = \frac{h \cos \phi}{1 + \sin \phi} \quad (32)$$

It can be demonstrated that the maximum value of the velocity occurs at $y^* = 0$ (for the symmetry) and

$$x_m^* = \sqrt{\frac{B-2}{B}} \quad (33)$$

The fully developed Poiseuille number can be calculated by means of the following equation:

$$fRe_{fd} = \frac{12(B+2)(1 - \tan^2 \phi)}{(B-2) \left[\tan^2 \phi + (1 + \tan^2 \phi)^{\frac{1}{2}} \right]^2} \quad (34)$$

The numerical results obtained by Shah and London [4] for $V_{\max}$, K_d , K_e , $K(\infty)$ and fRe_{fd} for this geometry are quoted in Table 6.

Because the manufacturing processes employed sometimes produces microchannels of triangular cross-section with one, two or three rounded corners instead of sharp corners, in Table 7 the numerical results obtained by Shah and London [4] for an equilateral triangular microchannel with rounded corners in terms of $V_{\max}$, K_d , K_e , $K(\infty)$ and fRe_{fd} are quoted. The radius of each rounded corner is considered as $a/3$, as shown in Fig. 5b, with $2a$ as the side length of the base triangle.

Trapezoidal Channels

It is well known that with a photolithographic-based process one can obtain microchannels having a cross-section fixed by the orientation of the wafer crystal planes; for example, the microchannels etched in (100) silicon using

Pressure-Driven Single Phase Liquid Flows, Table 6 Laminar fully developed values of x_m^* , $V_{\max}$, K_d , K_e and of the Poiseuille number (fRe_{fd}) for isosceles triangular microchannels as a function of the apex angle ϕ or the aspect ratio a/h

a/h	ϕ	$V_{\max}$	x_m^*	K_d	K_e	fRe_{fd}
8.003	3.58	2.593	0.896	1.545	2.805	12.352
5.715	5.00	2.521	0.871	1.526	2.731	12.474
4.000	7.13	2.442	0.839	1.505	2.64	12.636
2.836	10.00	2.368	0.805	1.482	2.546	12.822
2.000	14.04	2.302	0.766	1.459	2.454	13.026
1.866	15.00	2.29	0.758	1.455	2.438	13.065
1.500	18.44	2.259	0.732	1.443	2.392	13.181
1.374	20.00	2.249	0.722	1.439	2.377	13.222
1.072	25.00	2.228	0.692	1.431	2.347	13.307
1.003	26.50	2.225	0.684	1.43	2.342	13.321
0.866	30.00	2.222	0.667	1.429	2.338	13.333
0.750	33.69	2.225	0.650	1.43	2.342	13.321
0.714	35.00	2.227	0.645	1.431	2.345	13.311
0.596	40.00	2.241	0.626	1.436	2.366	13.248
0.500	45.00	2.264	0.608	1.445	2.4	13.153
0.289	60.00	2.38	0.565	1.489	2.571	12.744
0.251	63.34	2.416	0.559	1.499	2.617	12.622
0.134	75.00	2.587	0.530	1.543	2.815	12.226
0.125	75.97	2.605	0.527	1.548	2.835	12.196

Pressure-Driven Single Phase Liquid Flows, Table 7 Laminar fully developed values of $V_{\max}$, K_d , K_e and of the Poiseuille number (fRe_{fd}) for equilateral triangular microchannels with one, two, three rounded corners

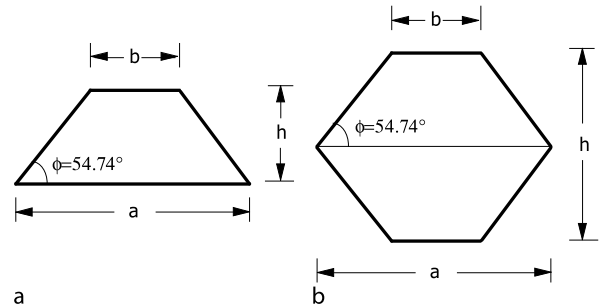
	$V_{\max}$	K_d	K_e	fRe_{fd}
No rounded corners	2.222	1.429	2.338	13.333
One rounded corner	2.172	1.406	2.254	14.057
Two rounded corners	2.115	1.379	2.163	14.899
Three rounded corners	2.064	1.353	2.074	15.993

a KOH solution have a trapezoidal cross-section with an apex angle of 54.74° imposed by the crystallographic morphology of the $\langle 100 \rangle$ silicon. By gluing together two KOH-etched trapezoidal silicon microchannels one can obtain microchannels having a double-trapezoidal cross-section (see Fig. 6). These cross-sectional geometries are investigated in this section.

Only numerical solutions of Eq. (11) have appeared in the open literature for this kind of geometry.

In Tables 8 and 9 the numerical results for KOH-etched trapezoidal and KOH-etched double trapezoidal microchannels respectively in terms of $V_{\max}$, K_d , K_e , $K(\infty)$ and fRe_{fd} are given.

Two different definitions of the aspect ratio are used in the literature for trapezoidal channels ($\beta = h/a$ or $\gamma = h/b$ with reference to the symbols quoted in Fig. 6a). In Table 8, both β and γ are quoted in order to simplify the



Pressure-Driven Single Phase Liquid Flows, Figure 6 (a) KOH-etched trapezoidal cross-section and (b) KOH-etched double-trapezoidal cross-section

use of the data. For a $\langle 100 \rangle$ silicon microchannel the aspect ratio β cannot exceed the value of $\tan(\phi)/2$ (equal to 0.707 for $\phi = 54.74^\circ$) corresponding to the degeneration of the channel cross-section to the triangular geometry ($\gamma = \infty$). The aspect ratio β of a double-trapezoidal channel is defined as the ratio between the height (h) and the maximum width (a); β can assume all the values between 0, the parallel plate configuration, and 1.414, the rhombic configuration.

Comparison Among Microchannel Cross-Sections

The data illustrated in the above sections can be used in order to minimize the pressure drop in a microfluidic

Pressure-Driven Single Phase Liquid Flows, Table 8 Laminar fully developed values of $V_{\max}$, K_d , K_e , $K(\infty)$ and of the Poiseuille number (fRe_{fd}) for KOH-etched trapezoidal microchannels as a function of the aspect ratio $\beta = h/a$ or $\gamma = h/b$

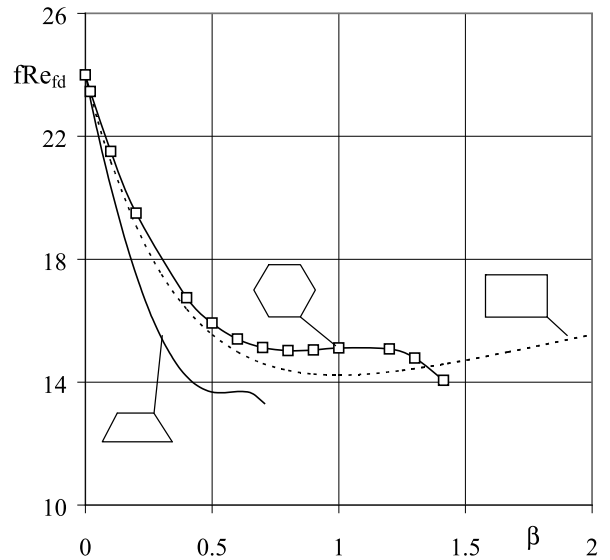
γ	β	$V_{\max}$	K_d	K_e	$K(\infty)$	fRe_{fd}
0	0.000	1.5	1.2	1.543	0.686	24
0.01	0.010	1.511	1.205	1.556	0.702	23.597
0.0277	0.027	1.530	1.212	1.580	0.736	22.931
0.05	0.047	1.555	1.222	1.610	0.776	22.174
0.1	0.088	1.610	1.243	1.674	0.862	20.737
0.2	0.156	1.719	1.279	1.793	1.028	18.650
0.3	0.211	1.822	1.310	1.897	1.174	17.244
0.4	0.255	1.908	1.334	1.984	1.300	16.263
0.5	0.293	1.976	1.354	2.054	1.400	15.565
0.6	0.325	2.027	1.368	2.108	1.480	15.060
0.7	0.352	2.065	1.379	2.150	1.542	14.690
0.8	0.375	2.094	1.388	2.182	1.588	14.417
0.9	0.396	2.116	1.394	2.207	1.626	14.215
1	0.414	2.132	1.399	2.225	1.652	14.063
1/0.9	0.432	2.146	1.403	2.241	1.676	13.940
1.25	0.452	2.158	1.407	2.254	1.694	13.832
1/0.7	0.473	2.168	1.410	2.266	1.712	13.745
1/0.6	0.497	2.177	1.412	2.275	1.726	13.685
2	0.522	2.183	1.414	2.282	1.736	13.654
2.5	0.551	2.189	1.415	2.287	1.744	13.654
1/0.3	0.583	2.196	1.417	2.294	1.754	13.679
5	0.620	2.206	1.420	2.307	1.774	13.694
10	0.660	2.219	1.426	2.330	1.808	13.622
∞	0.707	2.228	1.431	2.346	1.830	13.308

device with microchannels. In Fig. 7 a comparison among rectangular, KOH-etched trapezoidal microchannels and KOH-etched double-trapezoidal microchannels in terms of friction factor as a function of the aspect ratio β is shown. It is evident that, for a fixed value of the aspect ratio, the trapezoidal geometry allows the achievement of lower friction factors than the other cross-sections. The differences among the cross-sections in terms of friction factors decrease when the aspect ratio decreases.

A fifth-order polynomial approximation for calculating fRe_{fd} , K_d and K_e as a function of the channel aspect ratio (β) is given with the aim of offering a very simple but accurate tool for technicians and designers involved in microfluidic applications:

$$fRe_{fd} \text{ or } K_d \text{ or } K_e = \sum_{n=0}^5 g_i \beta^n \quad (35)$$

The values of the constants g_i are listed in Table 10 for microchannels having rectangular, trapezoidal and double-trapezoidal cross-sections; the maximum relative differ-



Pressure-Driven Single Phase Liquid Flows, Figure 7 Laminar fully developed Poiseuille number for KOH-etched trapezoidal microchannels, KOH-etched double-trapezoidal microchannels and rectangular microchannels as a function of the aspect ratio

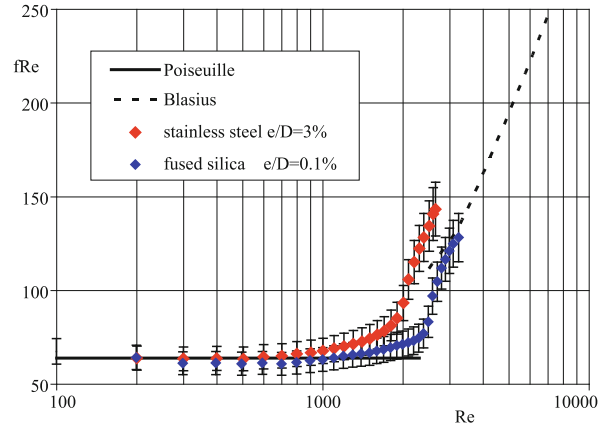
Pressure-Driven Single Phase Liquid Flows, Table 9 Laminar fully developed values of $V_{\max}$, K_d , K_e , $K(\infty)$ and of the Poiseuille number (fRe_{fd}) for KOH-etched double-trapezoidal microchannels as a function of the aspect ratio β

β	$V_{\max}$	K_d	K_e	$K(\infty)$	fRe_{fd}
0	1.500	1.200	1.543	0.686	24.000
0.02	1.518	1.207	1.563	0.712	23.459
0.1	1.598	1.233	1.648	0.830	21.507
0.2	1.717	1.267	1.758	0.982	19.501
0.4	1.931	1.321	1.950	1.258	16.746
0.5	1.989	1.337	2.007	1.340	15.923
0.6	2.019	1.346	2.039	1.386	15.403
0.7	2.030	1.350	2.053	1.406	15.126
0.8	2.033	1.351	2.057	1.412	15.027
0.9	2.036	1.351	2.060	1.418	15.043
1	2.044	1.353	2.068	1.430	15.111
1.2	2.083	1.369	2.123	1.508	15.081
1.3	2.106	1.381	2.164	1.566	14.785
1.414	2.119	1.388	2.188	1.600	14.055

ence Δ quoted in Table 10 is positive when Eq. (35) gives values greater than the rigorous calculation.

Laminar-to-Turbulent Transition

The laminar regime in a channel holds until the Reynolds number reaches a critical value above which the laminar motion becomes unstable and a transition to a turbulent flow will generally occur. It has been demonstrated experimentally that for a microchannel the critical value of the Reynolds number depends on the entrance conditions, on the cross-section geometry and on the wall roughness. The effect of the roughness on the laminar-to-turbulent transition is very important and it can be evidenced by observing Fig. 8 in which the experimental values of the Poiseuille number (fRe) are shown as a function of the Reynolds number for two microtubes made in stainless steel and in fused silica. The stainless steel microtube has an internal diameter of 133 μm and a relative roughness (e/D) equal to 3%, while the tube made in fused silica is characterized by an internal diameter equal to 100 μm and a relative roughness equal to 0.1%. The agreement with the Poiseuille law is confirmed for low values of the Reynolds number ($Re < 1600$). The stainless steel microtube exhibits an earlier deviation from the Poiseuille law at Re equal to 1600. On the contrary, for the smoother microtube in fused silica the Poiseuille law holds up to Re equal to 2300. In general, for smooth microtubes the critical Reynolds number is about 2000–2300, in agreement with the behavior of macro tubes. On the contrary, for rough microtubes it can be concluded that higher relative roughness tends to antic-



Pressure-Driven Single Phase Liquid Flows, Figure 8 Effect of the relative roughness on the laminar-to-turbulent transition in a microtube

ipate the transition and thus decreases the value of the critical Reynolds number down to 1700–1800.

Turbulent Regime

For Reynolds numbers larger than the critical Reynolds number the flow becomes turbulent and the fluid velocity at a point can be defined as the mean value of the velocity at this point with respect to time by taking a time interval t_0 larger than the typical time of one turbulent fluctuation and shorter than the typical time of pressure variations. Since mass and momentum must be conserved in turbulent as well as in laminar flow, the conservation equations (Eq. (1)) hold equally for turbulent flow, provided the velocities and pressures in these equations are interpreted as instantaneous velocities and pressures of the turbulent field. From a practical standpoint such equations are of little value because there is no interest in knowing the complete and complex history of instantaneous velocities and pressures but only their time-mean values that can be measured and observed (see also ► [turbulence in microchannels](#)). In microchannels with hydraulic diameters less than 500 μm the maximum value that the Reynolds number can assume drops to between 10,000 and 30,000. In fact, in order to reach these values of the Reynolds number a very large pressure difference has to be imposed at the ends of the microchannels. For this reason, the analysis of the turbulence in microchannels is bounded to Reynolds numbers less than 30,000.

Circular Cross-Section

The time-mean value of the dimensionless velocity $\langle V_z \rangle$ can be written in fully developed turbulent flow in a micro-

Pressure-Driven Single Phase Liquid Flows, Table 10 Polynomial coefficients appearing in Eq. (35) for rectangular, KOH-etched trapezoidal and double-trapezoidal silicon microchannels

Rectangular microchannel $0 < \beta < 1$							
	g_0	g_1	g_2	g_3	g_4	g_5	Δ (%)
fRe_{fd}	24	-32.5272	46.7208	-40.8288	22.9536	-6.0888	0.06
K_d	1.2	0.3563	0.1532	-0.8946	0.7953	-0.2312	0.05
K_e	1.543	1.0376	1.1632	-3.8892	3.1821	-0.8831	0.10
KOH-etched trapezoidal microchannel $0 < \beta < 0.707$							
	g_0	g_1	g_2	g_3	g_4	g_5	Δ (%)
fRe_{fd}	24	-42.267	64.272	-118.42	242.12	-178.79	-0.14
K_d	1.2	0.3962	0.9808	-1.4388	-2.5004	3.3655	-0.27
K_e	1.543	1.1296	3.7251	-2.6799	-14.585	15.496	-0.68
KOH-etched double-trapezoidal microchannel $0 < \beta < 1.414$							
	g_0	g_1	g_2	g_3	g_4	g_5	Δ (%)
fRe_{fd}	24	-27.471	26.117	-6.6351	-0.2956	-0.5974	0.05
K_d	1.2	0.2833	0.5441	-1.7965	1.5499	-0.4277	-0.1
K_e	1.543	0.8343	2.3859	-6.9184	5.8104	-1.5876	-0.25

tube as follows:

$$\langle V_z \rangle = (1 - 2r^*)^{\frac{1}{m}} \quad (36)$$

$$m = \begin{cases} 6 & 4000 < Re < 110,000 \\ 7 & 110,000 \leq Re < 3,200,000 \\ 10 & Re > 3,200,000 \end{cases}$$

in which the following dimensionless quantities are used:

$$\langle V_z \rangle = \frac{\langle v_z \rangle}{\langle W \rangle}; \quad r^* = \frac{r}{D} \quad (37)$$

The ratio $\langle V_{\max} \rangle$ can be computed by using the following equation:

$$\langle V_{\max} \rangle = \frac{\langle v_{z,\max} \rangle}{\langle W \rangle} = \frac{(m+1)(2m+1)}{2m^2} \quad (38)$$

Parallel Plate Cross-Section

For shallow rectangular microchannels, having a low value of the aspect ratio β , the rectangular cross-section can be modeled as a parallel plate geometry (see Fig. 4). In this case one obtains that the velocity distribution for fully developed turbulent flows becomes the following:

$$\frac{\langle v_z \rangle}{v^*} = \frac{1}{k} \left[\ln(1 - \sqrt{4y^*}) + \sqrt{4y^*} \right] + \frac{\langle v_{z,\max} \rangle}{v^*} \quad (39)$$

$$\left(0 \leq y^* \leq \frac{1}{4} \right)$$

in which $y^* = y/D_h$ with $D_h = 2h$, (τ_0 is the wall shear stress) and $k = 0.36$.

Friction Factor

For Newtonian fluids, the Fanning friction factor f is defined in terms of the wall shear stress as

$$f = \frac{\tau_0}{\frac{\rho W^2}{2}} \quad (40)$$

For turbulent fully developed flow ($Re > 10,000$) in a *smooth* microtube, the Fanning friction factor can be calculated from the *Karman–Nikuradse relation*:

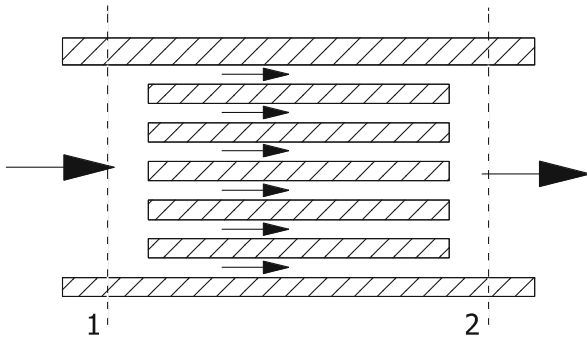
$$\frac{1}{\sqrt{4f_{fd}}} = -0.8 + 0.87 \ln(Re\sqrt{4f_{fd}}) \quad (41)$$

In the range $5000 < Re < 30,000$, in which in general microtubes could work, the *Blasius equation* may be used instead of Eq. (41) for smooth tubes:

$$f_{fd} = \frac{0.079}{Re^{0.25}} \quad (42)$$

In *rough* microtubes, like metallic microtubes, the effect of the inner roughness e on the friction factor cannot be neglected and the friction factor can be calculated by using the *Colebrook correlation*:

$$\frac{1}{\sqrt{4f_{fd}}} = -2.0 \log \left[\frac{\frac{e}{D_h}}{3.7} + \frac{2.51}{Re\sqrt{4f_{fd}}} \right] \quad (43)$$



Pressure-Driven Single Phase Liquid Flows, Figure 9 Model for pressure drop analysis of multiple microchannel systems

Since the *Colebrook correlation* is awkward to use, an approximate explicit relation for the friction factor was published by *Haaland*:

$$\frac{1}{\sqrt{4f_{fd}}} = -1.8 \log \left[\left(\frac{e}{D_h} \right)^{1.11} + \frac{6.9}{Re} \right] \quad (44)$$

This relation gives values within 2% of the *Colebrook correlation*.

Typical values of the relative roughness defined as the ratio e/D_h in microchannels vary between 0.0001 (glass microtubes) and 0.1 (stainless steel microtubes).

Pressure Drop Calculation

Pressure drop evaluation for flow through both single-microchannel and multiple-microchannel systems is very important in the design of microfluidic devices.

The model for the pressure drop analysis is depicted in Fig. 9 in which multiple microchannels with an abrupt entrance and exit are shown. The equations to be presented in this section apply to this model and are not restricted to a particular geometry of the microchannel cross-section.

They are applicable to single-microchannel flow by reducing the appropriate terms. The general relation for flow-stream pressure drop calculations is the following:

$$\Delta p = \frac{\rho_1 W_1^2}{2} \left[(K_c + 1 - \sigma^2) + 2 \left(\frac{\rho_1}{\rho_2} - 1 \right) + 4f \frac{L}{D_h} \frac{\rho_1}{\rho} - (1 - \sigma^2 - K_{ex}) \frac{\rho_1}{\rho_2} \right] \quad (45)$$

in which L is the microchannel length, W_1 is the average velocity defined by Eq. (9) and based on the minimum free flow area, σ is the free flow/frontal area ratio ($\sigma = 1$ for a single microchannel), K_c and K_{ex} represent entrance and

exit loss coefficients and ρ is given by

$$\frac{1}{\rho} = \frac{1}{L} \int_0^L \frac{1}{\rho} dz \quad (46)$$

The terms involved in Eq. (45) in order to calculate the total pressure drop represent:

- the entrance effects (see ►developing flow and ►entrance region);
- the flow acceleration effect for a compressible fluid;
- the usual friction pressure drop;
- the exit effects.

In order to investigate only the entrance pressure drop Eq. (45) can be reduced as follows:

$$\Delta p_{inlet} = \frac{\rho_1 W_1^2}{2} (1 - \sigma^2) + K_c \frac{\rho_1 W_1^2}{2} \quad (47)$$

in which the first term is the pressure drop which would occur due to a change in the cross-section alone, without friction, and the second term represents the loss due to irreversible free expansion that follows the abrupt contraction. The irreversible component of the pressure drop is contained in the abrupt contraction or entrance coefficient K_c . The exit pressure rise is similarly broken into two parts:

$$\Delta p_{exit} = \frac{\rho_1 W_1^2}{2} (1 - \sigma^2) - K_{ex} \frac{\rho_1 W_1^2}{2} \quad (48)$$

The first is the pressure rise which occurs due to the area change alone, without friction, and is identical to the corresponding term in the entrance pressure drop. The second term is the pressure loss associated with the irreversible free expansion following an abrupt expansion and this term in the present case subtracts from the other.

The values of K_c and K_{ex} have been established analytically for a number of simple entrance and exit geometries and are quoted in many textbooks [9]. These coefficients have been evaluated on the assumption of essentially uniform velocity in the manifolds linked to the microchannels but by considering a fully established velocity profile in the microchannels (that means $L \gg D_h$). These assumptions can be considered verified for long microchannels for which $L > 100D_h$.

In applying the data related to the values of K_c and K_{ex} it should be remembered that these coefficients already include the pressure change associated with change in velocity profile.

Thus the friction factor used in Eq. (45) to evaluate the overall pressure drop should be defined on the basis of the mean wall shear stress after the entrance region and not on the basis of the overall pressure drop.

For liquids for which the incompressible model holds Eq. (45) can be simplified as follows:

$$\Delta p = \frac{\rho_1 W_1^2}{2} \left[K_c + 4f \frac{L}{D_h} + K_{ex} \right] \quad (49)$$

where f is the friction factor based on the mean wall shear stress after the entrance region.

For long microchannels for which $L \gg D_h$ (this condition is very common in microchannels) one can use the fully developed value of the friction factor in the calculation of the pressure drop:

$$\Delta p = \frac{\rho_1 W_1^2}{2} \left[K_c + 4f_{\text{Re}_{fd}} \frac{L}{\text{Re} D_h} + K_{ex} \right] \quad (50)$$

To eliminate the uncertainties caused by estimation of minor losses (K_{ex} , K_c) many authors have suggested performing tests with microchannels of different lengths; in this manner it is possible to even out the inlet and outlet minor losses by subtracting the pressure drop of the shorter tube from that of the longer one for a fixed value of the average Reynolds number. By following this method, the value of the friction factor would be calculated as follows:

$$f_{fd} = \frac{\rho}{2\mu^2} \left(\frac{\Delta p(L_1) - \Delta p(L_2)}{L_1 - L_2} \right) \frac{D_h^3}{\text{Re}^2} \quad (51)$$

where L_1 and L_2 are the lengths of the longer and shorter tube respectively.

It is important to observe that this method is in general valid for liquid flows in microchannels but becomes unsuitable for gaseous flows when the effects of compressibility are strong, as the values of the minor losses at the inlet and at the outlet, even for a fixed average value of the Reynolds number, can be very different.

When the initial velocity profile is uniform and there are no abrupt contraction ($K_c = 0$) and expansion ($K_{ex} = 0$) one can take into account the effects of momentum change in the entrance region by using $K(\infty)$ which includes the increased wall shear in the entrance region but separates out the downstream fully developed flow pressure drop due to wall shear to give

$$\Delta p = \frac{\rho_1 W_1^2}{2} \left[4f_{\text{Re}_{fd}} \frac{L}{\text{Re} D_h} + K(\infty) \right] \quad (52)$$

Experimental Results

As mentioned above, the first experimental work in which water through glass microtubes with an inner diameter down to 15 μm was tested was due to Poiseuille in 1840 [2]. The experimental data obtained by Poiseuille were used to state the famous *Poiseuille law* for laminar

flows through circular tubes, according to which for Stokes flows the product of the friction factor times the Reynolds number is a constant (equal to 16 for circular tubes) that depends on the cross-section geometry only.

On the contrary, as evidenced in [10], many experimental works published in the last decade seem to be partially in disagreement with the conventional theory and if these results are used in order to establish the validity of the conventional theory for microchannels in laminar and turbulent regime the answer obtained is not unequivocal. In fact, some authors found that the predictions of the conventional theory agree with the experimental results on the friction factor; however, for the same range of hydraulic diameters, some other authors found the opposite result. It is possible to sum up the main results quoted in the open literature on the friction factor in microchannels highlighting the peculiarities, proposed by different authors, with respect to the conventional macrochannels:

- the friction factor for laminar fully developed flow is found to be smaller or larger than the conventional value;
- the Poiseuille number for laminar fully developed flow depends on the Reynolds number;
- the fully developed turbulent friction factor seems to be in disagreement with the *Blasius equation* for smooth microchannels and with the *Colebrook correlation* for rough microchannels;
- the friction factor depends on the material of the microchannel walls (metals, semi-conductors and so on) and/or on the test liquid (polar fluid or not), thus evidencing the importance of electro-osmotic phenomena at microscales;
- the friction factor depends on the relative roughness of the walls of the microchannels also in the laminar regime.

However, from a chronological analysis of the experimental results of the literature it is possible to extrapolate how the deviations of the behavior of fluid flow through microchannels from that through large-sized channels are decreasing. This fact can be explained by taking into account the dramatic improvement of the techniques of microfabrication with the consequent reduction of the issues of surface roughness of the microchannels and a more appropriate control of the channel cross-sections; for this reason the results of the older studies may not provide useful comparisons. Another possible explanation of the decrease of the observed deviations is related to the increase of the reliability/accuracy of the more recent experimental data quoted in literature. This seems to be confirmed by the increasing number of experimental results on the friction factor of the last few years, which confirm the success of conventional correlations in pre-

dicting pressure drop for Newtonian liquids in microchannels having inner hydraulic diameters down to 10 μm .

Future Directions for Research

The analysis of the behavior of single phase Newtonian liquid flow through microchannels in laminar and turbulent regime evidenced that the conventional theory validated for large channels holds for hydraulic diameters down to 10 μm . Further investigations are required for non-Newtonian flows, like polymer solutions and surfactants, for which slip velocity at the walls has been observed even for microchannels having a hydraulic diameter greater than 1 μm . In addition, an accurate analysis of the critical Reynolds numbers for which the laminar-to-turbulent transition occurs in microchannels of different cross-sectional geometry (for Newtonian and non-Newtonian liquids) could be important to improve the understanding of the rules of the development of turbulence at the microscale.

Cross References

- Entrance Region
- Developing Flow
- Pressure-Driven Single Phase Gas Flows
- Turbulence in Microchannels

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Pressure-Driven Two Phase Flows

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Synonyms

Two-phase flow driven by pressure

Definition

Pressure-driven ► **two phase flows** are flows of two immiscible fluids in the same micro- or nanofluidic conduit driven by a pressure gradient. The fluids can be of different chemical compositions – liquid/liquid, or in different physical states – gas/liquid.

Overview and Theory

In the past decade, micro- and nanofluidics has been studied in great detail because of the increasing availability of methods for fabricating complicated flow configurations and measuring transport phenomena on the micro and nanoscales. The micro- and nanofluidic channel systems with integrated pumps, valves, and detectors, are known as ► **Lab-on-a-Chip** (LOC) or ► **micro total analysis system** (μTAS) [1]. Ideally, each process – e. g., sampling, sample pretreatment, reaction, separation, detection and analysis in an analytical system – is realized on one integrated device. Fluid mechanics studies in these processes therefore are indispensable for the design and fabrication of LOC and μTAS .

Fluids that are of interest in nature, biotechnology and chemistry, are rarely simple single-phase fluids. Most compartmental chemistry, interfacial phenomena and applications employ two-phase flows. It is therefore not surprising that recently a surge of interest has arisen in two-phase microfluidic manipulation.

Compared to single-phase flow systems, when discussing two-phase flow in micro- and nanosystems, there are many questions that arise even in a simple configuration. When two phases flow in the same conduit or channel, the flow patterns change between droplet-base flows and stratified flows depending on the physicochemical properties of fluids and channel surfaces. At the same time, in two-phase flow systems, several flow phenomena, such as film flow, corner flow and slip flow play increasingly important roles with reduction of the dimensions. Both advantages and disadvantages originate from the reduction in dimension

and the coexistence of two phases. In all these phenomena, the fluid manipulation is no doubt the most important factor.

Manipulating fluids in a two-phase flow system is much more complicated than in single-phase systems. Fluid flows can be actuated spontaneously, non-mechanically or mechanically. The most frequently used flow actuation method, till now, is still pressure-driven pumps due to their availability and versatility. Especially, for a two-phase flow system, the naturally existing inhomogeneity makes other methods much more difficult to be applied. Electroosmosis for instance, resulting from the force exerted on a space-charge region in the liquid by an electric field, will not work if the conductivity continuum in the fluid is interrupted by an entrapped dielectric liquid droplet or gas bubble.

Fluid flow in channels of constant cross-sectional shape can be simply described by Poiseuille's law:

$$Q = \int v_i dA \approx \frac{\Delta P}{R_{hy}} \quad (1)$$

where Q ($\text{m}^3 \text{s}^{-1}$) is the volume flow rate, v_i (m s^{-1}) is the flow velocity at the position i , A (m^2) is the area of cross section. The hydrodynamic resistance, R_{hy} , can be approximately expressed as:

$$R_{hy} \approx \eta l C^2 A^{-3} \quad (2)$$

where η (Pas) is the fluid viscosity, l (m) is the channel length, and C (m) is the perimeter of the cross section. ΔP (Pa) is the pressure difference between two ends of the fluid, which is the total pressure acting on the fluids. It is the sum of externally applied pressure ($P_{ex.}$) and the internal (capillary) pressure ($P_{cap.}$):

$$\Delta P = P_{ex.} \mp P_{cap.} \quad (3)$$

If there is a positive capillary pressure in a channel, the externally applied pressure has to exceed it to pump fluid, and inversely fluid can be automatically imbibed in a microchannel if there exists a negative capillary pressure (capillary action). Therefore, pressure-driven flows can be manipulated not only by externally applied pressure but also by the internally generated capillary pressure.

The dynamical response of two-phase flows can be commonly characterized successfully in terms of the dimensionless numbers [2]. Table 1 lists some force-related dimensionless numbers. These dimensionless numbers demonstrate competing phenomena of forces: buoyancy, gravitational, inertial, viscous and interfacial forces. The Grashof number (buoyancy to viscous forces), the Bond number (gravitational to interfacial forces) and the

Reynolds number (inertial to viscous forces) scale with the third, the second and the first power of flow dimensions, respectively. The relative importance of the different forces that act on fluids therefore dramatically changes on downscaling. With downscaling, the buoyancy, gravitational and inertial forces become less and less important and viscous and interfacial forces become more and more predominant. In micro-systems, the relative importance of forces, calculated for a water/oxygen two-phase flow, is given by the following order: buoyancy forces < inertial forces < gravitational forces < viscous forces < interfacial forces.

Of these dimensionless numbers, the Reynolds number is the most frequently used number to describe microfluidics. It expresses the ratio of inertial forces to viscous forces:

$$Re = \frac{\rho v L}{\eta} \quad (4)$$

where ρ (kg cm^{-3}) is the effective density, v (m s^{-1}) is the average flow velocity, L (m) is the characteristic flow dimension, and η (Pas) is the effective viscosity. In a straight channel, the value of $Re \approx 10^3$ is accepted as a limit for the transition from a laminar flow to a turbulent flow. In a straight micro- or nanochannel with diameters between 100 nm and 100 μm , where $\eta(\text{H}_2\text{O}) = 1.025 \times 10^{-3}$ Pas, $\rho(\text{H}_2\text{O}) = 10^3 \text{ kg m}^{-3}$ and $v = 1 \text{ mm s}^{-1}$, the calculated Reynolds number lies between 10^{-1} and 10^{-4} . For a gas, oxygen for instance, $\eta(\text{O}_2) = 20.317 \times 10^{-6}$ Pas and $\rho(\text{O}_2) = 1.429 \text{ kg m}^{-3}$, the Reynolds number at these channel dimensions ranges from 10^{-5} to 10^{-2} . Therefore, inertial forces are over-

Pressure-Driven Two Phase Flows, Table 1 Important force-related dimensionless numbers in two-phase microfluidics and scaling with device dimension L

Dimensionless number	Definition	Equation	Scaling	Value*
Grashof	Buoyancy Viscous	$\frac{L^3 \rho^2 g \beta \Delta T}{\eta^2}$	3	10^{-4}
Bond	Gravitational Interfacial	$\frac{(\rho - \rho_f) g L^2}{G \sigma}$	2	10^{-5}
Reynolds	Inertial Viscous	$\frac{\rho v L}{\eta}$	1	10^{-1}
Capillary	Viscous Interfacial	$\frac{\eta v}{G \sigma}$	0	10^{-4}

* For a water-oxygen two-phase flow: $T = 20^\circ$, $\eta(\text{O}_2) = 20.317 \times 10^{-6}$ Pas, $\eta(\text{H}_2\text{O}) = 1.025 \times 10^{-3}$ Pas, $\sigma(\text{H}_2\text{O}) = 72.8 \text{ mN m}^{-1}$, $\rho(\text{H}_2\text{O}) = 10^3 \text{ kg m}^{-3}$, $\rho(\text{O}_2) = 1.429 \text{ kg m}^{-3}$, $\theta(\text{water and glass}) \approx 50^\circ$, $\beta(\text{water at } 20^\circ\text{C}) = 0.00021^\circ\text{C}^{-1}$ assuming that $L = 10 \mu\text{m}$, $v = 1 \text{ cm s}^{-1}$ and $g_c = 1$

L – channel dimension, ρ – fluid density, g – gravity acceleration constant, β – coefficient of expansion, ΔT – temperature difference, η – fluid viscosity, ρ_f – surrounding fluid density, f – subscript of density, G – geometrical constant, σ – interfacial force, v – flow velocity

whelmed by viscous forces in microfluidic devices. Laminar flows are thus expected in micro- and nanochannels for both gas and liquid, and not turbulent or random flows. The forces acting on two-phase fluids can be divided into tangential and normal components. On the micro- and nanoscales, as shown in Table 1, a pair of competitive forces – the viscous force and the interfacial force – become dominant. The viscous forces mainly act tangential to the interface elongating it, whereas the interfacial forces act normal to the interface inducing droplet or bubble formation (minimizing the interfacial area). The capillary number is the ratio of these two competitive forces, and can be expressed as:

$$Ca = \frac{\eta v}{G\sigma} \quad (5)$$

where η (Pa s) is the viscosity of continuous phase, G is a geometrical constant, and σ (Nm^{-1}) is the interfacial tension. The capillary number is independent of flow dimensions, and can be controlled by the properties of fluids and channels and the interactions between them. It is the most useful dimensionless number to describe two-phase microfluidic flow phenomena including flow formation, sizing, coalescing, mixing and splitting.

Keeping in mind that two-phase flows include both liquid-liquid systems, and gas-liquid systems, the different behavior of a gas and liquid must be emphasized before discussing two-phase flows. The average distance between molecules in a gas phase is one order of magnitude higher than the diameter of its molecules, while it approaches the molecular diameter in a liquid phase. Therefore, in some microdevices the gas is compressed when the pressure strongly changes. The Mach number Ma is a dynamic measure of fluid compressibility, and is defined as the ratio of flow velocity (v) to sound speed (a):

$$Ma = \frac{v}{a} \quad (6)$$

The flow of a compressible fluid like air can be treated as incompressible if the local Mach number is less than 0.3. Ma is very small in microfluidic systems due to the low flow velocity. Compression however might still occur in the case of density gradients, for instance by high local pressure changes, wall heating or cooling. In that case, equations for compressible flow must be applied [3]

Basic Methodology

Channel Geometries

Two-phase flows can be easily generated by combining flow of different phases in certain channel geometries. The

basic channel configurations used to generate multiphase flows are T, Y or cross junctions (Fig. 1).

Pumping by Externally Applied Pressure

In micro- and nanofluidic devices, a pressure is normally externally applied by syringe pumps or vacuum pumps. The pressure difference, generated at two ends of fluids, pushes or pulls fluid to move, and the resulting volume flow rate is expressed by Eq. (1).

The advantage of syringe or vacuum pumping is its versatility. It can be applied to drive all kinds of fluids (gas, liquid or gas-liquid two-phase fluids), and theoretically, it can also be used on all scales as long as the pressure is high enough to overcome the flow resistance. However, the hydrodynamic resistance in a channel is proportional to flow distance and inversely proportional to the fourth power of the channel size, and therefore, very high pressure is necessary to drive flows in smaller channels. For example, without capillary pressure, calculated from Eq. (1), to have water flow with 1 cm s^{-1} , in a $10 \mu\text{m}$ diameter channel of 1 mm length 10^3 Pa is needed, however, 10^7 Pa has to be applied for the same flow in a 100 nm diameter channel.

Syringe pumping and vacuum pumping remain predominant in microfluidic flows. Fluid flows driven by capillarity become more efficient on downscaling due to its favorable scaling behavior.

Pumping by Capillary Pressure

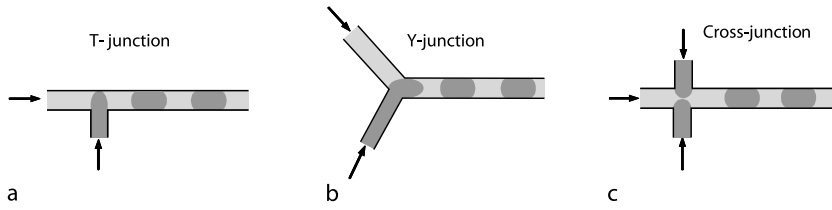
A hydrophilic liquid, water for instance, *pulls* itself into or through a hydrophilic micro- or nanochannel by capillary action which is induced by interfacial tensions (Fig. 2a). The capillarity-induced pressure pulling the liquid along the channel (Fig. 2a and 2b) is expressed by Young–Laplace’s equation:

$$P_{\text{cap.}} = G\sigma_{\text{lg}} \cos \theta d^{-1} \quad (7)$$

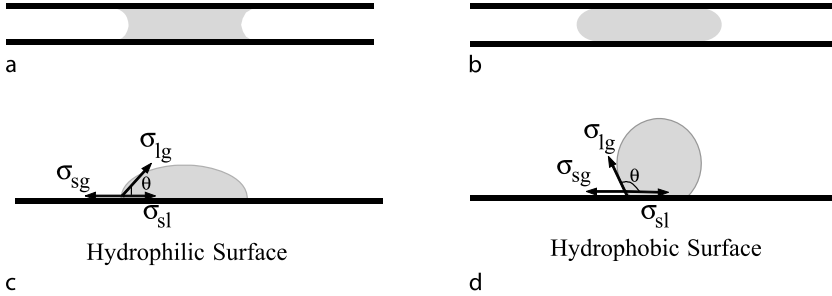
where G is a geometrical constant, d (m) is the characteristic diameter, σ_{lg} (Nm^{-1}) is the liquid-gas interfacial tension, and θ (degree) is the contact angle which is set up when a liquid droplet contacts both a solid and a gas (or a solid and a second immiscible liquid) (Fig. 2c and 2d). The liquid curvature of the droplet is caused by the differences among interfacial tensions. The force balance is described by Young’s equation:

$$\sigma_{\text{sg}} = \sigma_{\text{sl}} + \sigma_{\text{lg}} \cos \theta \quad (8)$$

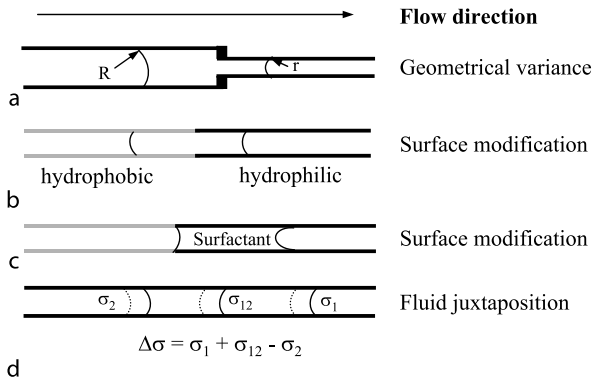
where σ_{sg} and σ_{sl} (Nm^{-1}) indicate the interfacial tensions of solid-gas and solid-liquid, respectively. Fluids,



Pressure-Driven Two Phase Flows, Figure 1 Configurations for two-phase flows: (a) T-junction, (b) Y-junction and (c) Cross-junction



Pressure-Driven Two Phase Flows, Figure 2 Water (a) in a hydrophilic channel, (b) in a hydrophobic channel, (c) on a hydrophilic surface and (d) on a hydrophobic surface.



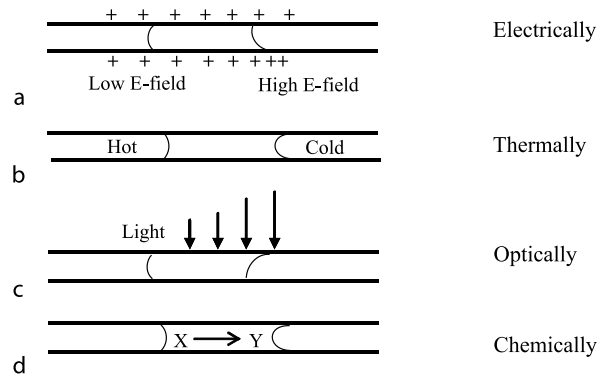
Pressure-Driven Two Phase Flows, Figure 3 Capillary pressure induced by (a) Geometrical variance, (b) Surface modification (solid), (c) Surface modification (liquid), (d) Fluid juxtaposition (bi-slug)

therefore, can be spontaneously propelled by a capillarity-induced pressure gradient.

In a channel, the motion of a fluid segment can result from the capillarity-induced pressure difference between the front and a rear of the fluid column. Such a pressure difference (ΔP) is described by:

$$\Delta P = P_{\text{cap},2} - P_{\text{cap},1} = G\sigma_{\text{lg}} \left(\frac{\cos \theta_2}{d_2} - \frac{\cos \theta_1}{d_1} \right) \quad (9)$$

According to the equation, the pressure difference can be obtained by changing the channel diameter (d), the interfacial tension (σ) or the contact angle (θ). Therefore, self-



Pressure-Driven Two Phase Flows, Figure 4 Interfacial forces modulated (a) Electrically, (b) Thermally, (c) Optically and (d) Chemically

propelling slugs can be realized by careful design, material selection, microfabrication, and surface control and processing. For example, a slug will move from the wide part to the narrow part when the channel diameter changes (Fig. 3a). The movement can also be realized by using different channel materials (Fig. 3b) or fluids (Fig. 3c) with different interfacial energy. Similar phenomena happen for two or more juxtaposed phases (Fig. 3d). The motion can even be inverted by juxtaposing a liquid of low surface tension.

Furthermore, interfacial tensions can be tuned electrically (Fig. 4a), thermally (Fig. 4b), optically (Fig. 4c), chemically (Fig. 4d), or by combinations of them such as optically-electrically.

Flow Phenomena

Flow patterns

Two-phase flow in microfluidic systems is broadly categorized into ► **segmented (droplet-based) flow** and ► **parallel (stratified) flow** architectures. Gas-liquid two-phase flows in microchannels have been studied in detail for heat-exchanger applications. Between segmented flows and stratified flows there can be several other flow patterns – bubbly flow, ring flow, lump flow and annular flow (Fig. 5a) [4]. The flow of immiscible liquids has been successfully manipulated to study chemical reactions and for biotechnology purposes. The two-phase microchannel flows change from the droplet-based flows to stratified flows depending on the flow rate and fluid composition, in a manner ultimately dependent on the capillary number (Eq. (5), Fig. 5b) [5].

Immiscible gas-liquid and liquid-liquid mixtures form droplet-based flows over a wide range of flow conditions and channel dimensions without additional control due to interfacial forces. Vice versa, two immiscible streams would flow alongside one-another if there were no interfacial forces between them. Theoretically, a stable elongated interface between two phases only occurs when tangential forces (viscous shear forces) are equal to or greater than normal forces (interfacial forces), i.e., at higher capillary number. Figure 5c shows a typical example of a two-phase flow transition according to the capillary number [6]. Two streams of water are added to a carrier liquid in a cross configuration. The two streams of water coalesce to form a big droplet-based flow at extremely small Ca (~ 0.0004). By increasing Ca , alternating droplet-based flow is generated with decreased droplet size. Droplet size (R) in a two-phase flow depends on the channel size (h) and the capillary number: $R \sim h/Ca$ [7]. Finally, stratified flow is obtained for $Ca > 0.15$ at a water fraction of 0.2.

Based on the definition of the capillary number, the two-phase flow patterns in microchannels can be influenced by interfacial forces (σ), fluid viscosity (η), flow velocity (v) and geometric features (G). The interfacial forces can be modulated by adding surfactants, surface coating and patterning techniques, but also by electrical, thermal or optical modulations of the interfacial tensions (see previous section). The fluid viscosity will be changed by changing temperature or adding polymers. The flow velocity is mainly tuned by actuation forces and geometrical tailoring. The geometric features include designing smart geometries and adding microtextures inside micro- and nanofluidic devices.

When two immiscible phases come into contact at a junction, an approximately parabolic flow profile can be obtained for a two-phase stratified flow in a common

pressure-driven microfluidic system. On the other hand, in a droplet-based flow, stimulated by the shear between the stationary fluid at the channel wall and the slug axis, there exists internal recirculation in both continuous and dispersed phases [8]. Complex flow patterns may appear under different conditions, depending on the forces acting on the fluids.

Whether a stable two-phase flow can be obtained in a fluidic system depends on both fluid dynamic conditions and surface chemistries. The aim is to open the way to control the flow and to achieve versatility and reproducibility in on-chip flow management. The design will require smart network topologies and interplay of multiple physical and chemical effects.

Slip Flow

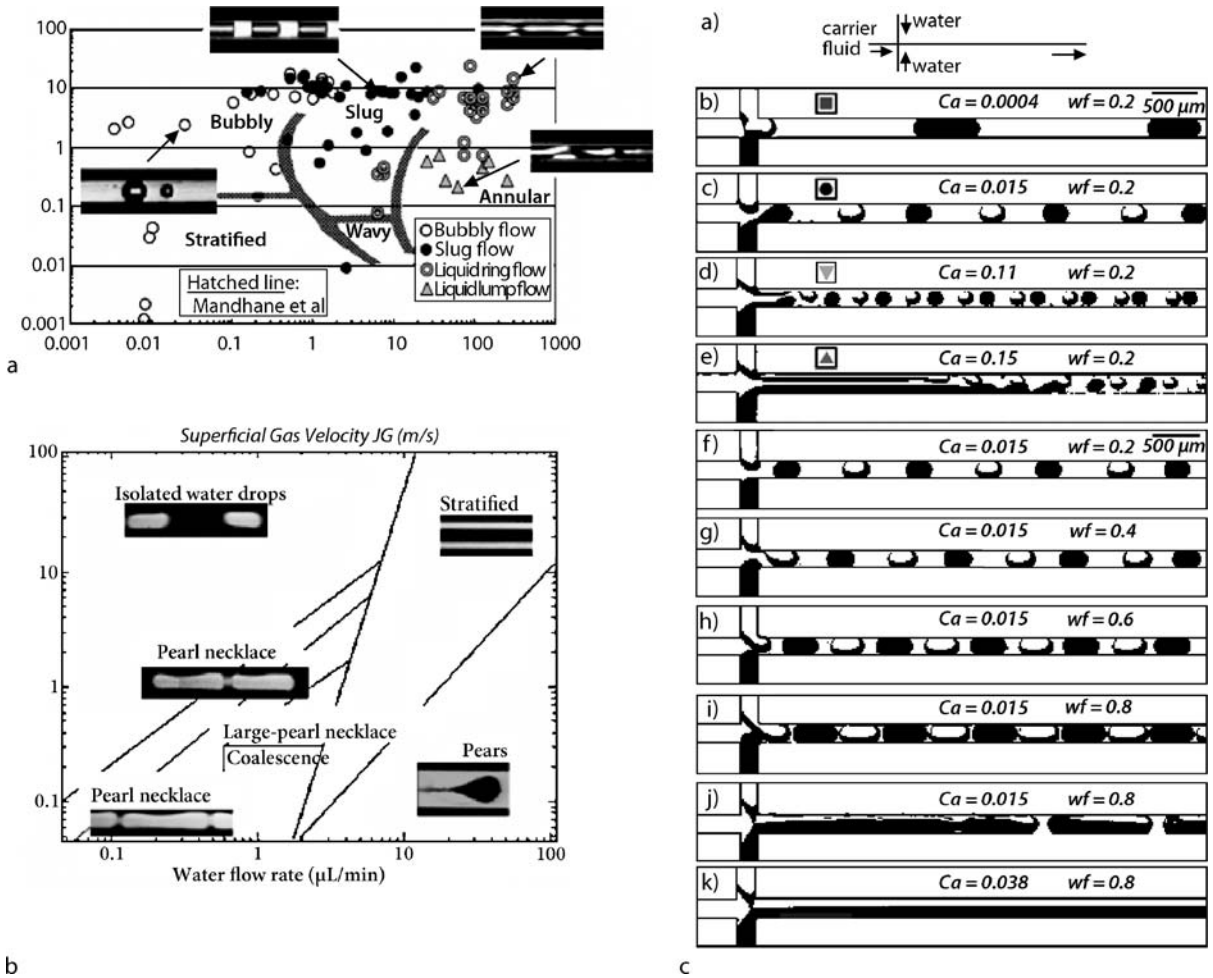
Traditional *non-slip* assumptions have been doubted for a long time in microfluidic systems. Recently, the results from molecular dynamics simulations and experiments on micro- and nanoscales strongly indicate the existence of slip flow. Theoretically, fluids will slip when the attraction between fluids molecules is stronger than the wall/fluid molecule attraction. The amount of slip is described as the slip length which is the notional distance inside the surface at which the velocity equals to zero (Fig. 6a).

Molecular dynamics (MD) simulations have shown that slip can occur only when a critical surface shear stress is reached, and that there exists a general relationship (linear or nonlinear given by different results) between the slip length and the local shear rate at a solid surface. Significant slip has been experimentally demonstrated by driving different fluids through micro- and nanochannels. Slip length depends on the interactional parameters between solid and fluid: shear stress, wetting properties, fluid properties (e.g., viscosity and density) and interfacial roughness.

Different fluids have different interactions with each other or with solid surfaces. The fluid/fluid intermolecular interactions can be stronger or weaker than those of fluid/solid surfaces. Therefore, in two-phase flows, boundary conditions are much more complicated. As diagnostic techniques improve our ability to probe the fluid surface at the molecular scale, it is expected that slip flows can be investigated by direct measurement of fluid velocity or the movement of contact lines.

Film Flow

When droplet-based multiphase flow is considered, there are two possible configurations. In the first, a drop forms a distinct contact line with the walls of the channel. In the other, a drop is separated from the boundaries by a thin wetting film. Because of the affinity of fluids and surfaces,



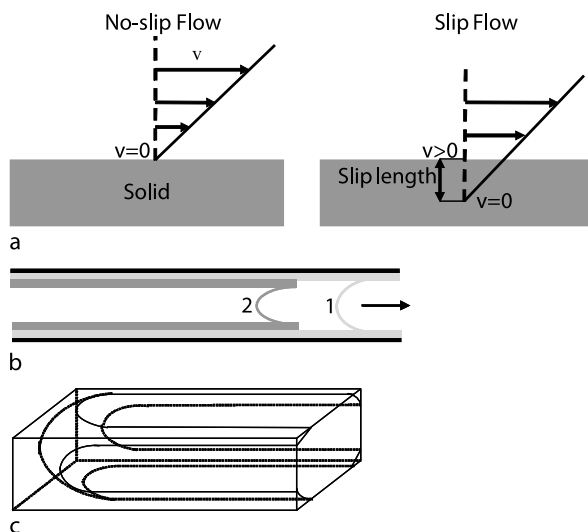
Pressure-Driven Two Phase Flows, Figure 5 (a) Phase diagram of a gas-liquid two-phase flow. Reprinted from [4] with permission of Elsevier. (b) Phase diagram of a liquid-liquid two-phase flow. Reprinted from [5] with permission of the American Physical Society. (c) Two-phase flow patterns transition according to the capillary number. (a) is the schematic illustration of the experimental setup, (b)–(k) are the microphotographs of the observed flow patterns as Ca and wf (water fraction) are varied. Reprinted from [6] with permission of the American Chemical Society

hydrophilic fluids prefer to flow along hydrophilic channel surfaces, and hydrophobic fluids like to flow along hydrophobic surfaces. As hydrophilic liquids are imbibed into hydrophilic channels by capillary force, hydrophobic fluids flow into hydrophobic channels much more easily. When fluid slugs flow forward in a channel, the trail left by the previous slug act as a lubricant for the following one (Fig. 6b). The film thickness can be theoretically calculated from the ratio of the loss of the fluid volume to the surface area traveled by the fluid. The thickness of a lubricating film experimentally depends on viscosity of neighboring slugs, fluid velocities and surface properties. Viscosity causes the deposition while capillary forces oppose the formation of a film. The film thickness (h_{∞}) deposited behind a wetting meniscus is therefore a function of Ca .

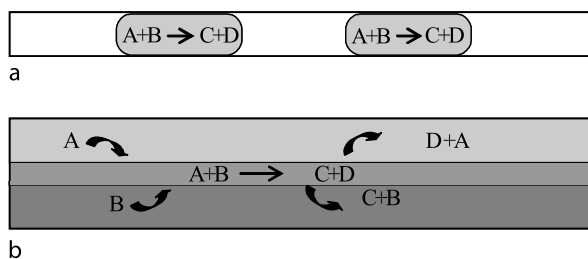
At $10^{-5} < Ca < 10^{-2}$ it can be expressed by: [9]

$$h_{\infty} = nRCa^{2/3} \quad (10)$$

where n is a prefactor determined by fluid properties, and R is the channel radius. Although the thickness decreases with flow dimension, the percentage of film flow will still increase with downscaling due to both the decrease of fluid amount and the increase of specific interfacial areas. The greater the capillary number, the larger the film thickness according to Eq. (10), which allows more liquid flow through the film. A stable thin liquid film between the tube wall and gas slugs can appear at high velocities on carefully treated clean surfaces, and vice versa, so that thin films can be decreased in thickness or eliminated by sur-



Pressure-Driven Two Phase Flows, Figure 6 Schematics of (a) No-slip flow and slip flow. (b) Film flow: (1) and (2) are immiscible fluids. (c) Corner flow in a rectangular channel



Pressure-Driven Two Phase Flows, Figure 7 Schematic applications of (a) Droplet-based flow and (b) Stratified flow.

face modification of the channel walls by changing the wetting properties.

Corner Flow

As we know, liquid surface tension is prone to minimize the liquid surface area to minimize the surface energy. Most of the channels fabricated on the micro- or nanoscale possess rectangular or slit-like cross sections. To satisfy the contact angle wetting condition the fluid will curve along the perimeter of the interface, thus minimizing interfacial area, and achieving a lower energy state (Fig. 6c). The polymorphism of the wetting liquid depends on two parameters: the channel geometry, and the interaction between substrate materials and liquids. The corner flow velocity is influenced by a flow resistance coefficient and geometric factors: the corner half-angle and the contact angle [10].

In a two-phase flow system, the different affinity of the two fluids for the wall makes one fluid preferentially flow along corners due to the capillarity. As the number of the polygon sides decreases and the corners sharpen, the cross-sectional area of fluids held in corners increases, and corner flow becomes more important. Therefore, corner flow must be considered carefully in such processes as evaporation and drying of micro- and nanochannels with small angular polygonal cross sections.

Key Research Findings

Applications

Two-phase systems are more complex than their single-phase counterparts, but are ubiquitous in process applications. Both the immiscible individual fluids separated by interfaces and the interfaces themselves provide us with mediums that are very suitable to study microreactions and interfacial phenomena.

Droplet-based Flows

Immiscible gas-liquid and liquid-liquid mixtures form droplet-based flows over a wide range of flow conditions and microchannel dimensions without additional control. There are several advantages for droplet-based flows to be applied in different fields. Specific interfacial areas as large as $9000\text{--}50,000\text{ m}^{-1}$ have been obtained by decreasing channel dimensions and droplet (bubble) sizes down to $50\text{ }\mu\text{m}$, which is 100 times larger than generally obtained in macroscopic channels. Thermal and mass transfer is therefore rapid due to the large specific interfacial area and consequently small molecular diffusion distance. Moreover, the process controllability is wide in droplet-based flows. The size, the number, and the composition of droplets can be controlled at each step of the process. By varying the initial composition of droplets, one can assess its effect on reaction yield and kinetics, providing a useful operational research tool for laboratories. Furthermore, in a droplet-based multiphase flow, droplets are separated from channel walls by a thin layer of immiscible (carrier) fluid, therefore the internal compositions of the droplets can not change due to contact with the channel walls, and the internal recirculation inside the droplets makes components mix efficiently in a very short time. This supplies us with more flexibility to employ each droplet as a separate microreactor.

Droplet-based two-phase flows offer controlled picoliter or femtoliter volumes of droplets (Fig. 7a) which are almost ideal chemical reactors and therefore have been utilized in emulsification and encapsulation, microreaction, synthesis, mixing, and bioassay [7].

Stratified Flows

In a multiphase stratified flow, the interfaces between immiscible fluids have several characteristics. Firstly, the specific interfacial area can be very large just as droplet-based flow. It can for example be about $10,000\text{ m}^{-1}$ in a microchannel compared with only 100 m^{-1} for conventional reactors used in chemical processes. Secondly, the mass transfer coefficient can be very high because of the small transfer distance and high specific interfacial area. It is more than 100 times larger than that achieved in typical industrial gas-liquid reactors. Thirdly, the interfaces of a stratified microchannel flow can be treated as nano-spaces. Simulation results show that the width of the interfaces of a stratified flow is in nanometers, and that diffusion-based mixing occurs at the interface. The interface width can be experimentally adjusted by adding surfactants. Finally, reactants only contact and react with each other at the interface. Therefore, the interfaces supply us with mediums to study interfacial phenomena, diffusion-controlled interfacial reactions and extraction.

The ideal 2D “nano-reactors” of the interfaces between immiscible fluids in stratified flows (Fig. 7b) have been used for chemical synthesis, triphase hydrogenation and biological enzymatic degradation, for extraction and separation, and for kinetic studies.

Future Directions for Research

Two-phase flows for a long time have been propelled by externally applied pressure, due to the inhomogeneity of two confined phases in the same system. High external pressure is needed to maintain flow in small devices due to the hydrodynamic resistance which increases with downscaling. This makes syringe pumping and vacuum pumping less suitable at smaller length scales. However, the capillarity-induced flow shows its importance at small flow dimensions. The capillary pressure is induced by interfacial forces which increase with downscaling. Moreover, more and more means have been found to modulate different interfacial forces of gas-liquid, gas-solid, liquid-liquid and liquid-solid. As device size decreases, the capillary pressure-driven flow therefore gains importance and may increasingly be used for systems where externally applied pressure would have to be extremely high.

Cross References

- ▶ Pressure-driven Flow
- ▶ Pressure-based Sample Injection
- ▶ Two-phase Flow

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Pressure Drop in Microchannels

- ▶ Turbulence in Microchannels

Pressure Injection

Definition

Pressure injection is another important technique for transferring sample to microfluidic chips.

Cross References

- ▶ Techniques for Transferring Samples to Chips

Pressure Measurements, Methods

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Synonyms

Pressure sensors

Definition

Experimentation and novel measurement techniques are crucial for the further development of microfluidic devices. Pressure is one of the basic parameters involved in microfluidic experiments. However, it is not realistic to apply conventional pressure measurement techniques to microsystems, since the characteristic dimensions of these measurement instruments are already comparable to those of the microdevices used. Therefore, novel pressure methods are needed for pressure measurement on the microscale.

Overview

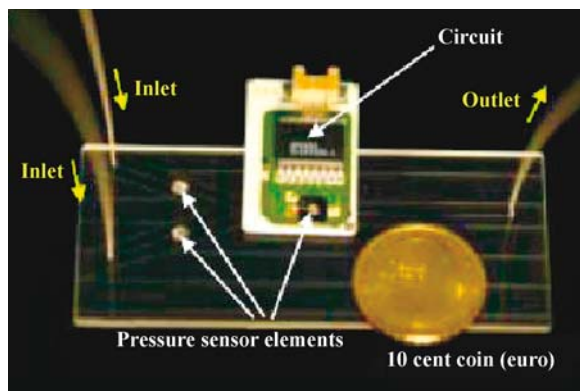
It is not practical to measure pressures inside microchannels and nanochannels by use of conventional sensors, since it is very difficult or impossible to implement these sensors in microsystems without disturbing the flow field. Therefore, some novel methods are required for pressure measurement on the microscale.

The common practice for measuring the pressure drop along a microchannel is to use pressure transducers in the inlet and exit reservoirs, which gives overall information about the pressure rather than the pressure distribution along the channel. This approach is used in many studies related to fluid flow in microchannels [1–4].

Sekimori et al. [5] have developed a pressure sensor for use in Lab-on-a-Chip (LOC) devices. It has an embedded miniaturized structure and high chemical resistance, and causes no interference in the microflow during the measurement. This pressure sensor element has a volume of 1 mm^3 and was fabricated by using MEMS technology. The pressure sensor element was installed in an LOC without any dead volume or disturbance to the microflow, and the pressure inside the microchannel was measured (see Fig. 1).

Wu et al. [6] proposed the use of single-walled carbon nanotubes (SWNTs) as nanoscale electromechanical pressure sensors. These authors demonstrated computationally a reversible pressure-induced shape transition in armchair SWNTs, which in turn induced a reversible electrical transition from metal to semiconductor. They also discussed the potential long lifetime of this pressure sensor due to the excellent mechanical durability of carbon nanotubes.

Li and Chou [7, 8] proposed the use of SWNTs as mechanical sensors to measure also mass, strain, and temperature, as well as pressure. The principle of sensing was based on the shift of the resonant frequency of a carbon nanotube resonator when it is subjected to changes in attached mass, external loading, or temperature. Li and Chou discussed the feasibility of such a sensor, with the aid of computer simulation using atomistic modeling together with molec-



Pressure Measurements, Methods, Figure 1 Installation of the pressure sensor to a microchip. Reprinted from [5] with permission from Dr. Kitamori

ular structural mechanics. They concluded that the sensing capability of this nanoscale sensor was superior to that of current microsensors, and that the sensitivity could be further enhanced by using smaller-sized carbon nanotubes.

Kim and Daniel [9] have recently suggested the use of atomic force microscopy (AFM) for measuring pressure profiles in micro/nanochannels. The method is based on measurement of the deflection of a thin plate over the channel surface by topographic imaging of the plate using AFM. Kim and Daniel verified this measurement technique numerically with artificially generated topographic data. Since the topographic imaging takes quite a long time, this technique is only applicable to steady-state processes. Moreover, special attention needs to be paid to providing vibration-free surroundings, since vibrations transmitted via the fluid can cause noise in the data, leading to loss of accuracy.

Matsuda et al. [10] have proposed the use of a pressure-sensitive paint (PSP) technique, which is based on the interaction of atoms or molecules with photons, to measure the pressure inside micro/nanochannels. This technique is limited to gaseous flow, and has drawbacks for high-pressure and low-speed applications. Moreover, the surface where the pressure is to be measured must be visible to the detector. The temperature sensitivity of the PSP technique must also be considered during the calibration. In conclusion, although several techniques have been suggested by many researchers, the development of measurement techniques for use on the microscale is a challenging topic, and this topic seems likely to remain open-ended in the near future.

Cross References

- [Nanofluidics in Carbon Nanotubes](#)
- [Control of Microfluidics](#)

- Fluidic Nanosensors
- Velocity Sensors

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Pressure Sensors

- Pressure Measurements, Methods

Pressure Wave

- Shockwaves in Microchannels

Primitive Model

Synonyms

Element model

Definition

Primitive model is a single Differential–Algebraic Equation (DAE) derived from basic conservation laws in different domains describing the dynamic response of a constituent element (e.g., capacitors, inductors, resistors, channels, etc.) in a complex system.

Cross References

- Model Order Reduction (MOR)

Probe Microscopy

- Scanning Probe Microscopy

Programmed Cell Death

- Microfluidics for Studies of Apoptosis

Proper Orthogonal Decomposition (POD) Based MOR

Synonyms

Karhunen–Loève (K–L) expansion; Karhunen–Loève (K–L) decomposition; Karhunen–Loève (K–L) approach

Definition

Proper Orthogonal Decomposition is a technique that extracts the orthogonal basis function spanning the reduced subspace using an ensemble of data from experiments or numerical simulation of the original full systems.

P

Cross References

- Model Order Reduction (MOR)

Protein Adsorption

- Biomolecular Adsorption in Microfluidics

Protein Array

- Biochip
- Bioprinting on Chip

Protein Array and Cell Array

- Microarray

Protein Chip

- Lab-on-Chip Devices for Protein Analysis
- Protein Microarrays

Protein Microarrays

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Synonyms

Microarrays; Protein chip

Definition

Protein microarrays are small chips made of glass, plastic or silicon that enable the study of protein–protein, protein–antibody and protein–drug interactions in a miniaturized, automated fashion.

Overview

Protein microarrays are chips made of glass, plastic or silicon spotted with an array of known antibodies or peptides onto which a biological sample of immobilized proteins is applied and binding determined (Fig. 1). These microarrays are used for identifying interactions between molecules fixed onto the surface (i.e. antibodies) and proteins in a solution. In a single experiment, these microarrays enable the simultaneous analysis of thousands of molecular parameters [1]. Typical applications of protein microarrays include determining protein–protein, enzyme–substrate, antibody–protein and drug–protein interactions, and determination of protein expression levels. In general, protein microarrays are classified under two broad categories: *analytical protein microarrays* for identifying the contents of an analyte and *functional protein microarrays* for studying the interaction between the protein of interest and other molecules [2]. Analytical protein microarrays often have microfluidic channels where a solution containing the proteins being investigated is pumped into a flow cell containing the microarrays. These types of protein microarrays are mainly used for basic molecular biology research, disease marker identification, toxicological response profiling and pharmaceutical target screening [3]. In contrast, functional

protein microarrays have the proteins being investigated attached to the chip itself.

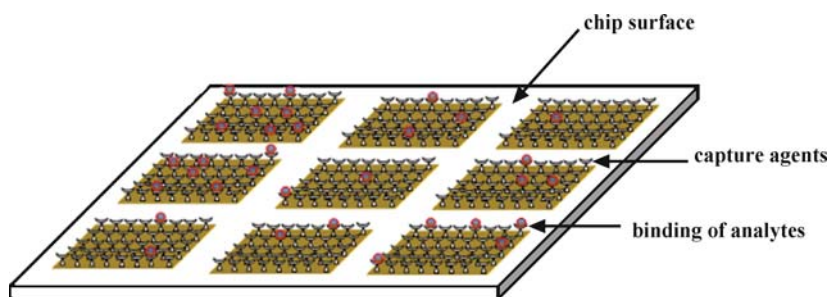
Protein microarrays are very similar to DNA microarrays; however, they are much more difficult to design [4]. DNA is an unstructured linear polymer that hybridizes due to information contained in the nucleic acid sequence. Proteins, however, are much more complex and possess a wide degree of secondary, tertiary and quaternary structures. In addition, protein function is highly dependant on protein structure. As a consequence, protein microarrays must be synthesized in a way that prevents denaturing of the molecule. Maintaining proteins in buffers at appropriate temperatures and pH is crucial in preserving protein shape. In manufacturing protein microarrays several sources of proteins and peptide have been arrayed onto a solid support including synthetic peptides, recombinant proteins, monoclonal antibodies and polyclonal antibodies. Out of these sources, antibody microarrays have been the most developed due to their good biochemical stability.

Basic Methodology

The design and fabrication of a protein microarray parallels that of a DNA microarray. Using a solid support that exhibits low non-specific binding is quite important in the design of a protein microarray. As shown in Fig. 1, typically glass and in some cases silicon and polymer are used as the support for protein microarrays. To increase signal-to-noise ratios, coating with nitrocellulose polymers can be used [4]. Some of the common protein microarray fabrication methods include robotic contact printing, ink-jetting, piezoelectric spotting and photolithography. A microfluidics system for automated sample incubation with arrays on glass slides and washing has also been developed.

In terms of attachment chemistry, covalent bond formation between the protein and the glass slide or affinity interactions are exploited as the means of attaching a protein to the solid support. In the covalent bond formation method, random regions of the protein are used as linkages to bind to the solid support resulting in different sides of the protein being exposed for probing interactions. In the affinity interaction method, the proteins to be plated are expressed as recombinant His-tagged proteins. The tagging of the protein enables the uniform placement of the protein onto the solid support, which can result in a more uniform signal. However, it has been suggested that the tag can artificially enhance some protein–protein interactions. Tagging both N- and C-terminals and comparing the results can resolve this problem.

Several companies have explored different means of attaching an array of proteins onto a solid surface. Schle-



Protein Microarrays, Figure 1 A protein microarray with capture agents spotted on a chip surface. Typically glass and in some cases silicon and polymer are used as the support. Robotic contact printing is often used for spotting the agents on the surface. A microfluidics system for automated sample incubation with arrays on glass slides and washing is included. The analytes of interest are retained by selective binding to the capture agents. Specific binding is monitored by fluorescence or mass spectroscopy.

cher & Schuell and the Pall Corporation have used nitrocellulose-coated FAST and Vivid slides using simple random adsorption. The sensitivity for their antibody microarray has been reported in the range of 1 pg/mL. Xenopore, NoAb, BioDiscoveries, Microsurfaces and TeleChem International have products with surface-reactive groups that can minimize protein denaturation. Perkin Elmer and Biocept have commercially available glass slides coated with a hydrogel that increases the protein loading capacity.

Protein Microarray Applications

Protein microarrays find several uses such as in gene expression, detection of signal transduction, genome mismatch scanning, inflammation, cancer, cell cycle, DNA replication, oxidative stress, hormone action, apoptosis, neurodegenerative disease, infectious disease and cytoskeleton [4]. In broad terms, protein microarrays have found unique and superior advantage in the following areas: expression profiling, protein-protein interactions, drug discovery, and diagnostics.

- **Expression Profiling.** This is one of the main application areas for protein microarrays. The microarrays essentially help in identifying proteins in a sample or mixture. This requires good binding specificity and good binding affinity. In other words, the interactions between antibody and protein must be very specific and strong. Today, this is not done quantitatively, rather qualitatively. As a result, we get a relative difference in expression rather than a quantitative amount. This difficulty in quantifying binding efficiency accurately is due to high non-specific binding such as antibodies binding to proteins at varying degrees. Typically, quantification is done with standard calibration curves on the array for calibration.

- **Protein-protein interactions.** This is another area of protein microarray applications where proteins come to contact with other elements in the cell by binding interactions. These interactions influence the behavior of those elements with which they come in contact (i. e. signal transduction). Studying protein-protein interactions allows deeper understanding of the functionality of cells. Protein microarrays offer a means of directly studying these interactions, unlike DNA microarrays which are indirect in that – for protein expression – extrapolation of genetic sequencing or gene expression is used.
- **Drug discovery.** Drugs are often small molecules and interact with proteins. These proteins become activated or repressed due to the drug-protein interaction. Microarrays are being used to gain more understanding of cross reactivity and toxicity due to a given drug-protein interaction. Microarrays can be used as a means to optimize drug targets and avoid toxicity or side effects from activation of unwanted receptors.
- **Diagnostics.** Two diagnostic formats are generally available. One involves antigens printed onto the microarray with the purpose of identifying antibodies in patients. The other format involves antibodies printed onto the microarray with the intent of identifying proteins expressed as a result of genetic defects or disease.

Major Protein Microarrays in the Market

Many microarrays are available on the market, one of which is Clontech's AB Microarray 500 [5]. This microarray is comprised of over 500 antibodies immobilized onto a glass surface at a high density. This device allows for direct comparison of protein expression in two biological samples. The device uses a fluorescence-based analysis which determines the relative measure of pro-

tein abundance in one sample compared to another. The device is typically used as a screening tool to identify proteins involved in physiological or pathological processes. For example, it can be used to detect proteins involved in signal transduction, cell-cycle regulation and apoptosis. Applications range from toxicity testing, to disease investigation and drug discovery [5].

Several companies are involved in developing protein microarrays for expression profiling and diagnostics. Examples include Zyomyx (Hayward, CA), Axcell Biosciences (Newtown, PA), BioSite (San Diego, CA), Lumi-Cyte (Fremont, CA), Phyllos (Lexington, MA) and ProteinOne (Bethesda, MD). Other players in the protein microarray market include BD Biosciences and Invitrogen who actually have leading positions in the market. Other chips on the market address the need for probing for a specific group of proteins rather than a diverse range. Zyomyx (Hayward, CA), for example, has developed a chip specific for human cytokines.

Key Findings and Future Directions for Research

Significant research and product development efforts in protein microarrays are being carried out in the following key areas: development of capture agents, surface attachment chemistry, and better detection methods.

- *Development of capture agents.* Monoclonal antibodies take a long time to develop and are labor intensive. Among some of the ongoing research in improving capture agents is the use of phage display methods that can be used to isolate high-affinity binding agents against protein targets [6]. Phage display is a system in which a protein is displayed on the surface of a phage as a fusion with one of the coat proteins of the virus.
- *Surface substrates and attachment.* The quality of data obtained from protein microarray experiments is severely affected by the nature of the surface substrate and attachment chemistry [7]. The immobilization of biotinylated proteins onto streptavidin surfaces has been demonstrated to lead to an increase in sensitivity over random attachment in a microarray assay. Poly(ethylene glycol) derivatives used as coatings have also proved successful in increasing sensitivity.
- *Detection methods.* There are several detection methods currently under development. These include laser scanning, ELISA, phosphorimaging, atomic force microscopy, nanoparticles and rolling circle amplification. Currently, research efforts continue in improving detecting binding events. One promising area is the use of sandwich immunoassay with labeled detection antibodies. For ELISA-based assays that have certain restrictions, a highly sensitive detection strategy that

uses rolling circle amplification (RCA) has been introduced [8].

Cross References

- Protein Microarrays
- DNA Microarray
- Lab-on-a-Chip (General Philosophy)
- Lab-on-Chip Devices for Protein Analysis

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Protein Profiling

- Lab-on-Chip Devices for Protein Analysis

Proteolysis

Synonyms

Enzymatic or chemical digestion

Definition

Proteolysis is the process of cleaving protein molecules into smaller pieces (peptides) via chemical or enzymatic reactions. Living systems carry out proteolysis to obtain amino acids from protein substrates or as part of the construction of new proteins. In proteomic analyses, it is often

necessary to perform proteolysis in order to facilitate identification of a protein of interest, as digest peptides form a characteristic “fingerprint” of the parent protein.

Cross References

- ▶ [Lab-on-Chip Devices for Protein Analysis](#)
- ▶ [Liquid Chromatography](#)
- ▶ [Mass Spectrometry](#)
- ▶ [Microfluidic Bioreactors](#)
- ▶ [Proteomics in Microfluidic Devices](#)

Proteomics

- ▶ [Lab-on-Chip Devices for Protein Analysis](#)

Proteomics in Microfluidic Devices

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Definition

Proteomics is the study of the collection of proteins (the proteome) expressed in a given system. The field of proteomics can be divided into three subtypes:

- *profiling* – identification and quantification of the proteins in a sample and drawing comparisons between samples,
- *functional* – the study of the behavior of specific proteins, and
- *structural* – the study of protein folding and the complexes it forms.

Microfluidics is characterized by devices that contain channels or features with dimensions measured in microns, and are capable of transporting and analyzing sample volumes in the nanoliter range.

Overview

Proteomics is a field that first gained prominence in the mid 1990s. The sheer number of proteins as well as the number of variations that comprise the proteome presents a huge challenge for analysis. Serum protein concentrations can vary by several orders of magnitude, and often

a single disease will affect both high- and low-abundance proteins to varying degrees. In addition to basic protein structures and sequences, a vast array of post-translational modifications exists that significantly impacts the functionality, structure, and activity of the native proteins. The primary applications of microfluidics in proteomics have been in the profiling and functional subtypes – structural proteomics has not yet been extensively studied on the microscale.

Profiling proteomics is used to identify all of the proteins in a sample or to compare changes in protein expression between several samples. Due to the complexity of the proteome, accurate analysis requires a great deal of sample processing, including proteolytic digestion, followed by multidimensional separations and mass spectrometry. Separations and mass spectrometry are covered in other chapters of the *Encyclopedia* ▶ [lab-on-chip devices for protein analysis](#), ▶ [chromatographic chip devices](#), ▶ [interfaces between microfluidics and mass spectrometry](#), but chemical processing for profiling proteomics will be discussed here. Readers are encouraged to see Friere and Wheeler [1] for a comprehensive discussion of proteome profiling in microfluidics.

The study of functional proteomics on the microscale comprises an assortment of assays to identify functional groups on proteins and evaluate binding events to determine functional properties of proteins. Functional proteomics is exploited in *enzyme assays* and *immunoassays* to characterize the behavior of a protein within a biological system.

There are a number of ways in which enzymatic studies contribute to the understanding of the proteome. Enzymes are commonly used in proteomics to investigate analytes that are difficult to measure by conventional means. Quantitation of small molecules with enzymatic methods provides insight into the concentration and activity of the proteins associated with those molecules. A great variety of these enzymatic assays have been carried out in microfluidic devices. Another function of enzymatic assays is in kinetics; measurements of properties of enzymes such as the Michaelis–Menten constant (the concentration of substrate when the reaction rate is half the maximum rate) and the turnover number (the number of moles of substrate that are converted to product per catalytic site per unit time) are vital to understanding the mechanics of the proteome, and are used to characterize the effects of known drugs and discover new ones.

The immunoassay is another widely used technique that has been applied to a diverse set of proteomic applications, including the identification and quantization of protein markers for diagnosis of diseases where laboratory facilities are not readily available. Immunoassays can be

designed to detect very small amounts of antigens and to measure the concentrations quite accurately; in addition, further information about the disease markers, such as the existence of structural changes and posttranslational modifications, can be elucidated. Since immunoassays typically comprise a number of repetitive steps and require expensive reagents, they are a ripe target for adaptation to the microscale format, where sample sizes, reagent quantities, and analysis times are all greatly reduced.

Basic Methodology

The methods used in microscale proteomics applications can be divided into two categories, related to the features of microfluidic devices, and the nature of the analyses and assays themselves.

Features of Microfabricated Proteomic Devices

In proteomics applications, the most important features of microfluidic systems are related to delivery of fluid reagents, immobilization of proteins, mixing and reactions, and geometric microarrays.

Movement of Fluids

A critical decision for any microfluidic system is how to generate and control the flow of fluid. The most common methods rely on external pumps or electrokinetic flow. Conventional pumps are too large for pumping $\sim$ nL volumes of solutions, requiring splitters to isolate a fraction of the pressure to apply to the device. Smaller pumps such as syringe pumps are used frequently, but can be difficult to interface with microchannels in a device. Another common option is using electroosmotic (or electrokinetic) flow (EOF) to move analytes through a chip. This method is advantageous because it doesn't require any pressure to be applied on the sometimes fragile chips, it can move fluids at fast rates (exceeding 33 mL/min), and electrodes can easily be integrated into the chip itself. The magnitude of EOF is a function of the density of charge on microchannel walls (or on the surface of packed media); thus, careful consideration must be made in determining the materials used to fabricate devices. If materials without adequate charge density are used (or if the native EOF is too weak or strong), there are a number of coatings that can be used to tune the direction and magnitude of EOF ([▶electroosmotic flow](#)). Electrokinetic effects can also be used to separate the constituents of solutions; however, this can be one of its biggest disadvantages. Any fluid pumped by this method will have its components at least partially separated from one another, creating an inherent bias in any injected sample plug, and flow rates will vary due to

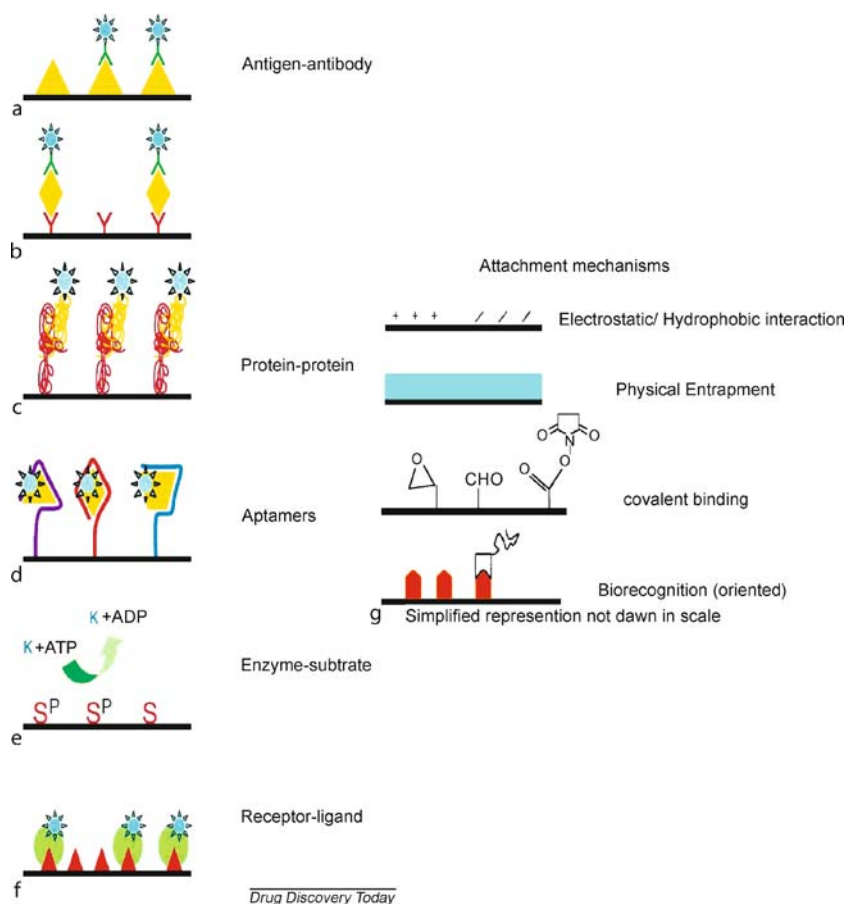
protein adsorption on the interior of the chip or with Joule heating in the device. A final disadvantage is that all solutions must contain buffer ions in a particular concentration range, which is not convenient for all applications.

There are several less common alternatives for controlling fluid flow in microchannels. [▶Digital microfluidics \(DMF\)](#) is an alternative fluid handling method that is beginning to be used in proteomic applications. In DMF, discrete droplets are manipulated on a surface (rather than as streams in enclosed channels) via electrowetting and/or dielectrophoretic forces generated when an electrical potential is sequentially applied to electrodes in an array. This technique can be used with a wide variety of different fluids and can be applied to some aspects of sample preparation and purification. Hydrodynamic pumping, in which flow is driven by siphoning from one fluid reservoir to another, is an option for driving flow that does not introduce a sampling bias. It can, however, be difficult to control; consistency and fine-tuning of flow rate are problematic. Centrifugal pressure flow is another attractive option for fluid movement in protein studies, as it is insensitive to pH, ionic strength, and other characteristics of the fluid that may need to be varied for different assays. Centrifugal pumping is also relatively unaffected by the deposition of analytes onto channel walls (unlike EOF, DMF, etc.); however, it typically relies on single-use valves and therefore flow can only be driven one-time, and in one direction.

Immobilized Proteins

There are two basic formats in which assays are incorporated in microfluidic devices: homogeneous, or solution-based, assays, and heterogeneous, or surface-based, assays. Both approaches significantly reduce sample and reagent consumption relative to their macroscale counterparts, but heterogeneous assays are emerging as the dominant method for proteomics, as they can be adapted to many different types of interactions (Fig. 1a–f) [2].

Heterogeneous assays require proteins or other substrates to be immobilized on surfaces, which allows for conservation and repeated use of the immobilized bed, as well as facilitating the analysis of a sample without need for a separate concentration or mixing step. Also, the binding event or interaction between analytes is not restricted by equilibrium conditions or adequate mixing/diffusion of reagents. Finally, immobilization makes it easier to incorporate more than one reaction-zone in a device, increasing the number of analytes that can be performed on a sample and taking greater advantage of the benefits of a microfluidic system. Several properties are required for useful surface-immobilized proteins, including: the sur-



Proteomics in Microfluidic Devices, Figure 1 (a)–(f) Several heterogeneous protein assay formats (reprinted from [2]). A variety of protein binding events can be studied, ranging from antibody/antigen to protein/protein to enzyme/substrate interactions. (g) Strategies for the immobilization of proteins on solid supports (reprinted from [4])

face must have a high binding capacity for the capture protein, the immobilized protein must retain its biological activity, the protein must be spatially accessible, and there must be a low level of *nonspecific* interactions between proteins in solution and the coated surface.

In microfabricated devices, protein immobilization is most often carried out on glass or silicon supports, using a wide variety of surface chemistries [3]. The most common methods for attachment of proteins to surfaces are depicted in Fig. 1g [4], ranging from simple (e.g., adsorption of proteins via electrostatic or hydrophobic interactions with the surface, entrapment in immobilized polymers, etc.) to more sophisticated techniques, relying on covalent bonding and biological recognition (e.g., biotin/streptavidin). In general, the most robust immobilized protein beds are formed by covalent attachment. One popular method requires two steps: a first layer is formed by silane chemistry such that the surface presents a reactive group (e.g.,

epoxy, amino, thiol, etc.), and a second layer (e.g., the protein to be immobilized or a linker molecule) is covalently attached via the immobilized reactive groups. Other strategies include covalent attachment to functionalized polymer coatings on the interior of the device or linkage to the free end of a self-assembled monolayer (SAM) on a gold surface. Most of these methods result in incomplete coating of the surface, and some reactive groups are left vulnerable to interaction with target proteins during analysis; this phenomenon is called non-specific interaction, and can negatively impact assay performance. A number of strategies exist for blocking surfaces to prevent non-specific binding of analytes, most of which involve coatings of polymers (e.g., polyethylene glycol) or proteins (e.g., albumin or casein) that will not interact with the sample. However, even with these precautions, interactions between analytes and the surface of the device are very difficult to eliminate.

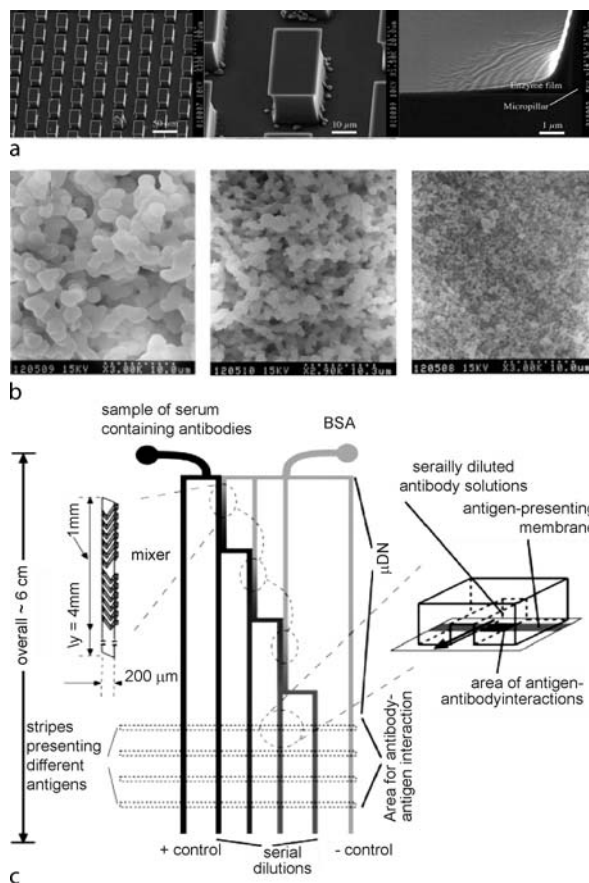
Mixers and Microreactors

In contrast to heterogeneous assays, in homogeneous assays, solutions of an enzyme and a substrate or an *antibody* and antigen are combined in a channel to evaluate the enzyme kinetics or properties of inhibitors, or measure the amount of displaced antigen, which requires thorough mixing. In low Reynold's Number microfluidic channels, turbulent mixing is not possible, and diffusive mixing dominates. Diffusive mixing is a reasonable technique for small molecules; however, enzymes and other large molecules diffuse much too slowly for adequate mixing in many proteomic applications. A number of passive mixing schemes in microfluidic systems have been developed – for example, a network of interconnecting channels of varying lengths and widths is capable of mixing solutions being pumped by electroosmotic flow, while patterned surface grooves enable mixing in a pressure-flow device.

One solution to the challenges of mixing proteins in microfluidic systems is to immobilize proteins in *microreactors* [5]. These systems typically consist of chambers of enzymes immobilized on beads, micropillars [6], or porous polymer monoliths [7] (Fig. 2a–b). Such systems have large surface area-to-volume ratios, which minimizes diffusion time for reactions with solution-phase reagents. Microreactors can be used either for the conversion of an analyte to another form that is more easily detected, or for direct studies of the properties of enzymes and substrates. One of the most common uses is for the digestion of proteins for proteome profiling, but such systems can also be used for the removal of amino acid residues from peptides or proteins or for enzyme kinetic studies.

Microarrays

Biological microarrays are two-dimensional arrangements of immobilized spots of molecules on a surface. When a microarray is exposed to a solution containing unknown analytes bearing an appropriate label, analytes binding to particular spots on the surface can be rapidly identified. DNA microarrays were introduced in the late 1990s, and have become a widely used tool for high-throughput genomic analysis; however the development of similar tools for proteome analysis has been hampered by the diverse properties of proteins. For example, all DNA molecules inherently possess specific capture properties through complementary base-pairing; proteins do not have such a universal property. Additionally, protein capture *ligands* tend to be more delicate in structure and function than the relatively robust DNA molecule. Finally, small amounts of DNA can be detected quite easily using amplification techniques such as the polymerase chain reaction (PCR); no such amplification technique exists for proteins.



Proteomics in Microfluidic Devices, Figure 2 (a) An enzymatic microreactor containing micropillars that have been coated with trypsin for protein digestion. Reprinted from [6]. (b) Porous monolith microreactors of various pore sizes. Pore size is determined by the ratio of monomer to porogen (solvent) during polymerization. Reprinted from [7]. (c) A microdilution network (μ DN) for the quantitation of multiple antigens. The device uses a series of T-intersections to sequentially dilute a serum sample. Herring-bone patterned areas are used as mixers in the dilution channels. Several antibodies have been deposited on a membrane using a microfluidic network, and the membrane is incorporated in the bottom of the μ DN device. Reprinted from [12]

Despite these disadvantages, the promise of multiplexed detection of analytes has made protein microarrays a popular research topic in microfluidic implementations of proteomic analyses.

The immobilized spots used in protein microarrays may contain protein molecules, or smaller molecules such as inhibitors or ligands with which a protein in solution may interact. Very small spots are advantageous in that higher detection throughput can be achieved with minimal amounts of printed material and sample. Spots are typically deposited either by contact methods, such as printing arrays that use needles to deposit droplets directly on the

surface, or by non-contact methods, including the use of capillaries or ink jet dispensers to deposit droplets with nanoliter-picoliter volumes.

An alternative to microarrays is the use of arrays of multiple parallel channels. Typically, such structures are used to carry out a number of duplicate assays on one or more samples, to perform a serial dilution of a single sample for quantitation of a given protein, or to detect multiple targets in a single sample. In the latter case, an array of channels may be used in conjunction with a printed array of spots or bands – the channel network is used for fluid delivery to the printed array (Fig. 2c).

Analyses and Assays

Microfluidic devices are used in a variety of profiling and functional proteomics applications; the goal in much of this work is to reduce sample and reagent consumption and increase throughput. The primary application in profiling proteomics is in the preparation of protein samples for separations and mass spectrometry, while the major functional proteomics tools are enzymatic and immunological assays.

Sample Preparation for Profiling Proteomics

The identification of unknown proteins often requires that they be digested (or divided) into a group of constitutive peptides, which are interrogated by mass spectrometry. Comparing the masses of the peptides to genomic databases enables absolute identification by means of peptide mass fingerprinting (PMF). Proteins are digested by exposure to proteolytic enzymes (e.g., trypsin), or other lytic reagents (e.g., cyanogen bromide, CNBr); this is often part of a multistep process requiring 12–18 h, including mixing the sample with buffer and denaturant, mixing and incubating the sample with reducing agent to reduce disulfide bonds, mixing and incubating the sample with alkylating agent to prevent disulfide bonds from reforming, and finally, mixing and incubating with lytic reagent. This process is extremely tedious and time-consuming, making it an attractive application for microfluidics.

Several device configurations have been used for proteomic processing in microfluidic devices, including open channels, immobilized beads, and other solid phase media. Open-channel devices for tryptic digestion and reduction of disulfides in proteins are capable of achieving complete tryptic digestion in timeframes ranging from 15 min to 5 s. Immobilized beads increase the surface-to-volume ratio further; for example, a bed of trypsin-modified beads immobilized in a microfluidic chamber can be used very effectively for *proteolysis* of relatively large volumes of

flowing sample. An alternative method for increasing the surface area-to-volume ratio in microfluidic devices is to use trypsin-modified monoliths, formed from polymer plugs cured in situ, or from membranes sandwiched between channels. Trypsin and other enzymes can also be copolymerized within hydrogel microstructures – the hydrogel is a biocompatible matrix that reduces the likelihood of damage to the enzyme during and after immobilization.

Microfluidics is promising for developing tools with integrated chemical processing of proteomic analytes, by virtue of fast reaction kinetics. Sample purification and concentration can be carried out on the microscale by methods such as solid phase extraction on hydrophobic surfaces ► [lab-on-chip devices for sample extraction](#) or affinity chromatography ► [chromatographic chip devices](#). Immunoassays on magnetic beads have been used not just for detection of antigens (described below), but also as a purification method for the isolation of a target for mass spectrometry. However, to our knowledge, there have been no reports of implementing a fully integrated process, including stepwise reduction, alkylation, and digestion. Microfluidic tools for chemical processing will likely not be widely adopted until this critical benchmark is achieved.

Enzymatic Assays

Traditionally, enzymatic assays are performed in microtitre plates (plastic trays containing arrays of isolated wells), where an enzyme and its substrate are mixed in an individual well, and the substrate turnover is measured to determine the activity of the enzyme. Microfabricated devices that can perform enzymatic studies without interference between individual elements are highly desirable, as the macroscale method consumes a great deal of enzyme. There are three basic applications of enzymatic assays in proteomic studies:

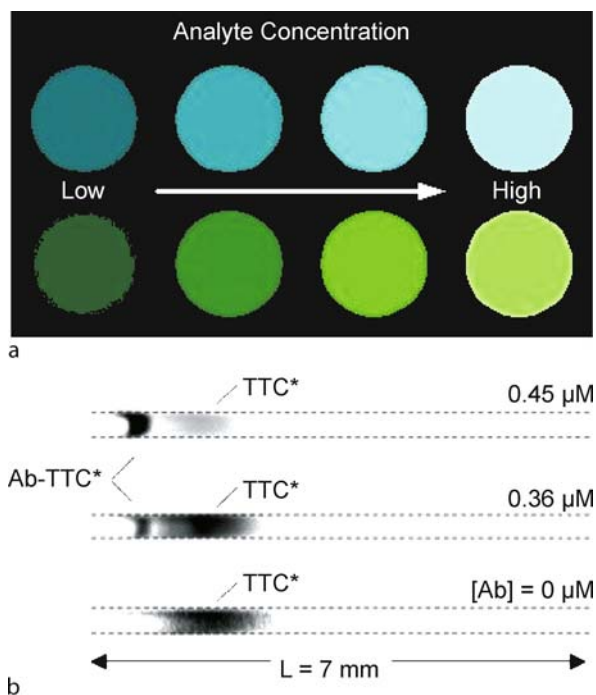
1. small molecule detection or sensing via an enzymatic reaction,
2. studies of enzyme–substrate, protein–ligand, and protein–protein interactions, and
3. kinetic studies.

The first type of enzyme application in microfluidics is chemical sensing. Sensors can be constructed in cases where an enzyme turns over a particular small-molecule substrate to produce a product quantifiable by fluorescence, chemiluminescence, absorbance spectroscopy, or electrochemical detectors. In cases where the substrate is not detectable itself, an enzymatic product can often be coupled to another enzyme that produces a detectable product. For example, there are a wide variety of small

molecules (such as nutrients, amino acids, and sugars) that can be coupled to the chemiluminescent reaction of luminol and peroxide in the presence of horseradish peroxidase. These enzyme-substrate assays were the first to be adapted to microfluidic devices – a great number of small-molecule sensors have been developed based on microfluidic channels with electroosmotic or hydrodynamic flow, and pre-loaded microfluidic cartridges containing nanoliter volumes of reagents have shown great promise as a replacement for 96-well plates for high-throughput screening [8].

The second type of enzyme application in microfluidics is protein interaction studies. In order to study interactions between enzymes and substrates, peptides, or other proteins, an alternative to microtitre plates is the microarray. These arrays are very often used in conjunction with microfluidic sample and reagent delivery systems; alternatively, microchannels can be used to deposit spots in an array. By immobilizing substrates or inhibitors on the array, it is possible to screen large numbers of molecules for their relative abilities to bind a particular enzyme. This is typically accomplished by tagging the enzyme with a fluorescent molecule and using the intensity of the array spots to determine the amount of enzyme bound (Fig. 3a). This is a promising approach for the identification of protein drug targets, or for the evaluation of compound libraries in drug discovery. Peptide fragments of proteins of interest have great potential as capture ligands, as they are easy to synthesize, stable, and able to mimic proteins. However, small molecules such as peptides are difficult to immobilize on a surface such that they are still accessible to the sample. Peptide fragments are also not as specific as complete proteins; for some applications, it is preferable to immobilize an entire protein to characterize its interactions with other proteins or enzymes of interest. Sample delivery to the elements of an array via microfluidics enables hundreds of different analyses to be performed on a single sample while reducing the size of that sample from microliter volumes to nanoliters.

A third type of enzyme application in microfluidics is kinetic studies, which may be carried out in either homogeneous or heterogeneous format. In homogeneous assays, adequate mixing of the enzyme, substrate, and any inhibitors that are being studied is crucial – this is done either via the merging of reagent streams at a T- or Y-shaped junction between channels or by means of a microfabricated mixer to ensure complete homogenization. A microfluidic mixer is advantageous – it ensures complete mixing of the enzyme and substrate and a smaller quantity of enzyme is consumed. Electrophoretically-mediated microanalysis, or EMMA, uses electrokinetic forces as the pumping method, which has the added ben-



Proteomics in Microfluidic Devices, Figure 3 Detection of binding of a protein analyte in a heterogeneous (a) or homogeneous (b) assay. In (a), the fluorescence intensity of spots on an array reflects the amount of the target protein in the sample solution. Different analytes are tagged with molecules that fluoresce at different wavelengths to simplify identification. In (b), the tetanus toxin-C-fragment (TTC) is fluorescently labeled and separated from the antibody-TTC complex in an electrophoretic channel for quantification. Reprinted from [11]

efit of separating product from substrate as the analysis proceeds. EMMA was first developed in capillary electrophoresis systems, but was readily adapted to microchannel setups [9]. Since a chip disperses Joule heat much more efficiently than a capillary does, the analysis can be run at higher potentials, increasing the speed of measurement. In most cases, a modified substrate is used that is converted to a fluorescent, chemiluminescent, or colored product by the enzyme, and optical methods are used to calculate factors such as the turnover number and Michaelis constant. Kinetic assays have also been carried out in centrifugal flow microfluidic devices [10], which are applicable for a wide variety of biological samples, including whole blood, plasma, and urine, that are incompatible with electroosmotic flow systems. Centrifugal devices are also easily adapted to perform multiple analyses on a single plastic device, reducing the costs of manufacturing. However, fresh enzyme is required for each determination, making enzyme recovery difficult. Heterogeneous enzyme microreactors eliminate the need for recovery

steps and dead time during the measurement. Microfluidic devices containing immobilized enzymes enable the study of kinetics with excellent enzyme recovery and decreased reagent consumption, increasing the overall efficiency of the method and greatly decreasing waste.

Immunoassays

Like enzymatic assays, immunoassays are conventionally implemented in microtitre wells, but they are even more easily adapted to the microscale. Homogeneous immunoassays are typically employed for the study of small molecules, while heterogeneous immunoassays are preferred for large-molecule detection.

A homogeneous assay is carried out by combining an antigenic sample and an antibody by means of electrokinetic flow in a microchannel. After the antigen binds the antibody, the complex can be separated from the remaining free antibody via electrophoresis to determine the amount of antigen present in the sample (Fig. 3b) [11]. On a microfluidic chip, this analysis can easily be multiplexed – a different assay can be performed in different channels on the same sample.

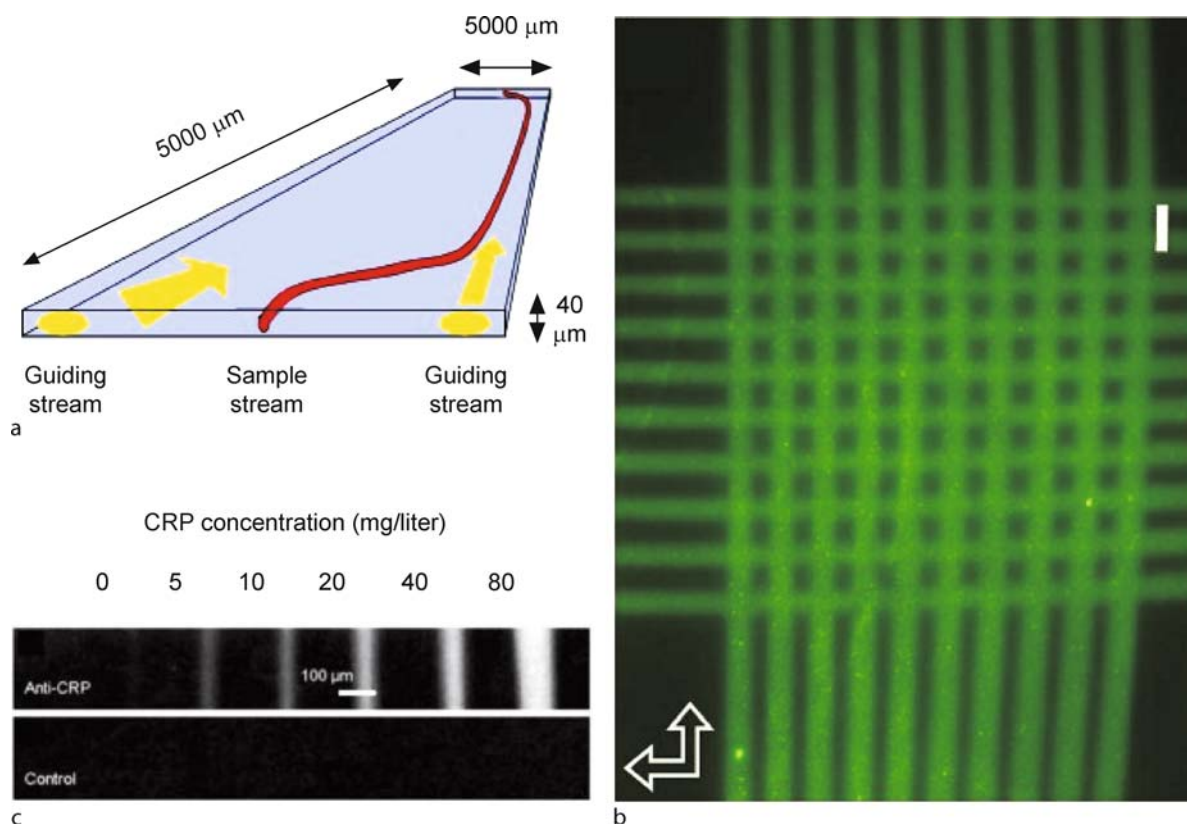
There are two common forms of heterogeneous assays used to detect and quantify disease markers: the direct-labeling method, in which the proteins in a sample are labeled with a detectable tag and isolated from a complex sample by means of a bed of immobilized antibodies, and the indirect-labeling method, in which two antibodies are used for each marker. Sandwich-type immunoassays, such as the enzyme-linked immunosorbent assay (ELISA), require two antibodies: an immobilized antibody used to capture a molecule of interest – a protein, a virus, or a small molecule (e.g., a hormone, pharmaceutical, or environmental pollutant) – and a second antibody in solution that is conjugated to a detectable tag. Immunoassays have also been carried out with detection via surface plasmon resonance microscopy, atomic force microscopy, and electrochemical sensors, eliminating the need for a detectable label. These assays are very useful because the antibody-antigen binding that must occur at each step is highly specific and thus has built-in redundancy – they are very sensitive and capable of quantitation of changes in concentration with a dynamic range of up to 2–3 orders of magnitude. As for enzymatic assays, microfluidic channels can be used either to deposit the first antibody onto a surface for immobilization or to direct the sample over the immobilized antibody (concentrating the antigen on the surface in the process) and any subsequent reagents – in some cases, microfluidic systems have been used for both purposes [12]. While basic immunoassays are well-suited for determining the pres-

ence of a given molecule in a sample and sometimes the quantity, they do not provide any other information about the target molecule. In recent work, immunoassays have been adapted to provide structural and post-translational modification information on marker proteins, however, these assays have not yet been widely applied to microfluidic devices and are still primarily performed in microtitre wells.

Key Research Findings

In the first report of on-chip proteolytic processing, Gottschlich et al. used open-channel devices for tryptic digestion and reduction of disulfides in proteins, which enabled complete processing in 15 min. More recently, Huang et al. [13] and Liu et al. [14] used surface adsorbed trypsin in channels to enable complete digestion in less than 5 s. In an alternative strategy, Yue et al. used a bed of trypsin-modified agarose beads immobilized in a weir in a glass microfluidic device to digest β -casein [15]. Peterson et al. used the related heterogeneous technique of forming methacrylate plugs cured in situ to increase the surface area-to-volume ratio to achieve tryptic digestion of myoglobin in ~ 11 s [16]. Sakai-Kato et al. developed similar methods using trypsin immobilized in a sol-gel matrix [17]. Gao et al. reported using a trypsin-modified polyvinylidene fluoride (PVDF) membrane in a PDMS microfluidic chip, enabling protein digestion in 3–10 min [18]. Each of these methods represents a vast improvement over conventional macroscale proteolysis, which typically requires hours-days. One promising approach for sample purification during processing is the use of digital microfluidics to remove hydrophilic impurities from proteomic samples [19].

Hadd et al. performed the first homogeneous assays to determine enzyme kinetics in a microfluidic chip – these assays relied on diffusional mixing of β -galactosidase and resorufin β -D-galactopyranoside (RBG) within a channel electrophoresis system [9]. This assay successfully characterized the kinetics of the enzyme, and reduced reagent consumption by four orders of magnitude over conventional assay methods. The same assay was later modified by Burke et al. by including a passive microfluidic mixer, resulting in a characterization of enzyme kinetics that more closely agreed with that obtained via conventional methods [20]. Puckett et al. developed a centrifugal microfluidic system in which homogeneous protein-ligand binding assays were performed to detect and characterize the binding interaction between phenothiazine antidepressants and calmodulin, a calcium-binding protein known to interact with this class of drugs [21]. This system utilized pre-measured aliquots of reagents,



Proteomics in Microfluidic Devices, Figure 4 (a) Use of hydrodynamic flow to create a microfluidic stream and direct it to specific areas of a larger channel. (b) The flow system is used to deposit biotinylated fluorescein in a grid pattern. The direction of flow during deposition is indicated by arrows. (c) Quantitation of C-reactive protein (CRP) in whole blood. Samples of blood were spiked with CRP and the microstream directed across a surface coated with anti-CRP before exposure with a Cy3-labeled secondary antibody. Reprinted from [24] under open access agreement with BioMed Central.

increasing the speed and reproducibility of the analysis for high-throughput applications. In an effort to integrate multiple analyses on one device, Wang et al. incorporated two homogeneous assays into one microfluidic channel: an enzymatic assay for glucose and an immunoassay for insulin [22]. Electrokinetic effects were used for both fluid movement and separation of the components before detection. Through the use of pre- and post-column reactors and an electrophoretic separation of analytes, two independent measurements were made in the same space.

Perhaps the greatest area of interest in microfluidics for proteomics applications is in multiplexed, heterogeneous assays. Kim et al. used microfluidic PDMS channels to pattern sol-gels on polyvinylacetate-coated glass slides under conditions mild enough to retain protein activity [23]. The use of a microfluidic channel system generated a well-defined, intricate pattern of immobilized protein that simplified data collection. Jiang et al. patterned antigens on a membrane via a microfluidic network, and

then positioned the membrane under a microdilution network to carry out serially-diluted immunoassays to detect multiple antigens on the patterned surface (Fig. 2c) [12]. This fluidic network enables the simultaneous quantitation of several molecules that vary widely in concentration – a major challenge in proteomics – in a single submicroliter sample. Brevig et al. developed a hydrodynamic microfluidic system that uses two streams of fluid to focus a sample and perform parallel immunoassays to quantitate levels of C-reactive protein, a well-known marker of inflammation (Fig. 4) [24]. As this method effectively creates a microscale stream within a larger chamber, it conserves sample but can also be carried out on whole blood without risk of clogging very small channels. In recent years, a great number of biochips have been made commercially available that use networks of microfluidic channels to perform analyses of over 100 different proteins using only 0.1 μ L of whole blood [25]. These chips are capable of separating plasma from blood cells, and then directing the plasma sample across an ELISA array – the heterogeneous

array format and simplified sample prep are advantageous in terms of both speed of analysis and reduced sample consumption. This is very promising, as the cleanup of whole blood is a major bottleneck to the use of microfluidics in medical applications.

Future Directions for Research

The future of miniaturization in proteomic applications lies in achieving greater sophistication of devices. A variety of arrays have been used to investigate post-translational modifications such as phosphorylation and glycosylation, and it is only a matter of time before these are incorporated into microfluidic systems. Factors such as antibody synthesis currently limit the range of proteins that can be incorporated on an array, and devices that are capable of simultaneous quantitation of a protein and the study of its post-translational modifications, activity, etc., are not yet fully realized. Enzymatic microreactors are evolving in directions that will enable rapid screening of enzyme inhibitors for pharmaceutical purposes, industrial-scale synthesis, and environmental applications such as conversion of used grease to diesel fuel. Tools such as biochips mark the beginning of a shift from central laboratory to point-of-care testing, one of the hallmark aims of microfluidic technology. Together, these innovations will enable the development of a true total analytical system capable of greatly increasing the efficiency and throughput of any number of diagnostic, prognostic, and industrial processes.

Cross References

- ▶ Biosample Preparation Lab-on-a-Chip Devices
- ▶ Chromatographic Chip Devices
- ▶ Digital Microfluidics
- ▶ DNA Micro-arrays
- ▶ Interfaces between Microfluidics and Mass Spectrometry
- ▶ Lab-on-Chip Devices for Immunoassay
- ▶ Lab-on-Chip Devices for Protein Analysis
- ▶ Lab-on-Chip Devices for Sample Extraction
- ▶ Microfluidic Bioreactors
- ▶ Droplet Microreactors

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Proton Exchange Membrane Fuel Cells

► Microstructured Hydrogen Fuel Cells

PRT

► Resistance Temperature Detectors

Pump Head

Synonyms

Backpressure; Differential pressure; Head pressure

Definition

Pump head is the pressure rise from pump inlet to pump outlet. The *characteristic curve* of a pump relates pump head to flow rate. The pump head at which the flow rate is zero is called the *shut-off head* or *maximum head*.

Cross References

- Magnetic Pumps
- Membrane Actuation for Micropumps
- Peristaltic Pumps
- Positive Displacement Pump

Pyroelectric

► Pyroelectric Flow Sensors

Pyroelectric Flow Sensors

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Synonyms

Pyroelectric sensors; Pyroelectric; Thermal flow sensors

Definition

A *pyroelectric* is substance that generates an electrical charge during a temperature change. The *pyroelectric coefficient* is the measured change in the polarization with a variation in the temperature in a single-domain pyroelectric material. *Pyroelectric sensors*, in principle, work as thermal transducers. The pyroelectric sensing element converts a non-quantified thermal flux into an output measurable quantity like charge, voltage or current.

Overview

Pyroelectric Effect

The pyroelectric coefficient (p) of a material under constant stress and electric field is defined by the following expression:

$$p = \left(\frac{\partial P}{\partial T} \right)_{E, \sigma} \quad (1)$$

where P is the polarization, T the temperature, E the electric field and σ the elastic stress.

Pyroelectrics can be classified into two main categories:

- non-pyroelectric pyroelectrics, those whose polarization cannot be switched by an application of external electric field (such as some semiconductors and biological materials); and
- pyroelectric pyroelectrics, those whose polarization is obtained by poling and also can be switched by an electric field.

The pyroelectric effect of pyroelectric pyroelectrics usually exists below a certain transition temperature called the *Curie point*, T_c , in proper pyroelectrics and is more temperature dependent than that of the non-pyroelectric pyroelectrics.

Primary and Secondary Pyroelectric Effects

Thermodynamic analysis of the pyroelectric effect yields the expression

$$P_i^\sigma = P_i^\varepsilon + d_{ijk}^T c_{jklm}^{TE} \alpha_{lm}^\sigma \quad (2)$$

where P_i^σ is the total pyroelectric effect measured at constant stress, and P_i^ε , the pyroelectric effect at constant

strain ε , is called the primary effect. The second term of Eq. (2) is the contribution caused by thermal deformation of the crystal and is known as the secondary pyroelectric effect. Parameters d , c and σ are the piezoelectric, elastic stiffness and thermal expansion coefficients of the crystal, respectively.

In pyroelectric ferroelectrics, generally, the primary effect is large and negative because the spontaneous polarization decreases with increasing temperature. The secondary effect is considerably smaller and may have either sign depending on the values of the constants. And so the experimentally observed pyroelectric coefficients are dominated by the primary effect and are negative in pyroelectric crystals. But in non-ferroelectric pyroelectric crystals there is no simple way to predict the sign and the magnitude of their coefficients. In practice the total pyroelectric effect can be measured and, also, the secondary effect calculated from the constants d , σ and c .

Pyroelectric Currents

Pyroelectric currents are proportional to the time derivative of the temperature and, in case of infrared (IR) detectors, to the changes of IR radiation. In order to assess stationary IR sources as well, a chopper is used to modulate the IR radiation. Signals thus exhibit a well-defined angular chopping frequency ω . The smallest detectable temperature change is limited to the temperature fluctuation of the detector system, the electronics or the intrinsic noise current j_n of the pyroelectric element, given here for a thin film by the expression

$$j_n = \sqrt{\frac{A}{t} 4KT(\sigma + \omega \varepsilon_r \varepsilon_0 \tan \delta \Delta f)} \quad (3)$$

where σ is the leakage conductivity of the pyroelectric thin film (in bulk detectors, the parallel resistor); ω the angular frequency; $\tan \delta$ the loss tangent; Δf the frequency bandwidth; A the surface; and t the thickness of the element.

The crossover frequency $\omega_{cl} = \sigma / \varepsilon_r \varepsilon_0 \tan \delta$ between resistor-type noise and dielectric noise is typically 0.1–10 Hz. At typical modulation frequencies of 30 Hz, the dielectric noise dominates. The figure of merit of dielectric and pyroelectric material parameters for an optimal detectivity can thus be formulated as

$$F_D = \frac{P}{\sqrt{\varepsilon_0 \varepsilon_r \tan \delta}} \quad (4)$$

Essential Materials

Pyroelectrics in Materials with Perovskite Structure

Perovskite oxides have the structural formula ABO in which A may be some large cation, such as Ba^{2+} or Pb^{2+} ,

and B may be some medium size cation, such as Ti^{4+} or Zr^{4+} . These cations are located in cages formed by the network of the oxygen anions. There is an abundance of perovskites, natural (minerals) and synthetic ones, because of a large number of cations having sizes that can fit into the oxygen cages. Pyroelectric perovskites are a sub-group in the perovskite family.

Lead Zirconate Titanate (PZT)

Because of its high piezoelectric constants, PZT is the most widely used pyroelectric ceramic in sensor applications. PZT is a solid solution of lead titanate and lead zirconate. The substitution of a small fraction ($\sim 1\%$) of the PZT cations by other cations can improve the properties. This method is always used to produce PZT that is optimized for different applications. Doping by cations of lower valence, e.g. Fe^{3+} replacing Ti^{4+} , Zr^{4+} and/or Na^+ for Pb^{2+} , results in oxygen vacancies that stabilize the domain walls. PZT ceramics with dopants of lower valence (acceptors) are very useful for underwater applications and also for high-power and high-voltage applications. Doping with a higher valence than the substituted cations (donors), e.g., Nb^{5+} for $\text{Ti}^{4+}/\text{Zr}^{4+}$ and/or La^{3+} for Pb^{2+} , increases the concentration of lead vacancies, enhancing the mobility of domain walls. These materials are used for medical transducers and for sensitive pressure sensors. PZT ceramics for pyroelectric applications are rich in titanium. The compositions rich in titanium (e.g., zirconium titanium) have a high pyroelectric coefficient and low permittivity, giving good pyroelectric figure of merit. Additionally, their coercive field is high, reducing the risk of depolarization during use and making this composition useful also in the form of thin films.

Relaxors and Relaxor Pyroelectrics

A large number of lead-containing perovskites, called relaxors, show anomalous characteristics typified by a strong frequency dispersion of the permittivity. Examples are lead magnesium niobate ($\text{PbMg}_{1/3}\text{Nb}_{2/3}\text{O}_3$ or PMN) and lead zinc niobate ($\text{PbZn}_{1/3}\text{Nb}_{2/3}\text{O}_3$ or PZN). These materials have many useful advantages:

- very high permittivity over a wide temperature range,
 - large electrostrictive strain and
 - strong pyroelectric response under a DC bias field.
- Solid solutions of relaxors and normal pyroelectrics, which are the subject of much interest, are called relaxor pyroelectrics, such as PMN– PbTiO_3 and PZN– PbTiO_3 .

Layer Perovskites

A large number of pyroelectrics crystallize in the so-called layer perovskite phases. These phases are made of layers of perovskites separated by other layers; a typical example is bismuth oxide. Many pyroelectric layer perovskites have a high T_c , making them interesting for high-temperature piezoelectric applications. In addition, many of them are lead free. A prototype material is bismuth titanate ($\text{Bi}_4\text{Ti}_3\text{O}_{12}$).

Lithium-niobate and Lithium-tantalate

Lithium-niobate (LN) and lithium-tantalate (LT) are uniaxial pyroelectrics, having trigonal structure, with spontaneous polarization arising from asymmetrical displacement of lithium relative to the other ions. These materials' T_c values are 1210 and 620 °C, respectively. They are always produced commercially in single-crystal forms. Both are much used for surface acoustic wave devices (e. g., high-frequency filters), while LT is used for pyroelectric detection due to its large pyroelectric coefficient and low permittivity.

Pyroelectric Polymers

Pyroelectricity is also found in polymers. Polyvinylidenefluoride (PVDF, $(\text{CH}_2\text{CF}_2)_n$) is used for sensor applications that can benefit from a large sensor area. The pyroelectric polymers have low pyroelectric coefficients d and high coefficients because of the low permittivity. The polymers are light and then have good acoustic impedance matching to water, to organic and biological matter.

Dimensional and Microstructural Aspects of Pyroelectric Materials

Pyroelectric sensors utilize a wide range of material forms: crystals, bulk ceramics, thick layers and thin films. Noteworthy among new compositions with improved performance is the class of relaxor pyroelectrics. Many novel pyroelectric sensors utilize thin and thick films. Bulk pyroelectric ceramics for room temperature, and pyroelectric sensors, are widely available. New compositions with higher permittivity are available now.

Poling of Pyroelectric Materials

After the manufacturing, pyroelectric materials usually do not possess pyroelectric properties, since the pyroelectric domains are randomly oriented. The poling process consists of the application of a DC electric field to align the domains. A remanent polarization persists after the removal of the field and the material is pyroelectrically

active. Poling is usually done at elevated temperatures (see below) using fields larger than the coercive field. During use of pyroelectric materials, care should be taken to avoid depolarization under high temperatures, electric fields or stresses.

In summary, pyroelectric materials have found applications in a wide range of fields. For device applications and the convenience of application engineers, several figures of merit criteria are defined. Most of these can be calculated using the thermal, electrical and optical characteristics of the materials available in the literature and various handbooks on the physical constants of materials.

Pyroelectric Flow Sensors

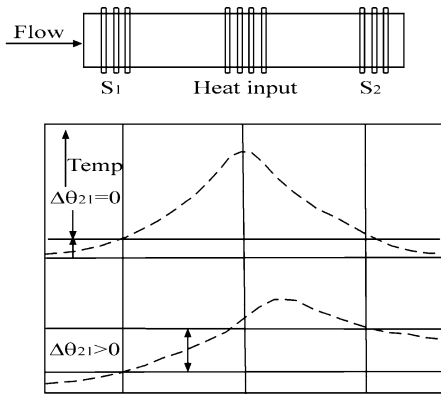
Pyroelectric flow sensors are a kind of thermal flow sensor. They can be classified according to the thermal transfer mechanism, i. e., calorimetric- or boundary-layer-type flow meters and here the latter will be introduced.

Thermal mass-flow meters directly measure the mass-flow rate utilizing the thermal transfer process between a heated element and the flowing mass. According to the thermal transfer mechanism, they can be classified as calorimetric- or boundary-layer-type flow meters. Calorimetric flow meters utilize the simple principle that as the mass flows through the device, power has to maintain the temperature of the incoming mass at a known level. By measuring the supplied power P and the temperature above ambient, ΔT , the mass-flow rate $\dot{M}$ can be calculated as the expression

$$\dot{M} = \frac{1}{c_{pg}} \frac{P}{\Delta T} \quad (5)$$

where c_{pg} is the specific heat of the mass at constant pressure. A typical example of calorimetric flow meters consist of a thin-walled capillary with a heater wire wound around its center. The wire resistor serves both to heat the center section and to provide a measure of the temperature, since the resistance of the heater wire is a function of temperature. Because the capillary has a very small diameter, the mass absorbs heat from the wall and quickly reaches the wall temperature. As the mass follows, additional power has to be supplied to compensate for the energy carried away by the mass. The mass-flow rate may be measured by keeping either the power or the elevated temperature constant.

Another type of capillary flow meter is shown in Fig. 1. A central heater (H) is wound about a thin walled capillary tube. Two resistive sensor elements are mounted on the capillary symmetrically, one upstream (S_1) and the other downstream (S_2) of the heater to monitor the temperature



Pyroelectric Flow Sensors, Figure 1 A capillary flow meter with external heater and transducers is depicted at the top with the corresponding temperature profile along the capillary for zero (upper) and nonzero (lower) gas flow

as depicted in the upper portion of Fig. 1. The difference in temperature between the two sensors

$$\Delta T_{21} = \Delta T(S_2) - \Delta T(S_1) \quad (6)$$

is used to determine the flow rate. In the absence of flow, the temperature profile in the capillary is symmetric with $\Delta T_{21} = 0$ as shown in Fig. 1. As mass flows, heat is transferred from the wall to the moving fluid at the upstream position, thereby lowering $\Delta T(S_1)$. At the downstream electrode, energy is injected into the wall at S_2 by the heated flowing fluid, thereby increasing the temperature $\Delta T(S_2)$. The net effect is an asymmetric temperature profile in the tube causing $\Delta T_{21} \neq 0$, as indicated at the bottom of Fig. 1. This differential measurement has the advantage of being directional and providing a zero signal at zero flow.

Basic Methodology

Methods for Determination of Pyroelectric Coefficients

There are several methods for measurement of pyroelectric coefficients, but most can be grouped into two kinds. The objective in the first case is to directly measure the pyroelectric current and at the same time change the temperature. In the second method measurements of polarization or charge can be estimated either by the integration of pyroelectric current during the continuous heating of the sample or recording the hysteresis loop at various temperatures by the Sawyer–Tower technique.

In the first approach the changing temperature can be generated by chopping the incoming radiation source or AC heating (known as the Chynoweth technique) of the sample and the pyroelectric response current i can be

expressed as follows:

$$i = \frac{p\eta WT_r}{Ct}$$

where η is the power absorbed by the sample, T_r is a transfer function depending on the thermal circuit of the sample and the chopping or modulation frequency, t is the thickness of the sample and C is the thermal capacity of the sample. Often a thin layer of highly absorbing material is applied on the sample in order to achieve $\eta = 1$.

The pyroelectric current $i(T)$ at a temperature T can also be measured by uniformly heating or cooling the sample at a constant rate (generally, 1 to 4 °C/min) and the pyroelectric coefficient, p , can be calculated using the following expression (Byer–Roundy technique):

$$i(T) = \frac{Ap}{(\Delta T/\Delta t)^{-1}}$$

where A is the electroded area of the sample and $\Delta T/\Delta t$ is the heating or cooling rate of the sample. Here, the charge release (in either of the two cases mentioned earlier) at a discrete temperature is determined and the pyroelectric coefficient is obtained graphically or by differentiation.

Methods for Thin-Film Deposition

Thin-film deposition techniques such as sol–gel method, hydrothermal route, sputtering, metalorganic chemical vapor deposition (MOCVD) and pulsed laser deposition have been developed and are being adapted in commercial devices. Table 1 lists some advantages and disadvantages of different film deposition techniques. Three major kinds are sol–gel, physical vapor deposition and chemical vapor deposition. In wet methods, the sol–gel technique uses fine powder in gel forming a liquid-like mixture of metal oxides and organic salts to grow transparent ceramic films. It is low energy consuming process at lower crystalline temperature and allows better control of chemistry by mixing of precursors in the liquid state at the molecular level. Most electroactive ceramics and polymers are amenable to this technique. Physical vapor deposition requires vacuum and uses DC or RF magnetron sputtering and pulse laser deposition to provide high quality, uniformity and stability but low deposition rates of materials. Chemical vapor deposition, i. e., MOCVD, on the other hand, uses volatile chemical precursors, which are vaporized on heated substrates. It is proving suitable for conformational deposition of high-quality three-dimensional structures on bottom electrodes and is characterized by high deposition rates.

Pyroelectric Flow Sensors, Table 1 The advantages and disadvantages of different film deposition techniques

Deposition technique	Advantages	Disadvantages
1 Sol-gel	Lower crystallization temperature, lower cost, better control of mixing of precursors, quickly produce new materials	Lack of phase and composition control, morphology, reproducibility
2 Physical vapor deposition (i) DC sputtering (ii) Pulsed laser deposition (iii) MBE	Compatibility with IC processing Uniformity and scalability, high quality Rapid sampling of materials, quickly produce new materials Composition control, precise atomic layering, extreme flexibility and scalability	High equipment cost Low deposition rate, high point defect concentration Lack of uniformity, high residual stresses Expensive development stage
3 Chemical vapor deposition	High deposition rate, good stoichiometry control, low point defect concentration, high conformity and scalability	Low precursor stability, expensive

Properties to be Considered of an Appropriate Material

While the pyroelectric, dielectric and thermal properties of pyroelectric materials are clearly fundamental in determining the basic performance of a device, there are many other properties to be considered when making a selection of an appropriate material. These are the following:

- **Electrical resistivity.** The circuit in Fig. 2 shows a resistor R_G connected across the active element. This serves three functions. First, it fixes the electrical time constant τ_E of the device. Second, it allows the gate leakage current of the FET to bias its operating point. If there was no bias resistor here, the FET amplifier would not be stable and would take a long time to settle. Third, it determines the voltage responsivity.
- **Piezoelectric properties.** At first sight it seems surprising that piezoelectric properties are important in pyroelectric devices. However, all pyroelectric detectors used in a high-vibration or acoustically noisy environment will produce a piezoelectric microphonic noise signal, mostly through flexure, which can be very significant if steps are not taken to suppress it.
- **Manufacturability.** The ability to make the selected material in a large area at low cost is a very important consideration. Pyroelectric ceramics have an important advantage here over some of the single-crystal materials that are available, as a consequence of their relative ease of manufacture using a range of techniques and their stability during other subsequent manufacturing processes.

Examples for Sensor Design and Fabrication

Many studies had demonstrated that pyroelectric anemometers based on single-crystal LiTaO_3 have a very wide range and the promise of interesting response char-

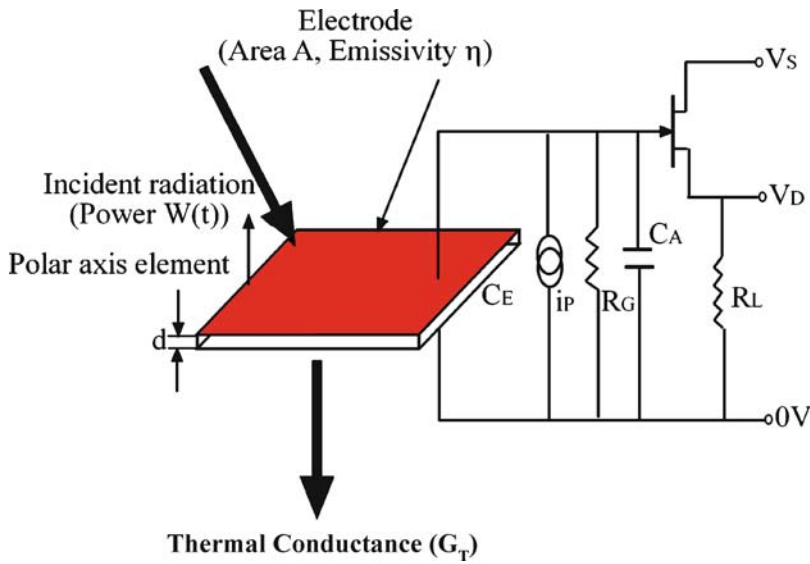
acteristics. The following example was well characterized by a conjugate gas-flow-heat-flow model.

A 200 μm thick, 30 mm diameter, Z-cut, single-crystal LiTaO_3 wafer, unpolished on both sides, was used to fabricate the pyroelectric flow sensor. The wafer was first degreased by immersing in trichloroethane, acetone and isopropyl alcohol for 10 min each and then it was rinsed in deionized (DI) water and blown dry with nitrogen. After cleaning, nichrome (NiCr) and gold films were sequentially deposited on the front surface of the wafer. This was done in a cryo-pumped electron-beam evaporator at a pressure of about 10^{-5} Pa without breaking vacuum between the NiCr and Au evaporations. The last step minimized contamination of the devices.

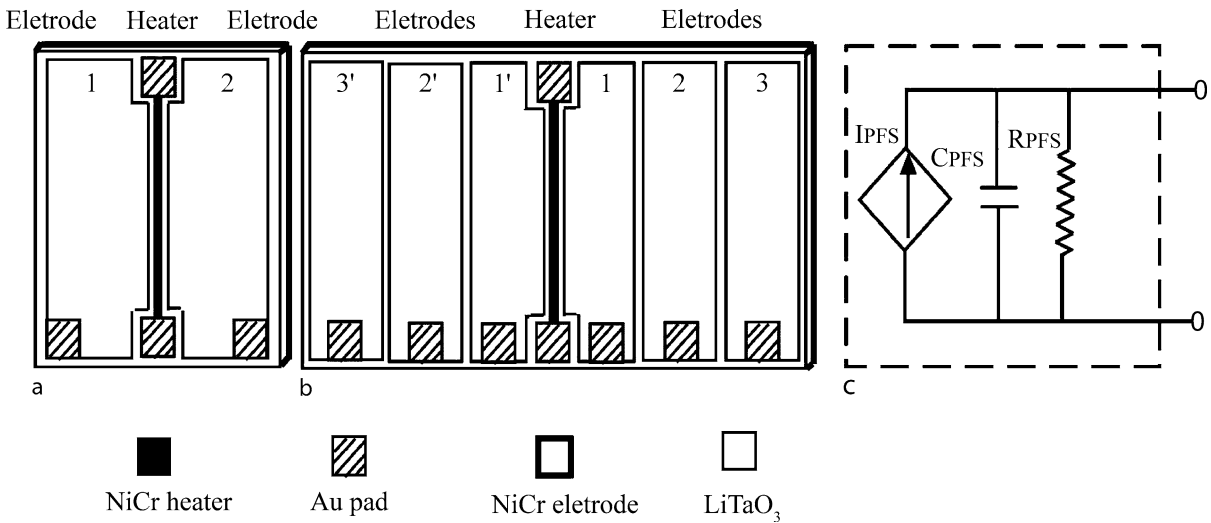
Two masks were prepared to define the nichrome and gold patterns. Using standard photolithography, the Au leads and contact patterns were formed in a photoresist layer. The residual gold was removed with a solution of 4 g KI : 1 g I_2 : 40 mL H_2O . The KI – I_2 solution was chosen because it etches the gold film but not the NiCr. The photoresist was then removed using photoresist stripper and rinsed in DI water.

After cleaning and processing, the NiCr pattern was formed in the photoresist by aligning a suitable mask to the Au pattern on the wafer. The NiCr on the wafer was etched using a solution of 310 g $(\text{NH}_4)_2\text{Ce}(\text{NO}_3)_6$: 1970 mL H_2O : 120 mL HNO_3 . With the completion of the front side patterning, a 200 nm thick NiCr film was evaporated onto the backside of the wafer. The wafer was then sliced into chips.

The two-electrode element, 3.8 mm $\times$ 3.8 mm $\times$ 0.2 mm in size, is shown in Fig. 3. The device was mounted on a ceramic carrier with one end of the heater connected to the NiCr common on the other side of the LiTaO_3 chip. The assembled unit was then inserted into a standard



Pyroelectric Flow Sensors, Figure 2 Schematic diagram of a pyroelectric detector element



Pyroelectric Flow Sensors, Figure 3 Layer of (a) a two-electrode element pyroelectric flow sensor and (b) a six-electrode one and (c) the equivalent circuit of the pyroelectric electrode

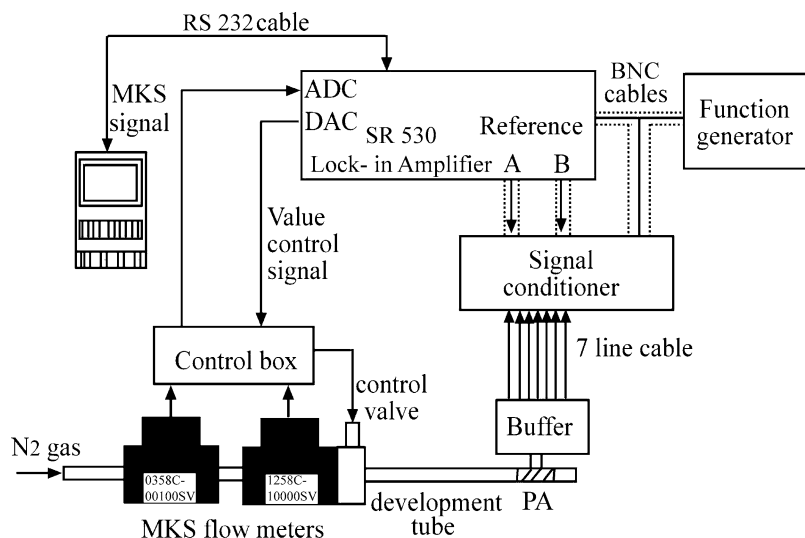
1/4' Union Tee. For completeness, the equivalent circuit of the pyroelectric flow sensors electrode is shown in Fig. 3. A schematic diagram of the measurement system is shown in Fig. 4. The basic component consists of a function generator that produces the heater current and a computer-controlled two-phase analog lock-in amplifier. Other parts of the electronic measuring system consisted of a function generator to supply a sinusoidal driving voltage to the heater of the sensor, a buffer, a signal conditioner for the output from the sensor, a two-phase lock-amplifier and a computer. The signals from the two electrodes of the sensor are sent directly to a buffer. The buffer section

transforms the high impedance of the sensor signal into a low-impedance signal, which is sent to a signal conditioner through a seven-line cable. Because of the extremely high resistance, R_{PA} , of the sensor chip, the loading by the electronics produces a voltage output signal across the input resistance of the buffer amplifier.

Key Research Findings

Modification of Pyroelectric Materials

As we know, there are many kinds of pyroelectric materials, but all of them have some advantages and disadvantages.



Pyroelectric Flow Sensors, Figure 4 Diagram of the measurement system

tages. For example, as a perovskite-type pyroelectric compound desirable for pyroelectric infrared detectors because of its relatively large pyroelectric coefficient and spontaneous polarization, PT has poor mechanical properties due to its large tetragonal strain which makes poling of PT materials difficult. Much attention has been focused on the modification of PT by doping with the purpose of obtaining improved mechanical and electrical properties. The incorporation of off-valent (La^{3+} , Sm^{3+} , Nd^{3+} , Gd^{3+}) and isovalent ions (Ca^{2+} , Ba^{2+} , Sr^{2+}) into PT ceramics is reported to enhance the pyroelectric properties. Substitution of these ions results in the reduction of lattice anisotropy leading to hard and dense ceramics with high mechanical strength. From the standpoint of thermal responsivity, the performance of a pyroelectric detector is enhanced when it is in the form of a thin film. In recent device applications, it has become increasingly necessary to consider the improvement of these films in their performance in addition to the large component area and connection to many circuit elements. Recently, calcium- and lanthanum-doped lead titanate (PCT and PLT) thin films have been studied due to their possible use in optical waveguides, infrared sensors, dynamic random access memories and non-volatile memories. Thin-film optical devices are especially attractive because of their potential for monolithic integration. In addition, thin films offer the potential for increased speeds, reduced operating voltages and enhanced efficiency.

In order to improve mechanical properties for different applications, many pyroelectric polymer composites have been prepared, such as BaTiO_3 -PVC compos-

ites, $(\text{Bi}_{0.5}\text{Na}_{0.5})_{0.94}\text{Ba}_{0.06}\text{TiO}_3/\text{P}(\text{VDF-TrFE})$, 65PMN-35PT/ $\text{P}(\text{VDF-TrFE})$ and DTGS:PVDF.

Preparation of Thin Films

The use of bulk pyroelectrics in pyroelectric devices inevitably leads to a situation where the material must be cut, lapped and polished to make a thin, thermally sensitive layer. If an array of detectors is required for a thermal goal, this must be metallized on both faces, processed photolithographically and bonded to a silicon read-out circuit to yield a complete hybrid array. Clearly, it would be desirable if the material could be deposited as a thin film, to remove the requirement for lapping and polishing, if possible directly onto a complete wafer of chips, where it could be processed to yield an array of thin, thermally isolated structures. Fortunately, many techniques have been researched for pyroelectric thin-film deposition, mainly for applications to non-volatile memories. These include chemical solution deposition (CSD), metalorganic deposition (MOD), RF magnetron sputtering, pulsed laser ablation (PLD) and metalorganic chemical vapor deposition (MOCVD).

Pyroelectric flow sensors and chips

Pyroelectric sensors have recently been increasingly employed for sensing purposes. Excellent piezoelectric and pyroelectric properties of pyroelectric materials, combined with design flexibility and miniaturization offered by thin-film geometry and also developments in micro- and nanoscale fluids technologies have generated great

interest in pyroelectric thin films. The multisensing chip utilizes the pyroelectric effect in the following sensing devices: gas-flow sensor infrared detector arrays, chemical reaction sensors and charge-coupled device imagers. A multipurpose integrated sensor chip that responds simultaneously to several different physical and chemical variables has been fabricated.

Also, multifunctional pyroelectric flow sensors have also appeared. As we know, any change in the temperature induces charges on the surface of the material, inducing an electric field, which generates transverse strain and bending deformation (see Fig. 3). A typical example is used for flow sensor and valving purposes.

Future Directions for Research

During recent years, the study of micro- and nanoscale fluids has shown significant opportunities for high detectivity of elementary devices with small size. For pyroelectric flow sensors, it is highly desirable to develop theoretical models, experimental methods for pyroelectric element preparation and sensor fabrication, and higher sensitivity with excellent mechanical properties.

Cross References

- Ferroelectric
- Pyroelectricity
- Piezoelectric Valves

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Pyroelectricity

Definition

Describes the phenomenon of materials that generate an electric field across them due to a change in their temperature. All pyroelectric materials are piezoelectric.

Pyroelectric Sensors

- Pyroelectric Flow Sensors

PZT

- Piezoelectric Materials for Microfluidics